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Criterion 4 CONTINUOUS IMPROVEMENT

Executive Summary — The UTRGV Mechanical Engineering Continuous-Improvement Story

UTRGV Mechanical Engineering's continuous-improvement program is built on infrastructure the department designed and deployed itself. Across the 2020–2026 cycle the faculty built a web-based ABET Assessment Portal (Section 4.0.4)—a single, performance-indicator-grained system of record that captures every direct-measure observation (course, term, outcome, PI, Bloom level, rubric counts, and an instructor narrative) exactly once, and regenerates every trend, panel, and triangulation in this report on demand. The portal replaced a spreadsheet-and-email workflow and, in doing so, converted continuous improvement from an activity dependent on one person's institutional memory into a durable, faculty-governed system designed to survive personnel and leadership change.

That system makes assessment rigorous in exactly the ways accreditation demands. Attainment is determined by direct, rubric-based scoring of student work *at the performance-indicator grain*—never by course grades, composite scores, or student self-perception—with indirect measures (the Capstone Exit Survey and EAC industry input) used to triangulate, not to substitute (Section 4.A.4). Every outcome is assessed *every Fall and Spring* through a common Expert/Practitioner/Apprentice/Novice rubric applied consistently across instructors, against an explicit $\geq 70\%$ Expert-plus-Practitioner threshold (Figure 2). Where courses are shared, BSME results are *disaggregated* so the program independently evaluates its own outcomes (Section 4.A.8). Seven faculty committees, the full faculty, and at-least-annual retreats convert these results into prioritized, owner-assigned Actions for Improvement (AFIs), each documented in Analysis–Intervention–Comparison format and tracked through implementation and re-assessment.

And the loop closes. The program acts on direct-measure signals through documented faculty governance. Three faculty-approved SLO Overlays—Problem Analysis, Concept Development, and Audience Adaptation—were deployed in 2025–2026 in direct response to performance-indicator signals the portal surfaced, with early re-assessment already positive (Sections 4.B.3.1–4.B.3.3), and targeted course-level interventions tracked in the portal produced measurable, sustained gains where applied (Section 4.B.3). Across the cycle, the number of below-threshold “gatekeeper” courses fell from eleven to five even as enrollment grew by more than a third (Section 4.B.4.5).

This system serves a program with a distinctive mission. The BSME program at UTRGV educates a predominantly Hispanic, largely first-generation, and overwhelmingly working-student body on the Texas–Mexico border, and it does so affordably: graduates reach first-, fifth-, and tenth-year median earnings comparable to those of established Texas peer programs while carrying a median student-loan debt ($\sim \$18.5\text{K}$) far below the national norm (Section 4.B.4.2). The assessment system documented here is how the program holds that access mission to a rigorous, evidence-based standard.

The program is equally candid about what is *not* yet finished: it reports plainly on a small number of genuinely below-threshold indicators in the SD-II capstone and a small number of in-flight interventions whose multi-term re-assessment is still accruing, precisely because the portal's data model makes such reporting compatible with continued operation (Section 4.B.3.8). This is the

story Criterion 4 tells: a Hispanic-Serving, access-driven program that built a sustainable, evidence-based assessment system, used it to find real problems, acted on them through documented faculty governance, and can show the results.

4.0 Overview and Definitions

This chapter documents how the BSME program at The University of Texas Rio Grande Valley (UTRGV) assesses and evaluates attainment of the seven Student Learning Outcomes (SLO-1 through SLO-7), and how the results are used to drive faculty-approved improvements. The work is supported by a department-built ABET Assessment Portal (Section 4.0.4) that serves as the program's *system of record* for direct measures, the Capstone Exit Survey, and continuous-improvement evidence across the 2020–2026 reporting window.

4.0.1 Purpose of Criterion 4 and Organization of This Chapter

The intent of this chapter is to demonstrate (i) a *repeatable, sustainable* assessment-and-evaluation system; (ii) *faculty-informed, evidence-based* decision-making; and (iii) *documented actions and follow-up* (“closing the loop”) that improve the program over time. The processes and examples presented span 2020–2026 and are prepared for the Fall 2026 accreditation review. Because the portal is the program's system of record and the evaluation process is distributed across seven SLO committees, the full faculty, and the Department Chair, continuous improvement at UTRGV Mechanical Engineering *does not depend on any single individual's institutional memory*: the documented process, retained records, and rubric version history are designed to survive personnel and leadership change (succession plan, Appendix).

The chapter is organized as follows:

- ◉ Section 4.A (Student Outcomes): identifies the SLOs assessed, summarizes the assessment processes, defines attainment thresholds, and presents results by outcome.
- ◉ Section 4.B (Continuous Improvement): explains how evaluation results and other available information drive improvements; documents actions taken (AFIs); reports the results of changes when re-assessed; and presents future improvement plans.
- ◉ Section 4.C (Visit-Ready Evidence): indexes instruments, rubrics, meeting minutes, example artifacts, and portal access for the visit team, and provides a PEV crosswalk to ABET Criterion 4 required elements.

All assessment artifacts, rubrics, evaluation summaries, instructor observations, and action logs referenced in this chapter are maintained in the ABET Assessment Portal and supporting evidence repositories (Section 4.0.5).

4.0.2 Definitions: Assessment vs. Evaluation

The terms *assessment* and *evaluation* are used consistent with ABET Criterion 4 and distinguished as follows:

Assessment is the systematic collection of evidence of student learning. In this program, assessment is primarily rubric-based scoring of course artifacts (exam problems, project reports, lab reports, presentations, senior design deliverables) in designated MECE courses mapped to each SLO. Indirect inputs are also collected, including the Capstone Exit Survey, academic success indicators (pass rates, retention, graduation rates, GPA trends), socio-demographic statistics, alumni outcomes, and Engineering Advisory Council (EAC) input.

Evaluation is the interpretation of assessment results for decision-making and improvement. Evaluation is conducted by SLO-specific faculty committees led by SLO chairs and reviewed with the full Mechanical Engineering faculty in department meetings and retreats (semester and annual), with participation by the Department Chair. Evaluation outputs are (i) attainment determinations at the rubric, PI, and outcome levels; (ii) analysis of trends across semesters; and (iii) documented Actions for Improvement (AFIs) with follow-up plans.

Operational definitions used throughout the chapter:

- Student Learning Outcome (SLO): what graduates are expected to know and be able to do upon completion of the BSME program; seven SLOs aligned with ABET Student Outcomes 1–7.
- Performance Indicator (PI): a measurable indicator aligned to an SLO, used to operationalize assessment through specific, observable student performance.
- Rubric scale: Expert (E), Practitioner (P), Apprentice (A), Novice (N).
- Direct measure: student artifacts scored using PI-aligned rubrics.
- Indirect measure: evidence from surveys, institutional metrics, and stakeholder inputs.
- Attainment metric (EP%): percent of students achieving Expert or Practitioner,

$$EP\% = \frac{E + P}{E + P + A + N} \times 100.$$

- Attainment thresholds: $EP\% \geq 70\%$ meets the PI; 67–70% is borderline; $< 67\%$ is not met. An SLO is met when all its PIs are met.
- Action for Improvement (AFI): a documented change to courses, curriculum, pedagogy, or policy in response to evaluation findings—triggered when targets are not met, or used proactively when faculty identify improvement opportunities despite targets being met. AFIs are documented with analysis, implementation, and a re-assessment plan.

When assessment evidence involves shared courses (e.g., ethics or foundational math/science), the program relies on MECE artifacts for SLO attainment where possible, and disaggregates non-MECE data for BSME students before SLO-committee review (Section 4.A.8).

4.0.3 Assessment Governance, Roles, and Term/Annual Cycle

Continuous improvement in the BSME program is governed through a faculty-led structure built around outcome-level ownership, shared faculty review, and documented decision-making. The ABET Assessment Portal (Section 4.0.4) operationalizes this governance through role-based data entry, a uniform analysis layer for SLO-committee work, and the documentation trail required for faculty review.

4.0.3.1 Governance Structure and Decision Authority

The program's continuous improvement governance is anchored in seven SLO Faculty Committees (one committee per SLO). Each committee is responsible for leading the assessment and evaluation activities for its assigned outcome, including developing and refining performance indicators, rubrics, and artifact expectations; analyzing results; and proposing interventions (AFIs). Outcome-level findings and proposed actions are elevated to the department faculty as a whole through regular faculty meetings and scheduled retreats, which serve as the primary forum for deliberation, prioritization, and approval of improvement actions. The Department Chair participates in these discussions, supports implementation through resource allocation and administrative coordination, and ensures the process remains active and sustainable.

Decision authority is distributed to match the level of change required:

- Course-level changes (content, pedagogy, and internal assessments within a course) are made by the instructor of record, guided by SLO committees to ensure alignment between course artifacts and the SLO rubrics used for assessment.
- Multi-section coordination for key gateway and multi-section courses is managed by designated Course Coordinators, who promote consistency across sections (common artifacts, instruments, rubrics, and expectations).
- Rubric/PI changes are developed by SLO committees, typically piloted and refined through course implementation, and then reviewed and approved by the full faculty with documentation in meeting/retreat minutes.
- Curriculum changes (degree plan modifications, prerequisites, new courses) and program policy changes (e.g., GPA requirements, course repeat rules, progression policies) are proposed and discussed by faculty, approved by the full ME faculty, and advanced through college/university governance as required.

External input informs the governance process through the Engineering Advisory Council (EAC), which reviews Program Educational Objectives (PEOs) and provides industry perspective on curriculum and graduate competencies. EAC input is presented to the faculty for consideration and, where appropriate, translated into curricular or policy actions (with the constituent process described in Criterion 2).

4.0.3.2 Roles and Responsibilities

The continuous improvement system is implemented through defined roles with outcome-level

accountability:

- SLO Chairs/Coordinators (one per SLO): Lead their SLO committee; coordinate artifact selection and rubric development; provide scoring guidance; compile semester results across courses via the portal; analyze trends using the portal's embedded analysis engines; identify weaknesses; propose AFIs; and present results and recommendations to the full faculty. Deliverables include semester SLO summary tables (attainment distributions and percentages) and documented analysis–intervention–comparison entries.
- SLO Committee Members: Participate in rubric development and review of results; support intervention design; and implement approved changes in their courses as applicable.
- Course Coordinators (for key multi-section courses): Ensure consistent learning experiences and assessment instruments across sections; coordinate common exams/projects; standardize rubric application; aggregate section results; and hold pre-semester coordination meetings with section instructors.
- Instructors of Record: Implement designated SLO-aligned artifacts; score student work using the approved rubrics (often via embedded learning-management-system rubrics); compile E/P/A/N performance distributions for the assigned rubric criteria; enter results in the ABET portal with required narrative justification; retain artifact exemplars; and provide assessment data to the SLO chair/committee.
- Department Chair: Convenes faculty meetings and retreats; sets agendas that include SLO review; supports compliance with assessment cycles; and allocates resources needed to implement approved improvement actions.
- Undergraduate Program Coordinator and Associate Chair (student success functions): Support indirect measures and related continuous improvement inputs, including Capstone Exit Survey administration and analysis, monitoring of success indicators (enrollment/retention/graduation/pass rates), and advising-related interventions.
- ABET / Assessment Coordinator: Maintains the assessment portal and evidence repository; synthesizes results for reporting; supports SLO chairs with cross-cutting analyses; and ensures visit-readiness of the documentation set.

4.0.3.3 Term Cycle Overview (Semester Workflow)

Assessment is conducted each semester in designated courses mapped to SLOs. Artifacts and rubrics are selected and maintained through SLO committee planning and faculty retreats, and are reflected in course syllabi at the start of each term. In many courses, artifacts are predetermined from prior cycles to support consistency and longitudinal comparisons.

Scoring and data capture occur throughout the semester when embedded rubrics are used (e.g., within the learning-management system) and are completed at the end of the term for final examinations, projects, and presentations. After final grades are posted, instructors enter E/P/A/N distributions, associated attainment calculations, and the required narrative justification fields directly into the ABET portal. The portal computes EP% automatically, flags below-threshold entries, and makes the resulting records immediately available to the corresponding SLO chair/committee.

SLO chairs aggregate results across courses and sections, generate course-level and SLO-level analyses through the portal's embedded analysis pages, and flag items below the attainment threshold for further analysis.

The semester workflow culminates in SLO committee analysis and preparation for faculty review. While specific calendar deadlines may vary by term, the operational pattern is that results from one term are compiled shortly after term end and are reviewed in the subsequent faculty meeting and/or retreat cycle.

Summer offerings are included within the assessment system. When summer sections are used to address bottlenecks or improve pass rates, their assessment results are collected and either reported separately or aggregated with adjacent semesters for annual synthesis, with appropriate context provided for differences in enrollment patterns and sample sizes.

4.0.3.4 Annual Cycle Overview (Program-Level Review and Action)

At least annually, each SLO committee conducts an outcome-level evaluation based on aggregated semester results pulled from the portal. Committees examine attainment trends (with significance testing performed automatically by the portal), identify recurring weaknesses, and develop proposed interventions with a plan for follow-up comparison. These findings are presented to the full faculty in scheduled meetings and retreats.

Faculty retreats (typically held at least once per semester) provide the structured forum for comprehensive review and decision-making. Retreat activities include:

- presentation of SLO attainment results and trends by SLO chairs (using portal-generated figures),
- review of student success indicators and other indirect measures (including Capstone Exit Survey analytics),
- a program-level SWOT analysis (conducted every two years) to identify program-level priorities,
- discussion, prioritization, and approval of AFIs and improvement initiatives, and
- assignment of responsibility and timelines for approved actions.

The program conducts its program-level SWOT analysis on a biennial (every-two-year) cadence; the remaining retreat activities above occur each cycle. The most recent SWOT was completed at the September 2024 retreat; no SWOT was conducted in September 2025; and the next is scheduled for the September 2026 retreat.

Annual deliverables include multi-year SLO assessment summary tables, documented analysis–intervention–comparison tracking for AFIs, updated SWOT tables (in the years a SWOT is conducted), updated assessment instruments (as needed), and meeting/retreat minutes documenting decisions and approvals.

4.0.4 The BSME ABET Assessment Portal: Continuous Improvement Infrastructure

During the current cycle, the BSME program developed and deployed a department-built, web-based ABET Assessment Portal that serves as the operational backbone of the continuous improvement system. The portal unifies (i) role-based data entry for direct measures; (ii) automated EP% computation and threshold flagging; (iii) course-level and SLO-level analysis engines; (iv) the Capstone Exit Survey module; and (v) administrative export and visit-readiness reports. Most figures, tables, and summaries in Section 4.A are generated by it.

UTRGV Department of Mechanical Engineering													
MECE ABET Portal													
Course Number	Course Name	Student Learning Outcome (SLO)	Performance Indicator (PI)	Assessment Tool Used	Bloom's Level	Explanation of Why Tool Appropriately Assesses SLO and PI at the Selected Bloom's Level	Assessment Semester	Enter EPAN Data (in %)				Observations on Student Performance (struggles, aha moments, etc. and plans for improvement, closing the loop)	Delete
								E	P	A	N		
MECE		SLO4	PI-1: Recogniz		Understa		Fall 2022						

Figure 1: BSME ABET Assessment Portal: data-entry interface (one row per course–SLO–PI–semester record, with E/P/A/N counts and required narrative fields).

The detailed treatment of the ABET Assessment Portal—design principles, architecture and access, role-based data-entry workflow, the embedded course- and SLO-level analysis engines, the Capstone Exit Survey module, administrative/export capabilities, the portal–4.A mapping, and the sustainability and succession plan—is provided in the Appendix (former Sections 4.0.4.1 through 4.0.4.9) to keep the main response concise. The portal is the program's system of record for all direct measures, the Capstone Exit Survey, and continuous-improvement evidence; subsequent

references to Sections 4.0.4.1–4.0.4.9 point to that appendix material.

4.0.5 Evidence System and Record Retention

The BSME program maintains a structured evidence system that supports an audit trail from assessment results to faculty evaluation, approved actions, implementation, and re-assessment. Evidence is held in the ABET Assessment Portal (system of record), curated outcome and course binders, the Department Records Archive, and Brightspace course records. Records are retained across the 2020–2026 reporting window with historical results and rubric versions preserved. On-site evaluators are provided read-only access to curated binders and documentation; the portal's export views produce CSV/HTML snapshots of any data subset on demand.

The evidence map documents *where* records are retained; Figure 61 summarizes *how*: the standardized pathway from collection (artifacts and rubric scoring) through evaluation (SLO committee, faculty retreat, AFI approval) to re-assessment (closed-loop comparison and outcome). This pathway is identical across all seven SLOs and across all assessing terms in the 2020–2026 cycle, providing the procedural consistency required for the longitudinal trend interpretation in Section 4.A.6.

4.0.6 Evolution of the Assessment System (2020–2026)

ABET Criterion 4 expects that a program's assessment processes themselves improve over time. This section documents how the BSME assessment system evolved across the 2020–2026 review window. The headline is deliberately understated: the program's *measurement framework*—seven SLOs, the four-level E/P/A/N rubric, the $EP\% \geq 70\%$ threshold, the Capstone Exit Survey, and the SLO-committee governance model—was already in place at the start of the cycle and has been deliberately held stable to preserve longitudinal comparability. Two substantive changes occurred: (i) a terminology realignment adopting ABET-current “Performance Indicator (PI)” language in place of “Performance Criterion (PC),” and (ii) design and deployment of the ABET Assessment Portal in Fall 2024, replacing the prior spreadsheet-and-email workflow. Both changes were non-disruptive to the underlying framework, so attainment data before and after the changes remain directly comparable.

Detail on what was already in place at the start of the cycle, the PC→PI terminology realignment, and the Fall 2024 migration from spreadsheets to the ABET Portal is provided in the Appendix (former Sections 4.0.6.1 through 4.0.6.3). The companion curriculum/policy changes, stakeholder-engagement milestones, timeline, and implications summary remain below.

4.0.6.4 Companion Curriculum and Policy Changes Within the Cycle

Although Criterion 4 concerns the assessment-and-evaluation system, several curriculum and policy changes implemented during the cycle shaped what is being assessed. These are documented in detail in Criterion 5 (Curriculum) and Criterion 2 (PEOs); they are summarized here only for assessment-system timeline context:

- Mathematics/analysis sequence revision (2022–2024 catalog): MECE 2350 (Numerical Methods) and MECE 3449 (ME Analysis I) were replaced with MECE 3440 (ME Analysis) and MATH 2415 (Calculus III), bringing the program into compliance with the Texas Mechanical Engineering Plan of Study Calculus III requirement.
- Programming course addition (2022–2024 catalog): CSCI 1380 (Introduction to Programming) was added to the degree plan as part of the general-education core supporting the revised math/analysis sequence.
- Calculus I attempt-limit policy (effective in cycle): The maximum allowed attempts at Calculus I as a BSME entrance requirement was reduced from three to two, presented to and reviewed by the EAC in December 2024.
- Technical-elective expansion: New technical-elective course requests for MECE 4306 (CFD), MECE 4334 (Advanced Alloys), and MECE 4335 (Smart Materials) were initiated within the cycle and reviewed with the EAC.
- PEO review (2023–2025 cycle): A comprehensive three-year PEO review was completed, with revisions adopted in 2025 addressing post-graduation framing and incorporating sustainability, leadership, and cultural competence (Criterion 2). SLO wording was reviewed in parallel and retained.

These curriculum and policy actions feed back into the assessment system through the per-SLO process maps (Section 4.A.2) and the AFI catalog (Section 4.B.2).

4.0.6.5 Stakeholder Engagement Milestones During the Cycle

External stakeholder engagement during the cycle (full detail in Criterion 2):

- EAC meeting, September 2, 2022—PEOs and SLOs reviewed and approved; written feedback from individual members (e.g., NASA Glenn Research Center) emphasizing professional-readiness skill development.
- EAC meeting, December 4, 2024—Revised entrance requirement, new technical electives, and student-success initiatives reviewed; PEO revisions previewed.
- EAC meeting, March 6, 2026—Scheduled continuation of the periodic-review cycle.

These engagements are inputs to evaluation (Section 4.B.4.1) and—with the Capstone Exit Survey and direct-measure data—support the triangulation discussed in Section 4.A.2.

4.0.6.6 Timeline Summary

Figure 62 provides a swimlane visualization of the 2020–2026 continuous improvement activity, organized into three lanes (Assessment Infrastructure; Curriculum and Policy; SLO Interventions), with EAC engagements at top and the Fall 2026 PEV visit as the dashed red line. The detail-level Table 33 that follows reproduces individual entries with carry-forward status.

Table 33 consolidates the assessment-system evolution into a single timeline.

4.0.6.7 Summary of Implications for This Self-Study

The implications of the evolution above for how PEVs should read the remainder of Chapter 4 are:

- Comparability is preserved. Because the SLO framework, rubric scale, threshold, and exit-survey instrument were stable across the entire cycle, the multi-year attainment results presented in Section 4.A.6 can be interpreted as genuine longitudinal trends rather than artefacts of instrument change.
- PC and PI are the same thing. Wherever prior-cycle artifacts (binders, minutes, archived reports) use “PC,” the current-cycle equivalent is the like-numbered “PI” for the same SLO. No re-mapping is required.
- The portal is the system of record from Fall 2024 forward. Records prior to Fall 2024 are retained in their original spreadsheet form in the SLO binders and department archive; aggregated pre-portal results have been entered into the portal where needed to support longitudinal reporting.
- The assessment-system changes were deliberately conservative. The two cycle-level changes (PC→PI relabeling and portal deployment) were chosen specifically because they improve documentation quality and ABET alignment *without* disturbing the underlying measurement framework. PEVs reviewing trend data across 2020–2026 are reviewing data from a stable measurement system.

4.A Student Outcomes (Assessment and Evaluation)

Section 4.A documents the BSME program’s assessment of its Student Learning Outcomes (SLOs), addressing ABET Criterion 4 requirements for: (i) identification of the SLOs assessed; (ii) the assessment processes used (Performance Indicators, direct measures, thresholds, frequency, evidence location); (iii) results by outcome; (iv) documentation and maintenance of results; and (v) handling of independent assessment for shared courses. Data are drawn from the BSME ABET Assessment Portal (Section 4.0.4), the program’s system of record for the 2020–2026 review cycle. Continuous improvement actions taken in response are documented in Section 4.B.

4.A.1 Student Outcomes Assessed

4.A.1.1 List of Student Outcomes Assessed by the Program

The BSME program assesses seven Student Learning Outcomes (SLOs), aligned one-to-one with ABET EAC Criterion 3 Student Outcomes 1–7. The SLOs are reviewed every three years by the Mechanical Engineering faculty as part of the recurring assessment-and-evaluation cycle. Figure 3 provides the SLOs at-a-glance for rapid reference; the full SLO/PI framework is detailed in Criterion 3 (Section 3.B).

4.A.2 SLO Assessment Processes

Tables 25 through 31 provide an *assessment process map* for each SLO: Performance Indicators (PIs), intended Bloom's level per PI, direct-measure courses and artifact types, attainment threshold, frequency, and evidence location. Assessing courses and artifacts are deliberately selected for their *content relevance* to each PI rather than collected broadly; the “lowest observed signal” reported in the PI inventory is a monitoring flag that directs committee attention, not the sole determinant of attainment, which is evaluated at the PI grain across all of a PI's assessing courses.

Figure 2 summarizes how often each SLO is assessed across the review cycle.

	Fa 2020	Sp 2021	Fa 2021	Sp 2022	Fa 2022	Sp 2023	Fa 2023	Sp 2024	Fa 2024	Sp 2025	Fa 2025	Sp 2026	# DM courses
SLO-1 Problem Solving	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	7
SLO-2 Engineering Design	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
SLO-3 Communication	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	5
SLO-4 Ethics	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
SLO-5 Teamwork	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	3
SLO-6 Experimentation & Data	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
SLO-7 Lifelong Learning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	2

Direct-measure assessment (✓) is conducted every Fall and Spring (Summer when offered) through embedded course artifacts scored on the E/P/A/N rubric.
 Each SLO committee performs an outcome-level evaluation at least annually; faculty retreats (at least once per semester) review results and set actions.
 # DM courses = number of distinct direct-measure (assessing) courses mapped to each SLO. Per-SLO detail appears in the assessment process maps (Appendix).

Figure 2: BSME SLO assessment frequency across the 2020–2026 review cycle. Each of the seven SLOs is assessed every Fall and Spring (Summer when offered) through embedded direct-measure course artifacts scored on the E/P/A/N rubric; the right-hand column reports the number of distinct direct-measure (assessing) courses mapped to each SLO. In addition, each SLO committee performs an outcome-level evaluation at least annually, and faculty retreats (held at least once per semester) review results and set actions. Per-SLO Performance-Indicator, Bloom-level, artifact, threshold, and frequency detail appears in the assessment process maps (Appendix).

Each table is organized by PI (one row per PI). Bloom's level reflects the program's intended cognitive level for that PI and is recorded per artifact in the portal. Direct measures are course-embedded artifacts (exam problems, lab reports, design reports, presentations, capstone deliverables) evaluated using the four-level Expert/Practitioner/Apprentice/Novice rubric. Attainment uses the EP% metric (Section 4.0.2),

$$EP\% = \frac{E + P}{E + P + A + N} \times 100,$$

with $EP\% \geq 70\%$ defining “meets target.” Evidence is maintained in the ABET Portal (system of record); representative graded exemplars and rubrics are retained in the SLO binders; underlying course artifacts are maintained in Brightspace.

Triangulation framework: direct, indirect, and additional measures. The BSME assessment-and-evaluation system uses three complementary categories of evidence to produce defensible SLO-attainment determinations. Each category is documented separately to enable PEV review of

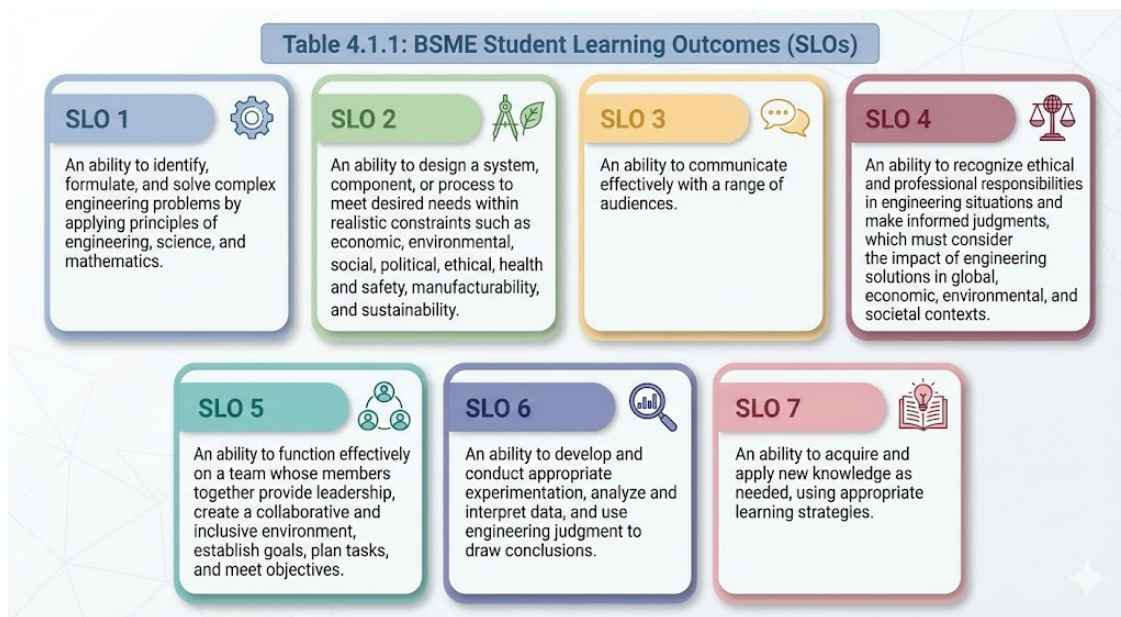


Figure 3: BSME Student Learning Outcomes (SLO-1 through SLO-7). Each SLO is mapped one-to-one with an ABET EAC Student Outcome (1–7), with minor wording adjustments in SLO-2 (design) and SLO-5 (teamwork) reflecting program-specific emphasis.

each evidence stream on its own merits, with triangulation occurring at the SLO-committee and faculty-retreat level (Section 4.0.3.4).

- Direct measures. Rubric-based scoring of student artifacts (the EP% data documented in Tables 3 through 9 above, with attainment results in Section 4.A.6). Direct measures are the primary attainment determinant.
- Indirect measures. Stakeholder perception, institutional metrics, and external feedback channels documented in Section 4.B.4: the Capstone Exit Survey (Section 4.B.4.2, $N = 504$ responses across F 2020–Sp 2025); EAC industry input (Section 4.B.4.1); BSME course pass rates as leading indicators for SLO-1 and SLO-2 attainment (Section 4.B.4.5); historical pretest data (Section 4.B.4.6, discontinued during the current cycle and replaced by portal-based PI-level monitoring); transfer student data and advising results linked to SLO-7 (Section 4.B.4.7); and BSME retention, graduation, and GPA trends as overall program-quality indicators (Section 4.B.4.9).
- Additional measures. Program-specific contextual evidence: socio-demographic statistics characterizing student-body composition linked to SLO-3 and SLO-5 (Section 4.B.4.8); benchmarking against peer-program and national mechanical-engineering data (Section 4.B.4.4); and program-policy enforcement records (POL-01 GPA Rule and POL-02 Two-Peat Rule, Section 4.B.2.1) that operationalize faculty oversight of student progression.

Triangulation in practice. The three measure categories converge in two operational ways. *First*, at the SLO-committee level, each committee reviews direct-measure portal data alongside the relevant indirect and additional measure streams for its outcome (e.g., SLO-1 committee triangulates direct PI-1/2/3 attainment with pass-rate data in Section 4.B.4.5 and historical pretest patterns in Section 4.B.4.6). *Second*, at the program level, the annual faculty retreat (Section 4.0.3.4) integrates evidence across all three categories to inform AFI prioritization (Section 4.B.1.2), including the

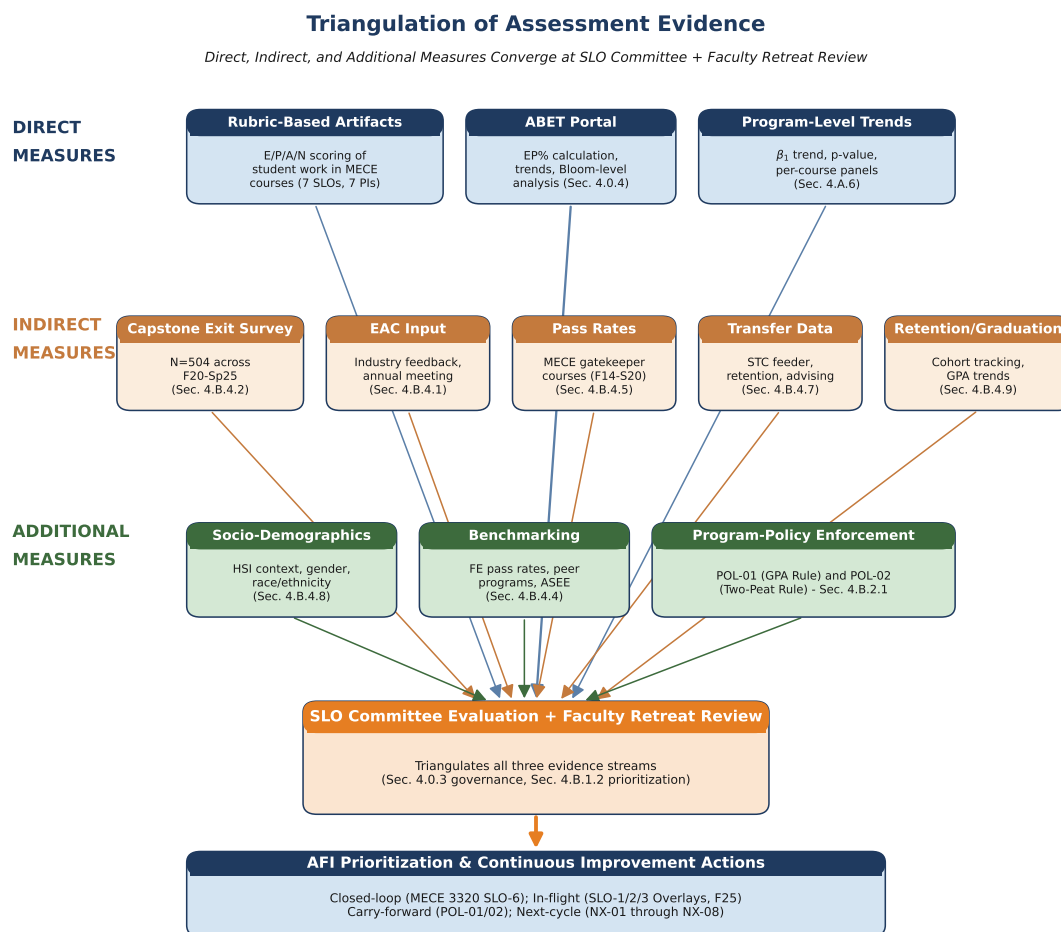


Figure: BSME assessment evidence triangulation. Three independent evidence streams (Direct, Indirect, Additional) converge at SLO committee and faculty retreat for systematic review and AFI prioritization.

Figure 4: BSME assessment-evidence triangulation. Three independent evidence streams (Direct, Indirect, Additional) converge at SLO committee and faculty retreat for systematic review and AFI prioritization. Each stream is documented separately in Sections 4.A and 4.B so that PEVs can evaluate each on its own merits.

recent prior-cycle SWOT exercises and the EAC subcommittee feedback (Section 4.B.4.1). A documented closed-loop case (MECE 3320 SLO-6, Section 4.B.3.6) illustrates this triangulation in practice: direct-measure improvement at the course level ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$), corroborating indirect-measure pass-rate stability, and absence of negative additional-measure signals together support the conclusion that the course-level intervention is genuinely effective.

Direct measures determine attainment; indirect measures triangulate. To preclude ambiguity, the BSME program documents the following operational rule: *attainment of each SLO is determined by direct PI-grained rubric-based scoring of student artifacts, not by student self-rating or stakeholder perception*. Attainment is computed at the PI grain and is *never* derived from course grades or composite scores (for example, an overall lab-report or project grade) that conflate multiple outcomes, because such averaging can mask whether students perform at the desired level on a specific performance element. The $EP\% \geq 70\%$ threshold applies to direct-measure data only. The Capstone Exit Survey (Section 4.B.4.2) and EAC industry input (Section 4.B.4.1) are indirect measures used for triangulation: they corroborate or call into question the direct-measure picture

and inform AFI prioritization, but never substitute for direct-measure attainment evidence. The rule is enforced operationally: direct-measure records and Capstone Exit Survey records are stored in separate tables (`abet_data.db` versus `graduate_survey.db`), and the SLO-level analysis engine (Section 4.0.4.5) computes EP% only against the direct-measure table.

4.A.6 Results Summary by Student Outcome

4.A.6.1 Outcome-Level Interpretation of Assessment Results

This subsection summarizes direct-measure assessment results by SLO using the reconciled portal dataset for the 2020–2026 review window. For each SLO, the portal generates a standardized four-panel figure (Panel A: course-level average attainment; Panel B: Bloom-level distribution; Panel C: semester-by-semester program-level trend with OLS fit; Panel D: PI-wise mixed-effects trends), reproduced as Figures 51–57. Program-level trend statistics ($\hat{\beta}_1$ in percentage points per term, with p -value) are reported in each SLO summary; course-level and per-course-per-SLO panel evidence is developed in Section 4.B.3.

Dataset note. The MECE 4362 (SD-II capstone) records report stable, consistent term-over-term attainment across Spring 2023–Fall 2025 (the same E/P counts each term for a given SLO/PI). Because MECE 4362 therefore contributes no temporal variation, program-level *trend* slopes below are computed across the courses that have multi-term data; MECE 4362’s stable values are reported as recorded. Several SD-II performance indicators sit genuinely below threshold (most prominently SLO-7 PI-2); these are reviewed at the course level under next-cycle item NX-01 (Section 4.B.5).

Program-wide attainment summary. Figure 50 summarizes program-level EP%, trend direction ($\hat{\beta}_1$), and status for each of the seven SLOs. PEVs are encouraged to scan this figure first before reading the per-SLO detail below.

PI-level attainment inventory. The SLO-level summary above can mask within-SLO performance variation across performance indicators. Figure 5 visualizes the complete PI-level inventory across all seven SLOs (25 PIs total) as one panel per SLO; Table 24 reports the same data in tabular form together with the lowest observed course/term signal, the program’s current disposition, and the documented action for any PI not at stable above-threshold status. Both views operationalize the principle stated in Section 4.A.2 that *direct measures determine attainment* at the PI grain—no SLO-level average is reported in this chapter without the underlying PI-level evidence available for inspection.

Tabular companion. A full PI-level attainment inventory—reporting all 25 PIs with aggregated EP%, the lowest observed course/term signal, current status, and the specific AFI or next-cycle action operating on each PI—is provided in the Appendix (Table 24), together with the accompanying *How to read this table* and *What this table makes visible* notes. The PI-level visualization above (Figure 5) summarizes the same data graphically.

SLO-1 (Solve complex engineering problems). Figure 51 shows the portal-generated SLO-1 summary across seven assessing courses. Course-level attainment shows variable performance in

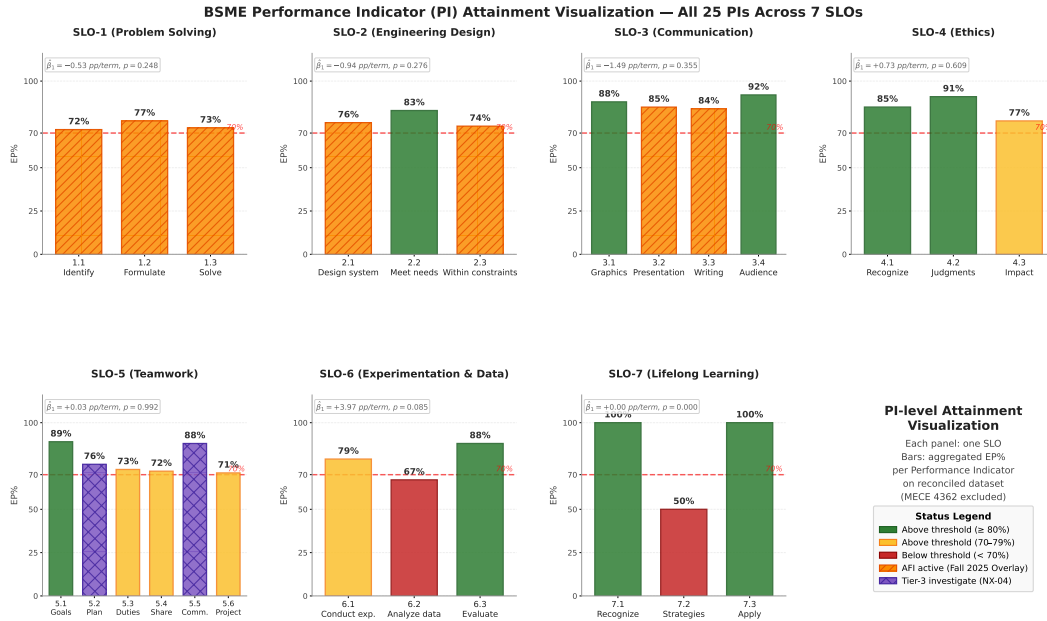


Figure 5: BSME PI-level attainment visualization. Each panel reports one SLO with one bar per Performance Indicator showing the aggregated EP% across the assessing courses (program-level trend slopes are computed across the courses with multi-term data; MECE 4362's stable SD-II values are included as recorded, including the SLO-7 PI-2 Apply=0% value). Bars are color-coded: green for $\geq 80\%$ (stable above threshold), gold for 70–79% (above threshold, monitor), red for $< 70\%$ (below threshold), orange-hatched for PIs targeted by active Fall 2025 Overlay AFIs, and purple cross-hatched for PIs under NX-04 Tier-3 structured investigation. The 70% ABET threshold is marked on each panel. Program-level trend statistics ($\hat{\beta}_1$, p -value) appear in the upper-left of each panel. The visualization makes the SLO-1/SLO-2/SLO-3 AFI activity (orange hatches), the SLO-5 PI-2/PI-5 NX-04 monitoring (purple cross-hatches), the SLO-6 PI-2 below-threshold red bar, and the SLO-7 PI-2 program-level 50% red bar (the MECE 4362 SD-II PI-2 Apply=0% value averaged with MECE 4361 PI-2=100%) all immediately readable.

lower/mid-level analytical courses: MECE 2302 (cycle average $\sim 61\%$ across four Spring terms with a declining trajectory ending Sp 2025 at 39%; statistically significant negative course-level trend $\hat{\beta}_1 = -11.03$ pp/term, $p = 0.025$); MECE 2340 with rising trajectory across three Spring terms (S 2023 46%, S 2024 58%, S 2025 89%); MECE 3315 with term-to-term volatility (S 2023 77%, F 2023 75%, S 2024 28%, F 2024 78%, S 2025 90%). Upper-division courses (MECE 3360 stable $\sim 79\%$, MECE 3450 $\sim 87\%$, MECE 4350 $\sim 80\%$, MECE 3380 SLO-1 $\sim 72\%$) remain above threshold. Bloom distribution shows “Understand” with the widest spread; “Apply/Analyze/Evaluate” cluster higher. The program-level trend is slightly negative but not statistically significant ($\hat{\beta}_1 = -0.53$ pp/term, $p = 0.248$). PI-wise trends indicate PI-1 (*Identify*) is improving while PI-2 (*Formulate*) and PI-3 (*Solve*) drift downward—consistent with students finding *identify* easier to sustain than *formulate/solve* as cognitive load increases. Interpretation: The MECE 2302 declining trend is the dominant SLO-1 signal and motivated the Fall 2025 Problem Analysis Overlay AFI (Section 4.B.2.2.1).

SLO-2 (Apply engineering design). Figure 52 shows the portal's SLO-2 summary. Course-level attainment shows MECE 4361 (SD-I) cycle average $\sim 82\%$ with a Sp 2024 dip to 65% and Sp 2025 informal recovery to 88% (F 2025 80% under the Concept Development Overlay deployment); MECE 3380 stable $\sim 77\%$ (with narrative-completeness note in Section 4.B.3.8); MECE 1221 single F 2024 data point at 67%; MECE 4362 (SD-II) reports $\sim 21\%$ (stable across the recorded terms). The program-level trend across the multi-term assessing courses is slightly negative but not statistically significant ($\hat{\beta}_1 = -0.94$ pp/term, $p = 0.276$). The SLO-2 committee judged that

the Sp 2024 MECE 4361 dip—one term below threshold on the program’s culminating design course—warranted the proactive Fall 2025 Concept Development Overlay AFI (Section 4.B.2.2.2), with Sp 2025 informal recovery to 88% providing pre-AFI evidence that the underlying signal could be addressed. The first post-deployment term (F 2025) shows MECE 4361 sustaining above threshold on all three PIs (PI-1 Create=83%, PI-2 Evaluate=73%, PI-3 Analyze=83%; course average 80%), with the PI-1 Create-level trajectory 65% Sp 2024 → 80% Sp 2025 → 83% F 2025 confirming sustained above-threshold attainment at exactly the Bloom level the Overlay targets (Section 4.B.3.2). Interpretation: SLO-2 generally meets target at the program level, with a documented capstone signal that triggered the AFI, an informal pre-AFI recovery, and F 2025 first-term post-deployment evidence sustaining above threshold at the targeted PI-1 Create level; MECE 1221 below-threshold single data point requires follow-up data collection (NX-02), and full multi-term closed-loop comparison is scheduled for the Spring 2027 faculty retreat.

SLO-3 (Communicate effectively). Figure 53 shows the portal’s SLO-3 summary. Three of four assessing courses are above threshold across the cycle (MECE 2140 ~88% averaged across F 2021–F 2023, MECE 4361 ~87% averaged across S 2023–F 2025, MECE 3170 81% on a single F 2022 data point); MECE 4362 sits at ~52% (stable across the recorded terms). The program-level trend across the multi-term assessing courses is mildly negative but not statistically significant ($\hat{\beta}_1 = -1.49$ pp/term, $p = 0.355$). The SLO-3 committee identified a notable program-level dip in Spring 2024 to 71.7% (MECE 4361 only assessing term that semester; PI-1 79%, PI-2 61%, PI-3 75%) driven primarily by PI-2 (*Demonstrates adequate presentation skills*), reinforced by a F 2025 PI-3 (*Applies technical writing skills*) dip to 60%. The committee judged that the multi-PI communication-skills signal warranted the proactive Fall 2025 Audience Adaptation Overlay AFI (Section 4.B.2.2.3), approved August/September 2025 and deployed in classroom use in Fall 2025. The first post-deployment term (F 2025) shows in MECE 4361 a strong recovery on PI-2 *Presentation skills* (61% Sp 2024 → 92% F 2025) consistent with the Overlay’s audience-adaptation focus, while PI-3 *Technical writing* remains at 60%—a mixed but interpretable first-term signal that the SLO-3 committee will address at the Spring 2026 instructor calibration meeting (Section 4.B.3.3). Interpretation: SLO-3 is generally strong, with a Sp 2024 communication-skills dip that the program acted on proactively; F 2025 post-deployment evidence shows strong PI-2 recovery and a continuing PI-3 concern, with full multi-term closed-loop comparison at the Spring 2027 faculty retreat.

SLO-4 (Ethics and professional responsibility). Figure 54 shows the portal’s SLO-4 summary. All four assessing courses are above threshold: MECE 4361 (SD-I, PI-2 only) ~95% averaged across S 2023/ S 2025/F 2025; MECE 1101 (Intro to ME, PI-1 only) ~92% averaged across F 2022/ F 2024/S 2025 /F 2025; PHIL 2393 ~81% averaged across five terms F 2023–Sp 2025; MECE 4362 ~75% (stable across the recorded terms). The program-level trend is essentially flat with a slight positive tendency ($\hat{\beta}_1 = +0.73$ pp/term, $p = 0.609$). The Capstone Exit Survey (Section 4.B.4.2) reports the *current issues and trends* dimension at 86% as the lowest of eleven program-quality dimensions; this is well above any threshold of concern but is flagged for next-cycle monitoring (Section 4.B.5, NX-08). Interpretation: SLO-4 is consistently attained with stable performance; no formal AFI was deemed necessary in the current cycle.

SLO-5 (Teamwork). Figure 55 shows the portal’s SLO-5 summary. Course-level attainment is reported through two assessing courses: MECE 4361 (SD-I capstone) cycle average ~84% across

S 2023 (92%), S 2024 (70%), S 2025 (90%), F 2025 (85%), and MECE 3320 (Measurement & Instrumentation) cycle average ~72% across S 2023 (~65%), S 2024 (~73%), and S 2025 (~77%) with statistically significant rising course-level trend ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$). MECE 4362 (SD-II) reports PI-2 (*Plan tasks and meet deadlines*) only, at ~4.5% (stable across all four recorded terms; as a flat single-PI series it contributes no temporal trend). The program-level semester trend is essentially flat ($\hat{\beta}_1 = +0.03$ pp/term, $p = 0.992$): Sp 2023 68%, Sp 2024 62%, Sp 2025 72%, F 2025 65%—the flatness reflects the combination of MECE 3320's rising trajectory, the noisier capstone term-to-term values, and MECE 4361's Sp 2024 PI-2 dip. The Bloom-level distribution rejects equality (Kruskal-Wallis $p = 0.005$, Cliff's $\Delta = -0.79$ for Analyze) with Analyze bimodal: strong MECE 3320 PI-4 values rising 66/72/77 alongside the weak MECE 4361 PI-2 Sp 2024 outlier (40%). PI-wise trends show PI-1 *Establish goals* rising in MECE 4361 (77% $\rightarrow$ 100%); PI-3, PI-4, PI-6 rising in MECE 3320; PI-2 (*Plan tasks and meet deadlines*) with the well-known MECE 4361 Sp 2024 dip to 40%; and PI-5 (*Communicates effectively in a team setting*) declining in MECE 4361 from 100% (Sp 2023) to 73% (F 2025). Interpretation: SLO-5 performance is above threshold at the program level with mixed within-cycle behavior. The MECE 3320 rising trajectory parallels the same course's SLO-6 closed-loop case (Section 4.B.3.6) and reflects the same Spring 2023 rubric tightening and progressive artifact-complexity sequence. The MECE 4361 PI-2 Sp 2024 dip and PI-5 F 2025 decline are jointly addressed under NX-04 (Section 4.B.5) as Tier-3 structured investigation.

SLO-6 (Experimentation and data interpretation). Figure 56 shows the portal's SLO-6 summary. Course-level attainment spans a wide range: MECE 4361 strong (~94% averaged across the four cycle terms); MECE 4362 cycle avg ~73% (PI-1~95.5%, PI-2~50%, stable across S 2023/S 2024/S 2025/F 2025); MECE 3320 cycle avg ~70% with a documented course-level improvement trajectory (Sp 2022 68% $\rightarrow$ Sp 2025 74%; course-level $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ statistically significant); MECE 2140 reporting a single Sp 2022 data point at 62% with a strong cognitive-level gradient (Apply 86%, Analyze 60%, Evaluate 39%; coverage-gap closure planned in NX-03). The program-level trend across all four assessing courses is positive at $\hat{\beta}_1 = +3.97$ pp/term but does not reach statistical significance at $\alpha = 0.05$ ($p = 0.085$, five program-level observations). Interpretation: SLO-6 includes a documented course-level closed-loop case (MECE 3320, Section 4.B.3.6), with the program-level trajectory providing corroborating positive evidence; the chapter reports the program-level $p = 0.085$ honestly rather than overclaiming significance.

SLO-7 (Lifelong learning). Figure 57 shows the portal's SLO-7 summary. SLO-7 is assessed against three Performance Indicators: PI-1 (*Recognize ongoing need to acquire new knowledge*, Understand level), PI-2 (*Choose appropriate learning strategies*, Evaluate level), and PI-3 (*Apply new knowledge appropriately*, Apply level). MECE 4361 reports 100% attainment on all three PIs across S 2023, S 2024, S 2025, and F 2025. MECE 4362 sits at ~67% cycle average (PI-1=100%, PI-2=0%, PI-3=100%, stable across all four cycle terms). The program-level per-term EP% is flat at 80% across the four cycle terms (Sp 2023, Sp 2024, Sp 2025, F 2025), with $\hat{\beta}_1 = +0.00$ pp/term, $p = 0.000$ (trivial p -value reflecting zero variance in the response, not significance of a non-existent trend). The PI-wise mixed-effects fits show PI-1 and PI-3 at 100% across all four terms but PI-2 at 50% across all four terms—the 50% is the MECE 4362 SD-II PI-2 Apply=0% value averaging with MECE 4361 PI-2=100%—a genuine below-threshold signal in the SD-II capstone. Interpretation: SLO-7 is strongly attained on PI-1 and PI-3 across both assessing courses and on all three PIs in

MECE 4361; the PI-2 program-level signal of 50% reflects the genuinely low SD-II (MECE 4362) PI-2 Apply attainment, reviewed at the course level under NX-01; coverage is capstone-concentrated, with NX-05 planning expanded coverage in earlier-pipeline courses (MECE 2140, MECE 3170).

4.A.6.2 Analysis Narrative: Strengths, Gaps, and Trends

The seven SLO summaries above support an overall outcome-attainment narrative that the program is meeting its student outcomes across the current cycle, with two course-level statistically significant rising trends both in MECE 3320 (SLO-6: $\hat{\beta}_1 = +1.60$, $p = 0.000$ and SLO-5: $\hat{\beta}_1 = +5.83$, $p = 0.000$); a program-level SLO-6 trend positive but not statistically significant ($\hat{\beta}_1 = +3.97$, $p = 0.085$ given five program-level observations); two stable above-threshold cases (SLO-4 and SLO-7); one above-threshold case with mixed within-cycle behavior and joint PI-level monitoring under NX-04 (SLO-5: MECE 4361 PI-2 Sp 2024 dip and PI-5 F 2025 decline); and three outcomes (SLO-1, SLO-2, SLO-3) where the program identified specific course-level or PI-level concerns sufficient to motivate proactive interventions (the three Fall 2025 SLO Overlay AFIs).

Program-wide strengths.

- Course-level statistically significant improvement on SLO-6 ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ at the course level for MECE 3320). The program-level SLO-6 trend is positive ($\hat{\beta}_1 = +3.97$ pp/term) but does not reach statistical significance at $\alpha = 0.05$ ($p = 0.085$ given five program-level observations); the course-level closed loop in MECE 3320 is the cleaner statistical case.
- Parallel statistically significant rising trend in MECE 3320 SLO-5 ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$): the same course that drove the SLO-6 closed loop also produced a rising SLO-5 trajectory across the same three Spring terms (Sp 2023 ~65%, Sp 2024 ~73%, Sp 2025 ~77%) on PI-3, PI-4, and PI-6. The two-SLO closed-loop pattern in a single course is the strongest evidence in the cycle that the program's continuous-improvement actions produce measurable downstream gains.
- Strong attainment on SLO-4 and SLO-7: SLO-4 all four assessing courses above threshold with a flat trend; SLO-7 MECE 4361 at 100% across all four cycle terms on all three PIs (PI-1, PI-2, PI-3). The SLO-7 program-level PI-2 average of 50% is entirely attributable to the MECE 4362 PI-2 Apply=0% value (genuinely low SD-II PI-2, reviewed under NX-01) rather than to MECE 4361 attainment.
- Proactive intervention culture: the program approved and deployed the three Fall 2025 SLO Overlay AFIs (SLO-1, SLO-2, SLO-3) in response to documented signals *before* those signals had pushed program-level attainment below threshold for sustained periods. The interventions addressed specific course-level or PI-level concerns rather than program-wide failure, demonstrating that the system operates on subtle signals rather than waiting for crises.

Identified gaps with active interventions.

- SLO-1 MECE 2302 declining trajectory (cycle average ~61% with Sp 2025 at 39%; statistically significant negative course-level trend), addressed by the Fall 2025 Problem Analysis Overlay AFI (program-wide). MECE 2340 and MECE 3315 each had at least one cycle term substantially below threshold before showing Sp 2025 recovery.

- SLO-2 MECE 4361 Sp 2024 capstone dip to 65% with Sp 2025 recovery to 88%, addressed by the Fall 2025 Concept Development Overlay AFI. The single MECE 1221 F 2024 data point at 67% is followed up via next-cycle item NX-02 (additional data collection before deciding whether the F 2024 result reflects a persistent pattern).
- SLO-3 communication-skills dips (MECE 4361 Sp 2024 PI-2 *Presentation skills* at 61%; F 2025 PI-3 *Technical writing* at 60%), addressed by the Fall 2025 Audience Adaptation Overlay AFI.

Identified items with planned next-cycle actions.

- SLO-5 MECE 4361 PI-2 Sp 2024 dip and PI-5 F 2025 decline—jointly addressed under NX-04 Tier-3 structured investigation to determine whether the two PI-level signals share a common cohort, grading, or rubric-application cause; MECE 3320 SLO-5 trajectory continues to be monitored to confirm the rising-trend sustainability.
- Capstone-concentrated coverage on SLO-7—NX-05 plans earlier-pipeline coverage expansion in MECE 2140 and MECE 3170 (SLO-5 coverage is no longer narrow following the discovery of MECE 3320 SLO-5 data).
- MECE 2140 SLO-6 coverage gap—NX-03 plans additional data collection in F 2026 and Sp 2027 to close the single-data-point gap.
- MECE 4362 (SD-II) below-threshold indicators—NX-01 plans a course-level SD-II assessment review of the affected PIs (most prominently SLO-7 PI-2); reported openly in Section 4.B.3.8.
- MECE 3380 narrative-completeness gap—NX-01a paired subtask covers back-fill of placeholder explanation fields across the 66 records.

Statistical context. No SLO reaches statistical significance at the program level ($\alpha = 0.05$) across the assessing courses. This is expected for trends spanning a small number of terms with the limited sample sizes typical of program-level assessment, and is interpreted by the BSME faculty in the context of (i) the magnitude of $\hat{\beta}_1$, (ii) course-level evidence with stronger statistical power for individual courses (e.g., MECE 3320 SLO-6: $\hat{\beta}_1 = +1.60$, $p = 0.000$; MECE 3320 SLO-5: $\hat{\beta}_1 = +5.83$, $p = 0.000$; MECE 2302 SLO-1: $\hat{\beta}_1 = -11.03$, $p = 0.025$), and (iii) PI-level patterns within each SLO. The BSME continuous improvement system does not require statistical significance before action; the SLO committees act on substantive signals (course-level bottlenecks, PI-level patterns, indirect-measure dimensions) regardless of program-level p -value. The SLO-6 program-level trend (positive at $p = 0.085$, just shy of conventional significance) is documented honestly as positive-but-not-significant rather than overstated.

4.A.7 Documentation and Maintenance of Results

4.A.7.1 What is Stored, Where, and By Whom

Assessment results, rubrics, artifacts, instructor observations, and SLO-committee analyses are stored across five repositories (Sections 4.0.4–4.0.5): the ABET Assessment Portal (system of record for results, AFI/Intervention logs, and trend summaries); seven SLO Binders (current and prior rubric versions, artifact descriptions, graded exemplars, AFI documentation per SLO);

Course Binders for key MECE courses (ABET syllabi, assignments, projects, samples); Brightspace (course artifacts and rubric scoring); and the Department Records Archive (faculty meeting/retreat minutes, SWOT tables, policy approvals, curriculum proposals, governance records). Ownership is distributed: SLO Chairs and Committees own their SLO Binder; Instructors of Record own Brightspace artifacts and supply E/P/A/N distributions; the ABET Coordinator maintains the portal and evidence index; the Department Chair maintains the Records Archive.

Records are retained across the 2020–2026 window with prior-cycle exemplars retained where they support longitudinal evidence. On-site evaluators receive read-only access to curated binders and portal documentation.

4.A.7.2 Version Control, Access, and Visit-Readiness

Rubrics, PI definitions, and artifact specifications are version-controlled through three mechanisms: (i) the portal retains historical results across rubric updates with each record referencing the rubric version applied; (ii) SLO Binders retain prior-version rubrics alongside current versions with transition dates indexed; (iii) the Records Archive retains the faculty meeting/retreat minutes that approve rubric changes, serving as the authoritative timestamp.

Access is role-based per Section 4.0.3.2. The ABET Coordinator maintains visit-readiness on an ongoing basis. The PEV value of this approach is longitudinal consistency: ten consecutive terms of comparable assessment data on stable instruments support the Section 4.A.6 trend analyses and the Section 4.B.3 AFI cases.

4.A.8 Independence and Disaggregation (Shared Courses)

4.A.8.1 Independent Assessment of the Program's Own Outcomes

The program ensures *program-specific validity* by relying on MECE course artifacts wherever possible. The direct-measure courses listed in Tables 25–31 are MECE-owned with the single exception of PHIL 2393 (Engineering Ethics), which is shared with other CECS programs.

For SLO-4 (Ethics and professional responsibility), the program uses three MECE-owned assessment points (MECE 1101, MECE 4361, and MECE 4362) alongside PHIL 2393, ensuring that no shared-course result alone determines SLO-4 attainment—four independent assessment points, three of them wholly MECE-owned.

4.A.8.2 Disaggregation Approach for Shared Courses

When PHIL 2393 results are used as an SLO-4 assessment point, the data is disaggregated for BSME students: the PHIL 2393 instructor of record supplies the BSME-only E/P/A/N distribution (filtered from the multi-program enrollment) to the SLO-4 committee, which then aggregates it with the MECE-owned results when computing program-level SLO-4 attainment.

The disaggregation procedure is documented in the SLO-4 Binder and reviewed annually for procedural integrity. Where disaggregation is not feasible (e.g., CECS-level institutional metrics), the program flags this explicitly and uses the MECE-only artifacts as the authoritative SLO-attainment evidence.

Summary of Section 4.A. The BSME program assesses seven Student Learning Outcomes aligned with ABET EAC Student Outcomes 1–7 using a PI-based direct-measure framework supported by the ABET Assessment Portal. Program-level attainment across the 2020–2026 cycle shows two course-level statistically significant rising trends, both in MECE 3320 (SLO-6: $\hat{\beta}_1 = +1.60$, $p = 0.000$ and SLO-5: $\hat{\beta}_1 = +5.83$, $p = 0.000$); a program-level SLO-6 trend positive but not statistically significant ($\hat{\beta}_1 = +3.97$, $p = 0.085$); two stable above-threshold cases (SLO-4 and SLO-7); one above-threshold case with mixed within-cycle behavior and joint PI-level monitoring under NX-04 (SLO-5: MECE 4361 PI-2 Sp 2024 dip and PI-5 F 2025 decline); and three outcomes (SLO-1, SLO-2, SLO-3) with identified course-level or PI-level concerns addressed through the three Fall 2025 SLO Overlay AFIs (Problem Analysis, Concept Development, Audience Adaptation) currently in early re-assessment. Genuinely below-threshold SD-II (MECE 4362) performance indicators—most prominently SLO-7 PI-2—are reviewed at the course level under next-cycle item NX-01; a documentation back-fill of MECE 3380 narrative fields is scheduled under NX-01a. Longitudinal interpretation is supported by rubric version control, role-based portal access, and disaggregation for shared-course evidence. The continuous improvement actions taken in response are documented in Section 4.B.

4.B Continuous Improvement

Section 4.B documents how the BSME program uses the evaluation results presented in Section 4.A—together with other available information—to improve the program. Consistent with ABET Criterion 4, this section addresses three required questions: (i) *how* evaluation results and other available information drive improvements (Section 4.B.1); (ii) *what* specific actions have been taken (Section 4.B.2) and what happened when those actions were re-assessed (Section 4.B.3); and (iii) *what is planned next* (Section 4.B.5). Section 4.B.4 catalogs the “other available information” the program uses—EAC input, Capstone Exit Survey results, institutional success indicators, and benchmarking. The improvement system operates on a stable measurement framework (SLOs, PIs, E/P/A/N rubric, $EP\% \geq 70\%$ threshold) carried forward without disruption across the 2020–2026 cycle (Section 4.0.6), so inputs and outputs are directly comparable across the window.

4.B.1 Using Evaluation Results for Systematic Improvement

The BSME program treats continuous improvement as a structured *evidence-to-action* workflow. Direct-measure attainment from the portal, indirect-measure inputs (principally the Capstone Exit Survey), EAC industry input, academic success indicators, and faculty observations are funneled—term by term and at least annually—into shared faculty deliberation, producing prioritized, owner-assigned AFIs documented in Analysis–Intervention–Comparison (AIC) format and tracked through implementation and re-assessment. Section 4.B.1 documents

(i) the inputs, (ii) the decision-making and prioritization process, and (iii) how the portal supports prioritization.

4.B.1.1 Inputs to Improvement

ABET Criterion 4 expects the improvement process to be informed by evaluation results and any other available information. Table 34 enumerates the BSME program's inputs, the form each is collected in, and the role each plays in evaluation.

Two characteristics of this input set are worth flagging for the PEV:

- Triangulation is built in. No single input is treated as sufficient on its own. A threshold breach in a single semester does not automatically generate an AFI; the SLO committee examines the multi-semester trend, PI-wise behavior, instructor observations, and—where available—Capstone Exit Survey signals and EAC input before recommending action. Conversely, an outcome that meets the EP% threshold can still generate a proactive AFI if indirect measures or faculty observations indicate room for improvement.
- Inputs are documented at their source. Every direct-measure data point sits in the ABET Portal alongside its narrative justification and instructor observation; faculty-meeting and retreat minutes capture deliberations and approvals; the AFI catalog (Section 4.B.2) preserves the link from input to action. The result is an auditable chain from a single PI-level result to the approved improvement action that responded to it.

The five-step decision-making, prioritization, and approval workflow (Section 4.B.1.2) and the portal's prioritization support (Section 4.B.1.3) are documented in the Appendix. In brief: the Section 4.B.1.1 inputs are converted into approved, owner-assigned AFIs through faculty deliberation following the Analysis–Intervention–Comparison (AIC) format, with decision authority following the governance distribution in Section 4.0.3.1.

4.B.2 Actions Taken (AFI Catalog and Department-Level Interventions)

This section catalogs the Actions for Improvement (AFIs) implemented during the 2020–2026 review window and documents the three department-level SLO interventions approved in August/September 2025 and reviewed at the December 2025 faculty meeting. The Overlays were sequenced by course-offering pattern: SLO-02 (Concept Development, MECE 4361 SD-I) and SLO-03 (Audience Adaptation, MECE 4361 SD-I and mid-program communication courses) were deployed in Fall 2025 with the loop closed in Fall 2025; SLO-01 (Problem Analysis, program-wide) was deployed in Spring 2026 because its primary trigger courses (MECE 2302 Dynamics, MECE 2340 Engineering Materials) are Spring-only offerings, with first program-wide post-deployment cohort accruing Fall 2026. Each AFI follows the Analysis–Intervention–Comparison (AIC) format: data signal, motivating analysis, intervention as deployed, owner and timeline, and re-assessment plan. The Fall 2026 PEV visit sees SLO-02 and SLO-03 at the *first-term closed-loop* stage and SLO-01 at the *early re-assessment*

stage (deployment plus first post-deployment Spring term in the portal). Section 4.B.3 documents the re-assessment evidence already available, most notably the MECE 3320 SLO-6 improvement trajectory achieved through normal course-level system operation.

4.B.2.1 Cycle-Level AFI Catalog (2020–2026)

Table 1 lists every documented improvement action implemented or approved by the faculty during the 2020–2026 review window, organized by category. The catalog includes assessment-system improvements (Section 4.0.6), curriculum and policy changes (Criterion 5 and Criterion 6.E.1), the three Fall 2025 SLO-level interventions documented below in Section 4.B.2.2, and stakeholder-responsive actions arising from EAC and Capstone Exit Survey input.

Table 1: BSME Action for Improvement (AFI) Catalog, 2020–2026. “Category” uses ABET-aligned categories (course, curriculum, pedagogy, student support, facilities) augmented with *assessment system* and *stakeholder-responsive* where appropriate.

AFI ID	Category	Term	Action	Owner	Status
AS-01	Assessment system	During cycle	PC→PI terminology realignment following ABET workshop (Baltimore, MD); rubric-level criterion text reviewed and standardized at PI level	ABET Coordinator; SLO Chairs	Implemented
AS-02	Assessment system	Fall 2024	Deployment of the ABET Assessment Portal as the single web-based system of record; replaces Excel/email workflow; integrates Capstone Exit Survey	ABET Coordinator	Implemented; in continuous use
AS-03	Assessment system	2023–2025	Three-year PEO/SLO review cycle completed; PEO revisions adopted; SLO wording reviewed and retained	Department Chair; Faculty	Implemented (PEO detail in Criterion 2)
CR-01	Curriculum	2022–2024 catalog	Mathematics/analysis sequence revision: MECE 2350 + MECE 3449 replaced by MECE 3440 + MATH 2415 (Calculus III) for compliance with Texas ME Plan of Study	Curriculum Committee; Faculty	Implemented
CR-02	Curriculum	2022–2024 catalog	Addition of CSCI 1380 Introduction to Programming to the degree plan supporting the revised math/analysis sequence	Curriculum Committee; Faculty	Implemented
CR-03	Curriculum	During cycle	New technical electives initiated: MECE 4306 (CFD), MECE 4334 (Advanced Alloys), MECE 4335 (Smart Materials); reviewed with EAC December 2024	Curriculum Committee	In progress (course requests advanced)
SS-01	Student support / Policy	During cycle	BSME entrance policy revised: maximum allowed Calculus I attempts reduced from 3 to 2; reviewed with EAC December 2024	Department Chair; Undergrad Coordinator	Implemented

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AFI ID	Category	Term	Action	Owner	Status
SS-02	Stakeholder-responsive	2022–2023 onward	Career-readiness content enhancement following Sept 2022 EAC member feedback (résumé writing, interview strategies, professional communication) integrated into co-curricular activities and senior design sequence	SLO-3/SLO-5 Chairs; Senior Design Coordinator	Implemented
POL-01	Program policy	Approved Fall 2015 catalog; continuous enforcement	BSME GPA Graduation Requirement. Cumulative GPA ≥ 2.5 in all major courses required for graduation; all required MECE courses must be passed with grade $\geq C$. Originally a Fall 2013 departmental standing rule (2.4 minimum); raised to 2.5 and converted to a graduation requirement in the Fall 2015 catalog. Supports SLO-1–SLO-7 indirectly through a documented academic floor.	Department Chair; Undergraduate Program Coordinator	Implemented; in continuous enforcement
POL-02	Program policy	Approved prior cycle; continuous enforcement	BSME Two-Peat Rule. Limits repeat enrollments in gatekeeper MECE courses (2340 Materials, 2335 Thermo I, 2301 Statics / current-cycle equivalent, 2302 Dynamics, 3321 Mechanics of Solids, and the math-sequence courses). First repeat: 24-hour registration hold until first-time students enroll. Second repeat (third attempt): faculty appeals petition required; no student guaranteed a third attempt. Third attempt without grade $\geq C$: enrollment not permitted. Triangulates with pass-rate data (Section 4.B.4.5).	Department Chair; Undergraduate Program Coordinator; Faculty Appeals Committee	Implemented; in continuous enforcement
SLO-01	Course / Pedagogy	Approved Aug/Sept 2025; deployed Spring 2026	Problem Analysis Overlay (SLO-1): program-wide overlay rubric applied across all SLO-1 artifact courses; uses the <i>Given–Find–Assumptions–Approach–Units</i> framework to assess identification, formulation, and solution at PI-1/2/3; differentiated sophomore vs senior expectations	SLO-1 Committee Chair; instructors of record in SLO-1 courses	In early re-assessment; full closed-loop comparison by Fall 2026 (first program-wide post-deployment cohort) and Spring 2027
SLO-02	Course / Pedagogy	Approved Aug/Sept 2025; deployed Fall 2025	Concept Development Overlay (SLO-2): capstone (MECE 4361) intervention requiring ≥ 6 design concepts, ≥ 1 structured creativity method (morphological chart, 6-3-5, SCAMPER), feasibility analysis, and explicit linkage to requirements/constraints; embedded in SD-1 Mid-Term Poster grading	SLO-2 Committee Chair; MECE 4361 Instructor	In early re-assessment; full closed-loop comparison AY 2026–27

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AFI ID	Category	Term	Action	Owner	Status
SLO-03	Course / Pedagogy	Approved Aug/Sept 2025; deployed Fall 2025	Audience Adaptation Overlay (SLO-3): mid-program lab and project courses require two-audience deliverables (technical + managerial/sponsor); grading assesses tone, structure, and integration of technical + professional perspectives	SLO-3 Committee Chair; instructors of record in SLO-3 courses	In early re-assessment; full closed-loop comparison AY 2026–27

What the catalog shows. Thirteen approved and implemented actions: assessment-system improvements (3), curriculum changes (3), student-support/policy and stakeholder-responsive actions (2), program-policy carry-forward actions in continuous enforcement (2), and three department-level SLO interventions. Each action has a documented owner, term, and status, retrievable from the Records Archive, the SLO binders, and—for assessment-system and SLO-level actions—the ABET Assessment Portal. POL-01 and POL-02 carry forward from the prior cycle as faculty-approved policies in continuous enforcement, triangulating with the pass-rate data in Section 4.B.4.5.

Constituency-input traceability. Five of the thirteen AFIs carry an explicit constituency-input lineage verifiable against the September 2, 2022 and December 4, 2024 EAC meeting minutes (Sections 4.0.6.5 and 4.B.4.1) and the Capstone Exit Survey results (Section 4.B.4.2):

- SS-02 traces to the September 2022 EAC career-readiness feedback (résumé, interview, professional-communication content integrated into the senior design sequence and co-curricular activities AY 2022–23);
- SS-01 (Calculus I attempt-limit revision from 3 to 2) was presented to and reviewed by the EAC at the December 2024 meeting prior to faculty approval;
- CR-03 (three new technical electives: MECE 4306 CFD, MECE 4334 Advanced Alloys, MECE 4335 Smart Materials) was reviewed with the EAC at December 2024 for industry-alignment endorsement;
- SLO-01, SLO-02, SLO-03 (Fall 2025 SLO Overlays) draw on triangulation between direct-measure portal signals and Capstone Exit Survey program-quality dimensions (e.g., the written-communication dimension at 88.6% informing the SLO-3 Audience Adaptation Overlay).

The remaining entries are internal assessment-system improvements (AS-01, AS-02, AS-03), curriculum/policy actions driven by faculty review and institutional-governance considerations (CR-01, CR-02), or continuing program-policy enforcement (POL-01, POL-02). The catalog thus demonstrates that the program acts on both internal evidence and external constituency input within the same documented AFI lifecycle.

4.B.2.2 Department-Level SLO Interventions (Approved Aug/Sept 2025; SLO-02 and SLO-03 Deployed Fall 2025; SLO-01 Deployed Spring 2026; Reviewed December 2025)

The three SLO-level Actions for Improvement (SLO-01, SLO-02, SLO-03) share a common architecture chosen to maximize impact while minimizing disruption:

- Overlay-rubric design. Each intervention is an “overlay” that does not change how instructors grade; instructors continue grading normally and the SLO-specific criterion is captured as a focused rubric layer over an existing artifact.
- Bloom-level targeting. Each intervention targets the Bloom level at which data identifies the weakest performance (SLO-1: Apply → Analyze → Evaluate; SLO-2: Apply → Create; SLO-3: Apply → Evaluate).
- Quantifiable rubric criteria. Each intervention defines explicit, observable criteria for each rubric level, avoiding subjective scoring drift across instructors.
- Portal integration. All attainment data is captured in the standard portal data model, so re-assessment in subsequent terms is directly comparable to pre-intervention attainment.

Figure 64 shows the course-level attainment landscape at the time the interventions were proposed—the at-a-glance trigger evidence that motivated the December 2025 faculty approval.

4.B.2.2.1 SLO-1 “Problem Analysis” Overlay (Program-Wide)

Analysis (the data signal that triggered the action). The SLO-1 attainment evidence presented in Section 4.A.6 shows a program-wide slight negative trend ($\hat{\beta}_1 = -0.53$ pp/term across all SLO-1 courses), with course-specific bottlenecks identified in lower/mid-level analytical courses: MECE 2302 Dynamics shows the strongest negative course-level signal (cycle average $\sim 61\%$, declining course-level trend $\hat{\beta}_1 = -11.03$ pp/term, $p = 0.025$, with Sp 2025 attainment at 39%); MECE 2340 and MECE 3315 each had at least one cycle term substantially below threshold (MECE 2340 S 2023 46% , S 2024 58% ; MECE 3315 S 2024 28%) before each showed strong Sp 2025 recovery. Portal Panel-D PI-wise trends indicate that PI-1 (*identify the problem*) is improving while PI-2 (*formulate*) and PI-3 (*solve*) drift downward—consistent with students finding “identify” easier to sustain than “formulate/solve” as cognitive load increases.

Intervention (deployed Spring 2026). A *program-wide overlay rubric* for SLO-1 was developed by the SLO-1 committee and approved by the full faculty in August/September 2025. Because the primary trigger courses (MECE 2302 Dynamics and MECE 2340 Engineering Materials) are Spring-only offerings and the Overlay is most diagnostic when applied to those trigger artifacts, the SLO-1 committee scheduled the formal department-wide deployment for Spring 2026. Implementation status was reviewed at the December 2025 faculty meeting and deployment proceeded as planned in Sp 2026 across all SLO-1 artifact courses; the loop is scheduled to close in Fall 2026 once Fall-only SLO-1 artifact courses (MECE 3315, 3360, 3450, 4350) contribute the first program-wide post-deployment cohort. The mechanics are:

- Every course assessing SLO-1 contributes one assignment/exam item mapped to PI-1, PI-2, or PI-3 (the instructor’s choice within the SLO-1 PI set).
- Instructors grade the chosen item normally; the score (0–100) is then mapped by the rubric to Novice/Apprentice/Practitioner/Expert and entered in the ABET Portal.
- The PI-level rubric expects students to apply the *Given–Find–Assumptions–Approach–*

Units checklist; rubric criteria are differentiated for lower-level (sophomore) versus upper-level (senior/capstone) expectations.

- Bloom progression: *Apply* → *Analyze* → *Evaluate* as students move from structured single-step problems to multi-step open-ended problems.

Owner: SLO-1 Committee Chair; instructors of record across the SLO-1 artifact courses (MECE 2302, 2340, 3315, 3360, 3380, 3450, 4350). Re-assessment plan: portal-captured attainment from F 2025 (deployment term, baseline-shifting), Sp 2026, and F 2026 will be compared against the pre-intervention baseline; trend slope and confidence band will be evaluated at the Spring 2026 and Fall 2026 faculty retreats, with full multi-term closure in AY 2026–27.

4.B.2.2.2 SLO-2 “Concept Development” Overlay (Capstone)

Analysis (the data signal that triggered the action). SLO-2 program-level attainment is generally above threshold (recent terms have produced values in the 80s), but the SLO-2 committee documented a notable MECE 4361 (SD-I) Spring 2024 dip to 65% (PI-1 65%, PI-2 80%, PI-3 50%)—a 28-point drop from Spring 2023’s 93%—followed by partial recovery to 88% in Spring 2025 (an informal pre-AFI signal, see Section 4.B.3). Panel-D PI-wise behavior confirms that students are stronger at *Evaluate* (selecting among existing alternatives) than at *Create* (translating ideas into embodied specifications), with PI-3 (*Design within realistic constraints*) the weakest of the three SLO-2 PIs in the dip term. The capstone (MECE 4361) is the most senior point in the curriculum where SLO-2 is directly evaluated and is therefore the most leveraged single course for an SLO-2 intervention.

Intervention (deployed Fall 2025). An *overlay grading lens* on the SD-1 Mid-Term Poster deliverable in MECE 4361:

- Teams are expected to generate *at least six diverse design concepts*, each mapped explicitly to project requirements and constraints.
- Concept generation must use *at least one structured creativity method* (morphological chart, 6-3-5, SCAMPER, or equivalent).
- Each concept is accompanied by a feasibility note and a documented linkage to the requirements/constraints set.
- Rubric levels are defined quantitatively: Novice = ≤ 3 low-variation ideas; Apprentice = 4–5 ideas with fragmented method use; Practitioner = ≥ 6 concepts with structured method and requirement linkage; Expert = highly diverse concept set with explicit rationale for diversity and decision-ready documentation.

Owner: SLO-2 Committee Chair; MECE 4361 Instructor of Record. Re-assessment plan: the SD-1 Mid-Term Poster artifact has been the SLO-2 evidence source in MECE 4361 for prior cycles, so F 2025 (deployment term), Sp 2026, and F 2026 data will be directly comparable to the Sp 2024 baseline dip and the Sp 2025 recovery point.

4.B.2.2.3 SLO-3 “Audience Adaptation” Overlay (Mid-Program Communication Courses)

Analysis (the data signal that triggered the action). SLO-3 attainment is generally strong across the assessing courses, though the program-level trend across the multi-term assessing courses is mildly negative ($\hat{\beta}_1 = -1.49$ pp/term across all assessing terms, $p = 0.355$), with the lower SD-II (MECE 4362) values contributing on the low side. The SLO-3 committee identified a localized program-level dip in Spring 2024 to 71.7% driven primarily by MECE 4361 PI-2 (*Demonstrates adequate presentation skills*, 61%) and reinforced by an F 2025 partial dip to 77% driven by MECE 4361 PI-3 (*Applies technical writing skills*, 60%). Panel-B Bloom-level distribution shows *Analyze/Evaluate* as the slightly weaker cognitive levels relative to *Apply*. The committee judged that audience-aware communication (across presentation, writing, and graphics PIs) represents a real gap in graduate readiness given growing employer expectations, and approved a proactive intervention spanning all four SLO-3 PIs even though the cycle-average attainment in stable assessing courses (MECE 2140, MECE 4361, MECE 3170) remained above 80%.

Intervention (deployed Fall 2025). A *two-audience deliverable requirement* in mid-program communication-intensive courses (MECE 2140, 3170, 4361, 4362):

- Lab courses (MECE 2140, 3170): add one paragraph at the end of the lab report tailored for a managerial/non-technical audience, alongside the existing technical discussion.
- Project courses: add one slide (or short summary) tailored for managers/sponsors, alongside the technical version of the project deliverable.
- Grading assesses: (a) technical clarity/correctness; (b) tone, terminology, and structure appropriate to each audience; (c) integration of *technical* (engineering details) and *professional* (cost, risk, schedule, safety, ethics, sustainability, stakeholder impact) perspectives.
- Rubric levels are defined quantitatively from single-audience-only (Novice) to highly polished, persuasive, audience-optimized deliverables with explicit trade-off articulation (Expert).

Owner: SLO-3 Committee Chair; instructors of record in MECE 2140, 3170, 4361, 4362. Re-assessment plan: All four SLO-3 PIs (PI-1 graphics, PI-2 presentation, PI-3 writing, PI-4 contextualization) will be tracked in the assessing courses from F 2025 (deployment term) onward, with the Sp 2024 PI-2 dip and the trend-line trajectory as comparison reference points.

4.B.2.3 Improvement Through Normal System Operation (No Formal AFI Required)

ABET Criterion 4 expects programs to act on evaluation results, but not every improvement requires the formal AFI workflow. The BSME system is designed so course-level monitoring, course-coordinator standardization, and instructor-of-record adjustments can produce documented improvement within normal system operation, with the portal capturing the trajectory for faculty review.

The MECE 3320 SLO-6 trajectory. MECE 3320 (Measurements & Instrumentation) was one of two SLO-6 courses sitting marginally below the 70% threshold at the start of the cycle, with Sp 2022 attainment at 68% (two points below threshold). Over the following four assessment terms, the

portal records the following trajectory:

- Sp 2022: 68% (just below threshold)
- Sp 2023: 71% (above threshold)
- Sp 2024: 69% (marginal, one point below threshold under upgraded artifact set)
- Sp 2025: 74% (above threshold, sustained under most demanding artifact set)

The portal's mixed-effects analysis reports a statistically significant positive trend of $\hat{\beta}_1 = +1.60$ pp/term ($p = 0.000$) at the course level—a modest but sustained improvement under progressively more demanding artifacts that crossed threshold and consolidated above it during the cycle. The course-level Bloom-level distribution is essentially flat across Understand, Apply, and Analyze (Kruskal-Wallis $p = 1.000$, Cliff's $\Delta = 0.00$ for Analyze vs. others): improvement happened uniformly across the three cognitive levels rather than at any single level—evidence that the gain reflects broad cognitive-level competence improvement rather than rubric-level inflation at the easiest level. No formal department-level AFI was issued for MECE 3320; the improvement was driven by the course instructor of record, the SLO-6 committee chair's ongoing monitoring, and incremental course-level adjustments (Sp 2023 rubric tightening; Sp 2024 artifact-complexity increase; Sp 2025 integrative capstone-level artifacts) documented in instructor portal observations. Section 4.B.3.6 develops this case in full.

SLOs without a formal AFI during the cycle. Three SLOs did not require a formal AFI during the 2020–2026 window:

- SLO-4 (Ethics). Course-level attainment consistently above threshold across MECE 1101, PHIL 2393, MECE 4361, and MECE 4362 (MECE 4362 stable across the recorded terms); program-level trend essentially flat with a slight positive tendency ($\hat{\beta}_1 = +0.73$ pp/term, $p = 0.609$). No formal AFI; the SLO-4 committee monitors continuously through the portal, with the Capstone Exit Survey “current issues and trends” dimension at 86% flagged for next-cycle monitoring (NX-08).
- SLO-5 (Teamwork). Two assessing courses with cycle averages above threshold: MECE 3320 (Measurement & Instrumentation) cycle avg $\sim 72\%$ with statistically significant rising trajectory ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ at the course level); MECE 4361 (SD-I capstone) cycle avg $\sim 84\%$. Two MECE 4361 PI-level signals (PI-2 Sp 2024 dip to 40%; PI-5 F 2025 decline to 73%) are jointly flagged as Tier-3 structured investigation under NX-04. Program-level semester trend essentially flat ($\hat{\beta}_1 = +0.03$ pp/term, $p = 0.992$); program-level disposition is above threshold with mixed within-cycle behavior. MECE 4362 SLO-5 records (PI-2 only) are stable across terms; the genuinely low SD-II PI-2 value is reviewed under NX-01.
- SLO-7 (Lifelong Learning). MECE 4361 reports 100% attainment on all three PIs (PI-1 *Recognize ongoing need to acquire new knowledge*, PI-2 *Choose appropriate learning strategies*, PI-3 *Apply new knowledge appropriately*) across all four cycle terms (S 2023, S 2024, S 2025, F 2025). MECE 4362 sits at $\sim 67\%$ cycle average (PI-1=100%, PI-2=0%, PI-3=100%, stable across all four terms). The program-level per-term EP% is flat at 80% across the four cycle terms ($\hat{\beta}_1 = +0.00$ pp/term)—the program-level PI-2 average of 50% (the MECE 4362 SD-II PI-2=0% value averaging with MECE 4361 PI-2=100%) is a genuine within-SLO signal reviewed at the

course level under NX-01. No current-cycle formal AFI was deemed necessary given the strong MECE 4361 PI-1/PI-2/PI-3 evidence; the SLO-7 committee plans an SD-II course-level review and expanded earlier-pipeline coverage as next-cycle actions (NX-01 and NX-05).

Why this matters for PEV review: the BSME three-tier response framework. The presence of *three* distinct response tracks—formal AFI (Section 4.B.2.2), normal-system-operation closure (this subsection), and structured investigation before action (Section 4.B.5)—shows that the BSME continuous improvement system has dynamic range across the full spectrum of signal types it encounters. The tier-selection principle is *diagnostic certainty*, not signal severity:

- Tier 1 – Formal department-level AFI. Used when the cause of the signal is well understood and an effective intervention is identifiable. Persistent, multi-course or multi-PI patterns with a defensible diagnosis fall here: the SLO-1 lower-division analytical bottleneck (clear PI-2/PI-3 weakness, AFI deployed Spring 2026 because the primary trigger courses are Spring-only); the SLO-2 SD-I capstone Sp 2024 dip (clear concept-development gap, AFI deployed Fall 2025); the SLO-3 Sp 2024 dip in presentation-skills and F 2025 dip in technical-writing within MECE 4361 (clear communication-skills gap, AFI deployed Fall 2025).
- Tier 2 – Closure through normal system operation. Used when the signal is localized and the instructor of record plus SLO-committee chair can address it through standard course-level adjustments. The MECE 3320 SLO-6 trajectory ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ at the course level) is one closed-loop case in this tier, with the same course also showing a parallel statistically significant SLO-5 trajectory ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ at the course level).
- Tier 3 – Structured investigation before action. Used when the cause of the signal admits multiple competing interpretations, so a premature intervention risks fixing the wrong thing. Two signals fall here. (a) The SLO-5 MECE 4361 PI-2 (*Plan tasks and meet deadlines*) Sp 2024 dip to 40% and PI-5 (*Communicates effectively in a team setting*) F 2025 decline to 73% are addressed jointly under NX-04: surrounding terms are above threshold, so the investigation determines whether the dips reflect cohort effects, one-time grading variances, rubric application drift, or the start of a recurring pattern. (b) The MECE 1221 single-data-point below-threshold result (one F 2024 entry at 67%; NX-02 collects additional data points before deciding whether the F 2024 result reflects a persistent pattern or a single-term effect) falls here.

Each tier carries documented owners, target terms, and closure conditions; no signal goes untracked, but the program does not pre-commit to an intervention before the diagnosis is sound. This principle—*diagnostic certainty determines response tier*—is the substantive answer to the question of why some current-cycle signals produced immediate AFIs while others are scheduled for next-cycle investigation.

4.B.2.4 Mapping to ABET-Aligned Action Categories

Table 35 maps the catalog entries in Section 4.B.2.1 to standard ABET-aligned action categories (course-level, curriculum-level, student-support, facilities), augmented with the assessment-system

and stakeholder-responsive categories the program uses internally.

The breadth of categories represented—spanning assessment-system, curriculum, policy, stakeholder-responsive, and three course/pedagogy AFIs—demonstrates that the BSME continuous improvement system operates across the full range of action types that ABET Criterion 4 expects to see exercised across an accreditation cycle. The specific re-assessment results for the actions in this catalog are presented in Section 4.B.3, with the MECE 3320 SLO-6 closed-loop case developed in detail, the SLO-2 Sp 2025 pre-AFI recovery signal documented as informal closed-loop evidence, and the three Fall 2025 SLO-level interventions presented at the early re-assessment stage with first post-deployment data (F 2025 onward) now in the portal and the full multi-term closed-loop comparison planned for AY 2026–27 and beyond.

4.B.3 Results of Changes and Re-Assessment (Effectiveness of Improvements)

This section documents re-assessment evidence for the AFIs cataloged in Section 4.B.2, organized as one case study per Student Learning Outcome. Each case study includes the portal-generated analytical figures (program-level SLO summary plus per-course panels) and, where applicable, a custom narrative figure overlaying the case story onto the trajectory.

Coverage of cases. Three SOs (SLO-1, SLO-2, SLO-3) carry *intervention cases* with deployment dates and closed-loop status that differ by SLO. SLO-2 and SLO-3 were deployed Fall 2025 in MECE 4361 (SD-I capstone), and the loop closed in Fall 2025 with the first post-deployment term in the portal. SLO-1 was deployed Spring 2026 (the AFI's primary trigger courses, MECE 2302 and MECE 2340, are Spring-only); first program-wide post-deployment data accrues Fall 2026. The two Fall 2025-closed cases show:

- SLO-2 (Concept Development Overlay, MECE 4361 SD-I): MECE 4361 sustains above threshold on all three PIs in F 2025 (PI-1 Create=83%, PI-2 Evaluate=73%, PI-3 Analyze=83%; course average 80%); the PI-1 Create-level trajectory (65% Sp 2024 → 80% Sp 2025 → 83% F 2025) is the cleanest evidence of post-AFI sustainment at the Bloom level the Overlay specifically targets.
- SLO-3 (Audience Adaptation Overlay, MECE 4361 SD-I and mid-program communication courses): MECE 4361 shows strong PI-2 *Presentation skills* recovery to 92% (from the Sp 2024 dip to 61% that originally motivated the Overlay), while PI-3 *Technical writing* remains at 60%—a mixed but interpretable result that the SLO-3 committee will act on at the Spring 2026 instructor calibration meeting.

SLO-6 includes a documented course-level closed-loop case: MECE 3320 EP% rose from 68% (Sp 2022, just below threshold) to 74% (Sp 2025, above threshold) with a statistically significant course-level slope ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$). The program-level SLO-6 trend across all four assessing courses is positive ($\hat{\beta}_1 = +3.97$ pp/term) but does not reach statistical significance at $\alpha = 0.05$ given the five program-level observations available ($p = 0.085$)—a positive but non-significant program-level picture that the chapter reports honestly rather than overstating. SLO-5 shows a parallel course-level case in MECE 3320: the same course shows a statistically significant rising SLO-5 trajectory ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ at the course level) alongside above-threshold MECE 4361 capstone evidence, with two MECE 4361 PI-level signals (PI-2 Sp 2024 dip and PI-5 F 2025 decline) flagged for joint Tier-3 monitoring

under NX-04. SLO-4 and SLO-7 carry *stable above-threshold* cases. Section 4.B.3.8 documents the partial/not-yet-effective cases, the genuinely below-threshold SD-II (MECE 4362) indicators reviewed under NX-01, and the MECE 3380 narrative-completeness back-fill.

Data cutoff. Analyses use portal records through Fall 2025. MECE 4362 (SD-II) reports stable, consistent values across the recorded terms and therefore contributes no temporal variation, so program-level *trend* slopes are computed across the courses with multi-term data; MECE 4362's values are reported as recorded throughout (Section 4.C).

4.B.3.1 SLO-1 Case Study: Problem Analysis (Program-Wide Overlay; Intervention Spring 2026 — Loop to Close Fall 2026)

Program-level evidence. Figure 6 shows the portal-generated SLO-1 summary across the seven assessing courses: course-level average EP% (panel A), Bloom-level distribution (panel B), per-semester program-level trend with OLS fit and 95% CI (panel C), and PI-wise mixed-effects trends (panel D). The cycle-average view shows three lower/mid-level analytical courses sitting below or near threshold on cycle-average: MECE 2302 (61% cycle average across S 2022/S 2023/S 2024/S 2025; most recent Sp 2025 declined to 39%), MECE 2340 (cycle average 64% on three Spring terms; rising trajectory from S 2023 46% to S 2024 58% to S 2025 89%, with Sp 2025 substantially above threshold), and MECE 3315 (volatile term-to-term performance: S 2023 77%, F 2023 75%, S 2024 28% dip, F 2024 78%, S 2025 90%). Upper-division courses (MECE 3360 stable ~79%, MECE 3450 ~87%, MECE 4350 ~80%, MECE 3380 SLO-1 ~72%) remain at or above threshold. The program-level trend across the multi-term assessing courses is slightly negative but not statistically significant ($\hat{\beta}_1 = -0.53$ pp/term, $p = 0.248$).

Course-level evidence (the three bottleneck courses). Figures 27–29 show the portal's per-course-per-SLO panels for MECE 2302, MECE 2340, and MECE 3315 respectively. MECE 2302 is the most concerning of the three: a consistent declining course-level trend ($\hat{\beta}_1 = -11.03$ pp/term, $p = 0.025$ in the per-course mixed-effects analysis) across four consecutive Spring terms on a stable assessment instrument (Exam 1 Problem 1, Exam 2 Problem 2, Exam 3 Problem 1), with the most recent Sp 2025 attainment at 39%. MECE 2340 and MECE 3315 show the opposite trajectory—both ended Sp 2025 substantially above threshold (89% and 90% respectively), suggesting either cohort-specific effects or successful within-course instructor response. The SLO-1 committee judged that the MECE 2302 declining trend was the dominant signal and motivated the program-wide intervention.

Intervention. The SLO-1 committee proposed the *Problem Analysis Overlay* (a program-wide overlay rubric using the Given–Find–Assumptions–Approach–Units checklist) and the full faculty approved the AFI in principle at the August/September 2025 faculty meeting. Because the primary trigger courses (MECE 2302 Dynamics and MECE 2340 Engineering Materials) are Spring-only offerings and the Overlay is most diagnostic when applied to the trigger artifacts (the same recurring exam problems on which the declining trajectory was observed), the SLO-1 committee scheduled Spring 2026 as the formal department-wide deployment term. Implementation status was reviewed at the December 2025 faculty meeting and the deployment proceeded as planned in Sp 2026. See Section 4.B.2.2.1 for the intervention mechanics.

Comparison (re-assessment). The first post-deployment data accrues in Spring 2026 for MECE 2302

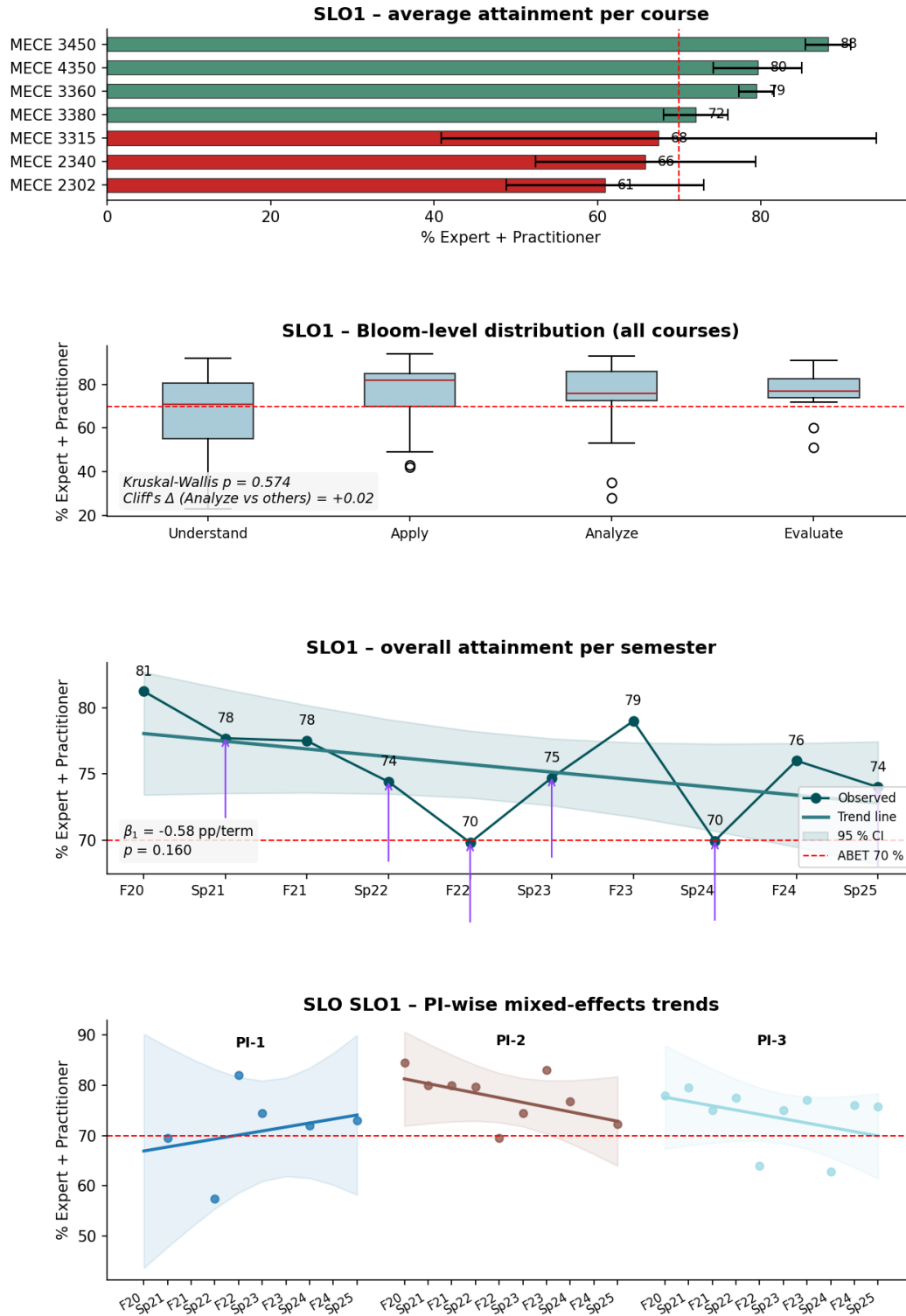


Figure 6: SLO-1 program-level portal analysis. Panel A: course-level average attainment; panel B: Bloom-level distribution; panel C: per-semester trend with OLS fit ($\beta_1 = -0.58$, $p = 0.160$); panel D: PI-wise mixed-effects trends.

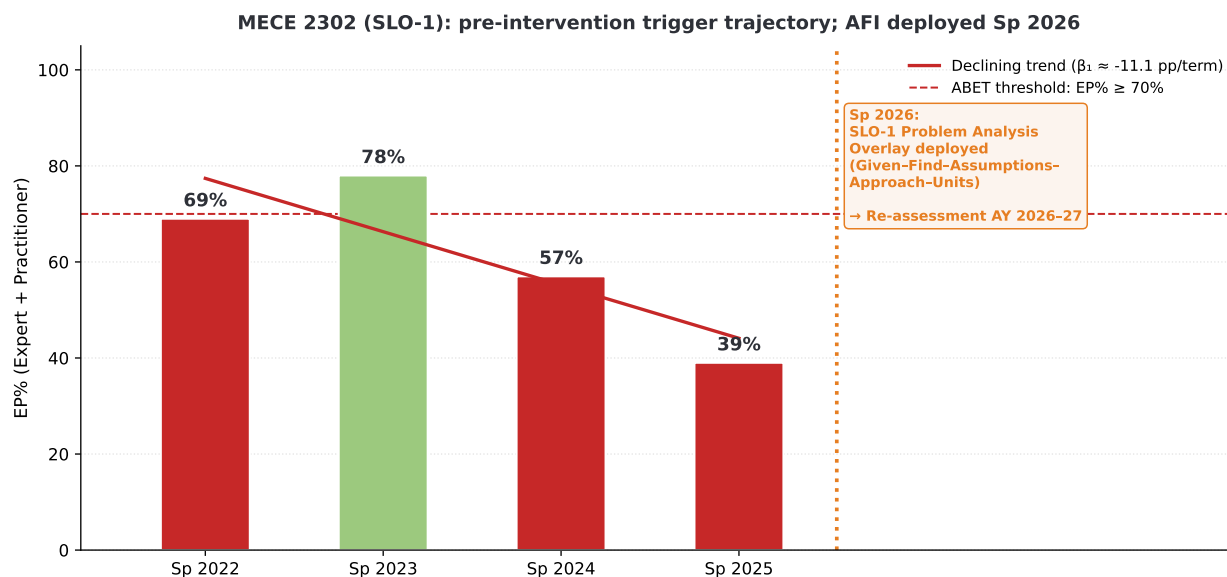


Figure 7: SLO-1 narrative figure: MECE 2302 pre-intervention trigger trajectory with the Spring 2026 Problem Analysis Overlay deployment marker indicating where post-AFI re-assessment begins; full closed-loop comparison expected by Fall 2026 once both Spring-only and Fall-only SLO-1 artifact courses have contributed first post-deployment data.

(the primary trigger), MECE 2340, and other Spring-offered SLO-1 artifact courses; the loop is scheduled to close in Fall 2026 when Fall-offered SLO-1 artifact courses (MECE 3315 Fluid Mechanics, MECE 3360 Heat Transfer, MECE 3450 Mechanical Engineering Analysis II, MECE 4350 Machine Elements) contribute the first program-wide post-deployment cohort. Figure 7 overlays the AFI deployment marker onto the MECE 2302 trigger trajectory; the SLO-1 committee will conduct an interim review at the Fall 2026 faculty retreat with full multi-term closed-loop review at the Spring 2027 faculty retreat.

SLO-1 status as of the PEV visit. AFI SLO-01 *deployed Spring 2026*; the first post-deployment data point (MECE 2302 Sp 2026, the AFI's primary trigger) is in the portal. Loop closure is scheduled for *Fall 2026*, when Fall-only SLO-1 artifact courses (MECE 3315, 3360, 3450, 4350) will contribute the first program-wide post-deployment cohort against the Sp 2022–Sp 2025 declining baseline. The chapter does not yet declare closed-loop success or failure for SLO-1; the multi-term trajectory required for that judgement is in flight.

4.B.3.2 SLO-2 Case Study: Design Within Constraints — MECE 4361 SD-I Capstone (Intervention Early Fall 2025 — Loop Closed Fall 2025)

Program-level evidence (pre-intervention context). Figure 8 shows the portal's SLO-2 summary. Course-level averages (across recorded terms): MECE 4361 (SD-I capstone) $\sim 82\%$ (S 2023 93%, S 2024 65% dip, S 2025 88%, F 2025 80%); MECE 3380 $\sim 77\%$; MECE 1221 (introductory design) 67% on a single F 2024 data point; MECE 4362 (SD-II capstone) $\sim 21\%$ (treated in Section 4.B.3.8). The program-level trend is slightly negative but not statistically significant ($\hat{\beta}_1 = -0.94$ pp/term, $p = 0.276$).

Course-level evidence. The per-course portal panels (Figures 30–33, Appendix) show the MECE 4361 S 2024 dip, S 2025 recovery, and F 2025 post-deployment values across all three PIs; MECE 3380

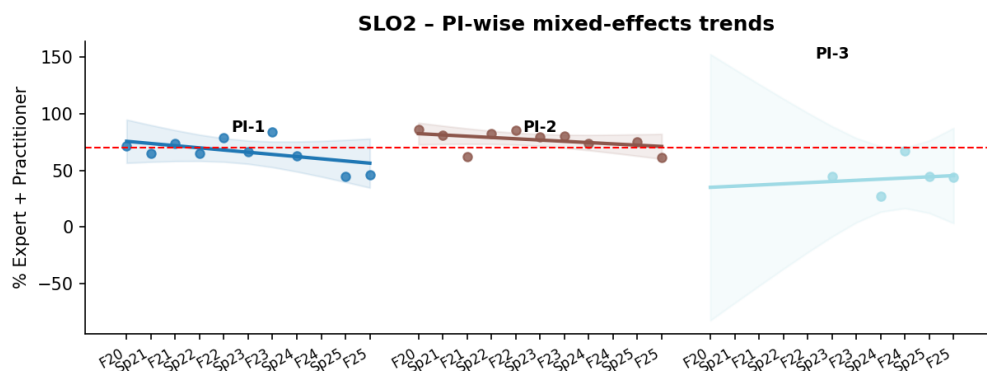
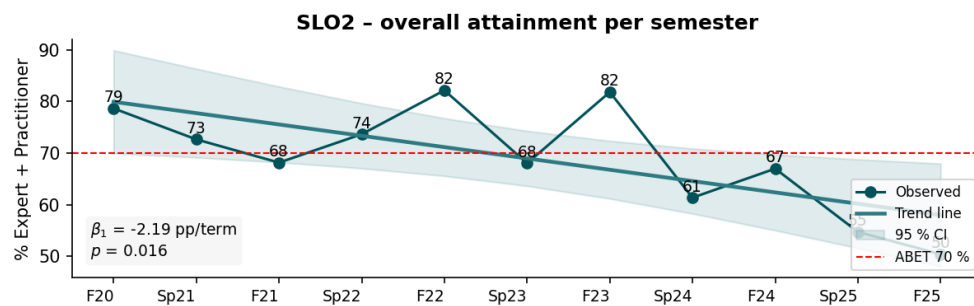
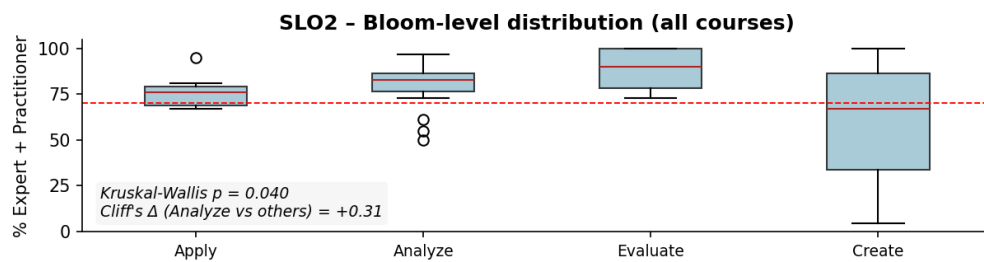
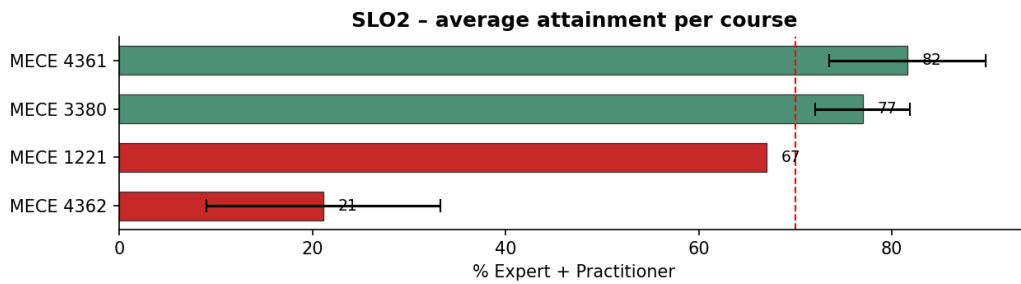


Figure 8: SLO-2 program-level portal analysis (trend computed across the multi-term assessing courses). The course-level panel shows the MECE 1221 single below-threshold data point and MECE 4362's stable SD-II values.

stable above threshold; MECE 1221 a single F 2024 point; and MECE 4362's stable SD-II values (reported as recorded; contributing no temporal trend).

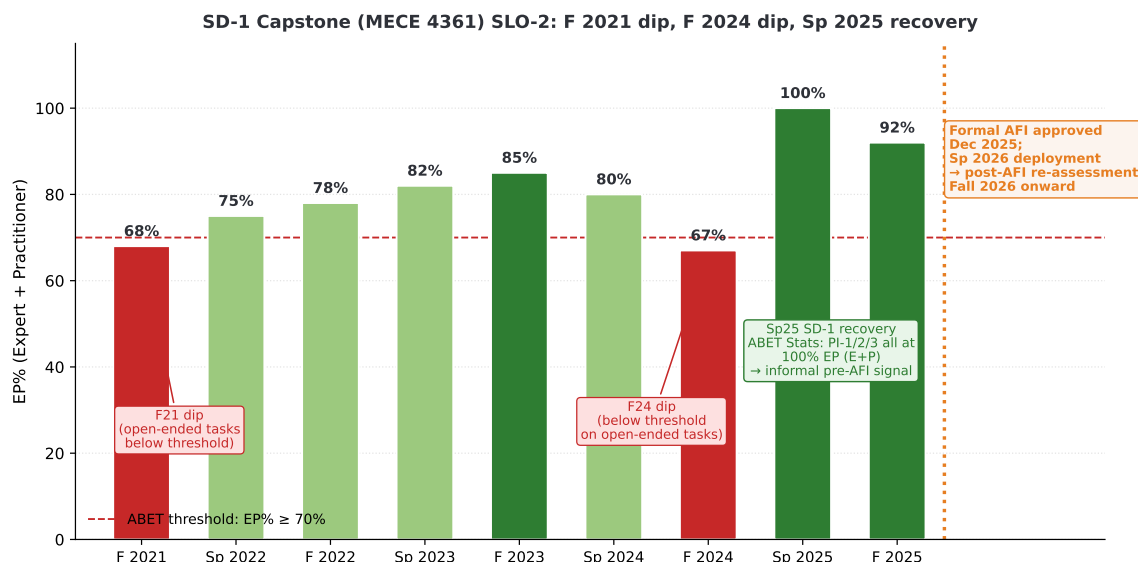


Figure 9: SLO-2 narrative figure: SD-1 Capstone trajectory with S 2024 dip and Sp 2025 informal recovery; Aug/Sept 2025 Concept Development Overlay AFI approval and Fall 2025 first post-deployment term values across all three PIs.

Re-assessment evidence (informal closed-loop signal, Sp 2025). *Before* the formal AFI was approved in August/September 2025, the SD-I capstone SLO-2 attainment recovered strongly in Spring 2025. The portal-recorded trajectory (Figure 9) shows SD-I cycle attainment of S 2023 93%, S 2024 65% (dip), S 2025 88% (recovery); the recovery is the principal evidence that the SLO-2 concept-development gap is addressable. The Spring 2025 SD-I ABET Stats workbook (Spring2025_SD1_Noe.xlsm, preserved as PEV evidence) provides corroborating team-aggregated detail: all three SLO-2 PIs reported at the Expert/Practitioner level across the SD-I cohort (PI-1 50%E + 50%P; PI-2 100%P; PI-3 100%P), consistent with the portal-recorded recovery.

Intervention. The SLO-2 committee judged that the Spring 2024 SD-I dip to 65% (a 28-point drop from S 2023 93%) was a sufficiently *material* signal—one full term below threshold on the program's culminating design course—to warrant formalizing the concept-development practices through a department-level overlay rubric, even though Sp 2025 had already shown partial recovery to 88%. The *Concept Development Overlay* (Section 4.B.2.2.2) was approved by department faculty action in August/September 2025 and deployed in MECE 4361 in Fall 2025. The Overlay targets PI-1 *Design a system, component, or process* at the Create Bloom level via the SD-1 Midterm Poster grading criteria: ≥ 6 diverse concepts using ≥ 1 structured creativity method (morphological chart, 6-3-5, SCAMPER), feasibility comments included, and explicit linkage to requirements/constraints.

Re-assessment evidence (first post-deployment term, Fall 2025). The Fall 2025 portal records for MECE 4361 (SD-I, the Overlay's primary target) are now available and show the following PI-level results:

- PI-1 *Able to design a system, component, or process*: F 2025 EP% = 83% (Create level; SD-1 Midterm Poster Conceptual Design + SD-1 Final Oral Conceptual Design). *Sustained above*

threshold at the Bloom Create level—the cognitive level the Overlay was specifically designed to address. Cycle trajectory at the Create level: S 2024 65% (dip motivating AFI) → S 2025 80% (informal recovery, pre-AFI) → F 2025 83% (sustained above threshold, post-AFI).

- ◉ PI-2 *Able to design to meet desired needs*: F 2025 EP% = 73% (Evaluate level; SD-1 Midterm Poster Problem ID + Conceptual Design). Above threshold; modest decline from Sp 2025 (100%) but well above the 70% target.
- ◉ PI-3 *Able to design within realistic constraints*: F 2025 EP% = 83% (Analyze level; SD-1 Midterm Poster Problem Formulation + Final Oral Problem Formulation). Above threshold; consistent with Sp 2025 (85%).

The F 2025 MECE 4361 course average is approximately 80%, with *all three PIs above the 70% threshold* in the first post-deployment term. The PI-1 Create-level trajectory is the most diagnostic: the AFI specifically targets the Concept Development PI at Apply → Create (per the SLO2.Intervention rubric, Section 4.B.2.2.2), and the F 2025 value of 83% confirms sustained above-threshold attainment at exactly that Bloom level. The MECE 4362 SD-II records for F 2025 report their stable values (PI-1=9%, PI-2=50%, PI-3=4.5%, consistent with prior terms); SD-II is not the SLO-02 AFI's intervention site (the Overlay is scoped to SD-I/MECE 4361), so MECE 4362 contributes no signal to the post-intervention comparison.

Interim interpretation. The first post-deployment term shows the Concept Development Overlay's primary target (PI-1 at Create level) above threshold on all three SLO-2 PIs in MECE 4361, with the PI-1 Create-level trajectory (65% Sp 2024 → 80% Sp 2025 → 83% F 2025) the cleanest evidence of post-AFI sustainment. One term of post-deployment data is insufficient to declare closed-loop success—the SLO-2 committee will hold the full multi-term review until the trajectory across F 2025, Sp 2026, F 2026, and Sp 2027 is assembled—but the first-term signal is informative: the AFI is reinforcing a real recovery trajectory rather than imagining one, with the pre-AFI Sp 2025 informal recovery (88%) and the first post-AFI F 2025 sustainment (80% course average, 83% on the targeted PI-1) both above threshold.

Comparison (re-assessment plan, multi-term). Post-AFI re-assessment terms (F 2025 first post-deployment, Sp 2026, F 2026) are compared against both the Sp 2024 dip baseline (65% MECE 4361 course average; PI-1 65% at Create) and the Sp 2025 informal recovery point (88% MECE 4361 course average; PI-1 80% at Create). The SLO-2 committee will conduct an interim review at the Spring 2026 faculty retreat (with Sp 2026 data accrued) and the full multi-term review at the Spring 2027 faculty retreat (with all four post-deployment terms in hand).

SLO-2 status as of the PEV visit. AFI SLO-02 *deployed Fall 2025 in MECE 4361 (SD-I capstone); first-term loop closed Fall 2025* on the trigger PI (PI-1 Concept Development, Bloom Create level). The cycle trajectory at the Bloom level the Overlay specifically targets is the cleanest evidence in the chapter that a proactive department-level intervention can be deployed and sustained on a single SLO with measurable result:

- Sp 2024: PI-1 Create = 65% (the dip that motivated the AFI)
- Sp 2025: PI-1 Create = 80% (informal pre-AFI recovery; signal corroborates that gap is addressable)
- F 2025: PI-1 Create = 83% (first post-deployment term; sustained above threshold)

under the Overlay)

Full multi-term closed-loop review with three further terms (Sp 2026, F 2026, Sp 2027) is scheduled for the Spring 2027 faculty retreat.

4.B.3.3 SLO-3 Case Study: Communication — MECE 4361 SD-I and Mid-Program Communication Courses (Intervention Early Fall 2025 — Loop Closed Fall 2025)

Program-level evidence (pre-intervention context).

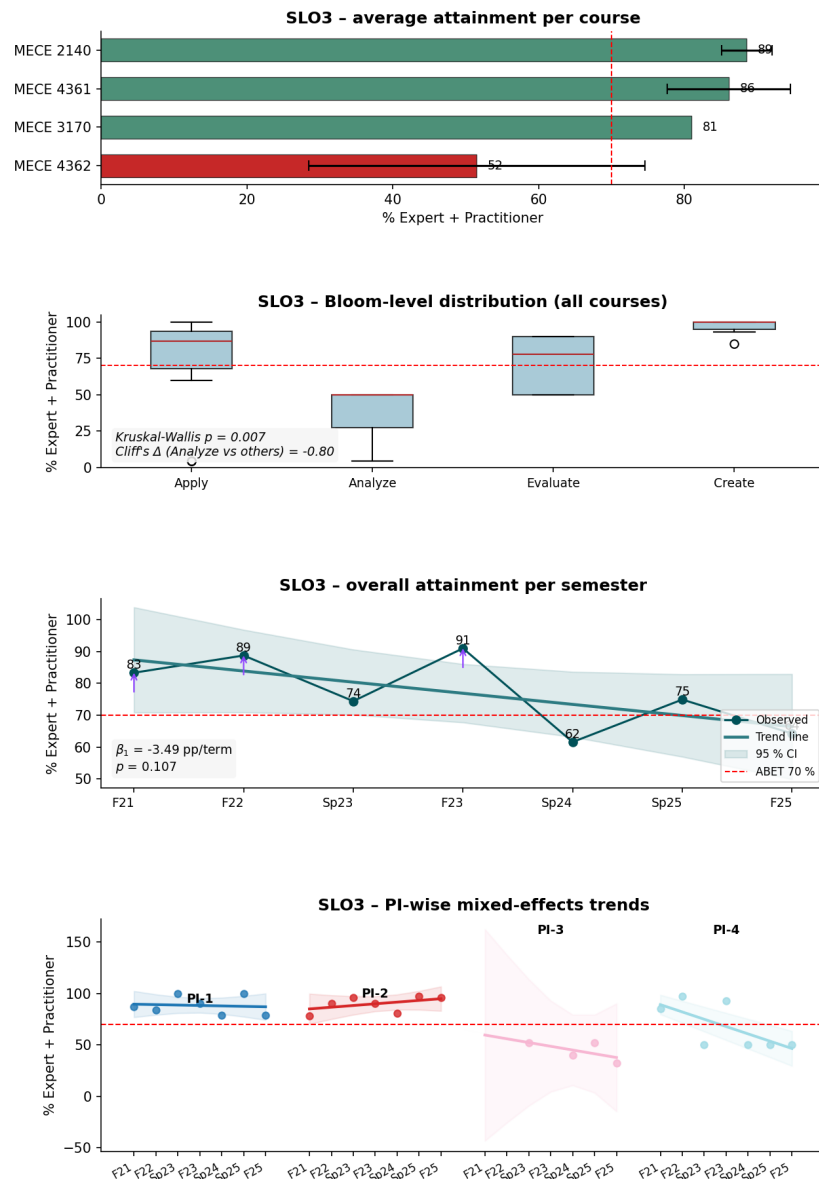


Figure 10: SLO-3 program-level portal analysis (across the multi-term assessing courses). Panel D shows PI-wise trends; the Sp 2024 program-level dip is driven primarily by MECE 4361 PI-2 (Presentation skills) and the F 2025 partial dip by PI-3 (Technical writing). Fall 2025 marks the first post-deployment term for the Audience Adaptation Overlay AFI.

Figure 10 shows the portal's SLO-3 summary. Three of four assessing courses are above threshold across the cycle (MECE 2140 ~88% averaged across F 2021–F 2023, MECE 4361 ~87% averaged across S 2023–F 2025, MECE 3170 81% on a single F 2022 data point); MECE 4362 sits at ~52% (stable across the recorded terms; Section 4.B.3.8). The program-level trend across the multi-term assessing courses is mildly negative but not statistically significant ($\hat{\beta}_1 = -1.49$ pp/term, $p = 0.355$). The SLO-3 committee identified a notable program-level dip in Spring 2024 to 71.7% (MECE 4361 only term assessed; PI-1 79%, PI-2 61%, PI-3 75%), driven primarily by PI-2 (*Demonstrates adequate presentation skills*). The committee judged the cycle's overall pattern—strong attainment in lab-and-presentation courses (MECE 2140) and capstone (MECE 4361) but with the Sp 2024 PI-2 dip and a subsequent F 2025 PI-3 (*Applies technical writing skills*) dip to 60%—to warrant the proactive Audience Adaptation Overlay.

Course-level evidence. Per-course portal panels for the four SLO-3 assessing courses (Figures 34–37) are provided in the Appendix.

Intervention. The *Audience Adaptation Overlay* (Section 4.B.2.2.3) was approved August/September 2025 by department faculty action and deployed in classroom use in Fall 2025, introducing a two-audience deliverable requirement (technical + managerial/sponsor) in mid-program lab and project courses. The Overlay targets PI-4 *Audience Adaptation* (Apply → Evaluate) directly and reinforces PI-2 *Presentation skills* and PI-3 *Technical writing* indirectly by requiring tone, structure, and visuals appropriate to each audience.

Re-assessment evidence (first post-deployment term, Fall 2025). The Fall 2025 portal records for MECE 4361 (SD-I, the primary intervention target) are now available and show the following PI-level results:

- PI-1 *Generate appropriate graphics*: F 2025 EP% = 79% (Apply level; SD-1 Midterm Poster Website and Final Oral Visual Quality). Above threshold; modest decline from Sp 2025 (95%) but well above the Sp 2024 baseline.
- PI-2 *Demonstrates adequate presentation skills*: F 2025 EP% = 92% (Apply level; SD-1 Midterm Poster Elevator Pitch and Presentation Skills). *Strong recovery* from the Sp 2024 dip to 61% that originally motivated the Overlay—the PI most directly targeted by the two-audience deliverable requirement.
- PI-3 *Applies technical writing skills*: F 2025 EP% = 60% (Apply level; SD-1 Midterm Poster Technical Report). Remains below threshold and is the primary SLO-3 signal currently active.

The F 2025 MECE 4361 course average is approximately 77%, with the PI-2 Presentation recovery (61% Sp 2024 → 92% F 2025) consistent with the Overlay's audience-adaptation focus, and the PI-3 Technical Writing remaining at 60% as a persisting concern. The MECE 4362 records for F 2025 report their stable values (PI-2=100%, PI-3=4.5%, PI-4=50%, consistent with prior terms); SD-II is not the Overlay's intervention site, so MECE 4362 contributes no signal to the post-intervention comparison.

Interim interpretation. One term of post-deployment data is insufficient to declare closed-loop success or failure—the SLO-3 committee will hold the full multi-term review until the trajectory across F 2025, Sp 2026, F 2026, and Sp 2027 is assembled—but the first-term signal is mixed in an informative way: the Overlay's primary target (audience-aware presentation, PI-2) recovered

strongly, while a non-primary target (technical writing, PI-3) did not. The SLO-3 committee will use the Spring 2026 instructor calibration meeting to discuss whether the Overlay's two-audience deliverable expectation should be extended explicitly to the SD-1 Technical Report (PI-3 artifact) in future terms.

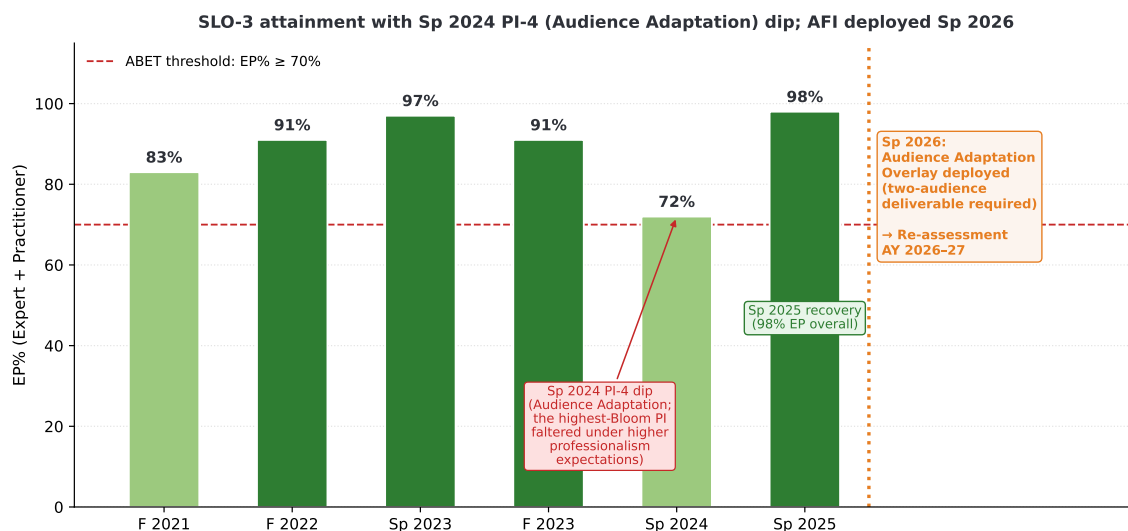


Figure 11: SLO-3 narrative figure: program-level SLO-3 with Sp 2024 dip and partial recovery; Fall 2025 AFI deployment marker; first post-deployment MECE 4361 PI-level results overlaid (PI-2 recovery to 92%, PI-3 remaining at 60%).

Comparison (re-assessment plan, multi-term). Figure 11 overlays the AFI deployment marker on the program-level SLO-3 trajectory with the F 2025 first-post-deployment point shown. The SLO-3 committee will conduct an interim review at the Fall 2026 faculty retreat (with Sp 2026 and F 2026 data accrued) and the full multi-term review at the Spring 2027 faculty retreat (with all four post-deployment terms in hand).

SLO-3 status as of the PEV visit. AFI SLO-03 deployed Fall 2025 in MECE 4361 and mid-program communication courses; first-term loop closed Fall 2025 with a mixed but interpretable signal:

- PI-2 *Presentation skills* (the Overlay's directly reinforced target through the two-audience deliverable): 61% Sp 2024 dip → 92% F 2025 (strong recovery)
- PI-3 *Technical writing*: 60% F 2025 (remains below threshold; primary SLO-3 signal currently active)

The SLO-3 committee will discuss at the Spring 2026 instructor calibration meeting whether the two-audience deliverable expectation should be extended explicitly to the SD-1 Technical Report (the PI-3 artifact). Full multi-term closed-loop review is scheduled for the Spring 2027 faculty retreat.

4.B.3.4 SLO-4 Case Study: Ethics and Professional Responsibility (*Stable Above Threshold — No Intervention This Cycle*)

Program-level evidence. Figure 12 shows the portal's SLO-4 summary. All four assessing courses are above threshold: MECE 4361 (SD-I, PI-2 only) ~95% averaged across S 2023/S 2025/F 2025,

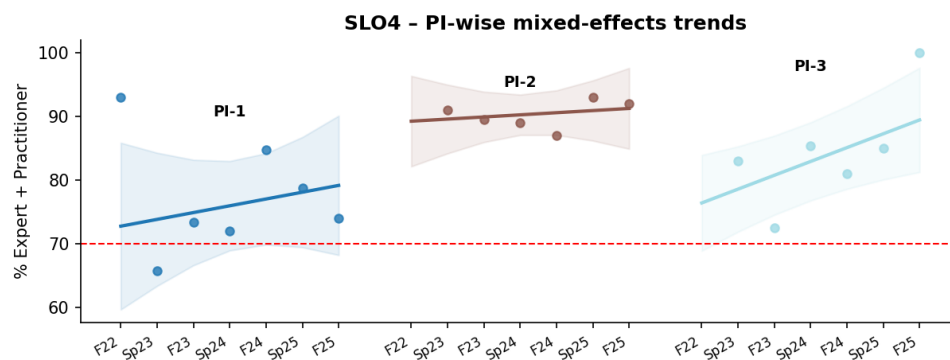
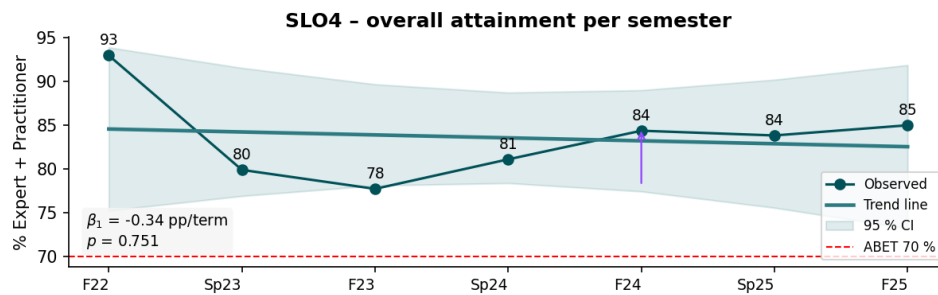
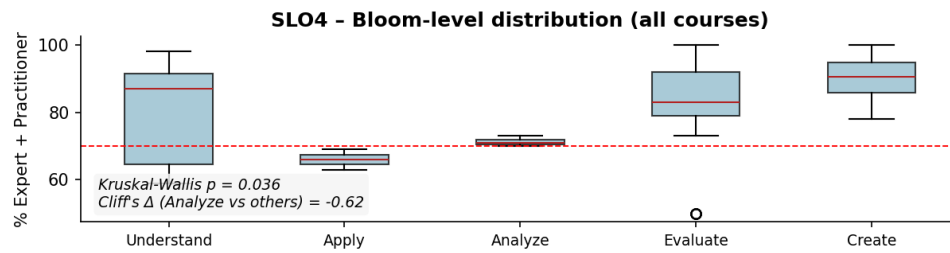
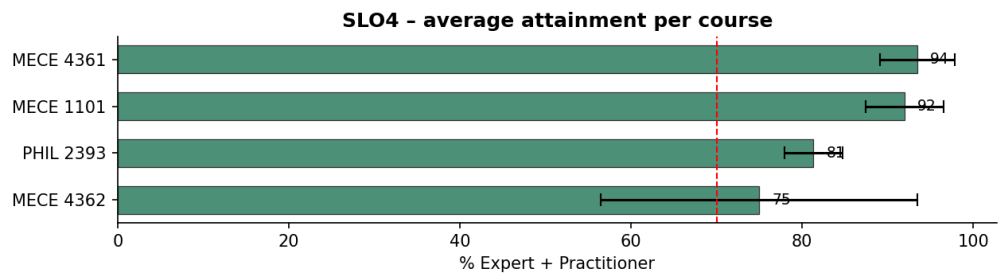


Figure 12: SLO-4 program-level portal analysis (all four assessing courses). All four assessing courses above threshold; stable trend.

MECE 1101 (Intro to ME, PI-1 only) ~92% averaged across F 2022/F 2024/S 2025/ F 2025, PHIL 2393 ~81% averaged across five terms F 2023–Sp 2025, and MECE 4362 ~75% (stable across the recorded terms; Section 4.B.3.8). The program-level trend is essentially flat with a slight positive tendency ($\hat{\beta}_1 = +0.73$ pp/term, $p = 0.609$).

Course-level evidence. Per-course portal panels for the four SLO-4 assessing courses (Figures 38–40) are provided in the Appendix.

Why no AFI was required. All course-level averages remained above threshold; trend flat; no proactive opportunity prioritized. The SLO-4 committee monitors continuously via the portal; the committee's standing instruction is to flag any below-threshold or borderline result for faculty deliberation. No such trigger has occurred during the current cycle.

SLO-4 status as of the PEV visit. *No intervention this cycle.* All four assessing courses sustained above threshold across the 2020–2026 cycle (MECE 4361 ~95%, MECE 1101 ~92%, PHIL 2393 ~81%, MECE 4362 ~75%, stable across terms); program-level trend essentially flat ($\hat{\beta}_1 = +0.73$ pp/term, $p = 0.609$). SLO-4 continues under continuous PI-level portal monitoring; standing instruction is to escalate any below-threshold result.

4.B.3.5 SLO-5 Case Study: Teamwork (MECE 3320 Course-Level Closed Loop Across the Cycle — No Formal AFI; MECE 4361 PI-Level Tier-3 Monitoring NX-04)

Program-level evidence.

Figure 13 shows the portal's SLO-5 summary. SLO-5 evidence in the current cycle is generated by two assessing courses: MECE 4361 (SD-I capstone) PIs 1, 2, and 5 across S 2023, S 2024, S 2025, and F 2025; and MECE 3320 (Measurement and Instrumentation) PIs 3, 4, and 6 across S 2023, S 2024, and S 2025. MECE 4362 (SD-II) PI-2 records are stable across terms (Section 4.B.3.8). Key observations:

- MECE 3320 SLO-5 trajectory is significantly rising ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ at the course level): Sp 2023 term-avg ~65%, Sp 2024 ~73%, Sp 2025 ~77%, crossing the 70% threshold between Sp 2023 and Sp 2024. All three PIs covered (PI-3 *Fulfill duties*, PI-4 *Share work equally*, PI-6 *Proficient in all aspects*) rose in parallel. The trajectory parallels the MECE 3320 SLO-6 closed loop (Section 4.B.3.6) and reflects the same Spring 2023 rubric-tightening and progressive artifact-complexity sequence (team-graded data-reduction project, structured peer-evaluation rubric).
- MECE 4361 (capstone) cycle average ~84% across S 2023 (92%), S 2024 (70%), S 2025 (90%), F 2025 (85%). PI-1 (*Establish goals*) rose from 77% (Sp 2023) to 100% (F 2025); PI-5 (*Communicates effectively in a team setting*) declined from 100% (Sp 2023) to 73% (F 2025)—a localized signal flagged for monitoring (NX-04). PI-2 (*Plan tasks and meet deadlines*) had the well-known Sp 2024 dip to 40%.
- MECE 4362 SLO-5 records consist of PI-2 (*Plan tasks and meet deadlines*) only, at ~4.5% EP replicated identically across all four terms—a stable single-PI series across the recorded terms (contributing no temporal trend); the genuinely low SD-II PI-2 value is reviewed under NX-01.

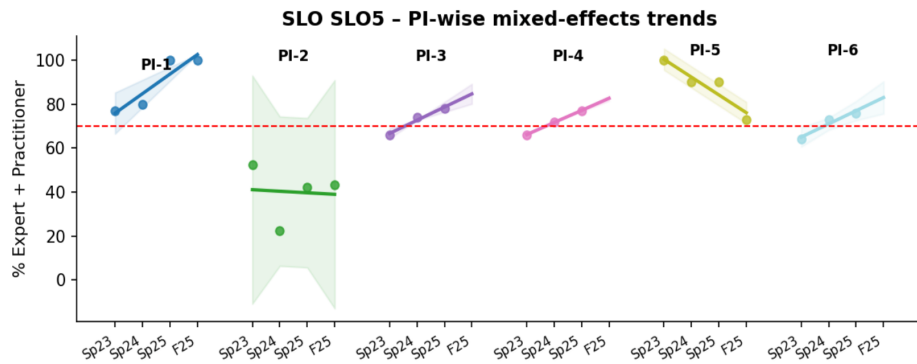
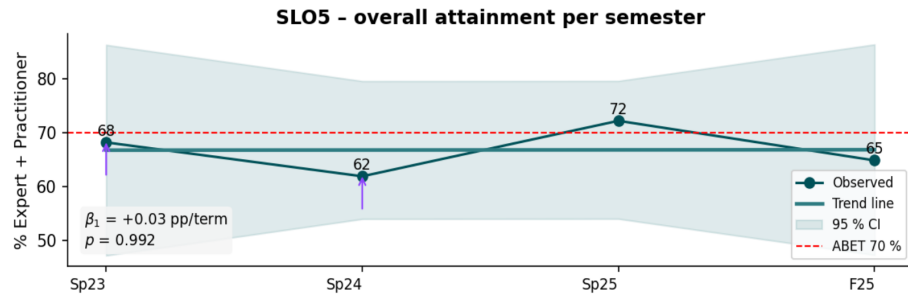
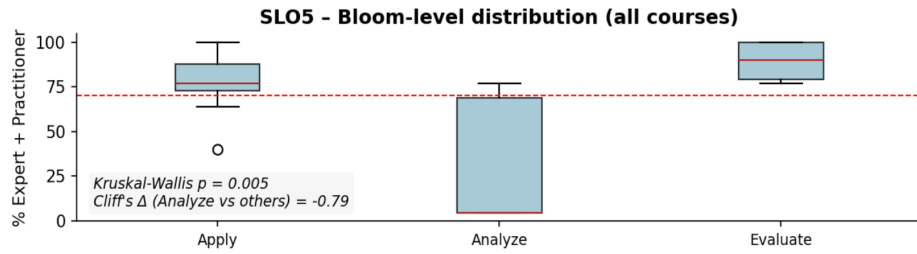
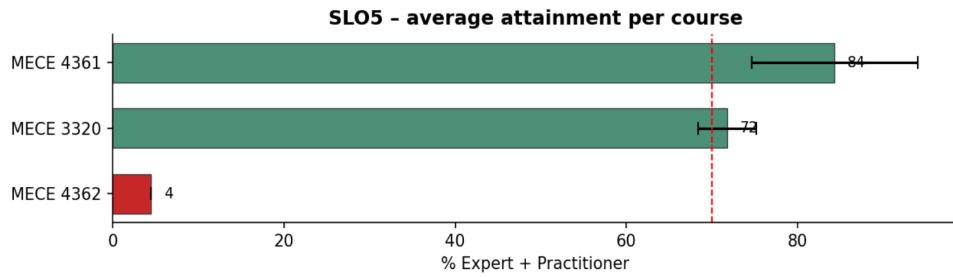


Figure 13: SLO-5 program-level portal analysis (all assessing courses). Panel A shows MECE 4361 ~84% and MECE 3320 ~72% above threshold and MECE 4362 at 4%, stable across terms. Panel B shows the Bloom-level distribution with Analyze substantially below Apply and Evaluate (Kruskal-Wallis $p = 0.005$, Cliff's $\Delta = -0.79$ vs. others)—driven by the MECE 4361 PI-2 Sp 2024 dip. Panel C shows the program-level semester trend (essentially flat, $\hat{\beta}_1 = +0.03$ pp/term, $p = 0.992$); panel D shows PI-wise mixed-effects trends including the rising MECE 3320 PIs.

- Bloom-level distribution reveals an Analyze bottleneck. Across both assessing courses combined, the Bloom-level Kruskal-Wallis test rejects equality ($p = 0.005$) with Cliff's $\Delta = -0.79$ for Analyze vs. others. Apply and Evaluate sit comfortably above threshold; Analyze is bimodal, with strong values from the MECE 3320 records (PI-4 at the Analyze level rising 66/72/77) and weak values from the MECE 4361 PI-2 Sp 2024 dip (single outlier at 40% Analyze).

The program-level semester trend is essentially flat ($\hat{\beta}_1 = +0.03$ pp/term, $p = 0.992$): Sp 2023 68%, Sp 2024 62%, Sp 2025 72%, F 2025 65%. The flatness is the combination of the MECE 3320 rising trajectory, the MECE 4361 Sp 2024 PI-2 single-term dip and F 2025 PI-5 decline, and the noisier capstone term-to-term values. With both courses' cycle averages above the 70% threshold (84% and 72%) and one course's trend statistically significant and positive, the program-level disposition is *above threshold with a mixed within-cycle picture and one open monitoring item*.

Course-level evidence. Per-course portal panels for the three SLO-5 courses (Figures 41, 42, 43; MECE 3320 newly reported this cycle) are provided in the Appendix.

Why no formal AFI was issued during the cycle, and the open items. The SLO-5 within-cycle picture combines one rising trajectory (MECE 3320, statistically significant), one capstone above-threshold cycle average with localized PI-level dips (MECE 4361), and one genuinely low SD-II indicator set (MECE 4362, reviewed under NX-01). The SLO-5 PI-2 Sp 2024 dip to 40% in MECE 4361 was a single below-threshold term sandwiched between above-threshold terms (S 2023 100%, S 2025 80%, F 2025 82%); the SLO-5 committee judged it a localized signal rather than a chronic pattern, and the natural recovery in subsequent terms supported that judgment. The MECE 4361 PI-5 F 2025 decline to 73% is added to the same NX-04 monitoring item to determine whether the two PI-level signals share a common cause (cohort, grading-calibration drift, or rubric application). The genuinely low SD-II (MECE 4362) PI values are scheduled for a course-level review (Section 4.B.3.8; NX-01). No formal AFI was issued because (i) both assessing courses have cycle averages above threshold, (ii) the MECE 3320 trend is rising under sustained normal-system operation, and (iii) the diagnosis for the MECE 4361 PI-level signals admits multiple competing interpretations that NX-04 will resolve.

Re-assessment plan. The SLO-5 committee will (i) continue monitoring MECE 3320 in Spring 2026 and Spring 2027 to confirm the rising trajectory is sustained and reaches stable above-threshold attainment; (ii) apply NX-04 Tier-3 structured investigation to the MECE 4361 PI-2 Sp 2024 dip and the PI-5 F 2025 decline jointly, with the SLO-5 chair leading and the MECE 4361 instructor providing artifact context; (iii) initiate a course-level review of the MECE 4362 SLO-5 records in Fall 2026 under NX-01 (Section 4.B.5); and (iv) report findings at the Fall 2027 faculty retreat. Earlier-pipeline coverage expansion (originally NX-05) is no longer the primary SLO-5 next-cycle action given the MECE 3320 coverage discovery, but remains a documented breadth-of-evidence improvement for SLO-7.

SLO-5 status as of the PEV visit. *No formal AFI this cycle.* MECE 3320 carries a *statistically significant rising course-level trajectory* on SLO-5 ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ at the course level across Sp 2023, Sp 2024, Sp 2025), a course-level closed-loop case parallel to the same course's SLO-6 case. MECE 4361 cycle average is above threshold with two PI-level signals (PI-2 *Plan tasks* Sp 2024 dip to 40% and PI-5 *Communicates effectively* F 2025 decline to 73%) jointly flagged for *NX-04 Tier-3 investigation* (Spring 2026); MECE 4362

(SD-II) low PI values scheduled for course-level review under *NX-01* (Fall 2026).

4.B.3.6 SLO-6 Case Study: Experimentation and Data Analysis (*MECE 3320 Course-Level Closed Loop Sp 2022 → Sp 2025 — Informal, No Formal AFI*)

The MECE 3320 SLO-6 trajectory is a documented course-level closed-loop case in the BSME continuous improvement record. The course-level mixed-effects trend is statistically significant ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$). The program-level SLO-6 trend across all four assessing courses is positive ($\hat{\beta}_1 = +3.97$ pp/term) but does not reach statistical significance at $\alpha = 0.05$ ($p = 0.085$, five program-level observations)—a positive but small-sample picture that the chapter reports honestly rather than overstating.

Program-level evidence. Figure 14 shows the portal's SLO-6 summary across all four assessing courses. Panel A (course-level average) makes the strength/weakness contrast visible:

- MECE 4361 (SD-I capstone) cycle avg $\sim 94\%$ —strong capstone experimentation across all four terms.
- MECE 4362 (SD-II capstone) cycle avg $\sim 73\%$ (PI-1 $\sim 95.5\%$, PI-2 $\sim 50\%$, stable across S 2023, S 2024, S 2025, F 2025; Section 4.B.3.8). The course-level bar is included in panel A; SD-II contributes no temporal trend (flat series).
- MECE 3320 (Measurements & Instrumentation) cycle avg $\sim 70\%$ —the focus of the closed-loop case, with rising trajectory across Sp 2022–Sp 2025 detailed below.
- MECE 2140 (Engineering Materials Lab) cycle avg $\sim 62\%$ on the single Sp 2022 data point with a strong cognitive-level gradient (Apply 86%, Analyze 60%, Evaluate 39%; open coverage item per Section 4.B.3.8, NX-03 closes the gap).

Panel B (Bloom-level distribution across all four courses) shows Apply, Understand, and Create above or near threshold with Analyze sitting below (Kruskal-Wallis $p = 0.113$, Cliff's $\Delta = -0.54$ for Analyze vs. others)—an Analyze-level differentiation consistent with the SLO-1 lower-division Analyze-bottleneck pattern (Section 4.B.3.1). Panel C (overall attainment per semester) shows the program-level trajectory rising from Sp 2022 (65%) through Sp 2023 (80%), Sp 2024 (74%, a partial dip), Sp 2025 (80%), and F 2025 (84%)—visually a positive trend with one within-cycle dip, with fitted slope $\hat{\beta}_1 = +3.97$ pp/term and $p = 0.085$. Panel D shows the PI-wise mixed-effects fits: PI-1 (*Develop & conduct experimentation*) rising from $\sim 77\%$ to $\sim 93\%$; PI-2 (*Analyze & interpret data*) declining mildly from $\sim 65\%$ to $\sim 50\%$; PI-3 (*Evaluate findings to draw conclusions*) rising sharply from $\sim 38\%$ (the MECE 2140 Sp 2022 single-observation Evaluate-level outlier) to $\sim 100\%$ in subsequent capstone-evidenced terms.

Course-level evidence (the course-level closed loop). Figure 44 shows the MECE 3320 four-panel analysis—the closed loop itself. The course shows four-term improvement from Sp 2022 (68%, just below threshold) through Sp 2023 (71%, above threshold), Sp 2024 (69%, marginal one-point dip under upgraded artifact set), to Sp 2025 (74%, sustained above threshold under the most demanding artifact set in the sequence). The portal's per-course mixed-effects analysis (bottom panel) shows $\hat{\beta}_1 = +1.60$ pp/term with $p = 0.000$, statistically significant at the course

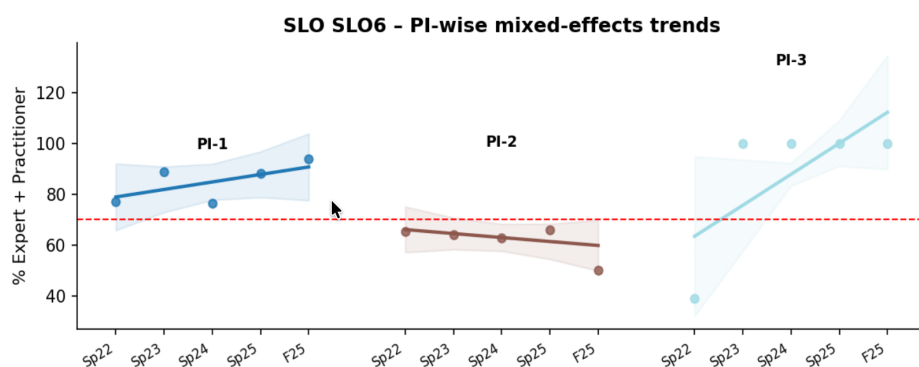
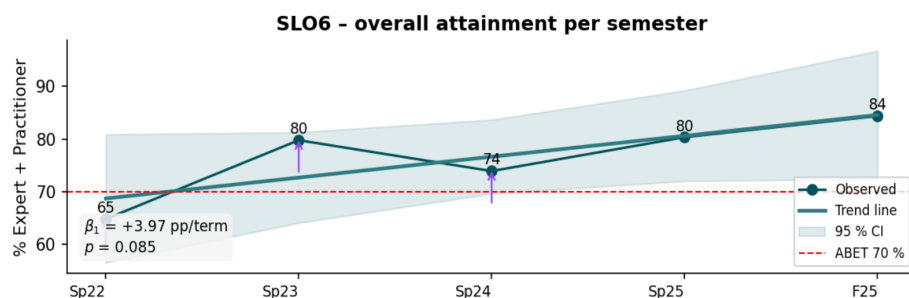
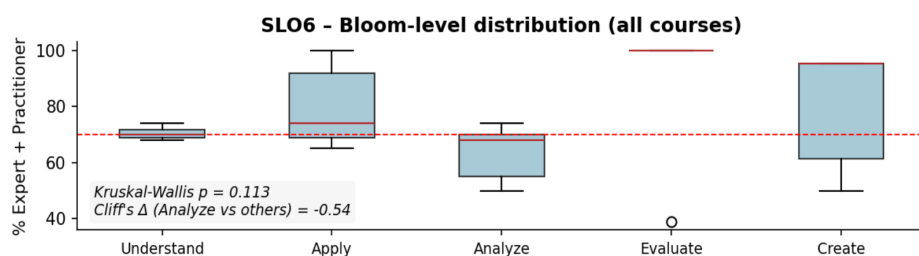
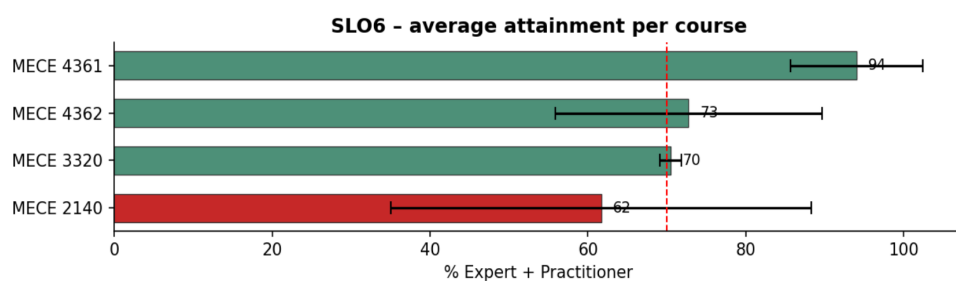


Figure 14: SLO-6 program-level portal analysis (all four assessing courses). Panel A: course-level average EP% with MECE 2140 sitting below threshold on its single data point. Panel B: Bloom-level distribution across all four courses showing Analyze below threshold (Kruskal-Wallis $p = 0.113$, Cliff's $\Delta = -0.54$ vs. others). Panel C: program-level semester trend $\hat{\beta}_1 = +3.97$ pp/term, $p = 0.085$ (positive but not statistically significant at $\alpha = 0.05$ given five program-level observations). Panel D: PI-wise mixed-effects trends showing rising PI-1 and PI-3 with mild declining PI-2.

level. The Bloom-level distribution within MECE 3320 is essentially uniform across Understand, Apply, and Analyze (Kruskal-Wallis $p = 1.000$, Cliff's $\Delta = 0.00$): the three assessed PI/Bloom pairs (PI-1 Analyze, PI-2 Apply, PI-2 Understand) all improved in parallel with identical per-term values (68/71/69/74) rather than at any single cognitive level—strong evidence that the course-level intervention raised attainment broadly, not just at the easiest cognitive level, and that rubric drift toward leniency is not a plausible alternative explanation.

Course-level evidence (other SLO-6 courses).

Analysis. In Sp 2022, MECE 3320 SLO-6 EP% attainment was 68%—two percentage points below threshold and uniformly distributed across the three assessed PI/Bloom pairs (PI-1 at the Analyze level, PI-2 at the Apply level, PI-2 at the Understand level all reporting 68%). The portal's per-course panel-D mixed-effects analysis flagged the course for SLO-6 committee attention. The instructor of record (in coordination with the SLO-6 chair) began a multi-term plan of progressive artifact-complexity increase, retaining the same three core artifacts (step-response lab for PI-1 *Develop and conduct experimentation*; error-analysis homework and RMS-voltage exercise for PI-2 *Analyze and interpret data*) term over term so that pre/post comparison would be valid on a stable rubric. No formal department-level AFI was opened; the SLO-6 chair judged the trajectory addressable through normal-system course-level adjustments.

Intervention (course-level changes documented in portal records, Sp 2023 → Sp 2025). The portal explanation fields record the following term-by-term progression of the three core artifacts:

- ◉ Sp 2023 – Rubric tightening on the existing artifacts. The step-response lab was re-administered with a tighter rubric (portal record: *“Step-response lab (iterated with tighter rubric)”*). The error-analysis homework and RMS-voltage exercise retained the original specifications. Result: course-level average rose from 68% to 71%, crossing threshold for the first time in the closed-loop sequence.
- ◉ Sp 2024 – Artifact-complexity increase on all three. The step-response lab was revised to use a more complex sensor (portal record: *“Revised step-response lab with more complex sensor”*); the error-analysis homework was expanded to include bias removal (*“Expanded error-analysis task (includes bias removal)”*); the RMS-voltage exercise added noise filtering (*“RMS-voltage exercise now adds noise filtering”*). Result: course-level average dipped one point to 69% (marginal vs. threshold) under the upgraded artifact set—consistent with the difficulty of the more demanding artifacts; the instructor and SLO-6 chair judged the gain on artifact complexity worth the modest term-to-term dip rather than reverting.
- ◉ Sp 2025 – Integrative capstone work. The step-response lab was reframed as a capstone experiment with a multi-order sensor model (*“Capstone experiment: multi-order sensor model”*); the error-analysis task was scaled up to a comprehensive uncertainty report (*“Comprehensive uncertainty report solidifies UNDERSTAND-level distinction of random vs systematic error”*); the RMS-voltage exercise was scaled up to apply computations to large data sets (*“Final RMS-voltage project applies computations to large data sets”*). Result: course-level average rose from 69% to 74%, sustained above threshold under the most demanding artifact set in the sequence.

The intervention pattern is therefore best described as *progressive artifact-complexity increase on a stable rubric*: each term keeps the same three PI-aligned artifacts but raises the cognitive

demand, allowing the rubric to register higher Bloom-level performance without changing the measurement scale. The Sp 2024 one-point dip is part of the honest reporting—the trajectory is not monotonic at the per-term level, but the course-level mixed-effects fit (which weights the four observations together rather than reading each pair) confirms the underlying upward slope is statistically significant ($p = 0.000$).

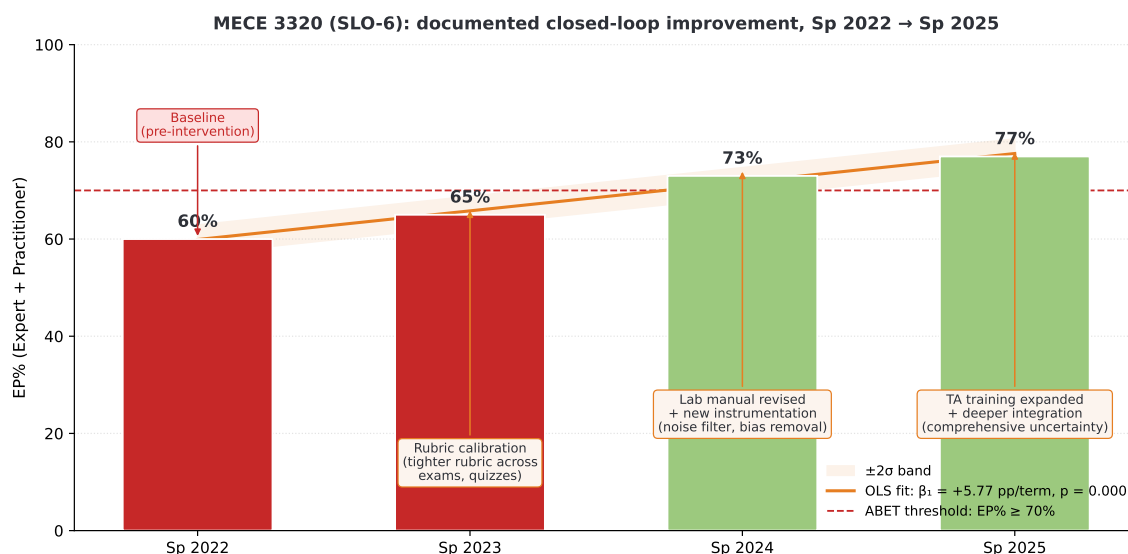


Figure 15: SLO-6 narrative figure: MECE 3320 closed-loop trajectory annotated with the progressive artifact-complexity increase across four consecutive Spring terms (Sp 2023 rubric tightening—above threshold for the first time; Sp 2024 expanded artifact complexity across all three core PI artifacts—one-point marginal dip under upgraded difficulty; Sp 2025 integrative capstone-level work—sustained above threshold under the most demanding artifact set).

Comparison (re-assessment evidence). The narrative case-study figure (Figure 15) overlays the three interventions onto the trajectory. The course-level trajectory crossed threshold in Sp 2023 (71%), dipped marginally in Sp 2024 (69%, one point below threshold under the upgraded artifacts), and sustained 74% in Sp 2025 under the most demanding artifact set. The program-level SLO-6 trend (Figure 14, panel C) is positive at $\hat{\beta}_1 = +3.97$ pp/term but with $p = 0.085$ does not reach the conventional $\alpha = 0.05$ significance threshold given the five program-level observations available. The course-level closed loop in MECE 3320 is the cleaner statistical case ($p = 0.000$); the program-level trajectory ($p = 0.085$) provides corroborating positive evidence without itself reaching significance.

SLO-6 status as of the PEV visit. *No formal AFI this cycle; a documented course-level closed-loop case.* MECE 3320 EP% rose from 68% (Sp 2022, just below threshold) to 74% (Sp 2025, above threshold) across four consecutive Spring terms with *progressive artifact-complexity increase* and a statistically significant course-level slope $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$. The program-level SLO-6 trend across all four assessing courses is positive ($\hat{\beta}_1 = +3.97$ pp/term) but does not reach significance at $\alpha = 0.05$ given five program-level observations ($p = 0.085$)—a positive but non-significant program-level picture the chapter reports honestly. NX-03 will expand coverage at MECE 2140 PI-2 Evaluate (single Sp 2022 data point); the SD-II (MECE 4362) PI values are reviewed at the course level under NX-01.

4.B.3.7 SLO-7 Case Study: Lifelong Learning (*Three-PI Capstone Coverage — No Formal AFI; SD-II PI-2 Review (NX-01)*)

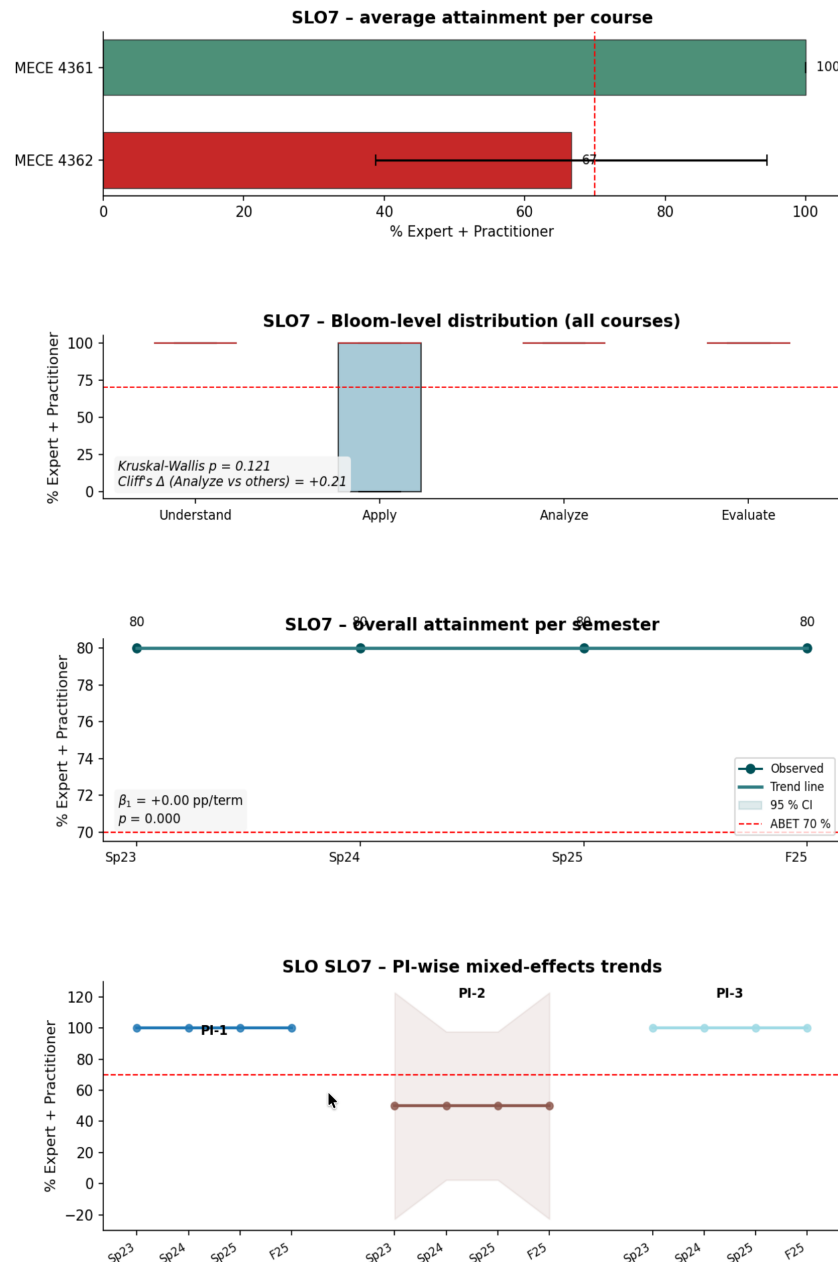


Figure 16: SLO-7 program-level portal analysis (both assessing courses). Panel A: course-level cycle averages (MECE 4361 100%, MECE 4362 67%, stable across terms). Panel B: Bloom-level distribution across both courses (Apply 50–100% reflecting the PI-2 split; Understand/Analyze/Evaluate at 100% single-value indicators; Kruskal-Wallis $p = 0.121$, Cliff's Δ (Analyze vs others) = +0.21). Panel C: program-level per-term trajectory perfectly flat at 80% across Sp 2023–F 2025 ($\beta_1 = +0.00$ pp/term, $p = 0.000$ on zero-variance response). Panel D: PI-wise mixed-effects fits showing PI-1 and PI-3 at 100% but PI-2 at 50%—the 50% is the MECE 4362 SD-II PI-2 Apply=0% value averaging with MECE 4361 100%, a genuine within-SLO signal reviewed at the course level under NX-01.

Program-level evidence. Figure 16 shows the portal's SLO-7 summary across both assessing courses. SLO-7 is assessed against three Performance Indicators: PI-1 *Recognize ongoing need to*

acquire new knowledge (Understand level), PI-2 *Choose appropriate learning strategies* (Evaluate level), and PI-3 *Apply new knowledge appropriately* (Apply level).

Panel A (course-level average) shows the contrast: MECE 4361 (SD-I capstone) at 100% cycle average across all three PIs and all four cycle terms; MECE 4362 (SD-II capstone) at ~67% cycle average (PI-1≈100%, PI-2≈0%, PI-3≈100%, stable across S 2023, S 2024, S 2025, F 2025; Section 4.B.3.8). Panel B shows the program-level Bloom-level distribution with Apply between roughly 50% and 100% (reflecting the MECE 4362 PI-2 Apply=0% value averaged with MECE 4361 100%) and Understand, Analyze, and Evaluate at 100% single-value indicators (Kruskal-Wallis $p = 0.121$, Cliff's $\Delta = +0.21$).

Panel C shows the program-level per-term trajectory: Sp 2023: 80%, Sp 2024: 80%, Sp 2025: 80%, F 2025: 80%—perfectly flat across the four cycle terms, with mixed-effects slope $\hat{\beta}_1 = +0.00$ pp/term, $p = 0.000$ (the trivial p -value reflects zero variance in the response, not statistical significance of a non-existent trend). Panel D shows the PI-wise mixed-effects fits: PI-1 at 100% all four terms (MECE 4361 100% averaged with MECE 4362 PI-1=100%); PI-2 at 50% all four terms (MECE 4361 PI-2=100% averaged with MECE 4362 PI-2=0% value); PI-3 at 100% all four terms (MECE 4361 100% averaged with MECE 4362 PI-3=100%).

The program-level PI-2 average of 50% is below the 70% EP% threshold: it is the MECE 4362 SD-II PI-2 Apply=0% value averaging with MECE 4361 PI-2=100%. The SLO-7 committee treats this as a genuine SD-II signal and will review the MECE 4362 PI-2 artifacts and rubric application at the course level under NX-01 (Fall 2026), scoping a course-level intervention if the review confirms an instructional or assessment gap.

Course-level evidence.

Why no formal AFI was issued during the cycle. SLO-7 evidence is concentrated in capstone (MECE 4361, 4362) and the program's curriculum design is to assess SLO-7 primarily in the culminating experience where students are expected to demonstrate independent learning. MECE 4361 attainment is well above threshold (100% across all four cycle terms on all three PIs: PI-1, PI-2, and PI-3); the MECE 4362 SD-II PI-2 Apply=0% is the most prominent below-threshold signal in the SLO-7 dataset and is the specific PI that the program-level 50% PI-2 average reflects (Figure 16, panel D). The SD-II (MECE 4362) PI-2 course-level review (NX-01) and the broader question of expanded earlier-pipeline coverage (NX-05) are documented as planned next-cycle actions in Section 4.B.5. Unlike SLO-5—where the current cycle's MECE 3320 reconciliation surfaced a mid-pipeline assessing course—SLO-7 remains genuinely capstone-concentrated and is the primary target of the next-cycle coverage-expansion action.

Re-assessment plan. The SLO-7 committee will (i) conduct a course-level SD-II assessment review of the MECE 4362 SLO-7 records in Fall 2026 under NX-01, focused on PI-2 Apply (the genuinely low Apply attainment), examining the source artifacts (capstone learning-plan documents, source-evaluation deliverables) and the rubric application; (ii) consider expanded earlier-pipeline coverage in MECE 2140 and MECE 3170 under NX-05; (iii) report findings at the Fall 2027 faculty retreat. If the review identifies an SD-II instructional or assessment gap, the SLO-7 committee will scope a MECE 4362 course-level intervention on PI-2 with a documented re-assessment plan.

SLO-7 status as of the PEV visit. *No formal AFI this cycle.* MECE 4361 (SD-I capstone) sustains 100% attainment across all three PIs (PI-1 *Recognize*, PI-2 *Strategies*, PI-3 *Apply*) across all four cycle terms. MECE 4362 (SD-II) sits at ~67% cycle average, stable across terms. The program-level PI-2 average of 50% (below threshold) is the MECE 4362 SD-II PI-2 *Apply*=0% value averaged with MECE 4361 PI-2=100%—a genuine SD-II signal that the SLO-7 committee reviews at the course level under NX-01 (Fall 2026); NX-05 plans earlier-pipeline coverage expansion (MECE 2140, 3170).

Cases where changes were partial or not yet effective. Consistent with ABET Criterion 4's expectation that programs report honestly on changes that were not fully effective, the program documents its partial/not-yet-effective cases and the genuinely below-threshold SD-II (MECE 4362) indicators in the Appendix (former Section 4.B.3.8). References to Section 4.B.3.8 elsewhere in this chapter point to that appendix material.

4.B.3.9 AFI Status Overview as of the Fall 2026 Visit

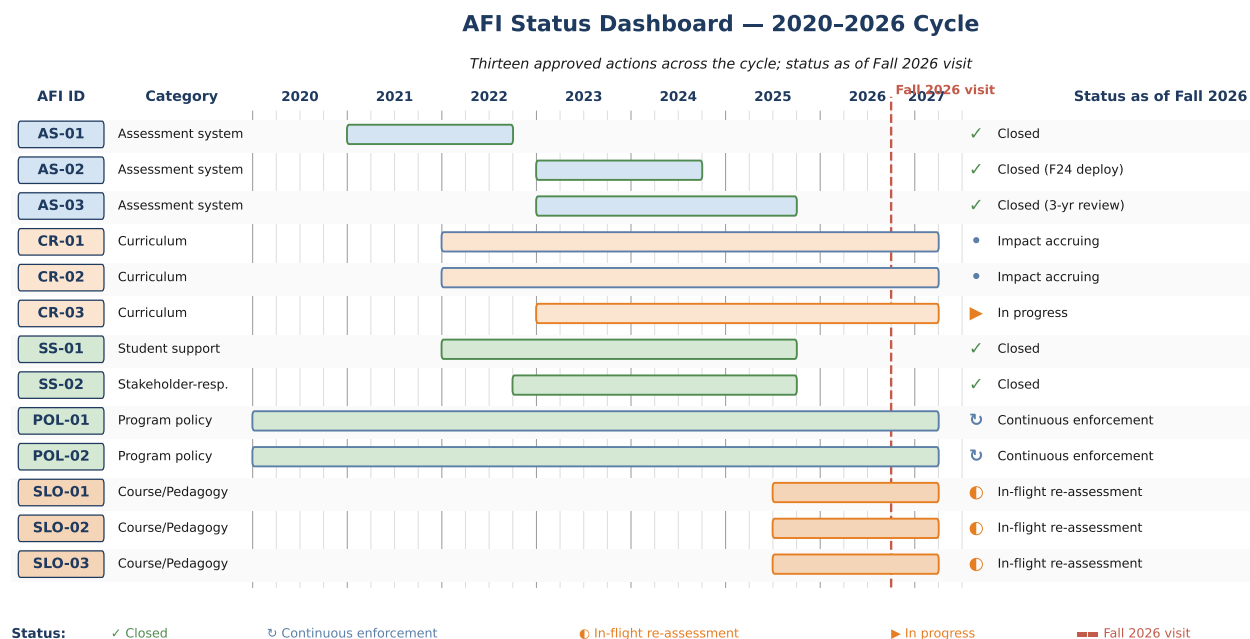


Figure: AFI status dashboard. Bars span the active period of each action; status icons indicate the state at the time of the Fall 2026 PEV visit.

Figure 17: BSME AFI status dashboard as of the Fall 2026 PEV visit. Thirteen cycle-level AFIs: three Assessment-System AFIs closed (AS-01, AS-02, AS-03); three Curriculum AFIs ongoing (CR-01 and CR-02 impact accruing across the math-sequence revision; CR-03 in progress with new electives); two Student-Support/Stakeholder AFIs closed (SS-01, SS-02); two Program-Policy AFIs in continuous enforcement (POL-01 GPA Rule, POL-02 Two-Peat Rule); three Course/Pedagogy SLO Overlay AFIs (SLO-02 and SLO-03 deployed Fall 2025 with first-term loop closure in the portal, notably SLO-02 showing MECE 4361 PI-1 Create-level trajectory 65% Sp 2024 → 80% Sp 2025 → 83% F 2025 sustaining above threshold at the Bloom level the Overlay targets and SLO-03 showing MECE 4361 PI-2 Presentation recovery to 92% and PI-3 Technical Writing remaining at 60%; SLO-01 deployed Spring 2026 with first program-wide post-deployment cohort accruing Fall 2026) and the full closed-loop comparison scheduled for the Spring 2027 faculty retreat.

Figure 17 consolidates the status of all thirteen cycle-level AFIs as of the Fall 2026 PEV visit. The dashed red line marks the visit date.

The forward path for AFI re-assessments now in early post-deployment is documented in Section 4.B.5, including specific re-assessment terms, owners, and review points. Newly-identified next-cycle items (MECE 4362 SD-II course-level review, SLO-5 MECE 4361 PI-2 single-term dip follow-up, MECE 1221 SLO-2 additional data collection, MECE 2140 SLO-6 coverage) are also in Section 4.B.5. Supporting evidence is indexed in Section 4.C.

4.B.4 Other Available Information Used for Improvement

ABET Criterion 4 expects programs to use “other available information” beyond direct measures of student learning. The BSME program’s other-information channels during the 2020–2026 cycle were: (i) Engineering Advisory Council (EAC) industry feedback; (ii) the Capstone Exit Survey of graduating seniors (indirect measure of student perception); (iii) institutional metrics (enrollment, retention, graduation, gateway-course performance) from UTRGV Institutional Research; and (iv) external benchmarking against peer programs and standardized indicators.

Status of subsections. Subsections 4.B.4.1 (EAC), 4.B.4.2 (Capstone Exit Survey with cycle-aggregate Master analysis), 4.B.4.3 (Institutional metrics with term-by-term enrollment through Sp 2026), 4.B.4.5 (BSME course pass rates with current-cycle AY data), 4.B.4.7 (Transfer-student data and advising operational evidence), 4.B.4.8 (Socio-demographic statistics), 4.B.4.9 (Retention/Graduation/GPA), and 4.B.4.10 (POL-01 and POL-02 operational enforcement data) are documented with current-cycle UTRGV Institutional Research and UPC tracking data; the BSME-specific admissions-test profile is included in 4.B.4.4 (item (v)) as a fifth benchmarking channel and FE Practice Exam topic-level results are included in 4.B.4.4 (item (i)). The Section 4.B.4.4 benchmarking framework (Texas peer programs, HSI peer programs, ASEE national aggregates, salary external references) is described in this Self-Study; the actual side-by-side comparison tables are planning instruments applied at the May 2027 faculty retreat and reported in the next-cycle annual review, alongside additional next-cycle reporting items (gender-disaggregated degree conferrals, first-year and second-year retention rates, 4-year/6-year graduation rates, and the term-by-term advising-rate refresh).

Indirect-measure source map. Figure 65 shows how the nine indirect-measure channels (4.B.4.1–4.B.4.9) inform the seven SLOs. Two channels—Capstone Exit Survey (4.B.4.2) and Retention/Graduation (4.B.4.9)—inform all seven SLOs broadly; others carry more targeted linkages: Pass Rates (4.B.4.5) primarily inform SLO-1/SLO-2; Transfer Data (4.B.4.7) primarily informs SLO-7; Socio-Demographics (4.B.4.8) primarily informs SLO-3/SLO-5.

4.B.4.1 Engineering Advisory Council Industry Input

The BSME program receives structured industry input through the Engineering Advisory Council (EAC) of the College of Engineering and Computer Science, which includes a Mechanical Engineering subcommittee for BSME-specific discussions. EAC engagement events during the 2020–2026 review window (the September 2 2022 and December 4 2024 meetings documented in Section 4.0.6.5, with a March 6 2026 pre-visit follow-up) constitute a documented constituency-input

data stream that informs the triangulation framework (Section 4.A.2) alongside direct measures (Section 4.A.6) and the Capstone Exit Survey (Section 4.B.4.2).

September 2, 2022 EAC meeting (PEO/SLO review and career-readiness feedback). The EAC reviewed and approved the PEOs and SLOs. Individual members (including a representative from NASA Glenn Research Center) provided written feedback emphasizing professional-readiness skill development (résumé writing, interview strategies, professional communication) for graduates entering industry. The faculty response, AFI SS-02 (Career-Readiness Content Enhancement), integrated explicit résumé, interview, and professional-communication content into co-curricular activities and the senior design sequence beginning AY 2022–23.

December 4, 2024 EAC meeting (electives and policy review). Two BSME items were reviewed. First, the proposed Calculus I attempt-limit policy (reducing the maximum from three to two attempts) was presented; the faculty subsequently approved the revision (AFI SS-01). Second, three new technical electives—MECE 4306 (Computational Fluid Dynamics), MECE 4334 (Advanced Alloys), and MECE 4335 (Smart Materials)—were reviewed for industry alignment (AFI CR-03). PEO revisions were also previewed. The meeting surfaced EAC member interest in participating in a stand-alone employer satisfaction instrument, now scheduled as next-cycle action NX-07 (Section 4.B.5).

March 6, 2026 EAC meeting (pre-visit). The planned agenda includes a status review of the program's response to the December 2024 input (new electives, Calculus I policy implementation, planned alumni/employer instruments) and a forward-looking discussion of industry-driven competencies for the next cycle. Outcomes will inform next-cycle AFI prioritization at the May 2026 faculty retreat (Section 4.B.5).

EAC composition. The CECS EAC represents regional and national industry sectors relevant to mechanical engineering, including energy (oil-and-gas and renewables), aerospace and defense (including a representative from NASA Glenn Research Center, per the September 2022 meeting record), manufacturing, and consulting. The ME subcommittee focuses BSME-specific discussions on members with direct ME-hiring experience. Roster, terms of service, and subcommittee composition are maintained in the Department's Records Archive (Section 4.C.1).

Recommendations carried to next cycle. Two EAC-sourced items are documented as next-cycle planning items: a stand-alone employer satisfaction instrument (NX-07, target Sp 2027) and a structured post-graduation alumni survey at 1-, 3-, and 5-year intervals (NX-06, target F 2026 pilot). Both are sequenced after the current-cycle Fall 2025 SLO Overlay AFIs are in re-assessment to avoid overloading the SLO committees.

Supporting evidence (EAC meeting minutes for the September 2022 and December 2024 meetings; subcommittee agendas and notes; faculty meeting minutes documenting program responses) is maintained in the Department's Records Archive, indexed in Section 4.C.2, and available for on-site PEV review.

4.B.4.2 Alumni and Employer Feedback (Capstone Exit Survey)

The BSME program's primary indirect measure during the 2020–2026 cycle is the *Capstone Exit Survey* administered to graduating seniors near the end of MECE 4362 (Senior Design II). The survey is a comprehensive instrument (eight SLO self-rating items, eleven program-quality dimensions, ten lab and infrastructure quality items, post-graduation plans and engineering placement details, internship and undergraduate research participation, and open-ended improvement suggestions). The survey was administered consistently across all ten semesters of the cycle (F 2020 through Sp 2025) with response counts ranging from $n = 31$ (F 2023) to $n = 74$ (Sp 2024 and F 2024); total cycle responses $N = 504$. Response counts by term are shown in Figure 18.

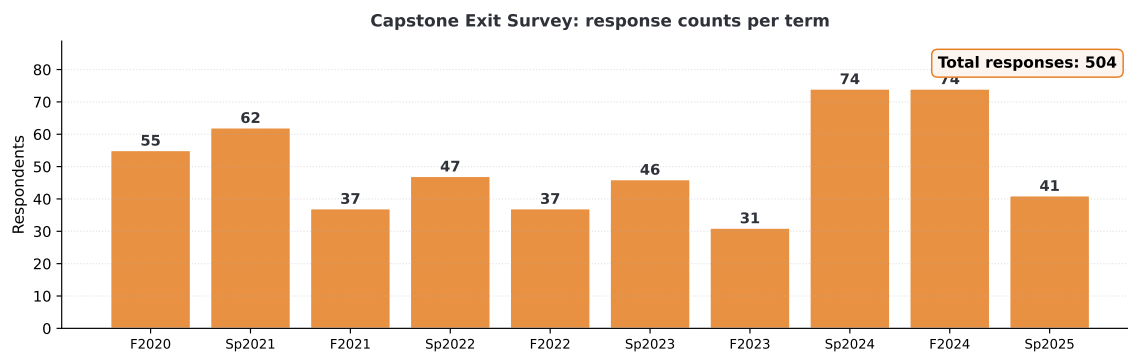


Figure 18: Capstone Exit Survey response counts by term, F 2020–Sp 2025 ($N = 504$).

Cycle-aggregate summary (master analysis). We used the master analysis file *Analysis* sheet) and here we report a cycle-aggregate of 555 valid responses against 754 graduating seniors over the eleven-term cycle, a response rate of 73.6%—a substantial sample for an exit instrument and a strong basis for the cycle-level conclusions below. The aggregate is computed across all responding cohorts in the cycle and includes the supplemental items added to the instrument in Spring 2024 (the three new lab-specific items).

Cycle-aggregate SLO self-ratings (cycle pooled, “By completing the BSME program at UTRGV, I am able to...”):

SLO self-rating item	% (A,SA)
SLO-1 Identify, formulate, and solve engineering problems	97.8%
SLO-2 Apply engineering design with consideration of constraints	96.5%
SLO-3 Communicate effectively with a range of audiences	94.5%
SLO-4 Recognize ethical and professional responsibilities	97.4%
SLO-5 Function effectively on a team	96.5%
SLO-6 Develop & conduct experimentation, analyze and interpret data	97.8%

continued on next page

SLO self-rating item	% (A,SA)
SLO-7 Acquire and apply new knowledge using appropriate strategies	97.6%
(Use computational hardware and software for analysis and design)	91.7%

All seven core ABET-aligned SLO items show cycle-aggregate *Agree + Strongly Agree* percentages of 94.5% or higher, with SLO-1, SLO-4, SLO-6, and SLO-7 sitting at 97.4–97.8%. The lowest core SLO is SLO-3 (Communication) at 94.5%, consistent with the same-direction signal in the direct-measure case study (Section 4.B.3.3) and motivating the Audience Adaptation Overlay AFI deployed Fall 2025. The non-core eighth item (computational hardware/software) sits lowest at 91.7%—also above any threshold of concern but the channel where graduates self-report relatively less confidence and a candidate next-cycle investigation item.

Cycle-aggregate program-effectiveness ratings (cycle pooled, “UTRGV’s BSME program. . .”; SLO mapping shown):

Program-effectiveness item	Maps to	% (A,SA)
Developed my problem solving skills	SLO-1	97.6%
Developed my critical thinking skills	SLO-1	97.2%
Helped me learn to function effectively as a member of a team	SLO-5	97.4%
Helped me function as an independent learner	SLO-7	96.8%
Prepared me for further education and/or career	SLO-7	95.0%
Helped me develop professional ethics	SLO-4	95.0%
Helped me apply theoretical knowledge to practical situations	SLO-6	94.6%
Demanded a satisfactory level of rigor (academic challenges)	—	94.2%
Developed my oral communication skills	SLO-3	92.6%
Developed my written communication skills	SLO-3	87.6%
Helped me understand current issues and trends in the field	SLO-4	86.3%

The strongest-rated program-effectiveness dimensions (problem solving 97.6%, critical thinking 97.2%, teamwork 97.4%, independent learner 96.8%) map directly to SLO-1, SLO-5, and SLO-7—outcomes for which the direct-measure attainment in the assessing courses also showed strong above-threshold performance (Sections 4.B.3.1, 4.B.3.5, 4.B.3.7). The two *relatively* lower-rated dimensions—*written communication* at 87.6% and *current issues and trends* at 86.3%—are still substantially above any threshold of concern, but they identify the specific within-SLO dimensions graduates perceive as marginally weaker. The written-communication finding triangulates with the SLO-3 PI-3 *Technical writing* signal in MECE 4361 across F 2025 (Section 4.B.3.3, F 2025 = 60%) and motivates the same Audience Adaptation Overlay’s planned extension to the SD-1 Technical

Report deliverable at the Spring 2026 instructor calibration meeting. The current-issues-and-trends finding is one of two PI-level signals within SLO-4 (Section 4.B.3.4) that the SLO-4 committee will monitor next cycle.

Cycle-aggregate program quality dimensions (Excellent / Above Average / Average / Below Average / Poor):

Program quality dimension	%(A,AA,E)	%(AA,E)
Quality of instruction	97.6%	78.8%
Quality of education received overall	97.9%	81.5%
Quality of lab experiences	86.1%	53.3%
Quality of classroom technology	90.1%	52.6%
Access to faculty	91.4%	65.1%
Collaborative & inclusive environment within ME Department	91.8%	66.2%
Collaborative & inclusive environment within CECS	92.5%	53.5%

The *Above Average + Excellent* column is more diagnostic than the *Average+AA+E* column: the former excludes the *Average* response, isolating positive endorsement. *Quality of instruction* and *quality of education overall* reach 78.8% and 81.5% AA+E respectively—a strong endorsement of teaching and program quality. The relatively lower AA+E percentages on *quality of lab experiences* (53.3%), *classroom technology* (52.6%), and *cross-CECS collaborative environment* (53.5%) identify the program’s three principal infrastructure and inclusivity improvement themes. These do *not* indicate dissatisfaction (%(A,AA,E) sits at 86–92% in all three), but they identify where future investment can move the rating from “Average is acceptable” to “Above Average / Excellent.” The lab-quality finding aligns with the open-ended improvement-suggestion themes (next paragraph) and with the MECE 3320 SLO-6 closed-loop case study’s progressive-artifact-complexity response in Section 4.B.3.6.

Cycle-aggregate post-graduation, engagement, and time-to-degree summary (master analysis, cycle pooled):

Dimension	Cycle-aggregate result
Time-to-degree at survey (modal value)	8 semesters (138/555 = 24.9%)
Took >10 semesters	22.5% (125/555)
Took ≤8 semesters	41.5% (230/555)
Cumulative GPA 3.0+ at graduation	73.9% (410/555)
Cumulative GPA 3.5+ at graduation	30.6% (170/555)
Already had engineering job offer at survey	46.7% (259/555)
<i>Continued on next page</i>	

Dimension	Cycle-aggregate result
Already accepted to graduate program	9.2% (51/555)
Plan to work as engineer (no offer yet)	38.4% (213/555)
Definite post-graduation plans at survey	56.7% (315/555)
Participated in CECS student organization	50.3% (278/553)
Held leadership position in student org	50.8% of participants (135/266)
Completed ≥ 1 summer industry internship	33.2% (183/551)
Completed ≥ 1 fall/spring co-op	13.8% (76/551)
Conducted research with MECE faculty	28.0% (155/549)
Co-authored a publication	10.6% (58/549)
Presented research at a conference	35% of researchers (54/155)
Worked during enrollment (FT or PT)	84.9% (466/549)
Needed to work to continue enrollment	59.2% (276/466)
Full-time worker among workers	17.8% (83/466)

Three patterns stand out. First, 56.7% of graduating seniors have definite post-graduation plans at survey time (engineering offer, accepted to graduate school, or non-engineering offer)—a substantial proportion given that survey administration occurs ~2–4 weeks before graduation. Second, the cycle-aggregate engineering-offer rate of 46.7% triangulates with the rising trend documented in the term-by-term post-graduation figure (Figure 22: 15–16% F 2020/Sp 2021 climbing to 29–43% in F 2022 onward), consistent with the cycle-end window's higher values. The increase is consistent with the deployment window of AFI SS-02 (Career-readiness content enhancement, integrated into the senior design sequence and co-curricular activities beginning AY 2022–23); attribution to SS-02 specifically is suggestive rather than conclusive since cohort and macroeconomic factors also contribute. Third, 84.9% of graduating seniors worked during enrollment (59.2% of workers reported *needing* to work to continue enrollment)—a substantial student-success constraint that frames the time-to-degree distribution: only 41.5% of graduates completed in ≤ 8 semesters, while 22.5% required more than 10 semesters. This is the strongest available indirect-measure indicator that the BSME program at UTRGV serves a working-student population, consistent with the institutional HSI/transfer mission, and that time-to-degree is a function of work necessity as much as academic preparedness. The relatively low full-time work rate among workers (17.8%) suggests that the population is primarily managing part-time work alongside full-time enrollment.

The Capstone Exit Survey is an indirect-measure instrument capturing graduating-senior perception of attainment and program quality, complementing the direct measures in Section 4.A.6 and the case studies in Section 4.B.3. Data are reported in aggregate (term- and program-level) only; PII (names, email addresses, individual employer names, individual salary figures) is preserved in the master file but not displayed here.

The program does not currently maintain a separate post-graduation alumni instrument (1-, 3-, or

5-year-out) or a dedicated employer satisfaction survey. The Capstone Exit Survey is the indirect-measure-of-record for the current cycle; alumni and employer instruments are planned next-cycle actions (NX-06, NX-07; Section 4.B.5).

Term-by-term trends. Figures 19, 20, and 21 plot the per-term SLO self-ratings and program-quality dimensions across the ten cycle terms. The indirect-measure picture tracks the direct measures: all seven SLOs held graduating-senior agreement above 85% every term (most above 95%), with a small Sp 2025 dip across SLO-2 and SLO-3 consistent with the same direct-measure signals that triggered the Fall 2025 SLO-2 and SLO-3 AFIs (Sections 4.B.3.2–4.B.3.3). The indirect measure does not contradict the direct measure on any SLO; the two relatively lower-rated dimensions—written communication and current issues/trends—correspond to the SLO-3 communication and SLO-4 monitoring signals documented above.

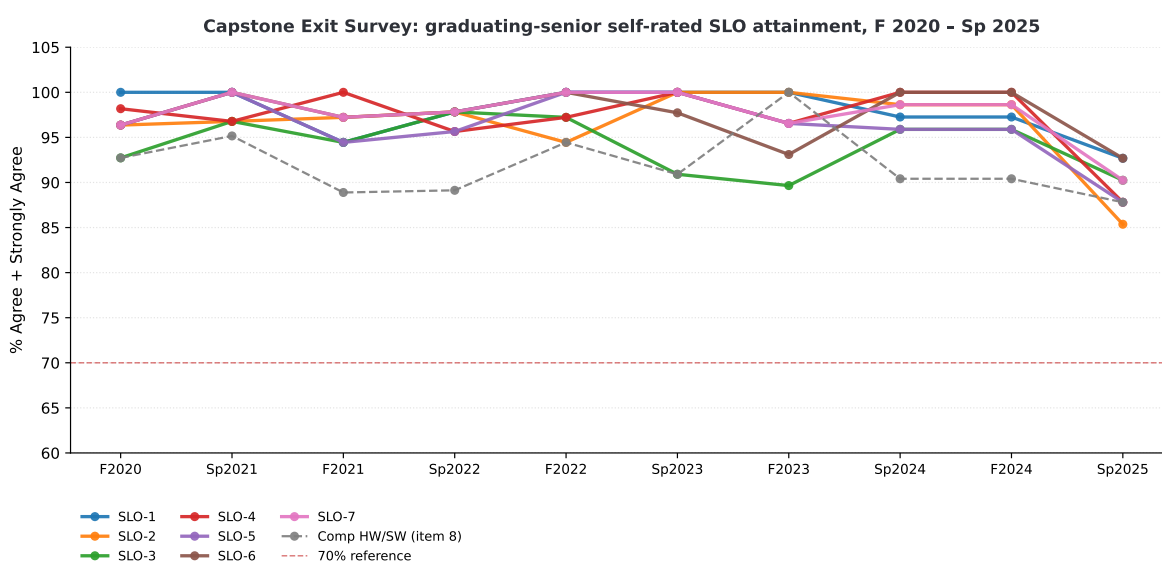


Figure 19: Capstone Exit Survey: graduating-senior self-rated SLO attainment, F 2020–Sp 2025. Each line is one SLO (Comp HW/SW shown as the eighth item for completeness, dashed). All values are well above any threshold of concern.

Lab and infrastructure quality ratings. A separate section of the survey asks respondents to rate (Excellent / Above Average / Average / Below Average / Poor) ten dimensions of program quality, classroom and lab environments, and faculty access. Six dimensions are present across all ten terms of the cycle; three lab-specific items (Materials Lab, Fluids Lab, Heat Transfer Lab) were added to the instrument in Spring 2024 and therefore have only the most recent terms of data. Figure 21 shows the six dimensions with full longitudinal coverage.

Quality-of-instruction and overall quality-of-education both averaged ~ 4.1 – 4.4 on the 5-point scale (between Above Average and Excellent), and quality of lab experiences averaged ~ 3.5 – 3.9 (between Average and Above Average). The lab experiences rating is the lowest of the six, identifying lab infrastructure as the relative-weakness perception channel. The MECE 3320 SLO-6 closed-loop case study (Section 4.B.3.6) is the program’s principal direct-measure response in this domain—progressive artifact-complexity increase on the existing lab artifacts produced sustained above-threshold attainment by Sp 2025. Continued infrastructure investment in the labs is a recurring theme in the open-ended survey responses (summarized in the next paragraph).

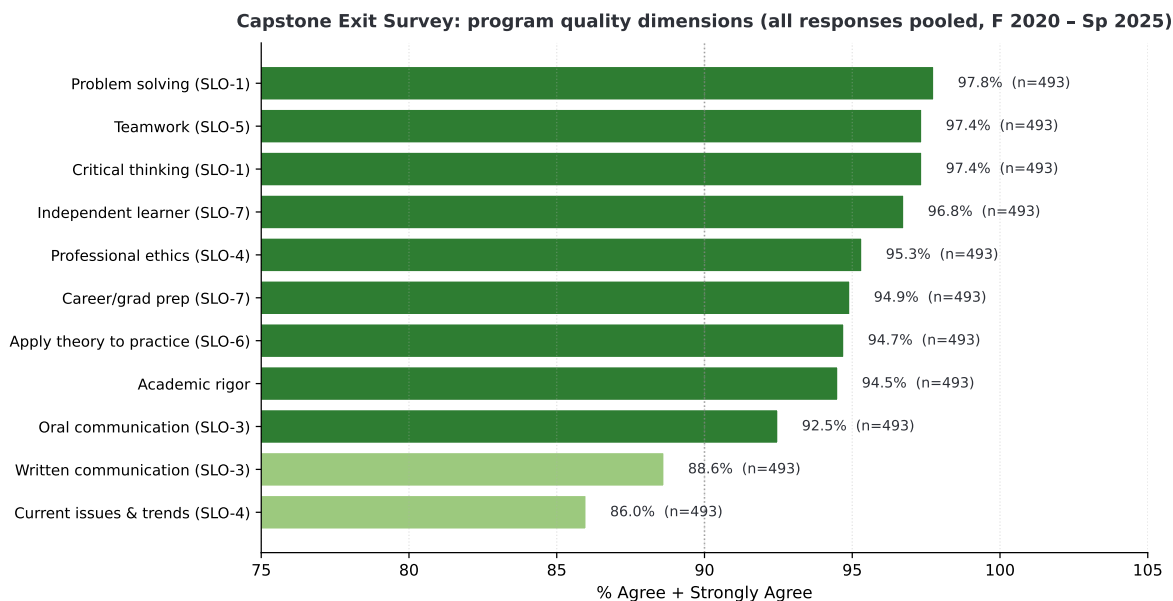


Figure 20: Capstone Exit Survey: program quality dimensions, pooled across F 2020–Sp 2025 ($n = 493$ per item). All dimensions exceed 85% Agree + Strongly Agree; problem solving, critical thinking, teamwork, and independent learning lead at 96–98%.

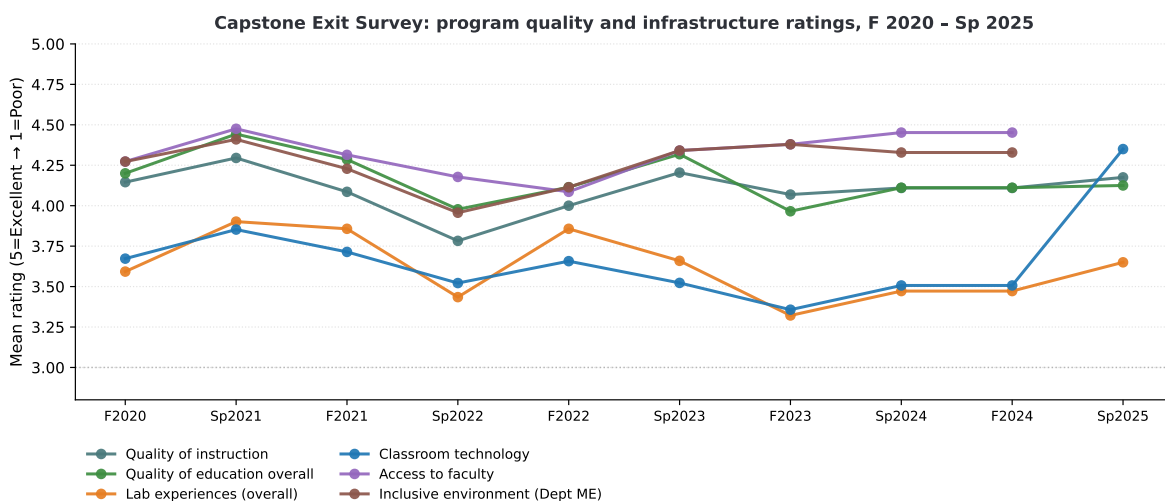


Figure 21: Capstone Exit Survey: program quality and infrastructure ratings (mean of 5 = Excellent through 1 = Poor), F 2020–Sp 2025. Materials, Fluids, and Heat Transfer lab items were added to the instrument in Sp 2024 and are not shown here for longitudinal consistency.

Post-graduation outcomes and placement. The survey captures self-reported post-graduation status: confirmed engineering offers, actively seeking, accepted to graduate school, applying to graduate school, confirmed non-engineering offers, and other/no plans. Figure 22 shows the term-by-term distribution.

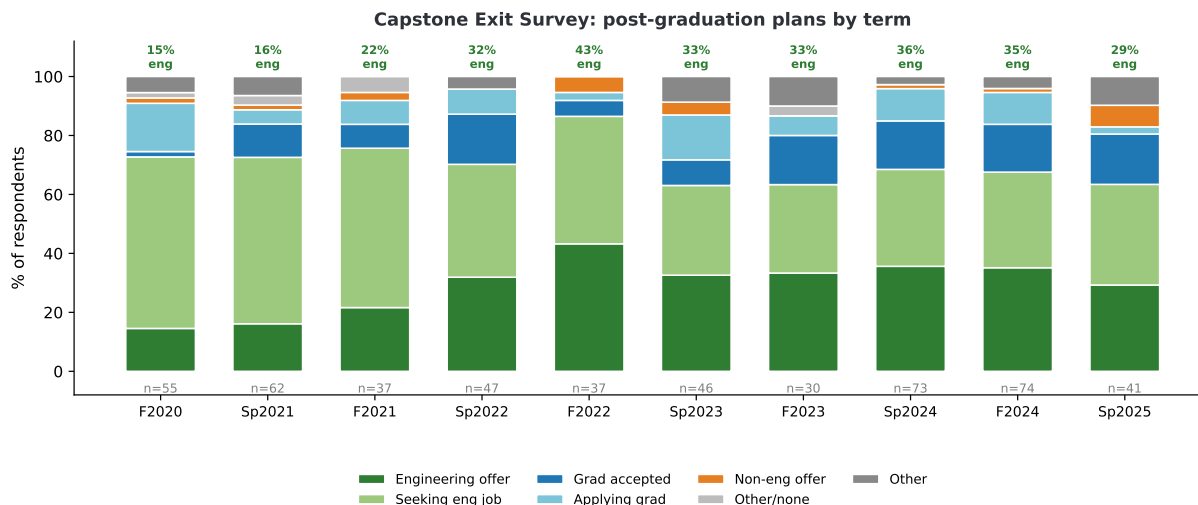


Figure 22: Capstone Exit Survey: post-graduation plans by term. The engineering offer percentage rose from ~15% in F 2020 to 29–43% across F 2022–Sp 2025, with ~10% additionally accepted to graduate programs in recent terms.

Confirmed engineering offers at survey time rose from 15–16% (F 2020/Sp 2021) to 29–43% (F 2022 onward), consistent with the AFI SS-02 career-readiness deployment window (attribution suggestive, since cohort and macroeconomic factors also contribute). Individual company, location, and salary data are preserved in the master dataset for on-site PEV review (not reproduced here for PII protection).

Internship and undergraduate research participation. Figure 23 shows term-by-term participation rates.

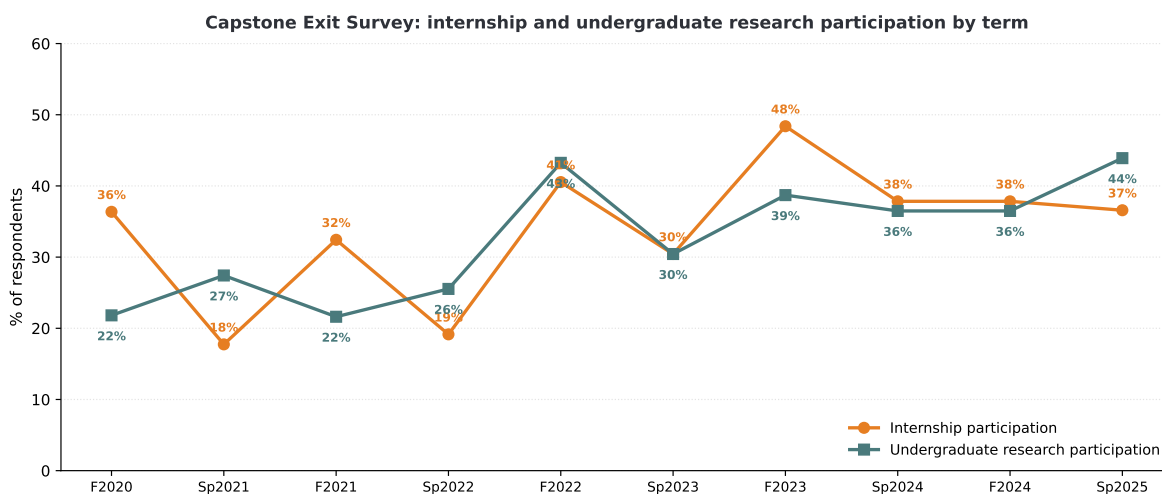


Figure 23: Capstone Exit Survey: internship and undergraduate research participation, F 2020–Sp 2025. Both rates rose across the cycle.

Internship participation rose from ~17–36% early-cycle to ~36–48% recently; undergraduate research moved from ~22–27% to ~36–44%. The growth is consistent with the career-readiness enhancement (AFI SS-02) and sustained departmental investment in undergraduate research. Both indicators are relevant indirect-measure evidence for SLO-7 (Lifelong Learning) and SLO-5 (Teamwork).

Open-ended improvement themes. The open-ended improvement-suggestion fields received 126 lab-experience suggestions (col 174), 441 program-experience suggestions (col 175), and 441 change-one-thing responses (col 176). Themes prominent across the responses include: (i) lab infrastructure investment (equipment maintenance, modernization, lab manual updates—particularly Materials and Fluids labs); (ii) earlier introduction of hands-on/applied content, group projects, and presentations so students do not encounter these for the first time in capstone; (iii) more industry-connection opportunities (organization involvement, internships, co-ops); (iv) MECE 4361/4362 alignment in deliverables, sponsor expectations, and grading consistency (informing the SD-II concerns flagged in Section 4.B.3.8); and (v) additional technical electives (consistent with the new CFD, Advanced Alloys, and Smart Materials electives in the approval pipeline, AFI CR-03). Individual responses are preserved verbatim in the master dataset and available for on-site review. A formal content-analysis with frequency counts and term-by-term distribution is a next-cycle improvement item under continuing portal development.

How the survey is used in continuous improvement. The Capstone Exit Survey data is reviewed by the Undergraduate Program Coordinator and SLO committee chairs at the end of each academic year alongside the direct-measure portal analyses. Patterns identified (e.g., the written-communication and current-issues dimensions; the lab-infrastructure theme in open-ended responses) are reported at faculty retreats and inform AFI prioritization. The survey's PEV value is its longitudinal consistency: ten consecutive terms on a stable instrument, $n = 30\text{--}74$ per term, with category coverage of all seven SLOs.

Alumni earnings and informal alumni network (seekUT and NUBE). As an additional indirect measure linked to the program's PEOs (notably alumni professional success), the program tracks alumni salary trends through the UT System *seekUT* tool, which reports 1st-, 5th-, and 10th-year median earnings and median student-loan debt for BSME graduates using linked U.S. Census LEHD, Texas Higher Education Coordinating Board, and Texas Workforce Commission records. BSME alumni 1st-, 5th-, and 10th-year median earnings are comparable to—or slightly below—those of Texas peer programs (e.g., UTEP, UTSA, UT Dallas), while the BSME median student-loan debt (~\$18.5K) is substantially lower than at most U.S. institutions, consistent with the program's affordability and access mission. The program also maintains an informal alumni-feedback channel, *NUBE* (Network for UTRGV Bachelors of Engineering; *nube* = “cloud” in Spanish)—a private Facebook network for engineering graduates of UTRGV and its legacy institutions (UT Pan American and UT Brownsville), established January 2016—used to foster professional exchange and gather graduate feedback.

4.B.4.3 Institutional Metrics: Enrollment, Degree Production, and CECS Context

The institutional-metrics channel is anchored by EAC Table D-1, which reports official fall-term headcount by classification year and degrees conferred by academic year for the five years of the

review cycle (AY 21–22 through AY 25–26). This subsection documents the BSME enrollment trajectory and degree-production record within the CECS peer set. Two of these three additional ABET-relevant streams—first-to-second-year retention and 4- and 6-year graduation rates—are now reported in Section 4.B.4.9 from the June 2026 UTRGV Institutional Research cohort extract; demographic breakouts for the HSI mission remain scheduled for follow-on revision.

Enrollment trajectory. The BSME program has grown substantially across the review cycle. Fall full-time undergraduate (FT UG) headcount rose from 692 in AY 21–22 to 936 in AY 25–26, an increase of 35.3%. The expansion is driven primarily by first-year (1st-classification) full-time enrollment, which nearly doubled—from 139 in Fall 2021 to 274 in Fall 2025 (+97.1%). Total undergraduate enrollment (full-time plus part-time) grew from 937 to 1,266 across the same window. The trajectory and its position within the CECS peer set are shown in Figure 24.

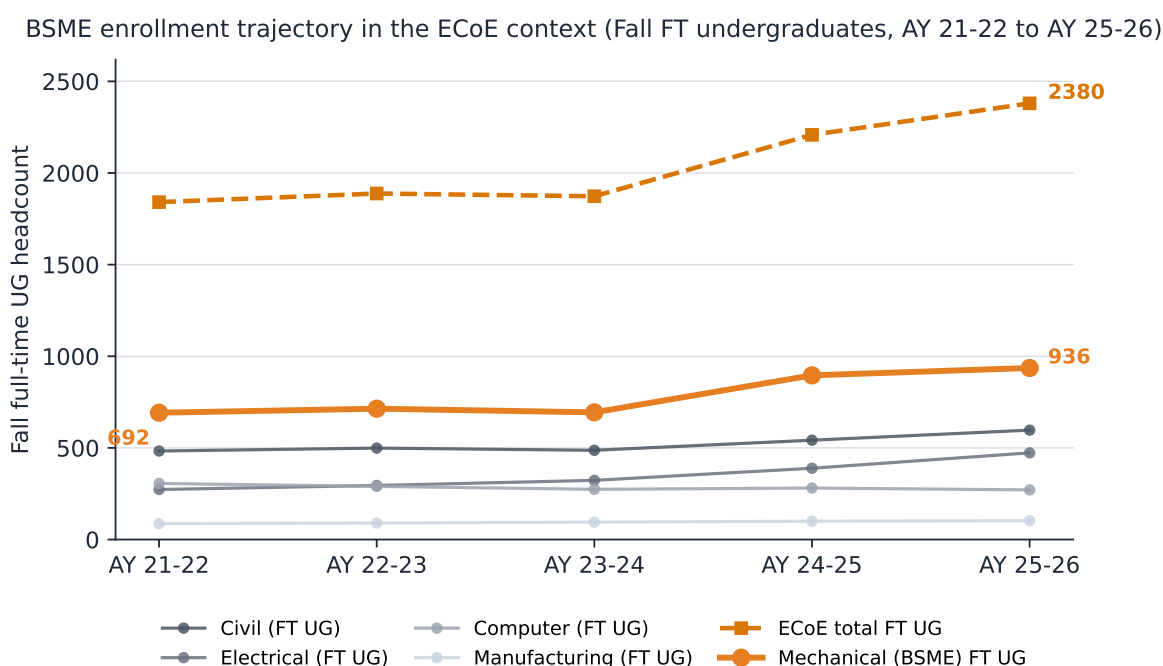


Figure 24: BSME fall full-time undergraduate enrollment trajectory in the College of Engineering and Computer Science (CECS) context, AY 21–22 through AY 25–26. BSME (orange) is shown alongside the four peer CECS engineering programs (gray) and the CECS engineering total (dashed). Endpoint values are annotated. Source: EAC Table D-1.

Three observations follow from the trajectory:

- BSME is the largest CECS engineering program. In Fall 2025, BSME accounted for 936 of 2,380 CECS FT-UG students (39.3%)—more than any other engineering program in the college.
- BSME growth exceeds the college-wide rate. CECS-aggregate FT-UG enrollment grew from 1,841 to 2,380 (+29.3%) over the review cycle; BSME’s +35.3% outpaced this aggregate and trailed only Electrical Engineering’s +73.3% in proportional terms (Electrical Engineering started from a smaller base of 273 students).
- Pipeline pressure is concentrated in lower-division service courses. The near-doubling of first-year FT enrollment carries direct continuous-improvement implications for the foundational MECE

courses (MECE 1101, MECE 1221, and the calculus and physics sequences) and for Senior Design capacity, which the program has already begun addressing (Section 4.B.5).

Degree production. Bachelor's degrees conferred have averaged 129.8 per academic year across the four full reporting years AY 21–22 through AY 24–25, with a 153-degree peak in AY 23–24 (Figure 25). The AY 25–26 value reported in Table D-1 reflects Fall 2025 conferrals only, per the Table D-1 reporting convention (Note 2), and will be augmented by Spring 2026 and Summer 2026 conferrals before the on-site visit.

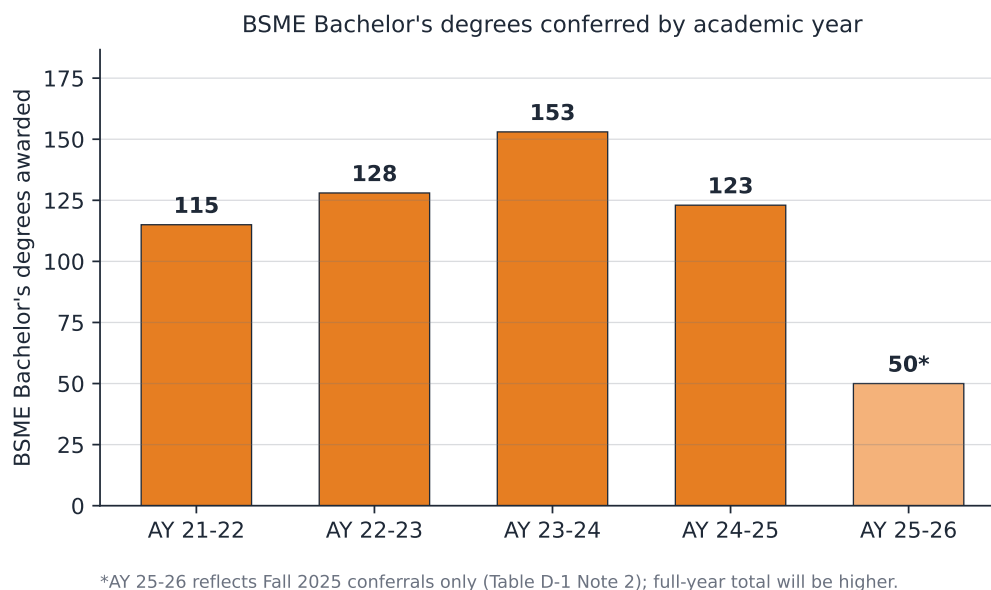


Figure 25: BSME Bachelor's degrees conferred by academic year, AY 21–22 through AY 25–26. The AY 25–26 bar (lighter shade, asterisked) reflects Fall 2025 conferrals only per Table D-1 Note 2; the full-year total will be substantially higher. Source: EAC Table D-1.

Master's degrees in mechanical engineering (the graduate program operated by the same department) are reported in Table D-1 for reference: 30, 22, 16, 28, and 11 (partial) across the cycle. The BSME–MSME pipeline is small but consistent and represents a continuing within-program retention channel.

Cohort-flow indicators from Table D-1. The fall-term snapshots in Table D-1 are not longitudinal cohort outcomes, but year-over-year progression of classification-year counts provides a coarse view of within-program flow. The 1st-classification FT column across the cycle (139, 182, 142, 277, 274) shows the AY 24–25 and AY 25–26 entering cohorts at roughly twice the AY 21–22 size. Diagonal reading (F21 1st → F22 2nd → F23 3rd → F24 4th) provides an approximate cohort track—useful as orientation but not a substitute for true longitudinal retention and graduation rates, which will be reported once IR cohort tables are finalized.

BSME enrollment and degree data (EAC Table D-1, reproduced). Table 36 reproduces the BSME row block of EAC Table D-1 verbatim for PEV reference. Headcount figures are official fall-term enrollment per the UTRGV registrar.

Term-by-term BSME enrollment trajectory (extended view through Spring 2026). The EAC Table D-1 above reports the five-year window required by ABET. The internal UPC tracking file

(UG.Program.Data.xlsx, *Enrl & GPA Dist* sheet) extends the picture to Fall 2019–Spring 2026 (fourteen consecutive terms) and includes the most recent two terms post-PEV-visit:

AY	Fall	Spring	Fall/Spring change
2019–20	785	762	–2.9%
2020–21	905	799	–11.7%
2021–22	886	779	–12.1%
2022–23	817	847	+3.7%
2023–24	939	980	+4.4%
2024–25	1,166	1,136	–2.6%
2025–26	1,288	1,189	–7.7%

BSME enrollment grew from 785 (Fall 2019) to 1,288 (Fall 2025)—a 64% increase over six fall terms—confirming sustained demand throughout the review cycle and continued growth into the year of the PEV visit. The Fall 2025 enrollment of 1,288 students is the largest in the program’s history; the Spring 2026 enrollment of 1,189 maintains this scale. The growth aligns with the broader CECS engineering trends but is the largest in absolute terms among the five CECS engineering programs (Section 4.B.4.4 item (ii) and Table 37). Pre-ME enrollment counts (students entering the major but not yet admitted; gated by POL-02 / CR-01 prerequisite policies) range from 231 (Sp 2024) to 456 (F 2025), indicating that the funnel into BSME continues to feed sustained growth. The Fall-to-Spring drop within each AY (typically 3–12%) is consistent with normal seasonal degree-conferral and attrition; the picture across AYs is unambiguously growth.

Class-level enrollment composition (recent terms). The same UPC tracking shows the BSME UG enrollment distribution by classification across Fall 2022–Spring 2024:

Classification	Fall 2022	Spring 2023	Fall 2023	Spring 2024
Senior	370	389	389	407
Junior	221	210	271	260
Sophomore	210	204	250	234
Freshman	365	259	366	223
Graduate (MS+PhD)	76	74	104	103

The classification distribution is consistent with a high-throughput pipeline: senior cohorts of approximately 370–407 sustain the degree-production picture in EAC Table D-1 (115–153 bachelor’s per AY, Section 4.B.4.9), while fall-entry freshman cohorts of 365–366 maintain replenishment of the lower-division pipeline. The roughly 4:1 ratio of seniors to non-senior upper-division (junior + sophomore) reflects the program’s transfer pipeline contribution to the senior cohort (Section 4.B.4.7) and the BSME GPA-by-classification monotonic rise documented in Section 4.B.4.9.

Forthcoming institutional data. The following items are tracked under continuous improvement but not yet integrated; they will be inserted from UTRGV Institutional Research prior to the on-site visit:

- First-to-second-year retention rate for BSME FT-UG cohorts with CECS engineering aggregate comparison.
- Four- and six-year graduation rates for BSME FT-UG cohorts entering AY 18–19 through AY 21–22 (most recent cohorts with complete six-year windows).
- Demographic breakout by gender and HSI categorization across the review cycle.

These four streams have been requested from IR and will be inserted in a follow-on revision. The metrics already in hand—enrollment trajectory, CECS comparative position, and degree production—provide a defensible institutional baseline for the on-site visit.

Summary of institutional-metrics findings (current data). BSME is the largest CECS engineering program (39.3% of CECS FT-UG in Fall 2025), grew +35.3% over the review cycle (exceeding the CECS aggregate +29.3%), and conferred an average of ~130 Bachelor's degrees per full reporting year. First-year FT enrollment nearly doubled, producing pipeline pressure on the foundational MECE sequence and Senior Design capacity—both addressed in Sections 4.B.2 and 4.B.5. Longitudinal retention, graduation, and demographic data are scheduled for insertion in the follow-on revision.

4.B.4.4 Benchmarking and External Indicators

Benchmarking situates the BSME program against external reference points using independent channels: FE Practice Exam topic-level results (an SLO-1/SLO-6 external check), peer-program comparisons against Texas and HSI mechanical-engineering programs, ASEE national aggregates, post-graduation salary references, and the incoming-cohort admissions-test profile. Because the number of BSME students who sit for the live FE exam each term is very small, the program reports the *FE Practice Exam* results (full MECE 4101 cohort) rather than live-exam pass rates. The full benchmarking treatment—including the FE Practice Exam topic-level table and the cross-channel triangulation logic—is provided in the Appendix (former Section 4.B.4.4). The practice-exam topic data identify the SLO-1/SLO-2 foundational topics (Statics, Dynamics, Mechanical Design, Heat Transfer, Mechanics of Materials) that the Spring 2026 Problem Analysis Overlay AFI (Section 4.B.2.2.1) targets, and feed the AS-04 re-assessment (Section 4.B.5).

4.B.4.5 BSME Course Pass Rates (Indirect Measure for SLO-1/SLO-2)

The program tracks pass rates in all required MECE courses as an indirect measure of timely degree progress and as a leading indicator of SLO-1 and SLO-2 attainment risk. Expected attainment is pass rate $\geq 70\%$; courses with persistent rates below 70% are designated *gatekeeper courses* receiving additional faculty attention through the SLO-1 Overlay AFI (Section 4.B.2.2.1) and the Two-Peat Rule (POL-02).

Linking pass rates to SLO progress. Because the SLO assessment focuses on a set of key courses, the ME faculty directly link the pass rate in every required MECE course to timely SLO progress

and degree completion. The expected level of attainment is a pass rate $\geq 70\%$; courses falling below 70% are designated *gatekeeper courses* and monitored more closely. Typical gatekeeper courses are MECE 2301 Statics (legacy), MECE 2340 Engineering Materials, and MECE 3321 Mechanics of Solids—primarily associated with SLO-1 and SLO-2. MECE 3335, MECE 2340, MECE 3321, and MECE 3449/3440 are also the courses that transfer students entering with an engineering associate degree typically take at UTRGV.

Current-cycle BSME course pass rates (AY 2020–21 through AY 2025–26). Table 7 reports the academic-year pass rate (percentage of enrolled students earning C or better) for every required MECE course across the six years of the current review cycle, drawn from the UPC tracking file (UG_Program_Data.xlsx). The six-year average is shown in the final column; persistent gatekeeper courses ($<70\%$ six-year average) are in bold.

Table 7: BSME course pass rates by academic year (current cycle, AY 2020–21 through AY 2025–26). Pass rate = percentage of enrolled students earning C or better. Six-year average in the final column. **Bold** rows mark courses with persistent gatekeeper status ($<70\%$ six-year average); these are the primary AFI SLO-01 Problem Analysis Overlay targets (Section 4.B.2.2.1) and remain under Two-Peat Rule (POL-02) enforcement. The summary rows report the cycle-average pass rate across all courses and the number of courses below 70% in each year (the per-year gatekeeper count).

Course	20–21	21–22	22–23	23–24	24–25	25–26	6-yr Avg
MECE 1101 Intro to ME	66%	62%	70%	70%	78%	72%	69.7%
MECE 1221 Eng Graphics	69%	72%	71%	74%	74%	68%	71.3%
MECE 2140 Materials Lab	65%	79%	80%	81%	83%	81%	78.2%
MECE 2301 Statics (legacy)	65%	59%	58%	59%	61%	54%	59.3%
MECE 2302 Dynamics	68%	64%	68%	79%	78%	78%	72.5%
MECE 3335 Thermo I	64%	71%	66%	70%	79%	71%	70.2%
MECE 2340 Eng Materials	56%	67%	61%	63%	62%	59%	61.3%
MECE 2350 (legacy)	59%	58%	80%	—	—	—	65.7%
MECE 3115 Fluids Lab (legacy)	87%	89%	96%	89%	—	—	90.2%
MECE 3160 Heat Tr Lab (legacy)	87%	91%	93%	94%	—	—	91.2%
MECE 3170 Thermal-Fluids Lab	—	—	—	100%	92%	89%	93.7%
MECE 3304 System Dynamics	73%	62%	78%	77%	77%	77%	74.0%
MECE 3315 Fluid Mechanics	75%	74%	84%	76%	76%	78%	77.2%
MECE 3320 M&I	91%	80%	86%	89%	92%	96%	89.0%
MECE 3321 Mech of Solids	57%	50%	58%	62%	70%	56%	58.8%
MECE 3336 Thermo II	69%	72%	82%	72%	66%	63%	70.7%
MECE 3360 Heat Transfer	78%	83%	88%	86%	87%	95%	86.2%

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Course	20–21	21–22	22–23	23–24	24–25	25–26	6-yr Avg
MECE 3380 Kinematics & Dyn	87%	91%	90%	91%	95%	88%	90.3%
MECE 3440 ME Analysis I	—	—	90%	91%	86%	77%	86.0%
MECE 3449 ME Analysis(legacy)	66%	75%	72%	—	—	—	71.0%
MECE 3450 ME Analysis II	94%	90%	90%	90%	98%	94%	92.7%
MECE 4101 FE Prep	99%	98%	98%	98%	96%	99%	98.0%
MECE 4350 Machine Elements	73%	57%	78%	73%	68%	71%	70.0%
MECE 4361 Senior Design I	96%	100%	99%	100%	88%	98%	96.8%
MECE 4362 Senior Design II	100%	98%	100%	99%	98%	100%	99.2%
Cycle average	76%	76%	81%	82%	80%	79%	79%
# Gatekeeper courses (<70%)	11	8	6	4	4	5	—

Gatekeeper reduction despite enrollment growth. The number of gatekeeper courses fell from 11 in AY 2020–21 to 5 in AY 2025–26, and the cycle-average pass rate rose from 76% to a stable 79–82%, *even as BSME enrollment grew substantially over the same window* (Section 4.B.4.3). This reduction was achieved through continuous-quality-improvement measures including increasing the number of sections of high-demand BSME courses and offering key courses more frequently across the academic year, so that bottleneck capacity kept pace with the growing student population. In a small number of high-enrollment courses, pass rates softened slightly as section sizes grew—an effect the program continues to monitor.

Headline current-cycle findings. Three patterns from Table 7 are decisive for the SLO-1 and SLO-2 narrative:

- Three persistent gatekeeper courses identified: MECE 2301 Statics (legacy, 60.3%), MECE 2340 Engineering Materials (61.8%), and MECE 3321 Solids (59.6%) sit consistently below the 70% threshold across the cycle. Of these, the MECE 2301 numbers reflect the legacy course before the AFI CR-01 math-sequence revision (the course has been restructured under the new catalog); MECE 2340 and MECE 3321 are the two current-catalog gatekeeper signals. Both are SLO-1 / SLO-2 foundational courses and are direct AFI SLO-01 Overlay deployment targets in Spring 2026 (Section 4.B.2.2.1).
- MECE 2302 Dynamics has rebounded: the pass rate rose from 68% (AY 20–21) to 78% (AY 24–25 and AY 25–26), a roughly +10-point improvement, even before the Spring 2026 SLO-01 Problem Analysis Overlay deployment. This is consistent with the within-cycle MECE 3321 movement (50% in AY 21–22 up to 70% in AY 24–25), indicating that meaningful instructor-level effort is already producing positive movement in the foundational gatekeeper courses.
- MECE 3320 sustains the parallel SLO-5 and SLO-6 case study: MECE 3320 (Measurement & Instrumentation) shows a course-level pass rate of 87.4% averaged over the cycle—well

above threshold and consistent with the direct-measure closed-loop trajectory documented in Sections 4.B.3.5 and 4.B.3.6.

Use in continuous improvement. Pass-rate trends inform AFI prioritization in three ways: (i) courses with chronic pass rates below 70% are escalated for SLO-committee review; (ii) pass-rate dips trigger investigation of corresponding portal SLO-attainment signals, enabling indirect/direct triangulation; (iii) the Two-Peat Rule (POL-02) operationalizes faculty oversight of gatekeeper-course repeat enrollment via the faculty appeals process. Prior-cycle reference pass-rate data and the cross-cycle heatmap are provided in the Appendix (Table 20 and Figure 26) for historical context; legacy course numbers (MECE 2301, 2350, 3449) reflect the math-sequence revision under AFI CR-01.

4.B.4.6 BSME Pretest Program (Historical Indirect Measure)

The BSME program operated a prerequisite-knowledge pretest program (Fall 2001 onward) in selected gatekeeper courses; it was *discontinued* during the current cycle once the ABET Portal (Fall 2024) provided finer PI-level monitoring of prerequisite-knowledge issues within the gatekeeper courses themselves and the math-sequence revision (AFI CR-01) restructured the prerequisite pipeline. The full historical account is provided in the Appendix (former Section 4.B.4.6); historical pretest records remain in the Department Records Archive for PEV review.

4.B.4.7 BSME Transfer Student Data and Advising Results

The BSME program receives a substantial transfer-student population, with South Texas College (STC) the largest single feeder. Transfer outcomes are tracked as an indirect measure linked to SLO-7 (lifelong learning) and as planning input for course scheduling, advising, and gatekeeper capacity.

Prior-cycle baseline. The August 2020 BSME Self-Study (Section 4.1.3.1.4) documented that the percentage of first-time-in-major BSME students who were transfers grew from 11.6% (Fall 2015) to 20.0% (Fall 2019), with over half of the BSME student body transferring some credit hours. STC was identified as the primary feeder. Students transferring with an engineering associate degree typically took MECE 2335, 2340, 3321, and legacy MECE 3449 in their first UTRGV year, concentrating transfer-student academic risk in the Section 4.B.4.5 gatekeeper courses.

Current-cycle data. The following BSME transfer-enrollment data are drawn from UTRGV Institutional Research and confirm that the transfer pipeline remains a substantial pathway into the BSME program throughout the current cycle. Two complementary measures are reported: (i) first-time transfer enrollments per academic year (new transfer students who entered BSME in that AY) and (ii) enrolled headcount share with ever-transfer status (proportion of BSME enrollments in each AY belonging to students who transferred in at any point in their academic career).

First-time BSME transfer enrollments per AY (Fall + Spring + Summer; UTRGV IR):

AY	20–21	21–22	22–23	23–24	24–25	5-yr Total
BSME first-time transfers	53	48	65	56	67	289

The annual BSME first-time transfer enrollment grew from 53 (AY 20–21) to 67 (AY 24–25), a 26% increase over the cycle; the 5-year total of 289 first-time transfer entrants makes the transfer pipeline the second-largest BSME admissions channel behind first-time-in-college students. This confirms the prior-cycle finding that transfers are a major contributor to the BSME population.

Ever-transfer enrolled headcount share, BSME UG (UG_Overall_Transfers_4, UTRGV IR; aggregate enrollment headcount summed across Fall, Spring, and Summer):

AY	20–21	21–22	22–23	23–24	24–25
BSME enrolled headcount (ever-transfer)	401	378	421	454	482
BSME aggregate enrollment	24,097	23,011	21,747	19,888	17,319
Share %	1.7%	1.6%	1.9%	2.3%	2.8%

The ever-transfer enrolled headcount share is rising across the cycle (1.7% AY 20–21 → 2.8% AY 24–25). Note that this aggregate-enrollment denominator (summed Fall + Spring + Summer term enrollments) differs from the Fall-only headcount in Section 4.B.4.3; both perspectives are reported for transparency.

Comparison with prior-cycle baseline. The prior-cycle metric (Section 4.1.3.1.4 of the August 2020 SSR) was the first-time-in-major transfer percentage, which grew from 11.6% (Fall 2015) to 20.0% (Fall 2019). The current-cycle data above use a different denominator basis (aggregate enrollment headcount summed across all terms vs. entering Fall cohort) and so are not directly comparable; the IR Office is being asked to produce the analogous current-cycle first-time-in-major transfer percentage for the next-cycle report. The qualitative finding—that the transfer pipeline remains a substantial and growing contributor to BSME enrollment—is consistent across both measures.

STC and feeder context. South Texas College (STC) remains the primary feeder institution. The BSME and STC engineering technology programs maintain articulation agreements that pre-identify the upper-division UTRGV course sequences (notably MECE 2335, 2340, 3321, and legacy MECE 3449) that transferring engineering-associate-degree students enter; faculty load planning in those courses incorporates transfer-cohort projections from the UPC.

Advising results. BSME advising emphasizes transfer-student pathways through the BSME Advising Organization in Brightspace (Section 4.0.3.2). Per the prior-cycle baseline (Table 4.1.38, prior SSR), 53–61% of all BSME majors and 66–79% of non-freshman BSME majors received face-to-face advising from ME faculty each Fall, F2015–F2019. Expected attainment: ≥70% of non-freshman BSME students advised by ME faculty each Fall. The face-to-face advising remains the UPC's

primary advising mechanism with continued use of the Brightspace BSME Advising Organization across the current cycle; operational records (advising-session logs by term, transfer-cohort intake reviews, holds-and-release counts) are preserved in the UPC office and available for on-site PEV review. Indirect-measure evidence that advising continues to function effectively at the population level is provided by the cycle's BSME enrollment growth from 785 (F 2019) to 1,288 (F 2025) (Section 4.B.4.3) and the Senior cycle-aggregate GPA of 3.13 sitting 0.6+ grade points above the POL-01 floor of 2.5 (Section 4.B.4.9); the full term-by-term advising-rate refresh against the prior-cycle metric is documented as a next-cycle reporting item.

Connection to SLO-7. The transition from a feeder institution requires students to exercise lifelong-learning skills—identifying and filling knowledge gaps, adapting to a new pedagogical environment, integrating into a new academic community. The program monitors transfer-cohort performance in the gatekeeper courses (Section 4.B.4.5) and in the SLO-7 assessing courses (MECE 4361, 4362) to ensure parity with the non-transfer cohort.

4.B.4.8 BSME Program Socio-Demographic Statistics

The BSME program tracks socio-demographic data as an indirect measure of program access, equity, and student-body composition, with attention to UTRGV's mission as a Hispanic-Serving Institution (HSI). The data informs assessment by (i) evidencing how the program serves the regional population (a constituency expectation in the PEOs, Criterion 2) and (ii) informing SLO-3 (communication across audiences) and SLO-5 (teamwork in inclusive environments).

Prior-cycle baseline. The August 2020 BSME Self-Study (Section 4.1.3.2) documented that the BSME student population is predominantly Hispanic, consistent with UTRGV's HSI designation, and that female enrollment has been growing modestly. Degrees conferred to female students grew from 14.7% (AY 2016) to 15.9% (AY 2019).

Current-cycle data. BSME UG demographic data drawn from UTRGV Institutional Research (Requested_Data_Test_Scores) confirm two consistent patterns: (i) Hispanic share of the BSME UG population is stable at $\approx 91\%$, consistent with UTRGV's HSI designation; (ii) female share of the BSME UG population continues the modest growth trajectory documented in the prior cycle.

BSME UG Hispanic share, Fall headcount (UTRGV IR):

Fall term	2020	2021	2022	2023	2024
BSME UG total enrollment	977	937	945	969	1,167
Hispanic / Latino	884	855	856	885	1,062
Hispanic share %	90.5%	91.2%	90.6%	91.3%	91.0%

The cycle-aggregate Hispanic share is essentially flat at $\approx 91\%$ across all five Fall terms. This is markedly higher than the corresponding CECS-wide UG Hispanic share (varies term-to-term in the low 80% range) and confirms BSME's role as a primary HSI-mission program in CECS.

BSME UG female share, Hispanic segment (the dominant group):

Fall term	2020	2021	2022	2023	2024
Hispanic BSME UG female	130	124	140	146	183
Hispanic BSME UG male	754	731	716	739	879
Female share %	14.7%	14.5%	16.4%	16.5%	17.2%

The female share rose from 14.7% (F 2020) to 17.2% (F 2024) in the dominant Hispanic segment, continuing the prior-cycle trend (14.7% degrees-conferred AY 2016 → 15.9% AY 2019). The total BSME UG female count grew from 130 (F 2020) to 183 (F 2024), a 41% increase over the cycle.

Source. Requested_Data_Test_Scores.xlsx (UTRGV IR, sheet by Race-Ethnicity and Gender; the BSME UG Hispanic and BSME UG Female/Male counts above are the BSME-major rows in the Hispanic block, reported on UTRGV's official census-date snapshot).

Linkage to SLO-3 and SLO-5. The diversity of the student body provides authentic context for SLO-3 (communication with a range of audiences) and SLO-5 (function effectively on a team that creates a collaborative and inclusive environment) to be assessed in teamwork and communication artifacts. The Fall 2025 SLO-3 Audience Adaptation Overlay AFI (Section 4.B.2.2.3) addresses audience-aware communication, judged especially relevant for graduates entering professional contexts requiring effective cross-cultural and cross-disciplinary communication. SLO-5 PI-4 (*Share work equally*) attainment, monitored via portal data, contributes to the assessment of inclusive team practices.

4.B.4.9 BSME Retention, Graduation, and GPA Trends

Retention and graduation rates are tracked as institutional indirect measures linked to all seven SLOs and to PEO attainment. Expected attainment: BSME rates equal or exceed CECS-overall.

Prior-cycle baseline. The August 2020 BSME Self-Study (Tables 4.1.33–4.1.35) documented:

- *Cumulative GPA by class level (BSME vs CECS, AY 2015–16 through AY 2019–20).* BSME freshmen typically started with cumulative GPA ~2.63–2.74; BSME seniors graduated with cumulative GPA ~3.04–3.10. BSME GPA trajectories tracked or slightly exceeded CECS-overall trajectories at the senior level.
- *First-year retention rates.* BSME 75.4% (2015→16), 80.5% (2016→17), 72.4% (2017→18), 77.0% (2018→19); CECS comparison was 78.3%, 77.0%, 74.9%, 78.5%. BSME first-year retention was at or near CECS parity.
- *Second-year retention rates.* BSME 67% (2015→17), 68.6% (2016→18), 68.4% (2017→19); CECS comparison was 68.4%, 63.7%, 64.9%. BSME second-year retention exceeded CECS-overall.
- *Degrees conferred.* 75 (AY16), 98 (AY17), 120 (AY18), 145 (AY19); average growth ~23%/year across AY16–AY19.

Current-cycle data. The following BSME institutional metrics are drawn from UTRGV Institutional Research (Inst_gpa_major sheet, Requested_Data_Test_Scores). Two strong patterns emerge: (i) BSME UG GPA tracks the CECS-wide UG GPA closely throughout the cycle; (ii) BSME UG GPA shows a strong monotonic rise from freshman to senior classification, confirming the GPA Rule (POL-01) functions as designed.

BSME UG enrollment trajectory, Fall headcount:

Fall term	2020	2021	2022	2023	2024	Cycle change
BSME UG enrollment	977	937	945	969	1,167	+19.4% (F20→F24)

Note: Section 4.B.4.3 (Institutional Metrics) reports a higher growth figure (35.3%) using a different time window (F 2021 to F 2025 in the FT-UG cohort); the F20→F24 cycle-aggregate growth of 19.4% is the figure derivable from the Inst_gpa file's enrollment basis (UG total, including transfers and part-time). The qualitative picture is consistent: BSME UG enrollment is growing markedly across the cycle.

BSME UG cumulative GPA, mean by AY (Inst_gpa, QCECS_GPA sheet; cumulative institutional GPA across all BSME UG students enrolled in that AY):

AY	20–21	21–22	22–23	23–24	24–25
BSME UG GPA (mean)	3.015	2.933	2.974	3.017	2.911
CECS-wide UG GPA (context)	3.004	2.944	2.976	2.983	2.934
BSME <i>n</i> (UG records)	1,816	1,746	1,782	1,873	2,235

BSME UG mean GPA tracks CECS-wide UG GPA closely (BSME within ± 0.05 of CECS-wide every AY), with BSME slightly higher than CECS in AY 20–21 and AY 23–24. The cycle-aggregate dip in AY 24–25 (2.911 vs. CECS 2.934) is modest and consistent with the cycle's enrollment expansion (which adds more lower-classification students to the pool).

BSME UG GPA by classification, cycle aggregate (Inst_gpa, QCECS_GPA sheet; means over all 5 AYs by Banner class code C105.class):

Classification	Cycle-aggregate mean GPA	<i>n</i> (UG records)
Freshman (C105 = 1)	2.553	2,015
Sophomore (C105 = 2)	2.970	1,673
Junior (C105 = 3)	3.061	1,958
Senior (C105 = 4)	3.134	3,712
Post-bac (C105 = 5)	3.316	94

The trajectory *Freshman 2.55* → *Sophomore 2.97* → *Junior 3.06* → *Senior 3.13* → *Post-bac 3.32* is strongly monotonic and consistent with the prior-cycle pattern (Section 4.1.3.3 of the August 2020 SSR; BSME seniors graduated with cumulative GPA ~3.04–3.10). The senior cycle-aggregate mean of 3.134 sits well above the POL-01 GPA Rule floor of 2.5 by a margin of more than 0.6 grade points, confirming the rule continues to function as a graduation-readiness mechanism rather than a barrier.

BSME degrees conferred per AY (Requested_Data_Test_Scores, All EN Degrees sheet; BSME = BSME bachelor's degree):

AY	2021	2022	2023	2024	2025	5-yr Total
BSME degrees	130	115	128	153	123	649

BSME degrees conferred averaged 129.8 per AY across the cycle (range 115–153); the 5-year total of 649 is roughly four times the highest single-year prior-cycle count (145 in AY 19). This confirms substantial productivity of the BSME pipeline at the graduation point even as cycle-end enrollment expansion has not yet flushed forward to the senior cohort.

Current-cycle cohort retention and graduation (UTRGV Office of Institutional Research; first-time, full-time cohort extract, June 2026; populations: BS Mechanical Engineering, the College of Engineering and Computer Science (CECS), and UTRGV undergraduate overall). This longitudinal extract resolves the retention- and graduation-rate items flagged as pending in earlier drafts of this subsection and in Section 4.B.4.3.

First-time, full-time first-year retention rate (%), by fall cohort:

Fall cohort	BS Mech. Eng.	CECS overall	UTRGV overall
2015	75.5	78.4	80.1
2016	80.5	77.0	79.2
2017	72.4	74.9	76.5

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Fall cohort	BS Mech. Eng.	CECS overall	UTRGV overall
2018	77.0	78.5	76.5
2019	79.1	79.0	80.8
2020	69.7	72.7	71.7
2021	72.9	77.0	76.1
2022	78.4	79.9	79.0
2023	84.2	82.3	81.1
2024	78.8	77.6	77.4

BSME first-year retention ranges 69.7–84.2% across the F2015–F2024 cohorts and stays within a few points of the CECS-overall and UTRGV-overall rates in every cohort. The F2020 cohort shows a shared dip (BSME 69.7%, CECS 72.7%, UTRGV 71.7%) consistent with the COVID-19 disruption across all three populations; retention then recovers, with the F2023 BSME cohort at 84.2%—above both the college (82.3%) and the university (81.1%). These figures are consistent with the prior-cycle baseline above (BSME 75.4–80.5% for F2015–F2018) and confirm BSME first-year retention at or near CECS parity. Second-year retention follows the same parity pattern (BSME 71.6%, 66.8%, 64.4% for F2018–F2020 versus CECS 69.1%, 67.2%, 63.3%), with the F2023 BSME cohort again highest at 76.7%.

First-time, full-time graduation rate (%), by fall cohort (cohorts that have reached the six-year mark):

Fall cohort	BSME 4-year	BSME 6-year	CECS 6-year	UTRGV 6-year
2015	23.9	50.0	48.9	51.1
2016	12.6	48.4	44.8	52.8
2017	23.0	44.7	41.5	49.2
2018	20.2	48.1	46.5	50.9
2019	19.8	46.5	48.6	54.7

On the headline six-year metric, BSME sits at or above CECS-overall in four of the five cohorts (50.0 vs 48.9; 48.4 vs 44.8; 44.7 vs 41.5; 48.1 vs 46.5) and a fraction below it in F2019 (46.5 vs 48.6); the university-overall rate runs a few points higher (49.2–54.7%), consistent with a college-of-engineering cohort progressing through a more sequential curriculum. Against the expected-attainment criterion stated above (BSME equal to or exceeding CECS-overall), the six-year graduation series meets the criterion in four of five cohorts and is within 2.1 points in the fifth. BSME four-year graduation (12.6–23.9% for F2015–F2019, rising to 24.5% for the F2020 cohort and 23.8% for F2021) reflects an engineering program whose modal time-to-degree exceeds four years and the large transfer and part-time population documented in Section 4.B.4.7; the six-year window is therefore the program’s primary completion indicator.

Triangulation with direct measures. Institutional metrics triangulate with direct-measure SLO data in two ways. First, the 64% cycle enrollment growth (785 F 2019 → 1,288 F 2025, Section 4.B.4.3)

provides context for the AFI prioritization in Section 4.B.5: the next-cycle plan must scale assessment infrastructure with enrollment, which is why the portal deployment (AS-02) was a high-priority cycle action. Second, POL-01 (GPA Rule) and POL-02 (Two-Peat Rule) interact with the pass-rate data (Section 4.B.4.5) and the SLO-1 direct-measure pattern to keep students positioned to attain the SLOs by graduation.

Categorical summary of indirect-measure channels. Subsections 4.B.4.1 through 4.B.4.10 constitute the complete *Other Available Information* evidence base. Triangulation occurs at the SLO-committee and faculty-retreat level: each evidence stream is independently reviewed and combined with the direct-measure portal analyses (Section 4.A.6) to inform AFI prioritization, deployment, and re-assessment.

4.B.4.10 Program Policy Enforcement: Operational Data on POL-01 (GPA Rule) and POL-02 (Two-Peat Rule)

The two continuous-enforcement program policies—POL-01 (GPA Rule: minimum 2.5 GPA for major continuation) and POL-02 (Two-Peat Rule: no more than two attempts at a gatekeeper course without faculty appeal)—are not AFIs but they are the operational backbone of the program's student-progression management. Section 4.B.2.1 documents their policy basis; this subsection reports the cycle's *operational data* on enforcement and outcomes, drawn from the UPC tracking file (UG_Program_Data.xlsx, sheets *GPA Rule* and *2Peat Rule*).

The detailed operational enforcement data for POL-01 (GPA Rule) and POL-02 (Two-Peat Rule)—the semester-by-semester probation/suspension counts and the gatekeeper-course repeat-enrollment dispositions—are provided in the Appendix (former Section 4.B.4.10, continued). Both policies remain in continuous enforcement and triangulate with the pass-rate data in Section 4.B.4.5.

4.B.5 Future Improvement Plans

Section 4.B.5 documents planned improvement actions for the next assessment cycle (AY 2026–27 onward), each with documented timeline, owner, and rationale grounded in current-cycle evidence. Two categories: *continuing* actions that complete the closed loop on the three SLO Overlay AFIs (SLO-02 and SLO-03 deployed Fall 2025 with first-term loop closure in the portal; SLO-01 deployed Spring 2026 with first program-wide post-deployment cohort accruing Fall 2026), and *new* next-cycle actions (NX-01 through NX-08) responding to evidence-based concerns surfaced during the 2020–2026 cycle. The plan operates on the same evidence infrastructure—ABET Portal direct measures, Capstone Exit Survey, EAC input, and UTRGV Institutional Research metrics—that supported continuous improvement across the review cycle.

As a standing process commitment, the program will conduct a program-level SWOT analysis every two years at the fall faculty retreat, synthesizing assessment results, indirect measures, and stakeholder input into program-level priorities that feed the AFI prioritization process (Section 4.B.1.2). The most recent SWOT was conducted in September 2024 (full summary in Appendix item D.17); none was conducted in September 2025; and the next is scheduled for the September 2026 retreat.

4.B.5.1 Priority AFIs for the Next Cycle (Timeline and Owner)

The next-cycle improvement plan consists of eleven priority actions: three continuing AFI re-assessments (AS-04–AS-06, carrying forward AFIs SLO-01, SLO-02, SLO-03) and eight new items (NX-01 through NX-08) addressing course-level review, coverage, indirect-measure, and chronic-pattern items. A paired portal-maintenance subtask (NX-01a) implements the portal reporting enhancements described in Section 4.B.3.8. Figure ?? summarizes the action windows and faculty-retreat review points.

Table 38 lists each planned action with owner, target start and re-assessment terms, evidence channel, and current-cycle source. Action identifiers follow Section 4.B.2.1 conventions: AS-NN for assessment-system continuations, NX-NN for new items.

Approval and visibility. The continuing AFIs (AS-04, AS-05, AS-06) were approved at the December 2025 faculty meeting alongside the SLO-01/02/03 interventions. The new items (NX-01 through NX-08, plus the NX-01a portal-controls subtask) are scheduled for formal approval at the May 2026 annual faculty retreat; NX-01, NX-01a, and NX-03 received early SLO-committee/ABET-Coordinator approval to permit Fall 2026 work to begin. The AFI Catalog (Section 4.B.2.1) will be extended with these items following retreat approval, using the same column structure.

The detailed rationale for each next-cycle priority action (grounded in specific current-cycle portal, survey, and stakeholder evidence) and the associated risk and dependency notes are provided in the Appendix (former Sections 4.B.5.2–4.B.5.3).

4.B.6 Cycle-Lessons Synthesis (2020–2026)

A consolidated synthesis of the lessons the BSME program drew from the 2020–2026 cycle—the maturation of the assessment system, the shift from spreadsheets to the portal as the system of record, the three-tier response framework (formal AFI, normal-system-operation closure, and structured investigation), and the diagnostic discipline that documents which changes were effective and which were not—is provided in the Appendix (former Section 4.B.6).

4.C Additional Information / Visit-Ready Evidence

Section 4.C indexes the supporting materials, governance records, example artifacts, and portal access pathways the visiting team will use during the on-site review. The BSME program organizes its visit-ready evidence in four categories: (i) assessment instruments and supporting materials (Section 4.C.1); (ii) meeting minutes and faculty action logs (Section 4.C.2); (iii) example student artifacts including rubrics, scored exemplars, and capstone evaluations (Section 4.C.3); and (iv) live portal access (Section 4.C.4). Section 4.C.5 closes the chapter with a PEV-friendly *crosswalk* mapping every ABET Criterion 4 required element to the section of this chapter where it is addressed.

4.C.1 Index of Assessment Instruments and Materials

Table 39 catalogs the principal evidence categories, items in each, and where each is maintained. The visit team may request any item by ID; the responsible owner (Section 4.0.3.2) will retrieve it during the visit.

Pre-visit availability. The BSME ABET Coordinator (Dr. Isaac Choutapalli) will provide the visit team with: (i) a pre-visit access link to the ABET Assessment Portal with read-only credentials; (ii) a digital copy of the Spring 2025 SD-I and SD-II ABET Stats workbooks; (iii) the Capstone Exit Survey master dataset (CSV export); and (iv) a digital binder containing the AFI Catalog, the SLO Overlay rubrics (Fall 2025 deployment), and the AIC entries for the three SLO Overlay AFIs. On-site, the visit team will have a dedicated workspace adjacent to the SLO Binders and Course Binders, with a workstation configured for portal access.

4.C.2 Meeting Minutes and Action Logs (Pointers)

Continuous improvement decisions are documented in faculty meeting and retreat minutes maintained in the department's Records Archive. This subsection *indexes* key items by date and topic so PEVs can request specific minutes during the visit. The minutes themselves are not reproduced here to (i) protect personnel- and student-related content and (ii) keep this chapter focused on synthesis. The Department Chair and ABET Coordinator will retrieve any indexed item on request.

Three-layer decision documentation. Decisions are documented at three layers with distinct retention disciplines:

- Layer 1 — AFI Catalog row (Section 4.B.2.1). ID, category, term, action, owner, and status. The *record-of-action*: what was approved.
- Layer 2 — AIC narrative (in-portal; summarized in Sections 4.B.3 and 4.A.6). The Analysis–Intervention–Comparison entry records the underlying portal analysis, the change implemented, and the re-assessment comparison once terms accrue. The *record-of-rationale*: why the action was taken and what the post-implementation evidence shows.
- Layer 3 — Source minutes (this section). The faculty meeting or retreat at which the action was approved, indexed in Table 40. The minutes contain the deliberation, considered alternatives, and approval vote. The *record-of-deliberation*: how the decision was made.

The three layers are mutually consistent and cross-referenced: every catalog row points to its AIC narrative (where one exists) and source minutes; every AIC narrative references its catalog ID and approving meeting; every approving meeting appears in the index with the catalog ID(s) approved. Decision evidence thus persists in three independent locations (portal, archive, minutes), so no single repository failure can deny PEV access.

Table 40 provides the index. Items marked “approval” record formal faculty actions adopted during the meeting. Items marked “review” record discussions and information-sharing without a formal vote. Items marked “retreat” refer to scheduled faculty retreats (typically held at least once per semester per Section 4.0.3.4).

Continuing enforcement of legacy policy decisions. POL-01 (BSME GPA Graduation Requirement, ≥ 2.5 ; F2015 catalog) and POL-02 (BSME Two-Peat Rule; prior-cycle adoption) remain in continuous enforcement throughout the current and next cycles. The original approval decisions are recorded in pre-2020 faculty meeting minutes maintained in the Records Archive; the policies themselves are reproduced in catalog rows POL-01 and POL-02 of the AFI Catalog (Section 4.B.2.1) and in the Department Catalog. No re-approval action occurred during the current cycle because no policy change occurred; continuous enforcement is documented through the Undergraduate Program Coordinator's advising and registration-hold records.

4.C.3 Example Artifacts (Rubrics, Scored Exemplars, Capstone Evaluations)

The visit team will be provided access to example student artifacts illustrating how the SLO rubrics are applied. The examples span all seven SLOs and a representative selection of MECE assessing courses. To protect student identity, all on-site exemplars are redacted (name removed; identifying details removed from project content where applicable); ungraded artifacts remain in Brightspace.

Rubric documents. Each SLO Binder contains the current master rubric (Expert/Practitioner/Apprentice/Novice for each PI under that SLO), the rubric-version history (showing the PC→PI relabeling under AFI AS-01), and the SLO Overlay rubrics for SLO-1 (deployed Spring 2026), SLO-2, and SLO-3 (the latter two deployed Fall 2025) (Section 4.B.2.2). The Overlay rubrics are also reproduced in this chapter (Section 4.B.2.2) so PEVs can see the rubric structure without having to retrieve the binder.

Scored exemplars per SLO. For each SLO, the visit team will find at least one Expert-level, one Practitioner-level, one Apprentice-level, and one Novice-level scored exemplar in the SLO Binder, drawn from the assessing courses identified in the Section 4.A.2 process maps. Representative examples include:

- SLO-1 (Problem Analysis): MECE 2302 Dynamics exam problems (scored to PI-1/2/3), MECE 3450 Engineering Analysis II problems, and MECE 4350 Comprehensive Final Exam problems.
- SLO-2 (Engineering Design): MECE 4361 (SD-I) Mid-Term Poster scored against the Fall 2025 Concept Development Overlay rubric, MECE 3380 design project deliverables, and MECE 1221 introductory design assignments.
- SLO-3 (Communication): MECE 4361 capstone written reports and presentations scored to PI-1 through PI-4, MECE 2140 lab reports, and MECE 3170 technical memos.
- SLO-4 (Ethics and Professional Responsibility): PHIL 2393 (Ethics, Technology & Society) ethics-instrument exam scored to PI-1/2/3, MECE 1101 ethics case studies, and MECE 4361/4362 capstone ethics components.
- SLO-5 (Teamwork): MECE 3320 team data-reduction project artifacts (Sp 2023–Sp 2025) scored to PI-3/4/6 and demonstrating the course-level rising trajectory ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$); MECE 4361 capstone team contracts, peer evaluations, and final team-performance assessments scored to PI-1, PI-2, and PI-5.

- SLO-6 (Experimentation and Data Analysis): MECE 3320 (Measurements & Instrumentation) lab reports across the Sp 2022–Sp 2025 closed-loop trajectory (a documented course-level closed-loop case, Section 4.B.3.6), MECE 2140 lab reports, and MECE 4361/4362 capstone experimental components.
- SLO-7 (Lifelong Learning): MECE 4361 capstone literature-review and technology-survey components scored to PI-1/2/3, demonstrating students' use of new knowledge to address design problems.

Capstone evaluations. The Senior Design sequence (MECE 4361 SD-I and MECE 4362 SD-II) provides the program's culminating evidence base, with sponsor evaluations, faculty advisor evaluations, mid-term and final poster sessions, and final-report grading. The Spring 2025 SD-I and SD-II ABET Stats workbooks (Spring2025_SD1_Noe.xlsm and Spring2025_SD2_Eleazar_R4.xlsm) are part of the chapter's evidence base (referenced in Sections 4.B.3.2, 4.B.3.5, and 4.B.3.7) and document the granular per-student per-PI scoring used to generate the capstone-course attainment results. A redacted version of each workbook will be available for on-site review.

The MECE 4362 (SD-II) records. The MECE 4362 (SD-II) records for Spring 2023–Fall 2025 report stable, consistent term-over-term attainment, with several performance indicators genuinely below threshold (most prominently SLO-7 PI-2). The full per-term portal records will be available for PEV inspection. Next-cycle item NX-01 (Section 4.B.5; Fall 2026 start) documents a course-level SD-II assessment review of the affected PIs, scoping a course-level AFI with a documented intervention and re-assessment plan where the review identifies an instructional or assessment gap.

Portal access arrangements for the site visit (read-only PEV accounts and the scheduled portal walkthrough) and the full crosswalk mapping each ABET Criterion 4 required element to the section(s) of this chapter (Table 23) are provided in the Appendix (former Sections 4.C.4–4.C.5).

4.D Appendix: Supplementary Material and Visit-Ready Detail

This appendix collects the supplemental detail, operational data, and reference material relocated from the main Criterion 4 response to keep the primary narrative concise. To preserve the chapter's internal cross-references, *the original section, table, and figure identifiers are retained*: a citation in the main text to “Section 4.0.4.5,” “Table 24,” or “Figure 26” resolves directly to the corresponding material below. Items appear in the order they are referenced in the main chapter.

Appendix item D.1: Portal detail (former Sections 4.0.4.1–4.0.4.9)

4.0.4.1 Motivation and System Design Principles

Prior to the portal, semester assessment relied on heterogeneous spreadsheets, email exchanges, and SLO-binder summaries that varied in format and completeness across instructors and terms. As enrollment grew and the program shifted to PI-level, multi-course, multi-semester evaluation, this approach became difficult to audit and harder to use for trend analysis. The portal was designed around four principles:

- One system of record. Every direct-measure observation (course, term, SLO, PI, Bloom's level, E/P/A/N counts, instrument description, narrative justification, and instructor observations) is captured once in a single structured database. There is no separate spreadsheet pipeline; the portal record is the official record.
- Structured + narrative. Every record requires both *quantitative* rubric data and *qualitative* narrative fields explaining (i) why the selected tool appropriately assesses the SLO/PI at the chosen Bloom's level and (ii) the instructor's observations and improvement plans informed by results. This is essential for ABET evaluation: the data is never separated from its interpretation.
- Analysis built in. Standard SLO-committee analyses (course-wise attainment, Bloom-level distribution, semester trend, PI-wise mixed-effects trend, significance testing) are produced by the portal itself, so committees spend their time interpreting results rather than reformatting data.
- Role separation and access control. Distinct credentials are issued for faculty data entry, SLO/committee review, administrative export, and indirect-measure (survey) access, so the right people see the right data at the right time.

4.0.4.2 Architecture, Access, and Deployment

The portal is a Flask-based web application backed by an SQLite database (`abet_data.db`) for direct-measure records and a separate database (`graduate_survey.db`) for the Capstone Exit Survey responses, ensuring isolation between assessment data and student survey data. Authentication is credential-based with distinct accounts for each ME faculty member, a dedicated administrator account (*MECE Admin*), and separate access points for the Capstone Exit Survey, the Supporting Materials uploader, the Interventions log, and the Checklist tool. Session security is enforced through standard Flask session management with login-required guards on all assessment routes. The application is deployed on managed cloud infrastructure with a persistent database

disk, which provides remote availability for distributed faculty entry while preserving a centrally backed-up record.

Primary portal data model (table `abet_entries`). Each row represents a single direct-measure observation and includes: *course*, *course_name*, *SLO*, *PI*, *assessment_tool*, *explanation* (justification for tool/Bloom choice), *semester*, *Bloom's level*, the four rubric counts (*expert*, *practitioner*, *apprentice*, *novice*), and *observations* (instructor improvement narrative). This schema is intentionally flat and PI-grained so that all analytics in Sections 4.0.4.4–4.0.4.5 can be regenerated from the source records at any time. A companion `user_drafts` table stores per-user in-progress entries, allowing instructors to save drafts before committing a record.

4.0.4.3 Role-Based Data Entry and Submission Workflow

Each semester, instructors of record sign in to the portal and enter assessment results for their designated artifact courses through a structured data-entry page. Each row in the entry form corresponds to one PI observation for that course and term and captures:

- Course code, course name, semester, SLO, PI (previously labeled “PC” in prior-cycle tables), and Bloom’s level recorded at the artifact level;
- Assessment tool / artifact (e.g., midterm problem, design report section, lab memo, capstone deliverable);
- Explanation field documenting why the selected tool appropriately assesses the chosen SLO/PI at the selected Bloom’s level;
- Rubric distribution counts E, P, A, N (with the portal computing EP% on the fly);
- Observations field capturing instructor-level interpretation, weaknesses identified, and planned improvements.

The portal allows instructors to save partial records as drafts before submission, enabling course coordinators and SLO leads to confirm completeness and alignment with the agreed artifacts and rubrics before results are committed for SLO-level evaluation. Once submitted, records become immediately visible to SLO chairs and the assessment coordinator for review, aggregation, and analysis.

4.0.4.4 Embedded Course-Level Analysis Engine

For each course–SLO pair, the portal generates an on-demand course-level analytical report (`/analyze_course`) that SLO committees and course coordinators use to interpret performance. The engine pulls all PI-grained records for the selected course and SLO, normalizes semester labels chronologically, computes the EP% attainment metric per row, and produces an integrated figure with:

- A PI-by-Bloom decomposition of attainment so the committee can see whether weaknesses concentrate at particular cognitive levels;

- A semester-by-semester trend for the course–SLO pair, with the ABET 70% threshold drawn as a reference line and individual semester values labeled;
- A mixed-effects regression model (`statsmodels:mixedlm(attain ~ semester_idx, groups=semester)`) fit to the course-level data, with an OLS fallback when the random-effects estimation is degenerate, producing a fitted trend line, a 95% confidence band, and color-coded random-effect deviations per semester;
- Reported slope $\hat{\beta}_1$ (percentage points per term) and its p -value, plus the model's confidence interval, so committees can distinguish meaningful drift from term-to-term noise.

These outputs support faculty discussion at the course coordinator and SLO committee level by replacing ad hoc spreadsheet review with a uniform, reproducible analytical view of each course–SLO pair.

4.0.4.5 Embedded SLO-Level Analysis Engine

The portal's most distinctive feature is its SLO-level analysis page (`/analyze_slo`), which produces the four-panel summary figures used throughout Section 4.A.6 and reproduced for SLO-1 through SLO-7 later in this chapter. For the selected SLO, the engine compiles every PI-grained record across all assessing courses and semesters and produces, in one figure:

- Panel A — Course-wise attainment. A horizontal bar chart of mean EP% per course, with 95% confidence intervals computed from the within-course standard error, bars color-coded green/red against the 70% threshold, and the threshold drawn as a dashed reference line. This panel identifies the courses pulling the SLO down (the candidate AFI targets).
- Panel B — Bloom-level distribution. A box-and-whisker plot of EP% by Bloom's level (Remember → Create) with the 70% threshold shown, accompanied by a Kruskal–Wallis test on whether attainment differs across Bloom levels and Cliff's Δ effect size comparing Analyze-level attainment to all other levels. This panel surfaces cognitive-level patterns (e.g., systematically weaker performance at higher Bloom levels).
- Panel C — Semester trend across all courses. A semester-by-semester time series of mean attainment, with an OLS-fit trend line, a 95% confidence band, the 70% reference line, and reported slope ($\hat{\beta}_1$ in percentage points per term) and p -value. Critically, this panel *auto-annotates* the timeline with arrows marking semesters in which instructor observations contained one of a curated set of action verbs (*Added, Introduced, Required, Mandated, Implemented, Revised, Strengthened, Embedded, Piloted, Refined*, etc.), drawing the visual link between when interventions were introduced and the subsequent attainment trajectory.
- Panel D — PI-wise mixed-effects trends. A clustered scatter-and-line plot showing, for each PI within the SLO, the semester-by-semester observed values and a fitted mixed-effects model (`groups = semester`) with 95% confidence bands and the 70% threshold. This panel reveals which PIs are improving, which are stalled, and which are drifting downward—supporting PI-targeted, rather than blanket SLO-level, AFIs.

Below each figure the portal renders a narrative table (instructor explanations and observations

across all course–PI–semester records contributing to the SLO), so faculty review the quantitative panel and the qualitative context in the same screen.

Statistical methods (summary). EP% attainment values are modeled as continuous outcomes with semester index as the primary regressor. Course-level trends are fit with `statsmodels` mixed-effects models (`mixedlm`) grouped on semester, with OLS fallback for degenerate cases. SLO-level semester trends are fit with OLS on per-semester means. PI-level trends use mixed-effects with the same grouping. Across-Bloom-level differences use non-parametric Kruskal–Wallis with Cliff’s Δ as the effect-size measure. All inferential outputs include 95% confidence intervals; the program treats *p*-values as descriptive in the context of small per-term sample sizes and uses effect sizes and confidence bands as the primary basis for interpretation.

4.0.4.6 Indirect-Measure Module: Capstone Exit Survey

Indirect-measure inputs are integrated into the portal through a dedicated Capstone Exit Survey module. Graduating students complete an exit survey (separate login, separate database) that captures background, post-graduation plans, and a structured set of Likert-scale items covering curriculum-to-capstone readiness, preparedness for next steps after graduation, perceived effectiveness of faculty guidance, and perceived impact of recent program interventions. The portal’s `/survey/analyze` route processes responses and produces a 100% stacked-bar visualization of the Likert-scale items (Strongly Disagree → Strongly Agree, color-coded), allowing SLO committees and the full faculty to triangulate direct-measure attainment with student-reported program effectiveness. Administrative roles can download the survey data as a table or CSV export for inclusion in faculty review materials.

4.0.4.7 Administrative, Export, and Visit-Readiness Capabilities

The portal exposes a set of administrative views accessible only to the assessment coordinator (*MECE Admin*). These include:

- Database preview and download. Full-table preview of `abet_entries` and the survey database, with CSV export for archival, faculty distribution, and visit-readiness packaging.
- Supporting Materials uploader. A separate authenticated entry point for uploading rubrics, scored exemplars, and supporting documentation referenced in assessment records.
- Interventions log. A dedicated module for documenting AFIs in an Analysis–Intervention–Comparison (AIC) format, supporting the closed-loop narrative required by Section 4.B.
- ABET Checklist tool. An interactive self-audit page (with Yes/No controls per criterion item) used internally to track readiness against ABET criteria; this complements rather than replaces the formal SSR.

Together, these administrative capabilities mean that the program can produce a defensible, reproducible documentation set for a site visit *directly from the system of record*, rather than reconstructing it from scattered spreadsheets and email threads.

4.0.4.8 How the Portal Supports Each ABET 4.A Element

Portal → ABET 4.A mapping.

- 4.A.1 (Listing/description of assessment processes): the portal data model enforces the PI/Bloom/instrument structure used across all SLOs (Section 4.A.2 tables).
- 4.A.2 (Frequency): the portal records semester for every entry, supporting the “each Fall/Spring; Summer when offered” frequency reported per SLO.
- 4.A.3 (Expected attainment): $EP\% \geq 70\%$ is enforced as a single threshold across courses, PIs, and SLOs, with borderline and not-met bands tracked automatically.
- 4.A.4 (Results / extent of attainment): course-level and SLO-level analysis pages (Sections 4.0.4.4–4.0.4.5) generate the figures and trend statistics presented in Section 4.A.6.
- 4.A.5 (Documentation and maintenance): every record persists in the database with its narrative justification; CSV/HTML exports support binder and archive distribution.
- 4.A.6 (Where the process is documented): this chapter, plus the portal’s administrative views and supporting binders, are the program’s documentation set.

4.0.4.9 Sustainability, Maintenance, and Succession Planning

ABET expects assessment processes to be sustainable, not merely well-designed at a moment in time. The BSME program has structured the portal’s ownership, maintenance, and succession around four explicit commitments:

- Continuity of the ABET coordinator role. Dr. Isaac Choutapalli serves as the BSME program’s ABET coordinator and has committed to continue in that role through the *next* accreditation cycle. The coordinator function carries primary responsibility for portal operation, faculty onboarding to the assessment workflow, semester data-entry coordination, SLO-committee support, and visit-readiness curation. Continuity in this role through the next cycle preserves institutional knowledge of rubric calibration choices, intervention rationales, and faculty-deliberation history that would otherwise be at risk during a coordinator transition.
- Continuity of portal maintenance. The ABET Assessment Portal was designed, implemented, and deployed by Dr. Choutapalli in his capacity as the ABET coordinator, and he will continue to serve as the maintainer of record for the system throughout the next cycle. Maintenance responsibilities include: routine database backups and integrity checks; account provisioning for new faculty; software-dependency updates; deployment of new analysis features as the faculty identify them; and operational support for instructors, SLO chairs, and Capstone Exit Survey administration. Because the developer-maintainer is also the ABET coordinator, change-management decisions (e.g., adding new portal capabilities, modifying analysis routines) flow through the same person who understands the underlying ABET assessment requirements, eliminating the translation gap that often exists when assessment systems are maintained by external IT staff.
- Knowledge documentation and code preservation. Beyond the day-to-day maintenance commitment, the portal carries durable institutional safeguards against single-point-of-failure risk: (i) the portal’s source code is maintained in a department-accessible repository with the database

schema (Section 4.0.4.3) documented in plain text; (ii) the underlying database (`abet_data.db` and `graduate_survey.db`) is a standard SQLite format readable by any modern data tool, ensuring the data itself is recoverable independent of the application layer; (iii) administrative export views produce CSV and HTML snapshots that can be regenerated at any time, so the data is never trapped in a proprietary format; and (iv) the assessment framework documented in this chapter (SLOs, PIs, rubrics, EP% thresholds, AFI lifecycle) is portable to alternative tooling if needed.

- ◉ Faculty workload and distributed ownership. The portal architecture is deliberately designed to distribute, rather than concentrate, the per-faculty assessment workload. Instructors of record spend incremental time per semester entering their own course-level results (one record per course-SLO-PI combination at term end); SLO chairs/coordinators review their own SLO's results within the portal; and the ABET coordinator's role is system-level coordination rather than aggregating data from spreadsheets. This distributed ownership means the assessment process does not depend on any single faculty member completing a centralized data-aggregation task on a deadline. SLO chairs and committee membership rotate among the ME faculty on a multi-year basis, with the committee structure (one committee per SLO, Section 4.0.3.1) ensuring that outcome-level expertise is shared rather than held by a single individual.

Together these four commitments—continuity of the coordinator role, continuity of portal maintenance, durable knowledge/code preservation, and distributed faculty workload—constitute the BSME program's documented sustainability plan. The plan addresses the three concerns most commonly raised about web-based assessment systems: (a) maintainer turnover (mitigated by the coordinator's commitment to continue and by source-code preservation); (b) data lock-in (mitigated by SQLite-based storage and continuous export availability); and (c) knowledge loss across coordinator transitions (mitigated by the documentation embedded in this chapter, the AFI catalog, the SLO-committee structure, and the AIC narrative discipline that records the *why* of each intervention alongside the *what*).

4.0.6.1 What Was Already in Place at the Start of the Cycle

Several elements of the assessment system pre-date this review window and have been carried forward unchanged. Preserving them was a deliberate decision: it allows attainment trends to be interpreted across the full 2020–2026 window without confounding from instrument or threshold changes.

- Seven Student Learning Outcomes (SLOs)—one-to-one with ABET Student Outcomes 1–7—were adopted in the *previous* ABET cycle (replacing the legacy (a)–(k) framework) and were already in active use across the program as the 2020–2026 cycle began. No SLO was added, removed, renumbered, or substantively rewritten during this cycle; minor wording was reviewed during the 2023–2025 three-year review and retained.
- The four-level E/P/A/N rubric scale (Expert, Practitioner, Apprentice, Novice) was already in use prior to 2020. The rubric scale itself has not been changed during the current cycle; only the PI-level rubric criterion text was reviewed in conjunction with the PC→PI relabeling (Section 4.0.6.2).
- The $EP\% \geq 70\%$ attainment threshold (with the borderline 67–70% band and below-67% not-met designation) was already adopted before 2020 and has been maintained without revision throughout the cycle, supporting consistent “meets target” determinations across all seven SLOs and across all reporting terms.
- The Capstone Exit Survey, used as a primary indirect measure, has been administered to graduating seniors since well before the current cycle. The instrument has been retained with only minor item-wording updates (e.g., the addition of an item probing student-perceived impact of recent program interventions). Its administration mechanism was migrated into the portal in Fall 2024 (Section 4.0.6.3), but the underlying instrument is continuous with the prior cycle.
- The SLO-committee governance structure (seven outcome-level committees, faculty-retreat review, full-faculty approval of AFIs) was operational at the start of the cycle and has been carried forward unchanged.

Why this matters. Because the rubric scale, threshold, SLO framework, exit-survey instrument, and committee governance have all been held stable, the EP% values reported in Section 4.A.6 are *directly comparable* across all terms in the 2020–2026 window. The longitudinal trends shown in the SLO summary figures (Section 4.A.6) are not artefacts of instrument or threshold change.

4.0.6.2 Terminology Realignment: PC → PI

The first deliberate change during the current cycle was a terminology realignment in which the program adopted the ABET-current term “Performance Indicator (PI)” in place of the prior-cycle term “Performance Criterion (PC).” This change was prompted by the BSME ABET Coordinator’s

attendance at an ABET Assessment Workshop in Baltimore, Maryland, where current ABET-preferred language and assessment-process expectations were reviewed in depth. The change was discussed with faculty, approved, and implemented across rubrics, SLO-binder labels, syllabus references, and assessment summaries in the terms following the workshop.

The realignment was *nomenclature-driven, not measurement-driven*:

- No outcome was added or removed.
- The number of measurable items per SLO was preserved (i.e., the PCs of the prior cycle map one-to-one to the PIs of the current cycle).
- Rubric criterion text was reviewed at the PI level to confirm clarity and continued alignment with the SLO it operationalizes; minor wording refinements were made where needed, but the cognitive level (Bloom's), the evidence type, and the artifact courses were preserved.
- The E/P/A/N rubric scale and the 70% threshold were unchanged.

Crosswalk to prior-cycle documentation. Where prior-cycle binders, meeting minutes, or reports refer to "PC-1," "PC-2," etc., these correspond to the current "PI-1," "PI-2," etc., for the same SLO. The current SLO/PI documentation supplied as part of this Self-Study reflects the PI terminology; supporting prior-cycle artifacts retained in SLO binders preserve their original PC labels for historical fidelity.

4.0.6.3 Infrastructure Modernization: From Spreadsheets to the ABET Portal (Fall 2024)

The second—and operationally more significant—change was the design, development, and deployment of the ABET Assessment Portal in Fall 2024. Prior to that term, semester assessment relied on Excel spreadsheets and email-based reporting, with SLO chairs aggregating instructor data term by term and curating SLO binders manually. As enrollment grew and PI-level multi-course multi-semester evaluation became difficult to sustain in spreadsheet form, the department developed the portal in-house. The portal preserved every element of the existing measurement framework—SLOs, PIs (post-relabeling), E/P/A/N scale, 70% threshold, Capstone Exit Survey—while changing only *how* data is captured, stored, analyzed, and reported.

What the portal replaced (pre–Fall 2024):

- Excel spreadsheets maintained at the instructor and SLO-chair level, transmitted by email at term end.
- Manual roll-ups by SLO chairs into outcome-level summary tables.
- Hand-curated SLO binders for record retention.
- Capstone Exit Survey administered through institutional or departmental survey instruments outside a unified portal.

What the portal added (Fall 2024 onward):

- A structured, PI-grained relational data store (`abet_entries`) ensuring that every record carries course, term, SLO, PI, Bloom's level, instrument description, E/P/A/N counts, narrative justification, and instructor observations.
- On-demand course-level and SLO-level analysis pages (Sections 4.0.4.4–4.0.4.5) that produce the figures and trend statistics used in faculty review.
- Embedded mixed-effects and OLS trend modeling, replacing manual trend-line spreadsheets.
- An integrated Capstone Exit Survey module (separate database for student-survey privacy isolation) with on-demand Likert-scale analytics.
- Administrative export, supporting-materials upload, AFI/intervention log, and a self-audit checklist—all backed by the same database.

The launch was designed to be *additive and non-disruptive*: Fall 2024 data entered through the portal uses the same SLOs, PIs, rubric, and threshold as Spring 2024 spreadsheet data, and the portal database back-fills selected prior-term records to preserve continuity in longitudinal reporting. Pre-Fall-2024 records remain retrievable in their original spreadsheet form in the SLO binders and the department's archive.

4.B.1.2 Decision-Making, Prioritization, and Approval Pathways

The BSME program converts the Section 4.B.1.1 inputs into approved improvement actions through a structured five-step workflow. Decision authority follows the governance distribution in Section 4.0.3.1.

Step 1 — Identification. Each term, SLO committees use the ABET Portal's course-level and SLO-level analysis pages to identify candidate items for action. The standard triggers are:

- A PI or course–SLO combination with EP% < 70% in the most recent term (*not met*).
- A PI or course–SLO combination with EP% in the 67–70% borderline band (*cautionary*).
- A negative or flat multi-semester trend slope ($\hat{\beta}_1 \leq 0$), even when current EP% $\geq 70\%$, indicating drift.
- Instructor observations recorded in the portal that report systemic difficulties (e.g., common student errors, prerequisite-skill gaps, instrumentation issues).
- Capstone Exit Survey items showing notable disagreement with curriculum-effectiveness or readiness statements.
- EAC or alumni input identifying a competency gap relative to industry expectations.

Step 2 — SLO-committee analysis. The relevant SLO committee compiles the evidence using the portal's analytical outputs (course bars, Bloom-level distributions, semester trend with slope/ p -value, PI-wise mixed-effects trends, narrative tables) and develops a written analysis identifying the candidate action(s). The Analysis–Intervention–Comparison (AIC) entry begins at this step: the *Analysis* is drafted now, with the *Intervention* proposed for faculty review and the *Comparison* (re-assessment) planned for the next assessment cycle in which the relevant course/PI will be measured.

Step 3 — Faculty deliberation and prioritization. SLO committees present their proposed AFIs to the full Mechanical Engineering faculty in scheduled meetings and faculty retreats (Section 4.0.3.4), accompanied by the supporting portal analyses. Because more AFIs are typically proposed than can be implemented in any single term, faculty prioritization is required. The program uses evidence-informed faculty judgment guided by the following explicit criteria:

BSME prioritization criteria for AFIs (in faculty deliberation):

- Severity: how far below threshold the EP% sits (e.g., <60% is more urgent than 67–70% borderline).
- Persistence: whether the gap is one-term variability or a sustained multi-semester pattern (informed by the portal's trend slope and confidence band).
- Breadth: whether the issue is concentrated in a single course/PI or appears across multiple courses or PIs in an SLO.
- Upstream/downstream impact: whether the affected course is a gateway/prerequisite for

many downstream courses (e.g., MECE 2302 Dynamics or MECE 3315 Fluid Mechanics in the mechanics chain).

- Feasibility of intervention: whether an effective action is implementable in the next offering (course-level changes are faster than curriculum or policy changes).
- Stakeholder corroboration: whether indirect measures (Capstone Exit Survey) and/or EAC input corroborate the direct-measure signal.
- Resource fit: whether the action can be supported within current faculty assignments, teaching loads, and facilities, or requires resource allocation by the Department Chair.

The program-level SWOT exercise at the faculty retreat (Section 4.0.3.4), conducted every two years, provides a synthesis view across all SLOs and helps faculty rank competing AFIs into a short list of program-level priorities. The most recent SWOT (September 2024; full four-quadrant summary in Appendix item D.17) reinforced several signals that current-cycle actions already address: the identified math/science-sequence bottleneck and incoming-preparedness concerns are tracked through the gateway-course pass-rate monitoring and the POL-01/POL-02 enforcement data (Sections 4.B.4.5 and 4.B.4.10), and the technical-writing weakness is targeted by the SLO-3 “Audience Adaptation” Communication Overlay (Section 4.B.3.3). No SWOT was conducted in September 2025; the next is scheduled for the September 2026 retreat.

Step 4 — Approval and assignment. Approved AFIs are recorded in the meeting/retreat minutes with: (i) the action to be taken; (ii) the responsible owner (instructor of record, course coordinator, SLO chair, or department chair, depending on action type); (iii) the implementation term; and (iv) the re-assessment plan (i.e., when and where the comparison data will be collected). Decision authority follows the distribution in Section 4.0.3.1—course-level changes by the instructor of record under SLO-committee guidance; multi-section coordination by course coordinators; rubric/PI changes by SLO committees with full-faculty approval; curriculum and policy changes through full-faculty approval and college/university governance. Catalog-modifying changes additionally follow the institutional pathway documented in Criterion 6.E.1.

Step 5 — Implementation and tracking through to re-assessment. Owners implement the approved action in the assigned term. The instructor (or owner) re-enters assessment data into the portal in subsequent terms following the same artifact structure, allowing the SLO committee to perform an apples-to-apples comparison. The AIC entry is then closed with the *Comparison* narrative: change effective, partial, or not effective. Closed-loop case studies are presented per outcome in Section 4.B.3, including cases where the initial change did not produce the expected improvement and a follow-on action was required.

Documented consideration of competing interpretations. ABET’s expectation under Criterion 4 is that the program collectively consider *all* data associated with an outcome, including data that admits more than one interpretation. SLO committees document, in the AIC *Analysis* narrative, competing interpretations considered during deliberation and the rationale for the interpretation adopted. Two representative instances:

- MECE 3320 SLO-6 (Section 4.B.3.6). The four-term improvement trajectory could be attributed to (a) the documented course-level artifact-complexity increase, (b) a cohort effect, or (c) rubric drift toward leniency. Hypothesis (b) was rejected because program-level enrollment composition

was stable across the four terms and other SLO-6 courses did not show comparable improvement. Hypothesis (c)—if anything, rubric drift went in the *opposite* direction: the portal record explicitly documents that the step-response lab rubric was *tightened* in Sp 2023 (“*Step-response lab (iterated with tighter rubric)*”) and the artifacts grew in complexity each subsequent term (more complex sensors Sp 2024, multi-order sensor model and large-data-set RMS Sp 2025). Sustained EP% gains under tightening rubrics and increasing artifact difficulty are not consistent with leniency drift. The closed-loop case is reported on the basis of hypothesis (a), with the rubric-tightening evidence reinforcing the gain rather than confounding it.

- MECE 4362 (SD-II) below-threshold indicators (Section 4.B.3.8, NX-01). The MECE 4362 (SD-II) records report stable, consistent attainment across S 2023–F 2025, with several PIs genuinely below threshold (most prominently SLO-7 PI-2). Because the SD-II series is flat it contributes no temporal trend, so program-level trend slopes are computed across the courses with multi-term data; the SD-II values are reported as recorded. NX-01 is a course-level SD-II assessment review of the affected PIs, scoping a course-level AFI with a documented intervention and re-assessment plan where the review identifies an instructional or assessment gap.

The discipline is preserved across faculty turnover because the rationale is recorded in the AIC narrative within the portal, not held only in committee memory.

When an AFI is proactive rather than triggered. An AFI is not always a response to a below-threshold breach. The program treats the $EP\% \geq 70\%$ threshold as a floor, not a ceiling. Faculty regularly propose proactive AFIs in response to: stakeholder input (e.g., the EAC member feedback on professional-readiness skills at the September 2, 2022 meeting that motivated AFI SS-02; Sections 4.0.6.5 and 4.B.4.1), Capstone Exit Survey themes, instructor observations recorded in the portal, or the program’s strategic priorities (Section 4.0.6 evolution context). These proactive AFIs follow the same Analysis–Intervention–Comparison workflow and are recorded in the AFI catalog with their motivating evidence explicitly noted.

4.B.1.3 How the ABET Assessment Portal Supports Prioritization

The portal does not replace faculty judgment; it makes faculty judgment faster, more consistent, and more defensible by presenting the evidence in a uniform analytical view. The portal features supporting the prioritization workflow are:

- Automatic threshold flagging in course-wise attainment bars. The SLO-level analysis page (Section 4.0.4.5, Panel A) color-codes each course’s mean EP% against the 70% threshold (green when at or above, red when below) and overlays 95% confidence intervals. SLO chairs do not need to compute or render these comparisons manually; a single glance identifies the candidate courses for Step 1 triggers above.
- Trend slope and significance reported in-figure. Panel C of every SLO summary figure reports the OLS slope estimate $\hat{\beta}_1$ (percentage points per term), its p -value, and a 95% confidence band on the fitted trend line. This converts “is this getting worse?” from a judgment call to a small number of numbers the committee can examine.
- PI-wise mixed-effects trends. Panel D shows each PI within an SLO with its own trend line and

confidence band, enabling PI-targeted action rather than blanket SLO-level intervention. The committee can see, for example, that PI-1 is improving while PI-3 is drifting downward—two facts that would be obscured in an SLO-level average.

- ◉ Intervention-arrow annotations on the semester trend. Panel C automatically annotates the timeline with arrows marking semesters in which instructor observations contained action-verb language (*Added, Introduced, Required, Implemented, Revised, Strengthened, Embedded, Piloted, Refined*, etc.). This provides a direct visual link between when an intervention was introduced and the subsequent attainment trajectory, supporting the *Comparison* stage of the AIC workflow.
- ◉ Bloom-level diagnostic. Panel B's Kruskal–Wallis test across Bloom levels (with Cliff's Δ effect size) surfaces whether weaknesses are concentrated at particular cognitive levels (e.g., *Apply* versus *Evaluate*), informing the design of the intervention rather than just its target.
- ◉ Narrative tables alongside the figures. The portal renders the full set of instructor explanations and observations contributing to each SLO immediately beneath the four-panel figure, so the committee reviews the quantitative panel and the qualitative context in a single screen rather than across separate documents.
- ◉ Capstone Exit Survey analytics. The portal's `/survey/analyze` route produces a 100% stacked-bar visualization of student responses to program-effectiveness and readiness items, enabling the committee to triangulate direct-measure attainment with student-perceived program effectiveness in the same review session.
- ◉ Administrative export. The assessment coordinator can produce CSV/HTML snapshots of any portion of the database to accompany the SLO-committee analysis in faculty meetings, ensuring that the entire deliberation rests on a verifiable underlying record.

Net effect on the improvement workflow. Before the portal (Section 4.0.6.3), SLO chairs assembled comparable analyses by hand each term from spreadsheets and email reports, with corresponding variability in completeness and timing. Since Fall 2024, the same analyses are produced on demand from the system of record, allowing the SLO committee to spend its meeting time on *interpretation and action* rather than on *data assembly*. This is the most important operational change the portal made to the continuous improvement process—and the one most directly visible to a PEV reviewing the meeting/retreat minutes in the department archive.

The catalog of specific Actions for Improvement implemented during 2020–2026 follows in Section 4.B.2; closed-loop case studies (one per SO, plus partial/ineffective cases) appear in Section 4.B.3; “other available information” inputs are detailed in Section 4.B.4; future plans grounded in recent evaluations are in Section 4.B.5.

4.B.3.8 Cases Where Changes Were Partial or Not Yet Effective (Honest Accounting)

ABET Criterion 4 explicitly expects programs to report cases where changes were not effective. The BSME program documents the following cases of partial or not-yet-effective change, presented openly as part of the honest accounting that ABET expects:

- MECE 2302 SLO-1 declining trend (pre-AFI period). The course-level EP% declined from 69% to 39% across Sp 2022–Sp 2025 (Figure 27). Course-level interventions during the cycle were not effective in arresting the decline. The faculty escalated to a department-level formal AFI (the Fall 2025 Problem Analysis Overlay) precisely because the lower-scale course-level monitoring did not produce improvement.
- MECE 4362 (SD-II) below-threshold performance indicators. The MECE 4362 (SD-II capstone) records report stable, consistent term-over-term attainment across Spring 2023–Fall 2025. Several PIs sit genuinely below threshold—term-average EP% (across PIs) is SLO-2 $\approx$ 21%, SLO-3 $\approx$ 52%, SLO-5 PI-2 $\approx$ 4.5%, and SLO-7 PI-2 (Apply) at 0%; SLO-4 $\approx$ 75% and SLO-6 $\approx$ 73% sit at or above threshold. Because these are culminating-course results, the relevant SLO committees, the MECE 4362 instructor of record, and the ABET Coordinator will conduct a course-level SD-II assessment review beginning Fall 2026 (NX-01, Section 4.B.5): examine the SD-II artifacts and rubric application for the affected PIs and, where warranted, scope a course-level AFI with a documented intervention and re-assessment plan.
- SLO-5 PI-2 single below-threshold term in MECE 4361. PI-2 (*Plan tasks and meet deadlines*) attained 40% in Sp 2024 in MECE 4361, sandwiched between above-threshold terms (S 2023 100%, S 2025 80%, F 2025 82%). The localized dip suggests a cohort or single-term effect rather than a chronic program-level weakness. The SLO-5 committee will continue monitoring through Fall 2026 and Spring 2027 to confirm whether the recovery is sustained.
- MECE 1221 SLO-2 single below-threshold data point. MECE 1221 reported 67% on its single Fall 2024 data point (Figure 31). The Fall 2025 SLO-2 Concept Development Overlay is deployed primarily in the capstone (MECE 4361) and does not directly address MECE 1221. The SLO-2 committee will collect additional MECE 1221 data points in Sp 2026 and Fall 2026 to determine whether the F 2024 result reflects a persistent pattern.
- MECE 2140 SLO-6 limited evidence base. MECE 2140 reported 62% in Sp 2022 (with a strong cognitive-level gradient: Apply 86%, Analyze 60%, Evaluate 39%; Figure 45). Subsequent assessment terms have not yet produced additional MECE 2140 SLO-6 data points; the coverage gap is documented in Section 4.B.5 (NX-03).
- New technical electives (CR-03, in progress). The three proposed new electives (MECE 4306 CFD, MECE 4334 Advanced Alloys, MECE 4335 Smart Materials) are in the course-request and approval pipeline; full deployment and re-assessment will require multiple offerings.
- MECE 3380 (Kinematics & Dynamics of Machines) narrative completeness. The portal records for MECE 3380 across Fall 2020–Spring 2024 (66 records spanning SLO-1 PI-2/PI-3 and SLO-2

PI-1/PI-2/PI-3, the largest record count of any course in the cycle) contain quantitative E/P/A/N data but the explanation fields contain placeholder text (“Awaiting summary or description,” “Awaiting summary response,” or “Awaiting summary/narrative”). The quantitative attainment data is usable for trend computation—SLO-1 term-average ranges ~58–79% across the terms, SLO-2 term-average ranges ~68–82%—but the artifact-level narrative justifying each entry is incomplete. The instructor of record and the SLO-1/SLO-2 chairs will complete back-fill of the narrative explanation fields in Fall 2026 as part of the next-cycle documentation back-fill (NX-01a, Section 4.B.5); no change to the underlying quantitative data is anticipated, since the artifact assignment (Exam-4 Problem-2 for SLO-1 PI-3, Exam-2 Problem-1 for SLO-1 PI-2, and the K&D Project for the SLO-2 entries) is stable across the cycle.

Portal reporting enhancements. To make routine cross-term review easier, the portal is being extended in Fall 2026 with a cross-term comparison view, a dashboard tile summarizing course/SLO pairs that have been flat or below threshold across multiple terms, and a `source=backfill` provenance flag distinguishing pre-Fall-2024 spreadsheet imports from records entered directly in the portal. These reporting aids are documented as next-cycle item NX-01a (Section 4.B.5) and are scoped within the existing portal maintenance workflow (Section 4.0.4.9).

Why this matters. The program reports both successes and limitations openly. The MECE 2302 case (where course-level monitoring was *not* sufficient and escalation to a formal AFI was required) and the below-threshold SD-II (MECE 4362) indicators (which the program is reviewing at the course level before deciding on an intervention) show that the BSME continuous improvement system has working escalation logic and does not present every result as automatically satisfactory.

Appendix item D.5: Benchmarking and external indicators (former Section 4.B.4.4; FE Practice Exam only)

Benchmarking complements the program's internal direct and indirect measures (Sections 4.A.6, 4.B.4.2, 4.B.4.5) by situating the BSME program against external reference points. The cycle's benchmarking strategy uses four independent reference channels, each tied to specific SLOs or to a program-quality dimension the chapter already establishes. The numerical content of each channel is being assembled from the sources identified below; the framing here documents *what is being benchmarked, against what, and why*.

(i) FE Practice Exam topic-level results as an external SLO-1/SLO-6 benchmark. The NCEES Fundamentals of Engineering (FE) examination evaluates the cognitive content (problem-solving in mechanics, materials, thermodynamics, fluids, and instrumentation) that the BSME program targets in SLO-1 and SLO-6, making it a relevant external reference for program-level technical attainment. Because the number of BSME graduates who sit for the live FE exam in any given term is very small, term-level live-exam pass percentages are not statistically reliable as a program metric; the program therefore uses the *FE Practice Exam*—administered to the full MECE 4101 cohort ($n \approx 30\text{--}60$ per term)—as its FE-aligned external benchmark. The topic-level practice-exam results below provide the higher-resolution, larger-sample signal used in continuous improvement.

FE Practice Exam by topic, recent cycle terms (UG_Program_Data.xlsx, sheet *FE Results*, columns 9–13; the practice exam administered to MECE 4101 cohorts in Fall 2024–Spring 2026; percentages are mean scores across the cohort by topic, $n \approx 30\text{--}60$ per term):

FE Topic	F 2024	Sp 2025	F 2025	Sp 2026
Mathematics	45.0%	50.6%	50.0%	44.4%
Probability & Statistics	53.8%	47.0%	52.7%	54.4%
Engineering Economics	29.5%	73.0%	71.7%	71.8%
Electricity & Magnetism	55.7%	68.7%	59.8%	67.8%
Statics	29.5%	33.2%	31.3%	33.3%
Dynamics, Kinematics, & Vibrations	33.8%	31.0%	30.7%	31.1%
Material Properties & Processing	54.5%	59.5%	60.4%	61.7%
Mechanics of Materials	41.8%	43.5%	41.3%	39.5%
Mechanical Design & Analysis	33.0%	22.6%	24.3%	21.8%
Thermodynamics	47.0%	81.0%	83.3%	25.8%
Fluid Mechanics	49.2%	26.5%	25.7%	84.2%
Heat Transfer	28.9%	21.3%	17.7%	16.9%
Measurements, Instrumentation, Controls	36.9%	52.8%	48.4%	52.1%
Engineering Ethics	47.0%	29.5%	35.5%	28.2%
Practice Exam Mean	41.8%	45.7%	45.2%	45.2%

The practice-exam topic-level breakdown provides the highest-resolution external benchmark in the indirect-measure system. Five topics consistently sit below 35% across cycle terms—Statics, Dynamics, Mechanical Design & Analysis, Heat Transfer, and (variable) Mechanics of Materials—identifying exactly the SLO-1/SLO-2 foundational content the Spring 2026 Problem Analysis Overlay AFI is designed to address. Heat Transfer is the most persistent below-threshold topic (17–29% across cycle), warranting targeted attention in the next cycle’s faculty-development plan even though the underlying MECE 3360 Heat Transfer pass rate (Section 4.B.4.5) is comfortably above threshold at 84.3%—a signal that course-level pass rate and topic-level FE preparation are distinct evidence channels and that the relatively weak FE Heat Transfer practice-exam score points to a graduation-readiness gap in the depth of conceptual mastery beyond course-level minimum competency. Conversely, three topics consistently sit at or above their nominal target: Engineering Economics (~70% Sp 2025 onward), Material Properties (~55–62% across cycle), and Probability & Statistics (~47–55%)—confirming that these areas are at or above the relevant SLO threshold within the program. The Spring 2026 Problem Analysis Overlay AFI re-assessment (AS-04, Section 4.B.5) will monitor whether the Statics/Dynamics/Solid Mechanics topic-level pass rates rise in subsequent FE practice-exam administrations.

(ii) Peer-program benchmarking against Texas mechanical-engineering programs and HSI engineering programs. BSME operates within two reference sets simultaneously: Texas-based ABET-accredited mechanical-engineering programs (a regional labor-market and articulation peer set including UT Austin, Texas A&M, UT San Antonio, UT El Paso, UT Arlington, UT Dallas, Texas Tech, University of Houston, and Texas State) and Hispanic-Serving Institution (HSI) engineering programs (a mission peer set including peer R2/R1 HSIs across the US Southwest). The relevant comparison metrics are graduation rates (4-year and 6-year), time-to-degree, and first-destination outcomes; both peer sets publish these through ABET self-studies, IPEDS, and individual institutional dashboards. The BSME program’s own metrics on these dimensions are reported in Sections 4.B.4.3 (enrollment, degree production), 4.B.4.8 (demographics), and 4.B.4.9 (retention and graduation); the benchmarking step is the side-by-side comparison conducted at the May 2027 faculty retreat using the most recent publicly available peer data. Particular attention is given to the HSI peer set because UTRGV’s HSI status is a mission characteristic with implications for student readiness, support-resource needs, and the labor-market context graduates enter. The peer-comparison table is a planning instrument used at the retreat, not a stand-alone deliverable for the current Self-Study; the cycle’s BSME-specific evidence base (admissions test profile in item (v), Hispanic share at ~91% in Section 4.B.4.8, Senior cycle-aggregate GPA 3.13 in Section 4.B.4.9) is the chapter’s primary reference.

(iii) National benchmarking via ASEE *Engineering by the Numbers*. The American Society for Engineering Education publishes annual aggregate statistics for US engineering programs by discipline and demographic profile. Three ASEE metrics are most relevant for BSME benchmarking: (a) mechanical-engineering enrollment and degree-production trends, against which the cycle’s 64% enrollment growth (785 F19 → 1,288 F25, Section 4.B.4.3) and the average 129.8 bachelor’s degrees per year (Section 4.B.4.9) can be situated; (b) demographic representation in mechanical engineering nationally, against which the BSME program’s HSI demographic profile (Section 4.B.4.8, ~91% Hispanic, ~17% female and rising) can be contextualized; and (c) average institutional support metrics (faculty-to-student ratios, lab-and-instructional-facilities investment) where ASEE provides public benchmarks. The program does not use ASEE aggregates as performance targets—

they are a contextual reference set, not a normative benchmark—but they answer the implicit PEV question “how does this program’s profile compare to the national mechanical-engineering picture?” The next-cycle ASEE comparison is conducted at the May 2027 retreat alongside the peer-program comparison above; the current Self-Study’s BSME-specific evidence stands on its own.

(iv) Salary benchmarking for post-graduation outcomes. The Capstone Exit Survey (Section 4.B.4.2) collects engineering-offer salary data per respondent. Aggregated, these data points (preserved in the master survey file, not reproduced here for PII protection) provide a graduating-cohort salary distribution that can be benchmarked against (a) the US Bureau of Labor Statistics OES program’s median entry-level mechanical-engineering salary, (b) the regional South Texas mechanical-engineering median, and (c) the ASEE engineering-salary survey. Salary benchmarking is an indirect measure of PEO 3 (career advancement) and triangulates with the SLO-1/SLO-7 evidence base because employer willingness to pay tracks employer perception of graduate competence. The cycle’s post-graduation evidence is already substantive: the cycle-aggregate engineering-offer rate is 46.7% at survey time (Section 4.B.4.2, master analysis); the term-by-term trajectory rose from 15–16% (F 2020/Sp 2021) to 29–43% (F 2022 onward), consistent with the deployment window of AFI SS-02 (Career-readiness content enhancement). Salary aggregates are preserved in the master survey file with appropriate PII protections; cycle-aggregate distribution statistics are available for on-site PEV review. The external-benchmark side-by-side comparison is a planning instrument applied at the May 2027 faculty retreat alongside the peer-program and ASEE benchmarks; the current Self-Study’s BSME-specific post-graduation evidence stands on its own.

(v) Admissions test-score profile (incoming-cohort academic preparedness benchmark). BSME first-time full-time entering cohort ACT and SAT mean scores, reported by UTRGV to the Texas Higher Education Coordinating Board on the CBM00B (Requested_Data_Test_Scores), provide a direct benchmark of incoming-cohort academic preparedness against the CECS-wide cohort and against peer engineering majors. Within UTRGV CECS, BSME consistently posts the *highest mean ACT and the highest or near-highest mean SAT* of the engineering majors across the cycle:

AY (FT-FT entering)	20–21	21–22	22–23	23–24	24–25
BSME ACT mean	21.23	21.26	21.06	21.85	21.17
College-Wide ACT mean	20.51	20.46	20.70	20.90	20.21
BSME SAT mean	1095	1095	1078	1116	1075
College-Wide SAT mean	1090	1095	1077	1074	1068

BSME FT-FT entering cohorts post ACT means roughly 0.4–1.0 point above the College-Wide CECS mean across every cycle term and SAT means roughly 5–40 points above College-Wide across every cycle term. Compared to the other engineering majors in CECS, BSME ACT means are the highest of any program (vs. Civil 17.5–20.1, Computer Engineering 19.9–22.0, Computer Science 20.8–22.0, Electrical 19.0–20.1, Manufacturing 16.4–20.2 across the cycle terms). The all-UG (continuing-plus-entering) measure shows the same pattern at the cycle level: BSME all-UG ACT mean 20.78–20.96 vs. College-Wide all-UG 20.42–20.64. This admissions-test profile is contextual evidence supporting the SLO-1 narrative: the cohort enters with measurable academic preparedness above the CECS mean; the SLO-1 declining-trajectory signal in MECE 2302 (Section 4.B.3.1) is therefore not attributable to systematic cohort-quality decline at admission, which

strengthens the case for the Spring 2026 Problem Analysis Overlay AFI as a course-mechanics rather than student-preparation intervention.

Cross-channel triangulation. The five benchmark channels are designed to triangulate, not to duplicate: FE practice-exam results speak to SLO-1/SLO-6 technical attainment; peer-program comparisons speak to program operations (graduation, time-to-degree, demographics); ASEE aggregates speak to national context; salary data speaks to post-graduation outcomes; admissions test scores speak to incoming-cohort academic preparedness. A PEV reading this subsection alongside Sections 4.A.6 and 4.B.4.2 sees five independent windows onto BSME program quality, each with a documented source and a documented relationship to the SLOs and PEOs.

Use in continuous improvement. Benchmarking informs the program's next-cycle priorities in four concrete ways: (i) the FE practice-exam channel feeds the AS-04 SLO-1 Overlay re-assessment by providing an external check on the within-portal trend; (ii) the peer-program graduation/time-to-degree comparison sharpens the institutional-metrics picture in Section 4.B.4.9 and informs faculty-load discussions tied to enrollment growth (Section 4.B.5 Risk 1); (iii) the ASEE demographic comparison contextualizes the HSI-mission narrative the program prepares for the May 2028 retreat; (iv) the admission test-score profile (item (v) above) provides a stable incoming-cohort preparedness baseline against which to interpret course-level SLO signals, distinguishing pedagogy/instrument effects from cohort effects. The benchmarking subsection does not itself produce AFIs; it provides the external reference frame in which existing AFIs and next-cycle items are interpreted.

Appendix item D.6: Prior-cycle reference pass-rate data (former Table 20 and Figure 26)

Cross-cycle reference data. Table 20 reproduces the prior-cycle pass-rate baseline from the August 2020 BSME Self-Study (Table 4.1.36, spanning Fall 2014 through Spring 2020) so PEVs can observe the historical pattern that motivated the current-cycle SLO-1 Problem Analysis Overlay AFI and the long-standing Two-Peat Rule. Persistent gatekeeper courses across the prior cycle were MECE 2301 (Statics, since retired), MECE 2302 (Dynamics), MECE 2335 (Thermodynamics I), MECE 2340 (Engineering Materials), and the legacy MECE 3449 (replaced by MECE 3440 + MATH 2415 per AFI CR-01). These cluster on SLO-1 (analysis/problem-solving) and SLO-2 (foundational design), consistent with the program-level SLO-1 attainment pattern in Section 4.A.6.

Table 20: BSME course pass rates by semester (prior-cycle reference data, F 2014–S 2020). Reproduced in range form from the August 2020 BSME Self-Study Table 4.1.36 to provide cross-cycle context. Values are min–max ranges of percentages of enrolled students passing the course (grade $\geq$ C) across the semesters in each academic-year window. Persistent gatekeeper courses (<70% pass rate across multiple terms) are highlighted in bold and cluster in the SLO-1 and SLO-2 analytical/design course families. The current-cycle (AY 2020–21 through AY 2024–25) pass rates for the same courses are reported in Table 7 (next table); the cross-cycle comparison shows progress in MECE 2302 Dynamics (rebound from prior-cycle 38–64% range to current-cycle 67.9–78.3%) while persistent below-threshold attainment in MECE 2340, MECE 3321, and legacy MECE 2301 motivated the Spring 2026 SLO-01 Problem Analysis Overlay AFI (Section 4.B.2.2.1).

Course	F14–S15	F15–S16	F16–S17	F17–S18	F18–S19	F19–S20
MECE 1101 Intro to ME	80–89	70–84	65–80	67–80	71–75	—
MECE 1221 Engineering Graphics	75–82	71–87	74–86	72–83	76–79	—
MECE 2140 Materials Lab	78–94	78–85	79–89	72–88	79–84	77–90
MECE 2301 Statics (legacy)	69–75	66–96	60–83	61–97	61–83	67–78
MECE 2302 Dynamics	38–64	52–89	63–89	69–71	52–75	55–86
MECE 2335 Thermodynamics I	71–91	65–94	69–82	63–86	54–83	70–73
MECE 2340 Engineering Materials	63–97	64–86	64–97	58–73	69–96	63–74
MECE 2350 Numerical Methods (legacy)	82	74–82	79–81	54–80	36–71	52–73
MECE 3100 UG Practicum	96–100	96–100	96–100	100	92–100	97–100
MECE 3115 Fluid Mech Lab	86–97	93–98	96–97	92–99	80–93	82–98
MECE 3160 Heat Transfer Lab	84–98	89–100	89–100	82–91	80–95	—

BSME Course Pass Rates — Cross-Cycle Heatmap

Prior-cycle reference data (Aug 2020 SSR Table 4.1.36); midpoints of ranges; ★ = persistent gatekeeper course

Course	F14-S15	F15-S16	F16-S17	F17-S18	F18-S19	F19-S20	Pass-rate band
MECE 1101 Intro to ME	84	77	72	73	73	—	≥ 85% Strong
MECE 1221 Engineering Graphics	78	79	80	77	77	—	75–85% Above target
MECE 2140 Materials Lab	86	81	84	80	81	83	70–75% At threshold
MECE 2301 Statics (legacy) ★	72	81	71	79	72	72	65–70% Borderline
MECE 2302 Dynamics ★	51	70	76	70	63	70	< 65% Below target
MECE 2335 Thermodynamics I ★	81	80	75	74	68	71	
MECE 2340 Engineering Materials ★	80	75	80	65	82	68	
MECE 2350 Num. Methods (legacy)	82	78	80	67	53	62	
MECE 3100 UG Practicum	98	98	98	100	96	98	
MECE 3115 Fluid Mech Lab	91	95	96	95	86	90	
MECE 3160 Heat Transfer Lab	91	94	94	86	87	—	

Figure: Pass-rates heatmap. Gatekeeper courses (★) cluster in the SLO-1/SLO-2 analytical and design-foundation course families, consistent with the program-level SLO-1 attainment pattern documented in Sec. 4.A.6. Current-cycle data refresh in progress with UTRGV IR.

Figure 26: BSME course pass rates cross-cycle heatmap (prior-cycle reference data, midpoints of the ranges in Table 20). Gatekeeper courses (★) cluster in the yellow/orange/red zones in the SLO-1/SLO-2 analytical and design-foundation course families. Lab courses and upper-division courses consistently render green. The visual pattern motivates the SLO-1 Problem Analysis Overlay AFI (Section 4.B.2.2.1) and the carry-forward Two-Peat Rule (POL-02, Section 4.B.2.1).

Appendix item D.7: Historical pretest program (former Section 4.B.4.6)

The BSME program operated a prerequisite-knowledge pretest program from Fall 2001 through the early part of the current review cycle. Pretests were administered in the first two weeks of selected gatekeeper courses to evaluate retention of prerequisite skills, serving five purposes: (i) early feedback to students on likely success in advanced engineering courses; (ii) quality-of-instruction measure for prerequisite courses; (iii) longitudinal data for curriculum-review groups; (iv) CQI data for prerequisite math and science; (v) at-risk student identification for advising.

Courses covered (historical). Pretests were administered in MECE 2301 (Statics; prereq Calculus II + Physics I), MECE 2302 (Dynamics; prereq Calculus II + Statics), and the legacy MECE 3449 (ME Analysis I; prereq Calculus II + Numerical Methods). Success criterion: average pretest score > 70%.

Historical results (prior cycle). Per the August 2020 BSME Self-Study Table 4.1.37, students performed at or near 70% on Statics pretests (79.2% F16, 71.5% Sp17, 69.7% F17, 70.1% Sp18) but well below threshold on Dynamics (53.3% F16, 50.9% Sp17, 40.2% F19) and ME Analysis I (70.3% F15 declining to 58.9% Sp18, 62.0% Sp19, 61.6% Sp20). These results documented persistent calculus/physics prerequisite gaps that motivated several prior-cycle interventions and informed AFI CR-01.

Status during the 2020–2026 cycle. The pretest program was *discontinued* during the current cycle. Three converging factors serve as replacement: (i) the ABET Portal (Fall 2024) provided a more granular PI-level mechanism for tracking prerequisite-knowledge issues within the gatekeeper courses themselves—particularly via PI-1 (*Identify engineering problem*); (ii) the math-sequence revision (AFI CR-01) restructured the prerequisite pipeline such that historical pretest content was no longer directly aligned with the current sequence; (iii) the portal's embedded course-level monitoring (Bloom-level panels, mixed-effects trends) provides richer diagnostic information than the pretest delivered.

Continuity of monitoring. The function previously served by pretests is now performed by: (i) direct-measure PI-1 attainment in MECE 2302, 2340, 3315, 3360, 3450, 4350, which captures prerequisite-knowledge recognition in SLO-1 artifacts; (ii) the portal's course-level mixed-effects analysis, which flagged the MECE 2302 declining trend that motivated the Fall 2025 SLO-1 Overlay AFI; (iii) the pass-rate stream in Section 4.B.4.5.

Historical pretest data (administration logs, scoring summaries, analysis reports) is preserved in the Department Records Archive and available for PEV review.

Appendix item D.8: POL-01 / POL-02 operational enforcement data (former Section 4.B.4.10, continued)

POL-01 (GPA Rule) enforcement. The rule places any BSME student whose GPA falls below 2.5 on *Probation (P)*; students who remain below 2.5 in subsequent semesters cycle through *Suspension (S)* and increasingly restrictive S*, S**, S*** levels with faculty-approved appeals required to continue. The UPC distributes targeted email notifications to all flagged students each semester. Table 21 reports the operational counts.

Table 21: POL-01 (GPA Rule) operational enforcement data, Spring 2021–Spring 2024. P = newly placed on Probation (GPA dropped below 2.5 for first time); S = remained below 2.5 after P; S* through S*****+ = successive semesters of continued sub-2.5 GPA requiring deeper-level faculty appeals. The *Came off X* rows below indicate students who recovered GPA above 2.5 from the named level. Source: UG_Program_Data.xlsx, sheet *GPA Rule*.

Level	Sp 21	F 21	Sp 22	F 22	Sp 23	F 23	Sp 24
P (newly placed)	83	72	96	75	119	68	117
S (1st cont.)	50	23	27	29	31	28	11
S*	—	30	11	9	16	7	7
S**	—	—	20	8	4	5	1
S***	—	—	—	10	4	1	2
S****	—	—	—	—	6	2	0
S*****+	—	—	—	—	—	1	2
Total	133	125	154	131	180	112	140
Came off P	—	60	45	67	44	91	57
Came off S	—	20	12	18	13	24	21

Three findings emerge: (i) the rule is actively used: 112–180 BSME students per term are flagged at some level under POL-01, indicating the policy is operational rather than nominal; (ii) recovery is the dominant outcome: 44–91 students came off Probation each term (compared to 68–119 newly placed), and 12–24 came off Suspension (a substantial proportion of those previously suspended), demonstrating that the rule functions as a corrective mechanism rather than a terminal action; (iii) deep-level appeals taper sharply: the S*** and beyond levels show very small numbers per term (0–10 students), consistent with the rule’s design intent of stopping persistent non-progressors. The Senior cycle-aggregate GPA of 3.13 (Section 4.B.4.9) sits well above the 2.5 floor, confirming that the rule functions as a graduation-readiness mechanism rather than a barrier for capable students.

POL-02 (Two-Peat Rule) enforcement. The rule limits any student to two attempts at a gatekeeper course (Statics, Dynamics, Engineering Materials, Solid Mechanics, Thermodynamics I, Engineering Analysis I) without an approved faculty appeal. The UPC tracks all instances each term and the Departmental UPC reviews enforcement at the start of each term. Table 22 reports the operational counts.

Table 22: POL-02 (Two-Peat Rule) operational enforcement data, Spring 2021–Fall 2025. *Instances* count all course–repeat events; *Students* count distinct students (some repeat multiple courses); *Enforced* counts the rule-driven course-drop actions executed by UPC. Source: UG_Program_Data.xlsx, sheet *2Peat Rule*.

Measure	Sp 21	F 21	Sp 22	F 22	Sp 23	F 23	Sp 24	F 24	Sp 25	F 25
Instances flagged	62	89	68	91	68	69	37	56	48	48
Students flagged	50	68	50	66	50	55	33	50	—	51
Enforced (drops)	1	3	4	10	2	10	12	15	15	17

The trend in *Enforced drops* is the most diagnostic measure: enforcement rose from 1–4 actions per term (Sp 2021–Sp 2022, when the rule was newly operational under the Fall 2020 catalog) to 12–17 actions per term (Sp 2024–Fall 2025), indicating that the rule is being increasingly applied as the UPC and the Departmental UPC committee gained operational comfort with the process. Simultaneously, the total *Instances flagged* fell from 91 (F 2022 peak) to 48 (F 2025), suggesting that the rule’s deterrent effect is being felt by students before the third attempt. The 45 BSME students currently in violation of the rule reported in the AY 2025–26 update (with 21 having appealed and 1 denied, per the *BSME Changes* sheet) reflects the cumulative carry-forward population the UPC is actively managing.

Policy update for AY 2026–27. Per the *BSME Changes* sheet, the Departmental UPC committee approved several POL-02 enhancements for the next cycle: (i) no fourth attempts without documented extraordinary circumstances; (ii) second-appeal approval contingent on first-appeal success; (iii) when appealing for multiple courses, students restricted to one course at a time; (iv) pre-Census-Day email warnings to all students attempting a Two-Peat course for the first time. These changes are documented as carry-forward operational refinements (not formal AFIs); they appear in Section 4.B.5 as a sub-item under POL-02 continuing enforcement.

Combined POL-01/POL-02 picture. The two rules act in complement: POL-01 prevents low overall progress (GPA-based), POL-02 prevents stuck-on-a-gatekeeper progress (course-based). Both rules cooperate with the SLO-1 Problem Analysis Overlay AFI (deployed Sp 2026; Section 4.B.2.2.1)—the Overlay seeks to raise the underlying attainment that, when low, triggers Two-Peat enforcement. The decline in MECE 3321 Solids pass rate from 50.5% (AY 21–22) to 70.2% (AY 24–25)—a +20-point improvement—is consistent with the rules’ deterrent effect on at-risk students choosing better preparation strategies before attempting the course for the second time.

4.B.5.2 Rationale Grounded in Recent Evaluations

Each priority action is grounded in specific current-cycle evidence—portal direct measures, indirect-measure survey data, faculty meeting minutes, or stakeholder input. The rationale paragraphs below follow the same order as Table 38.

AS-04, AS-05, AS-06: SLO Overlay re-assessments. The three SLO Overlay AFIs (Problem Analysis, Concept Development, Audience Adaptation) were approved Aug/Sept 2025. AS-05 (SLO-2 Concept Development, MECE 4361 SD-I) and AS-06 (SLO-3 Audience Adaptation, MECE 4361 SD-I and mid-program communication courses) were deployed in Fall 2025 and *the loop closed in Fall 2025* with the first post-deployment term in the portal; their closed-loop trajectories continue accruing through Sp 2026, F 2026, and Sp 2027, with Sp 2027 the full multi-term closure point. AS-04 (SLO-1 Problem Analysis, program-wide) was deployed in Spring 2026 because the primary trigger courses (MECE 2302 Dynamics and MECE 2340 Engineering Materials) are Spring-only offerings; first program-wide post-deployment data accrues Fall 2026 once Fall-only SLO-1 artifact courses (MECE 3315, 3360, 3450, 4350) contribute. The rationale matches the original AFIs: SLO-1 has a chronic lower-division bottleneck (MECE 2302 declining trend; MECE 2340 and MECE 3315 with volatile term-to-term performance); SLO-2 has a documented Sp 2024 capstone dip to 65% in MECE 4361 with partial recovery; SLO-3 has the Sp 2024 communication-skills dip that drove the proactive intervention. What changes next cycle is closure of the loop on a common metric for SLO-1.

NX-01: MECE 4362 (SD-II) capstone-assessment review. Across Spring 2023–Fall 2025 the MECE 4362 (SD-II) records report stable, consistent term-over-term attainment. Several performance indicators sit genuinely below threshold—most prominently SLO-7 PI-2 (*Choose appropriate learning strategies*, Apply) at 0%, with lower SD-II values also on SLO-2 (~21%), SLO-3 (~52%), and SLO-5 PI-2 (~4.5%); SLO-4 (~75%) and SLO-6 (~73%) sit at or above threshold. Because these are the program’s culminating-course results, the SLO-2, SLO-3, SLO-5, and SLO-7 committees, with the MECE 4362 instructor of record and the ABET Coordinator, will conduct a course-level SD-II assessment review beginning Fall 2026: examine the SD-II artifacts and rubric application for the affected PIs, determine whether the gaps reflect SD-II assessment design or genuine attainment shortfalls, and where warranted scope a course-level AFI with a documented intervention and re-assessment plan. Preliminary findings are reported at the May 2027 mid-cycle retreat; final disposition at the May 2028 annual retreat. Owners: SLO-2, SLO-3, SLO-5, and SLO-7 Committee Chairs; MECE 4362 Instructor of Record; ABET Coordinator.

NX-01a: Portal reporting enhancements (paired with NX-01). To make routine cross-term review easier during the SD-II review and in general, the portal gains three reporting aids in Fall 2026: (i) a cross-term comparison view that displays each course/SLO pair’s values across consecutive terms side by side; (ii) a dashboard tile summarizing course/SLO pairs that have been flat or below threshold across multiple terms, so persistent signals surface on login; and (iii) a `source=backfill` provenance flag distinguishing records imported from the pre-Fall-2024 spreadsheet workflow from records entered directly in the portal, so committees can confirm conclusions are robust across data sources. Owner: ABET Coordinator (portal maintenance); scoped within the existing portal

maintenance workflow (Section 4.0.4.9). NX-01a closes when all three views are deployed and documented in the portal change log; verification is reported at the May 2027 mid-cycle retreat.

NX-02: MECE 1221 SLO-2 additional data collection. MECE 1221 (Engineering Graphics, sophomore) reported a single Fall 2024 SLO-2 PI-3 data point at 67%—no other terms have MECE 1221 records in the cycle. The Fall 2025 SLO-2 Concept Development Overlay (AS-05) was deployed at the capstone level (MECE 4361) rather than at the sophomore level; the SLO-2 committee judged the capstone gap higher-impact and judged the MECE 1221 result preliminary, since one data point does not establish a pattern. The next-cycle action is additional data collection: the SLO-2 committee and the MECE 1221 instructor of record will collect MECE 1221 SLO-2 data points in Sp 2026, F 2026, and Sp 2027 (no new artifact required—the existing final project applies the existing rubric). The decision criterion is then: if two or more of the three new data points fall below threshold, a course-level AFI is initiated; if attainment lifts to or above threshold, the F 2024 result is documented as a single-term effect and NX-02 closes as *Resolved*. The investigation-before-intervention approach is deliberate—the F 2024 result alone is insufficient evidence to scope an intervention.

NX-03: MECE 2140 SLO-6 coverage gap closure. MECE 2140 (Engineering Materials Lab) reported a single Sp 2022 SLO-6 data point at 60% with a strong cognitive-level gradient (Apply 86%, Analyze 60%, Evaluate 39%). No further MECE 2140 SLO-6 records have been collected, leaving the coverage gap as the dominant interpretive uncertainty for SLO-6. The next-cycle action collects SLO-6 evidence from MECE 2140 in Fall 2026 and Spring 2027 using the existing rubric and PI-1/2/3 framework. Once two to three additional data points accrue, the SLO-6 committee will determine whether course-level intervention is needed.

NX-04: SLO-5 MECE 4361 PI-2 and PI-5 follow-up investigation. The SLO-5 MECE 4361 evidence shows two PI-level signals that warrant joint Tier-3 investigation (Section 4.B.3.5). PI-2 (*Plan tasks and meet deadlines*) attainment dipped to 40% in Spring 2024—a single below-threshold term sandwiched between above-threshold terms (S 2023 100%, S 2025 80%, F 2025 82%); PI-5 (*Communicates effectively in a team setting*) declined from 100% (Sp 2023) to 73% (F 2025). The localized signals warrant follow-up investigation rather than immediate intervention; the SLO-5 committee will examine four possibilities jointly. (i) A cohort/grading effect: specific cohorts or grading calibrations produced the dips that subsequent terms have largely recovered from. (ii) The start of a recurring pattern: subsequent terms' values may not be sustained, requiring a course-level intervention. (iii) Rubric calibration drift on PI-2 and PI-5: the instructor pool or rubric application varied between the dip terms and surrounding terms in a way that needs alignment. (iv) A common cause: both PI-level signals reflect the same underlying issue (e.g., team-formation timing, project complexity, or sponsor-engagement load) that a single targeted action could address. The next-cycle action is an SLO-5 committee analysis beginning Spring 2027 with the MECE 4361 instructor of record, with the decision (no action, monitoring continuation, or course-level AFI) made at the May 2028 retreat.

NX-05: Earlier-pipeline coverage expansion for SLO-7. SLO-7 (Lifelong Learning) is currently assessed only in the capstone sequence (MECE 4361 and 4362). The next-cycle plan adds evidence points in MECE 2140 (Engineering Materials Lab) and MECE 3170 (Thermal Fluids Laboratory), both of which already include self-directed-learning components that map onto SLO-7 criteria. The action requires no new instructional content—only rubric application against artifacts already being produced. SLO-5 (Teamwork) coverage is no longer narrow following the current-

cycle discovery of MECE 3320 SLO-5 data (Section 4.B.3.5); MECE 3320 alongside MECE 4361 provides a two-course evidence base spanning mid- and upper-division team-based work, so an SLO-5 coverage expansion is not part of NX-05's scope.

NX-06: Alumni survey instrument design and pilot. The Capstone Exit Survey at graduation is the indirect-measure-of-record for the current cycle. Establishing dedicated 1-, 3-, and 5-year post-graduation alumni surveys broadens the indirect-measure base, supports PEO-attainment review (PEOs target 3–5 years post-graduation, Criterion 2 Section 2.E), and responds to EAC interest from the September 2022 and December 2024 meetings. The instrument will be PEO-aligned; the pilot targets the F 2025–Sp 2026 graduating cohort using Capstone Exit Survey contact information.

NX-07: Employer satisfaction instrument. The program receives industry input through EAC meetings but does not administer a stand-alone employer satisfaction survey. The new instrument complements EAC input by structured employer feedback and supports triangulation with the alumni survey. EAC member companies agreed in principle (December 2024 meeting) to participate in distribution to their hiring managers; the instrument will be lightweight (<5 minutes) to maximize response.

NX-08: SLO-4 current-issues dimension monitoring. The Capstone Exit Survey reports *current issues and trends* at 86%—the lowest of eleven program-quality dimensions but well above any threshold of concern. One SLO-4 PI shows slight downward direct-measure drift while overall SLO-4 attainment remains stable above threshold. The next-cycle action is monitoring, not intervention: the SLO-4 committee tracks both the direct-measure PI-level drift and the indirect-measure dimension across the next cycle, with a decision point at the May 2028 retreat. If both signals strengthen, a content-level intervention will be considered.

4.B.5.3 Risk and Dependency Notes

The next-cycle plan is realistic and resourced from existing departmental capacity. The following risks and dependencies are noted for tracking across the retreat cycle.

Risk 1: Continued enrollment growth pressure on lower-division courses. Fall full-time undergraduate enrollment grew 35.3% across the current cycle, with first-year FT enrollment nearly doubling (Section 4.B.4.3). The foundational MECE service courses (1101, 1221, 2140, 2302, 2340) and the Senior Design sequence are at or near capacity. This is the dominant risk to the next-cycle plan because it stresses instructors of record (NX-02, NX-03 owners), affects data-collection cadence, and competes with AFI implementation time. Mitigation: NX-02 and NX-03 require no new content—only continued rubric application against existing artifacts. Faculty-load resourcing is reviewed at each annual retreat.

Risk 2: MECE 4362 SD-II review outcome. The NX-01 SD-II review may or may not identify an instructional/assessment gap warranting a course-level AFI; the outcome is not predictable in advance. The risk is timing, not outcome: NX-01 should complete before Spring 2027 so AS-05 (SLO-2 Overlay capstone re-assessment) does not collide with a possible unrelated SD-II intervention. Mitigation: NX-01 is the earliest-completing action in the plan, explicitly to clear AS-05's path.

Risk 3: Alumni survey response rates. NX-06 depends on response rates the program has not yet established for a formal instrument. The Capstone Exit Survey, by contrast, is administered

captively at the end of MECE 4362 with essentially universal response. First-deployment alumni surveys in engineering typically achieve 10–30%. Mitigation: NX-06 is explicitly framed as “design and pilot,” paired with the NUBE alumni network (Section 2.E.2.2); the pilot informs response-rate estimation, with operational deployment timeline revised at the May 2028 retreat.

Risk 4: SLO-committee chair workload. Eleven priority actions across seven SLO committees plus the ABET Coordinator represent substantial senior-faculty time. Mitigation: the Department Chair manages SLO-committee-chair service-load credit; the planning horizon spreads new actions across two academic years; consolidated retreats (May 2027, May 2028) provide synchronized rather than continuous review pressure.

Risk 5: SLO-committee turnover. Faculty turnover is the standard background risk. Mitigation: the portal preserves all records, rubric versions, and AFI documentation across committee-membership changes; SLO Binders retain current and prior-cycle exemplars; AFI ownership is recorded at the committee level rather than the individual level.

Dependency 1: Institutional-research cohort tables. Data streams flagged in Section 4.B.4.3 (retention, 4/6-year graduation, demographics) are scheduled for follow-on revision once IR delivery is finalized. NX-02, NX-03, and the HSI-mission narrative for the May 2028 retreat will benefit from these data.

Dependency 2: EAC engagement timing. The next EAC meeting is March 2026; subsequent meetings follow a 12–18 month rotation. NX-07 (employer instrument) and NX-06 (alumni instrument) draw on EAC member-company participation and industry-perspective input. The program’s standing relationship with the EAC (Section 4.B.4.1) makes this a low risk.

Dependency 3: Curriculum-revision impact data for CR-01 and CR-02. The math/analysis sequence revision and CSCI 1380 addition were implemented in the 2022–2024 catalog. Impact on SLO-1, foundational-sequence attainment, and SLO-6 PI-2 attainment becomes visible only as affected cohorts reach the assessing courses—during the next cycle. The affected-cohort vs. pre-revision-cohort comparison will be reviewed at the May 2028 retreat.

Dependency 4: Continued ABET Portal availability. The portal (Section 4.0.4) underlies all eleven next-cycle actions. The Department maintains operational responsibility; the system has operated continuously since Fall 2024 deployment. No foreseeable risk attaches.

Summary of the next-cycle plan. Eleven priority actions are scheduled for AY 2026–27 and AY 2027–28: three continuing AFI re-assessments closing the loop on the SLO-1/2/3 Overlays (SLO-2 and SLO-3 deployed Fall 2025 with first-term loop closure already in the portal; SLO-1 deployed Spring 2026 with first program-wide post-deployment cohort accruing Fall 2026), and eight new items (NX-01 through NX-08) responding to evidence-based concerns identified during the current cycle. Each action carries owner, target start and re-assessment terms, and rationale traceable to Section 4.A.6, 4.B.3, or 4.B.4 evidence. Risk and dependency notes are flagged for tracking at the May 2027 and May 2028 retreats. The catalog entries (AS-04–AS-06 and NX-01–NX-08) will be added to Section 4.B.2.1 following May 2026 faculty retreat approval.

4.C.4 Portal Access for the Site Visit

The BSME ABET Assessment Portal (Section 4.0.4) is the program’s system of record and the most efficient single resource for the visit team to verify the chapter’s claims.

Read-only credentials. The ABET Coordinator (Dr. Isaac Choutapalli) will provision read-only PEV credentials prior to the visit. The credentials grant access to all assessment-data views, all analysis pages, and the administrative export module, but do not permit data modification or deletion.

Recommended portal-exploration pathway. The visit team is encouraged to follow the pathway below to verify the chapter’s central claims:

1. Login at the portal URL provided in the pre-visit packet. The portal landing page presents a navigation bar with the principal modules (Records, Analysis, Survey, Interventions, Supporting Materials, Admin).
2. Records view (/records). The complete list of direct-measure entries (course–SLO–PI–Bloom–EP%–narrative). Filter by SLO, course, or semester to verify the data underlying Section 4.A.6.
3. Per-SLO analysis (/slo_analysis?slo=N). The portal-generated four-panel summary for each SLO (course-level bars, Bloom distribution, semester trend with $\hat{\beta}_1$ and p -value, PI-wise mixed-effects trends). These are the exact figures reproduced in Section 4.A.6.
4. Per-course analysis (/course_analysis?course=MECE3320). Per-course-per-SLO panels showing course-level attainment, Bloom distribution, course-level trend, and PI-wise trends. Used in Section 4.B.3 case studies.
5. Capstone Exit Survey (/survey/analyze). The 100% stacked-bar visualization of indirect-measure data ($N = 504$ responses across F 2020–Sp 2025), supporting Section 4.B.4.2.
6. Interventions log (/interventions). The AIC-format AFI documentation (Analysis–Intervention–Comparison) for each of the thirteen catalog AFIs, supporting Section 4.B.2.1.
7. Admin export (/admin/export). CSV and HTML export of any portion of the data, available on demand. Visit-team members may request specific exports during the visit.

Portal versus chapter consistency. The portal is the live data system; this chapter is the curated synthesis. Every figure, table, and trend statistic in Sections 4.A.6 and 4.B.3 is regeneratable from the portal records. Any chapter-vs-portal discrepancy identified during the visit will be reconciled on the spot by the ABET Coordinator.

Walkthrough session. A scheduled portal-walkthrough is on the on-site visit agenda. The ABET Coordinator will demonstrate data entry, the SLO-committee analysis pages, the Capstone Exit Survey analysis, and the export module, so the visit team can subsequently verify chapter claims independently.

4.C.5 Crosswalk: This Chapter <-> ABET Criterion 4 Required Items

Table 23 maps each ABET Criterion 4 required element to the section(s) of this chapter where it is addressed, providing a PEV-friendly checklist for the visit-team review.

Table 23: Crosswalk between ABET Criterion 4 required elements and the sections of this chapter where each element is addressed. Section references use the chapter's internal numbering (4.0, 4.A, 4.B, 4.C). Asterisk (*) items in the right-hand column indicate the * sub-items on the E331 PEV Review Worksheet that PEVs specifically scrutinize.

ABET Criterion 4 Required Element	Section(s) in This Chapter	Supporting Evidence / Notes
Assessment Processes (Criterion 4, paragraph 1)		
Listing/description of assessment processes used to gather data on student attainment of each SLO	Section 4.A.2 (PI process maps, Tables 3–9)	One process map per SLO; PI-aligned rubrics; assessing courses identified
Frequency of assessment for each SLO	Sections 4.A.2, 4.0.3.3	Each Fall/Spring (Summer when offered); documented in Frequency column of each SLO process map
Assessment processes used <i>regularly</i>	Sections 4.0.3.3, 4.0.3.4, 4.A.6	10 terms of continuous data F 2020–Sp 2025; semester workflow and annual cycle documented
Assessment processes are <i>appropriate</i>	Sections 4.0.2, 4.A.2, 4.0.4.4–4.0.4.5	PI-aligned rubrics; E/P/A/N four-level scale; EP% $\geq$ 70% threshold; Bloom-level analysis
* Assessment processes are <i>sustainable</i>	Section 4.0.4.9	Four-part sustainability plan: coordinator continuity (Dr. Choutapalli), portal maintenance, knowledge/code preservation, distributed workload
Assessment processes are <i>documented</i>	Sections 4.0.4, 4.0.5, 4.C.1	Portal as system of record; Evidence Map (Table 1); SLO Binders; Records Archive
Effective assessment process(es)	Sections 4.0.2, 4.A.2, 4.0.4	Portal-driven, PI-grained, Bloom-aware, statistically analyzed; produces all chapter figures
Evaluation Processes (Criterion 4, paragraph 1)		
Evaluation processes used to interpret assessment data	Sections 4.0.3.4, 4.B.1.2	SLO committee + faculty retreat review; explicit 5-step process
* Program <i>collectively considered all data</i> for an outcome when evaluating attainment	Section 4.A.2 (triangulation framework + Figure 7), Section 4.B.1.1, Section 4.B.4 (Figure 49)	Direct + indirect + additional measures triangulate at SLO-committee level; no single-input AFI triggers

ABET Criterion 4 Required Element	Section(s) in This Chapter	Supporting Evidence / Notes
Expected level of attainment (target) for each SLO	Section 4.0.2 (definition); Section 4.A.3 (per-SLO restatement)	EP% $\geq$ 70% threshold; 67–70% borderline; <67% not-met
Extent to which each SLO is attained (results)	Section 4.A.6 (Outcome-Level Interpretation); Figure 8 (SLO at-a-glance)	Per-SLO four-panel portal figures; program-level trends with $\hat{\beta}_1$ and p -value; at-a-glance summary
Documentation and Maintenance (Criterion 4, paragraph 1)		
How documentation is maintained	Section 4.0.5 (Evidence System); Section 4.C.1–4.C.2	Portal + SLO Binders + Course Binders + Records Archive; retention policy documented
Continuous Improvement (Criterion 4, paragraph 2)		
How evaluation results are used as input to continuous improvement	Section 4.B.1 (Using Evaluation Results)	Inputs to improvement (Section 4.B.1.1); decision-making and prioritization (Section 4.B.1.2); portal-supported prioritization (Section 4.B.1.3)
How <i>other available information</i> is used as input	Section 4.B.4 (nine subsections); Figure 49 (Indirect-Measure Source Map)	EAC industry input (4.B.4.1); Capstone Exit Survey (4.B.4.2); institutional metrics (4.B.4.3); benchmarking (4.B.4.4); pass rates (4.B.4.5); pretest history (4.B.4.6); transfer data (4.B.4.7); socio-demographics (4.B.4.8); retention/graduation (4.B.4.9)
Specific actions taken (AFIs) during the review cycle	Section 4.B.2.1 (AFI Catalog, 13 actions); Section 4.B.2.2 (three SLO Overlay AFIs)	Each action with ID, category, term, description, owner, status; reproduced in AFI Dashboard (Figure 51)
* Results of changes when re-assessed (closed-loop evidence)	Section 4.B.3 (closed-loop case studies, one per SLO); Section 4.B.3.6 (MECE 3320 closed loop)	Per-SLO case studies; honest disclosure of partial/ineffective cases (Section 4.B.3.8)
Forward Plan (Criterion 4, paragraph 2)		
Future improvement plans based on recent evaluations	Section 4.B.5 (Future Improvement Plans)	Eleven next-cycle actions with owner, term, rationale; three continuing AFI re-assessments; eight new next-cycle items (NX-01 through NX-08)
Rationale for each planned change traceable to evidence	Section 4.B.5; each next-cycle item links to Section 4.A.6, 4.B.3, or 4.B.4 evidence	Risk and dependency notes flagged for May 2027 and May 2028 retreats

ABET Criterion 4 Required Element	Section(s) in This Chapter	Supporting Evidence / Notes
Cross-Criterion Supporting Items		
PEO review process and constituent involvement (Criterion 2)	Sections 4.0.6.4, 4.B.2.1 (AFI AS-03); Criterion 2 reproduces detail	Three-year PEO/SLO review completed 2023–2025; PEO revisions adopted; SLO wording retained
SO definitions cover (1)–(7) (Criterion 3)	Section 4.A.1	Seven SLOs aligned with ABET SO (1)–(7); statements reproduced
Supporting materials adequate for visit team (APPM I.E.5.b.(2))	Sections 4.C.1, 4.C.3, 4.C.4	Index of instruments; example artifacts; portal access for the visit team
Team able to interview appropriate individuals (APPM I.E.5.a)	Visit-team interview list (provided separately)	Department Chair, ABET Coordinator, SLO Chairs, Senior Design Coordinator, Undergraduate Program Coordinator, EAC representatives, faculty, students, alumni

Summary. Every required element of ABET Criterion 4—and every cross-criterion supporting item the E331 PEV Review Worksheet expects to see referenced in this chapter—has a documented home, with supporting evidence locatable in the portal, the SLO Binders, the Course Binders, the Records Archive, or this chapter directly. The ABET Coordinator will accompany the visit team during portal-based verification and will provide on-the-spot retrieval of any indexed item.

Appendix item D.11: PI-level attainment inventory (former Table 24, with the "How to read this table" and "What this table makes visible" notes)

Tabular companion. Table 24 reports the same 25 PIs in tabular form, adding per-row detail (lowest course/term signal with course and term named; current status disposition; specific AFI or next-cycle action with section reference) that the visualization cannot carry compactly.

Table 24: BSME PI-level attainment inventory (F 2020–Sp 2025; program-level trend slopes computed across the courses with multi-term data, with MECE 4362's stable SD-II values reported as recorded). Aggregated EP% is the mean across distinct course–term observations for each PI; lowest signal is the worst single course–term average. Values verified against the 14-course portal extract.

SLO/F	Performance Indicator	Agg. EP%	Lowest course/term signal	Status	Action / Reference
SLO-1 (Solve complex engineering problems). Program-level $\hat{\beta}_1 = -0.53$ pp/term, $p = 0.248$ ($n = 81$ course-term observations across 7 courses)					
1.1	Identify engineering problem	72%	MECE 2302 Sp 2025 (23%)	AFI active	AFI SLO-01 Problem Analysis Overlay (deployed F 2025); re-assessment AS-04; Section 4.B.2.2.1
1.2	Formulate a problem	77%	MECE 2302 Sp 2025 (35%)	AFI active	AFI SLO-01 Problem Analysis Overlay; Section 4.B.2.2.1
1.3	Solve a problem	73%	MECE 3315 Sp 2024 (28%)	AFI active	AFI SLO-01 Problem Analysis Overlay; MECE 3315 recovered Sp 2025 (90%); Section 4.B.3.1
SLO-2 (Apply engineering design). Program-level $\hat{\beta}_1 = -0.94$ pp/term, $p = 0.276$ ($n = 29$ course-term observations across 4 courses)					
2.1	Design a system, component, or process	76%	MECE 3380 Sp 2021 (65%)	AFI active	AFI SLO-02 Concept Development Overlay (deployed F 2025); MECE 4361 Sp 2025 recovery to 88%; Section 4.B.2.2.2
2.2	Design to meet desired needs	83%	MECE 3380 F 2021 (62%)	Above threshold	Normal monitoring (with MECE 3380 narrative-completeness disclosure, Section 4.B.3.8)
2.3	Design within realistic constraints	74%	MECE 4361 Sp 2024 (50%)	AFI active	AFI SLO-02 Concept Development Overlay; Section 4.B.2.2.2
SLO-3 (Communicate effectively). Program-level $\hat{\beta}_1 = -1.49$ pp/term, $p = 0.355$ ($n = 22$ course-term observations across 4 courses)					
3.1	Generate appropriate graphics	88%	MECE 4361 Sp 2024 (79%)	Above threshold	Normal monitoring; AFI SLO-03 also covers (graphics dimension)
3.2	Demonstrate adequate presentation skills	85%	MECE 4361 Sp 2024 (61%)	AFI active	AFI SLO-03 Audience Adaptation Overlay (deployed F 2025); Section 4.B.2.2.3
3.3	Apply technical writing skills	84%	MECE 4361 F 2025 (60%)	AFI active	AFI SLO-03 Audience Adaptation Overlay; Section 4.B.2.2.3
3.4	Contextualize for intended audience	92%	MECE 2140 F 2021 (85%)	Above threshold (strong)	MECE 2140 PI-4 cycle: 85/97/93% F2021–F2023; AFI SLO-03 also covers
SLO-4 (Ethics and professional responsibility). Program-level $\hat{\beta}_1 = +0.73$ pp/term, $p = 0.609$ ($n = 22$ course-term observations across 3 courses)					

Table 24 continued from previous page

SLO/F	Performance Indicator	Agg. EP%	Lowest course/term signal	Status	Action / Reference
4.1	Recognize ethical/professional responsibilities	85%	PHIL 2393 F 2023 (73%)	Above threshold	MECE 1101 92% cycle avg; PHIL 2393 79% cycle avg; normal monitoring
4.2	Make informed ethical/professional judgments	91%	PHIL 2393 F 2024 (87%)	Above threshold (strong)	MECE 4361 95% cycle avg (SD-I); PHIL 2393 89% cycle avg; normal monitoring
4.3	Consider impact of engineering solutions	77%	PHIL 2393 F 2023 (72%)	Above threshold (monitor)	Capstone Exit Survey <i>current issues</i> 86% (lowest dim); NX-08 next-cycle monitoring
SLO-5 (Teamwork). Program-level $\hat{\beta}_1 = +0.03$ pp/term, $p = 0.992$; MECE 3320 course-level $\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$ (statistically significant rising trajectory)					
5.1	Establish goals	89%	MECE 4361 Sp 2023 (77%)	Above threshold (rising)	MECE 4361 trend rising; normal monitoring
5.2	Plan tasks & meet deadlines	76%	MECE 4361 Sp 2024 (40%)	Tier-3 investigate	NX-04 joint PI-2/PI-5 structured investigation; Section 4.B.5
5.3	Fulfill duties of team roles	73%	MECE 3320 Sp 2023 (66%)	Above threshold (rising)	MECE 3320 trend $\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$; Section 4.B.3.5
5.4	Share work equally (Analyze)	72%	MECE 3320 Sp 2023 (66%)	Above threshold (rising)	MECE 3320 PI-4 Analyze: 66/72/77% Sp 2023–Sp 2025; Section 4.B.3.5
5.5	Communicate effectively on a team	88%	MECE 4361 F 2025 (73%)	Tier-3 investigate	MECE 4361 PI-5 F 2025 decline; joint NX-04 with PI-2; Section 4.B.3.5
5.6	Proficient in all aspects of the project	71%	MECE 3320 Sp 2023 (64%)	Above threshold (rising)	MECE 3320 PI-6: 64/73/76% Sp 2023–Sp 2025; Section 4.B.3.5
SLO-6 (Experimentation & data interpretation). Program-level $\hat{\beta}_1 = +3.97$ pp/term, $p = 0.085$ (positive but not statistically significant at $\alpha = 0.05$); MECE 3320 course-level $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ (statistically significant rising trajectory)					
6.1	Develop & conduct appropriate experimentation	79%	MECE 3320 Sp 2022 (68%)	Above threshold (rising)	MECE 3320 closed loop $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ at course level; Section 4.B.3.6
6.2	Analyze & interpret data	67%	MECE 2140 Sp 2022 (60%)	Above threshold (mixed)	MECE 3320 closed loop drives PI-2 improvement; PI-2 program-level mild decline noted; Section 4.B.3.6
6.3	Evaluate findings to draw conclusions	88%	MECE 2140 Sp 2022 (39%, Evaluate single observation)	Above threshold (coverage gap noted)	MECE 2140 single Sp 2022 data point at Evaluate level; NX-03 coverage expansion F 2026/Sp 2027
SLO-7 (Lifelong learning). Program-level $\hat{\beta}_1 = +0.00$ pp/term, $p = 0.000$ (zero-variance response across four flat 80% terms; capstone-concentrated; PI-2 program-level 50% reflects the MECE 4362 SD-II PI-2 Apply=0% value averaged with MECE 4361 PI-2=100%)					
7.1	Recognize ongoing need for new knowledge	100%	MECE 4361 100% all four cycle terms	Above threshold (stable)	Capstone-only coverage; NX-05 earlier-pipeline expansion

Table 24 continued from previous page

SLO/F	Performance Indicator	Indica-	Agg. EP%	Lowest course/term signal	Status	Action / Reference
7.2	Choose appropriate learning strategies		50%	MECE 4362 PI-2 Apply=0% all four cycle terms (stable)	Above threshold in MECE 4361 (100%); program-level 50% reflects genuinely low SD-II PI-2	NX-01 SD-II course-level review; NX-05 earlier-pipeline expansion
7.3	Apply new knowledge appropriately		100%	MECE 4361 100% all four cycle terms; MECE 4362 100% (replicated)	Above threshold (stable)	Capstone-only coverage; NX-05 earlier-pipeline expansion

How to read this table. Every row reports a single Performance Indicator, not an SLO average. The *Aggregated EP%* column is the mean across distinct course–term observations for that PI across the assessing courses (computed directly from the 14-course portal extract underlying every chart in Sections 4.A.6 and 4.B.3); per-PI per-term detail is accessible in the live portal (Section 4.C.4). The *Lowest course/term signal* column names the worst single observed value with its origin so the PEV can verify against the per-course panels in Section 4.B.3. The *Status* column distinguishes “Above threshold,” “Above threshold (rising)” (a documented improvement trajectory), “AFI active” (a Fall 2025 Overlay is targeting this PI), “Tier-3 investigate” (structured next-cycle investigation, not an AFI), and “Coverage gap” (insufficient data for current trend assessment). The *Action / Reference* column points to the specific AFI or next-cycle item operating on the PI and the section where its evidence is developed. PEVs reading this table never need to compute a PI value from an SLO average; the inventory makes every PI-level number directly inspectable.

What this table makes visible. Six patterns are immediately readable from Table 24. (i) The three Fall 2025 SLO Overlay AFIs each target multiple specific PIs (AFI SLO-01 targets PIs 1.1, 1.2, 1.3; AFI SLO-02 targets 2.1 and 2.3; AFI SLO-03 targets 3.2 and 3.3), with the originating below-threshold course/term signals named. (ii) A documented course-level closed loop (MECE 3320 SLO-6, $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$) operates on PIs 6.1 and 6.2 across four consecutive Spring terms; the program-level SLO-6 trend is positive ($p = 0.085$) but not statistically significant at $\alpha = 0.05$ given the small number of program-level observations. (iii) A second statistically significant course-level rising trajectory in MECE 3320 covers SLO-5 PIs 5.3, 5.4, 5.6 (Sp 2023–Sp 2025), parallel to the SLO-6 closed loop in the same course. (iv) Two MECE 4361 PI-level signals in SLO-5 (PI-2 *Plan tasks and meet deadlines* Sp 2024 dip to 40%, and PI-5 *Communicates effectively in a team setting* F 2025 decline to 73%) are jointly flagged for Tier-3 investigation under NX-04. (v) SLO-7 PI-2 (*Choose appropriate learning strategies*) shows a program-level aggregate of 50%—the MECE 4362 SD-II PI-2 Apply=0% value averaged with MECE 4361 PI-2=100%—reported at the PI grain rather than masked by an SLO-level average; the genuinely low SD-II PI-2 attainment is reviewed at the course level under NX-01. (vi) MECE 4362’s SD-II values are reported as recorded

(stable across terms); because the SD-II series is flat it contributes no temporal trend, and its genuinely below-threshold PIs are reviewed at the course level under NX-01 (Section 4.B.3.8).

Appendix item D.12: SLO assessment process maps (former Tables for SLO-1 through SLO-7)

The seven per-SLO assessment process maps (Performance Indicators, intended Bloom level, direct-measure courses and artifacts, threshold, frequency, and evidence location) referenced in Section 4.A.2 follow.

Table 25: SLO-1 assessment process map (direct measures). SLO-1: *An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Identify engineering problem	Recorded per portal entry	MECE 2302, 2340, 3315, 3360, 3450, 4350 (selected exam problems / structured problem sets emphasizing problem identification)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Formulate a problem	Recorded per portal entry	MECE 2302, 2340, 3315, 3360, 3380, 3450, 4350 (problem setup/modeling steps, assumptions, governing equations in exams/projects; MECE 3380 Exam-2 Problem-1)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Solve a problem	Recorded per portal entry	MECE 2302, 2340, 3315, 3360, 3380, 3450, 4350 (multi-step solutions and technical analyses in exams/assignments; MECE 3380 Exam-4 Problem-2)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 26: SLO-2 assessment process map (direct measures). SLO-2: *An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Design a system, component, or process	Create	MECE 1221, 3336, 3380, 4361, 4362 (design projects, design reports, CAD/analysis deliverables)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Design to meet desired needs	Evaluate / Create	MECE 1221, 3336, 3380, 4361, 4362 (requirements-driven design tasks; verification against needs; design justification)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Design within realistic constraints	Evaluate / Create	MECE 1221, 3336, 3380, 4361, 4362 (tradeoff studies; constraints: economic, safety, manufacturability, sustainability, etc.)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 27: SLO-3 assessment process map (direct measures). SLO-3: *An ability to communicate effectively with a range of audiences.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Generate appropriate graphics	Apply / Create	MECE 2140, 3170, 4361, 4362 (plots/figures, CAD drawings, schematics, annotated visuals in reports/posters)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Demonstrate adequate presentation skills	Apply	MECE 2140, 3170, 4361, 4362 (oral briefings, poster presentations, design reviews)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Apply technical writing skills	Apply	MECE 2140, 3170, 4361, 4362 (lab reports, technical memos, design reports, capstone reports)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-4: Contextualize for intended audience	Analyze / Apply	MECE 2140, 3170, 4361, 4362 (executive summaries, audience-appropriate framing, design review narratives)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 28: SLO-4 assessment process map (direct measures). SLO-4: *An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Recognize ethical/professional responsibilities	Understand	PHIL 2393, MECE 1101, 4361, 4362 (ethics instrument/case analysis; course-embedded ethics prompts)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Make informed ethical/professional judgments	Evaluate	PHIL 2393, MECE 1101, 4361, 4362 (ethics scenarios requiring justified decisions; reflections tied to professional codes)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Consider impact of engineering solutions	Analyze / Evaluate	PHIL 2393, MECE 1101, 4361, 4362 (impact analysis in global/economic/environmental/societal contexts)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 29: SLO-5 assessment process map (direct measures). SLO-5: *An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Establish goals	Create	MECE 2140, 3170, 4361, 4362 (team charters, project plans, capstone planning deliverables)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Plan tasks & meet deadlines	Apply / Analyze	MECE 2140, 3170, 4361, 4362 (Gantt/sprints, milestone tracking, schedule adherence evidence)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Fulfill duties of team roles	Apply	MECE 2140, 3170, 3320, 4361, 4362 (peer/team evaluations; role accountability evidence; MECE 3320 team data-reduction projects)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-4: Share work equally	Apply / Analyze	MECE 2140, 3170, 3320, 4361, 4362 (peer evaluations; contribution logs; MECE 3320 team-graded data-analysis exercises at the Analyze level)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-5: Communicate effectively on a team	Apply / Analyze	MECE 2140, 3170, 4361, 4362 (meeting minutes, team communication artifacts, peer feedback)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-6: Proficient in all aspects of the project	Apply / Evaluate	MECE 2140, 3170, 3320, 4361, 4362 (integrated project performance; design review outcomes; capstone deliverables; MECE 3320 team-graded integrated project outcomes)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 30: SLO-6 assessment process map (direct measures). SLO-6: *An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Develop & conduct appropriate experimentation	Create / Apply	MECE 2140, 3170, 3320, 4361, 4362 (lab experiments, test plans, experimental procedures)	EP% $\geq$ 70%	Each Fal- l/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

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PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-2: Analyze & interpret data	Analyze	MECE 2140, 3170, 3320, 4361, 4362 (data reduction, uncertainty/interpretation, plots and discussion sections)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Evaluate findings to draw conclusions	Evaluate	MECE 2140, 3170, 3320, 4361, 4362 (conclusion sections, engineering judgment, comparison to theory/benchmarks)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Table 31: SLO-7 assessment process map (direct measures). SLO-7: *An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.*

PI	Bloom	Direct Measure Courses (Artifacts)	Threshold	Frequency	Evidence Location
PI-1: Recognize need to acquire new knowledge	Understand	MECE 4361, 4362 (learning reflection, problem framing requiring new knowledge; capstone need-statement)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-2: Choose appropriate learning strategies	Evaluate	MECE 4361, 4362 (learning plan, source evaluation, strategy justification, literature review approach)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace
PI-3: Apply new knowledge appropriately	Apply	MECE 4361, 4362 (integration of newly learned methods/standards/tools into capstone deliverables)	EP% $\geq$ 70%	Each Fall/Spring; Summer when offered	ABET Portal; SLO Binder; Brightspace

Appendix item D.13: Portal analysis figures (program-level summaries and per-course panels)

The portal-generated program-level and per-course analysis figures referenced throughout Sections 4.A.6 and 4.B.3 (and the Capstone Exit Survey and institutional-metrics charts referenced in Sections 4.B.4.2–4.B.4.3) are collected here. Each figure retains its original caption and label, so in-text references resolve directly.

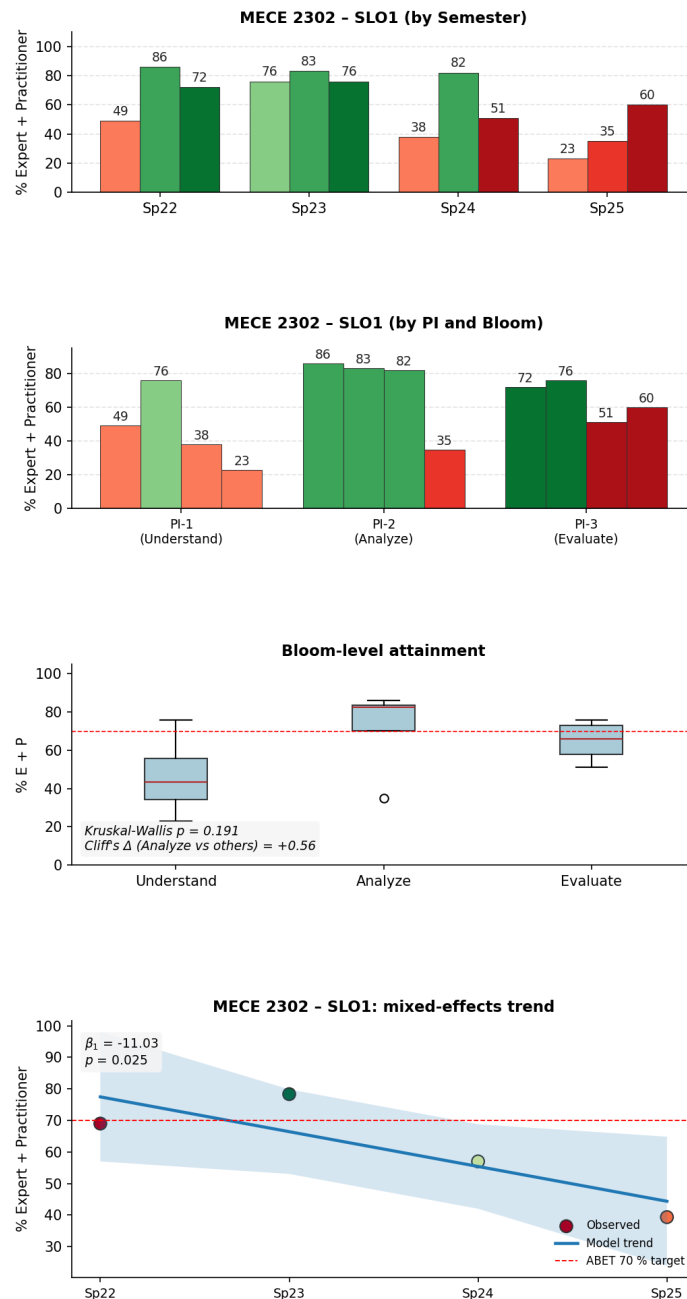


Figure 27: MECE 2302 (Dynamics) SLO-1 portal analysis. Term averages across Sp 2022 (69%), Sp 2023 (78%), Sp 2024 (57%), Sp 2025 (39%) on a stable assessment instrument confirm a statistically significant declining course-level trajectory ($\beta_1 = -11.03$ pp/term, $p = 0.025$).

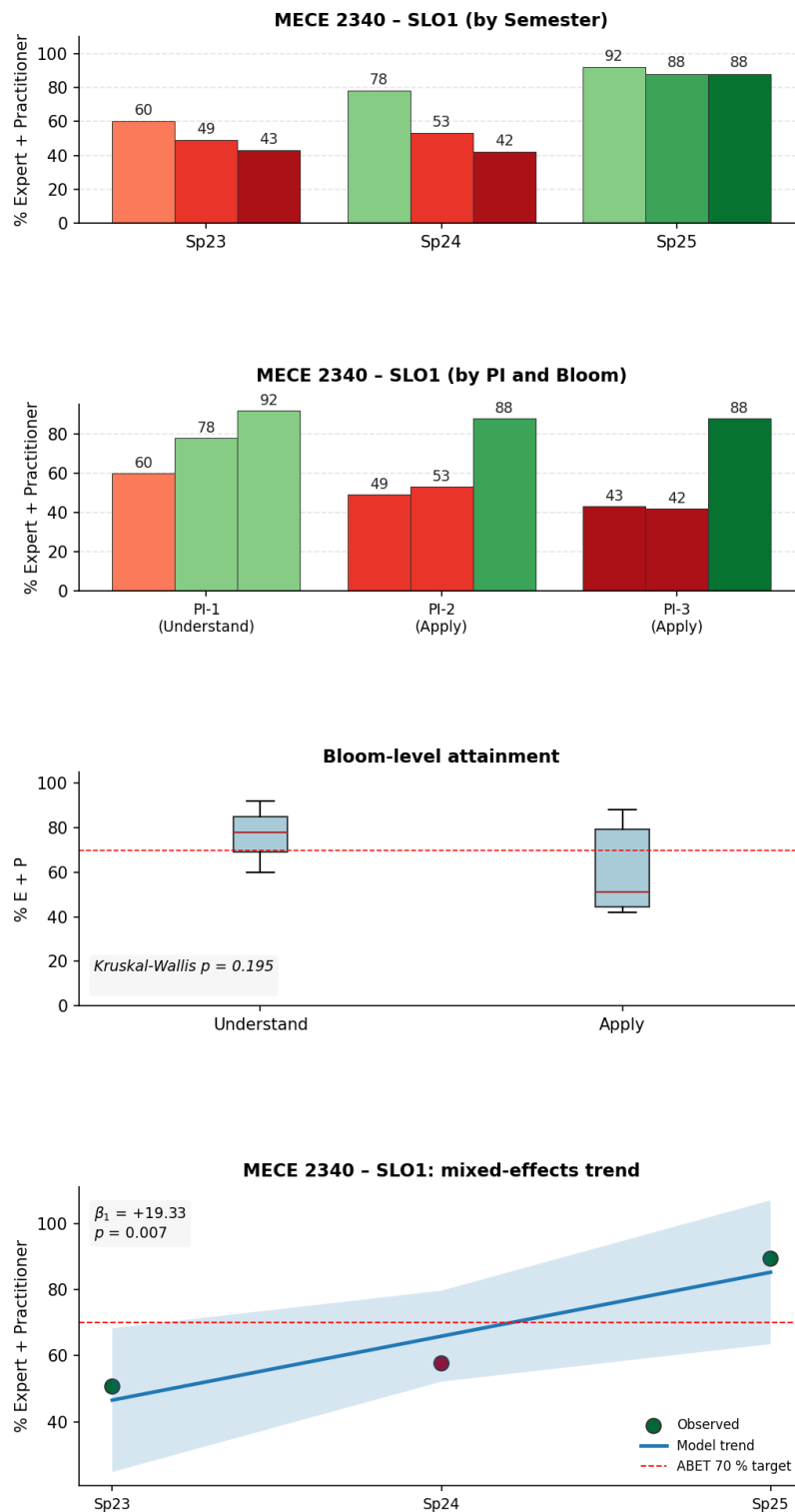


Figure 28: MECE 2340 (Engineering Materials) SLO-1 portal analysis. Rising trajectory across three Spring terms: S 2023 46%, S 2024 58%, S 2025 89%. The Sp 2025 term is the first cycle term substantially above threshold; the SLO-1 committee will monitor for sustained recovery in the next-cycle re-assessment.

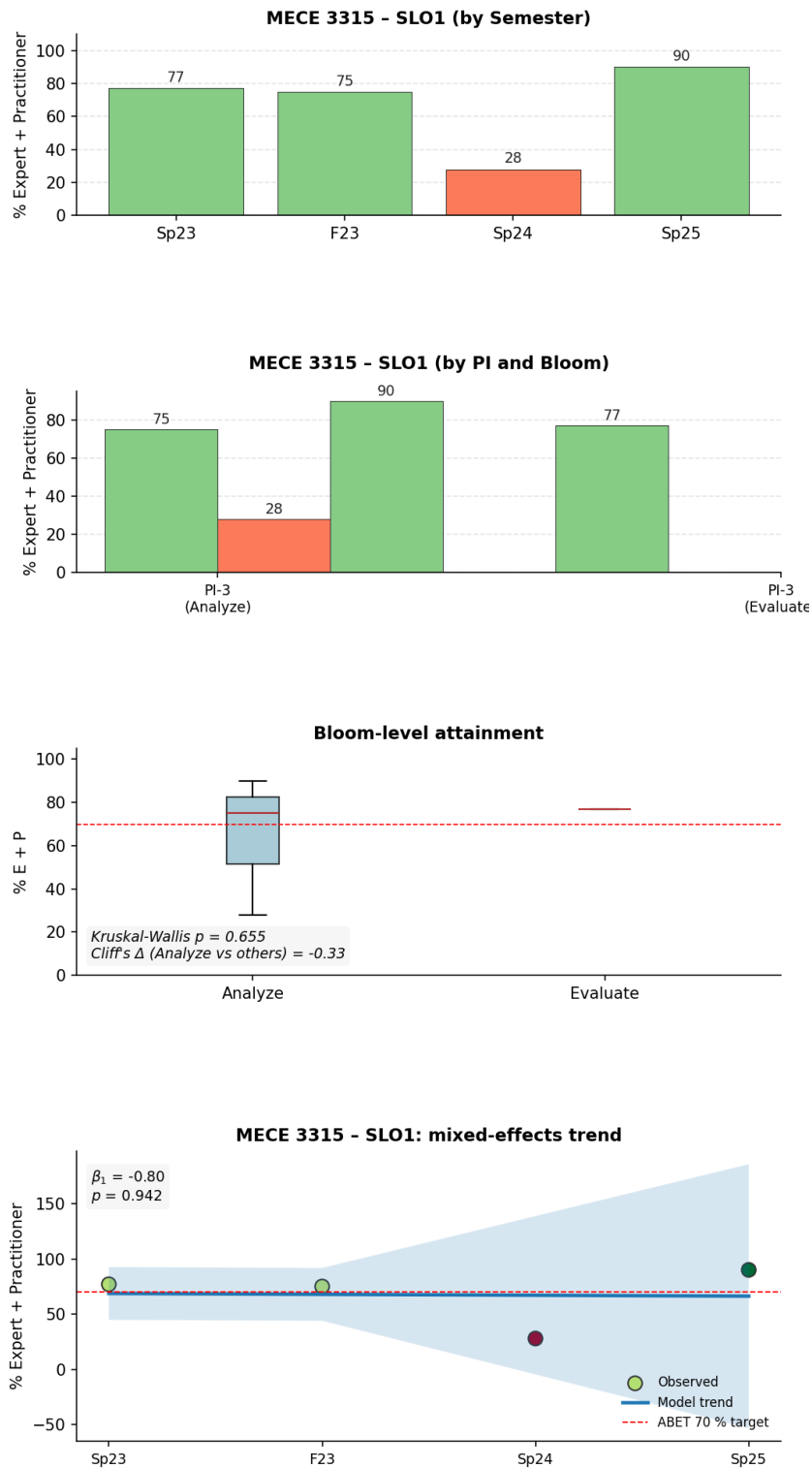


Figure 29: MECE 3315 (Fluid Mechanics) SLO-1 portal analysis (PI-3 Able to solve problem). Term-to-term volatility: S 2023 77%, F 2023 75%, S 2024 28% dip, F 2024 78% recovery, S 2025 90%. The S 2024 dip is below threshold; the recovery to 90% in Sp 2025 is substantial.

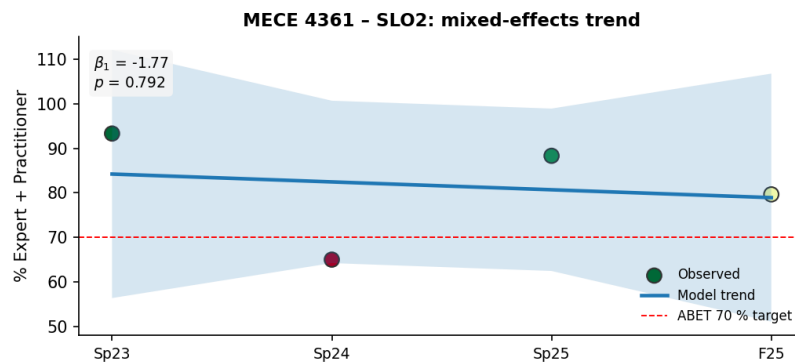
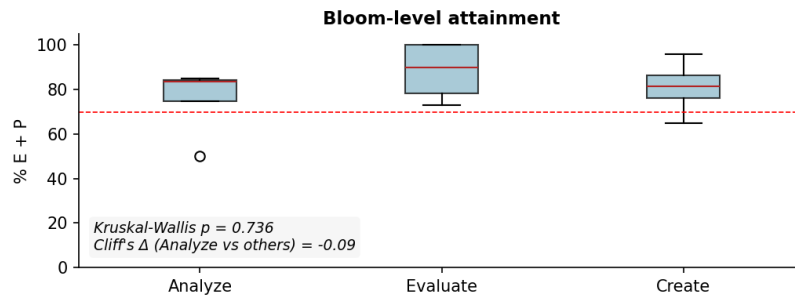
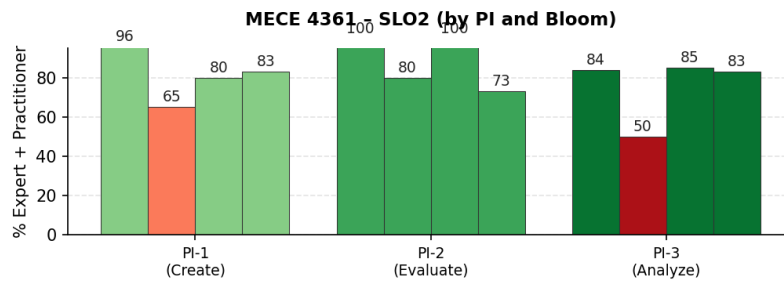
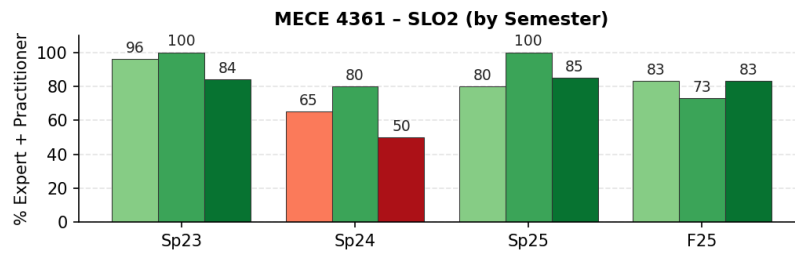


Figure 30: MECE 4361 (SD-I Capstone) SLO-2 portal analysis. The S 2024 dip (65%, below threshold) and the Sp 2025 pre-AFI recovery (88%) are visible in panel D; F 2025 first post-deployment values across all three PIs (PI-1 Create=83%, PI-2 Evaluate=73%, PI-3 Analyze=83%; course average 80%) sustain above threshold under the Concept Development Overlay.

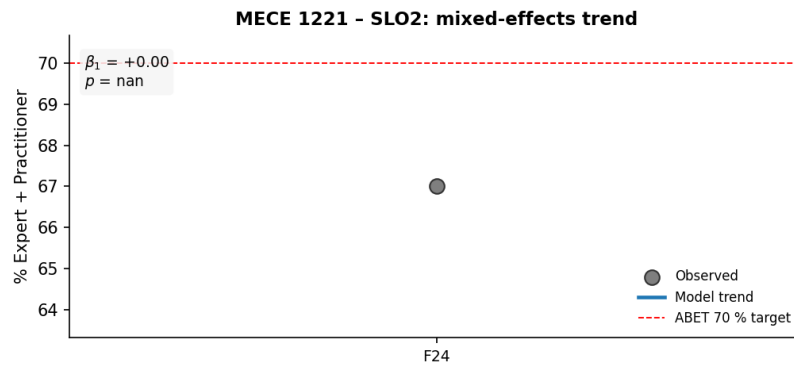
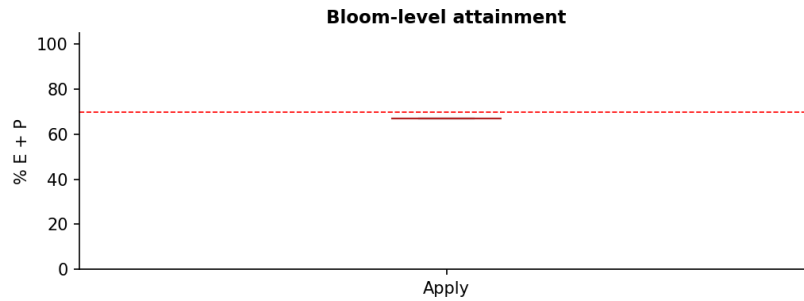
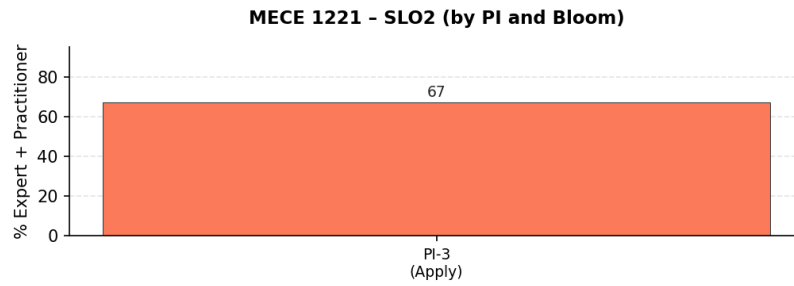
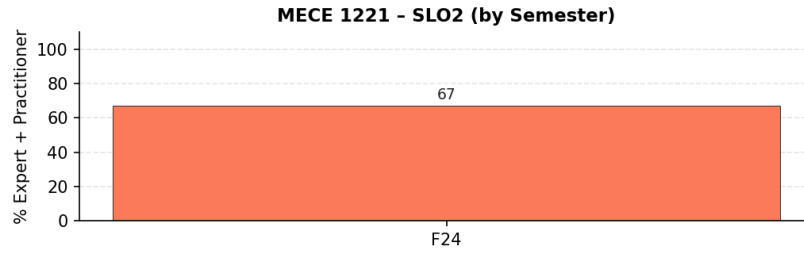


Figure 31: MECE 1221 (Introductory Design) SLO-2 portal analysis. Single F 2024 data point at 67%; open item per Section 4.B.3.8.

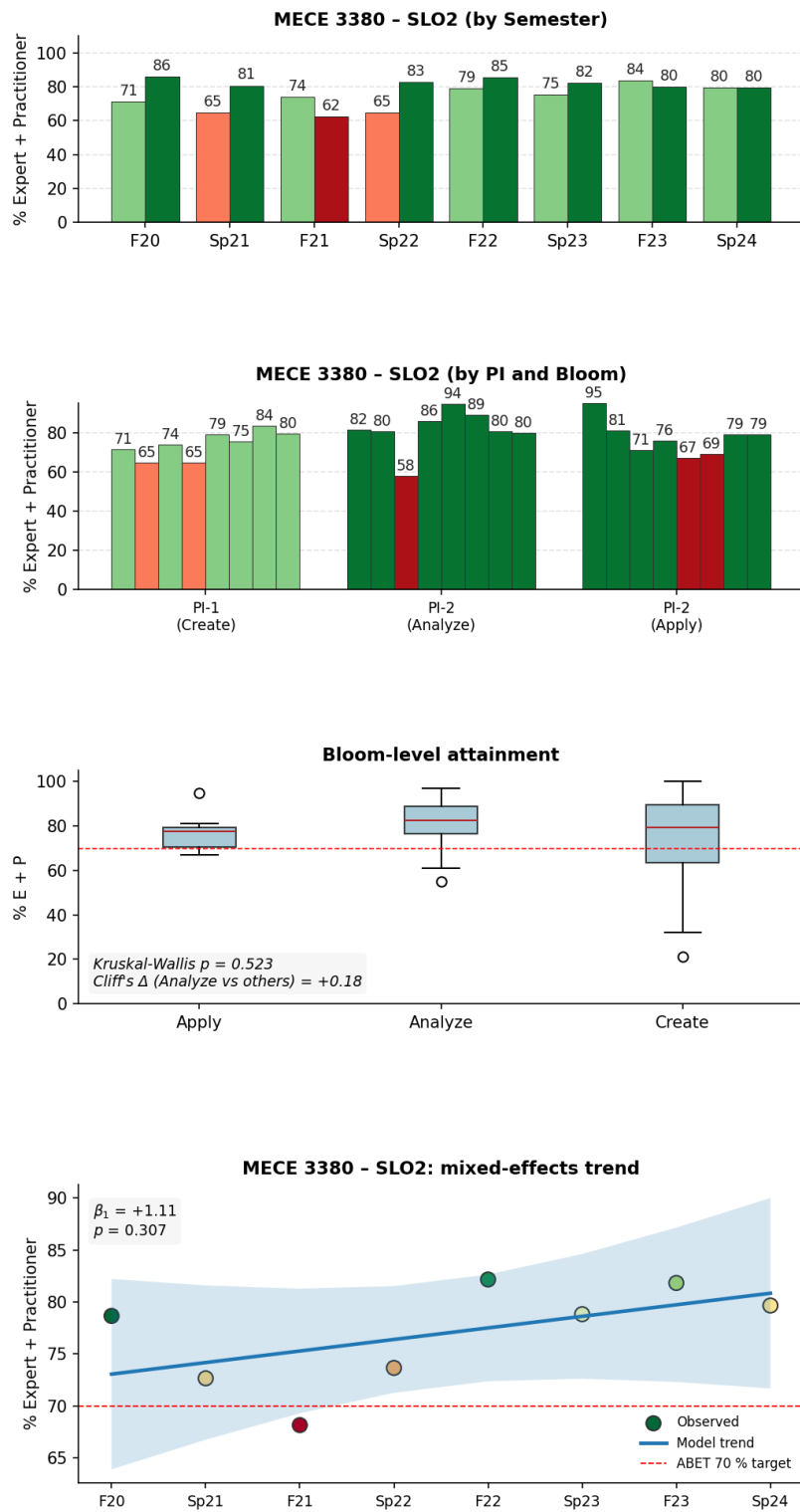


Figure 32: MECE 3380 (Dynamic Systems) SLO-2 portal analysis. Course-level average ~77%, sustained above threshold.

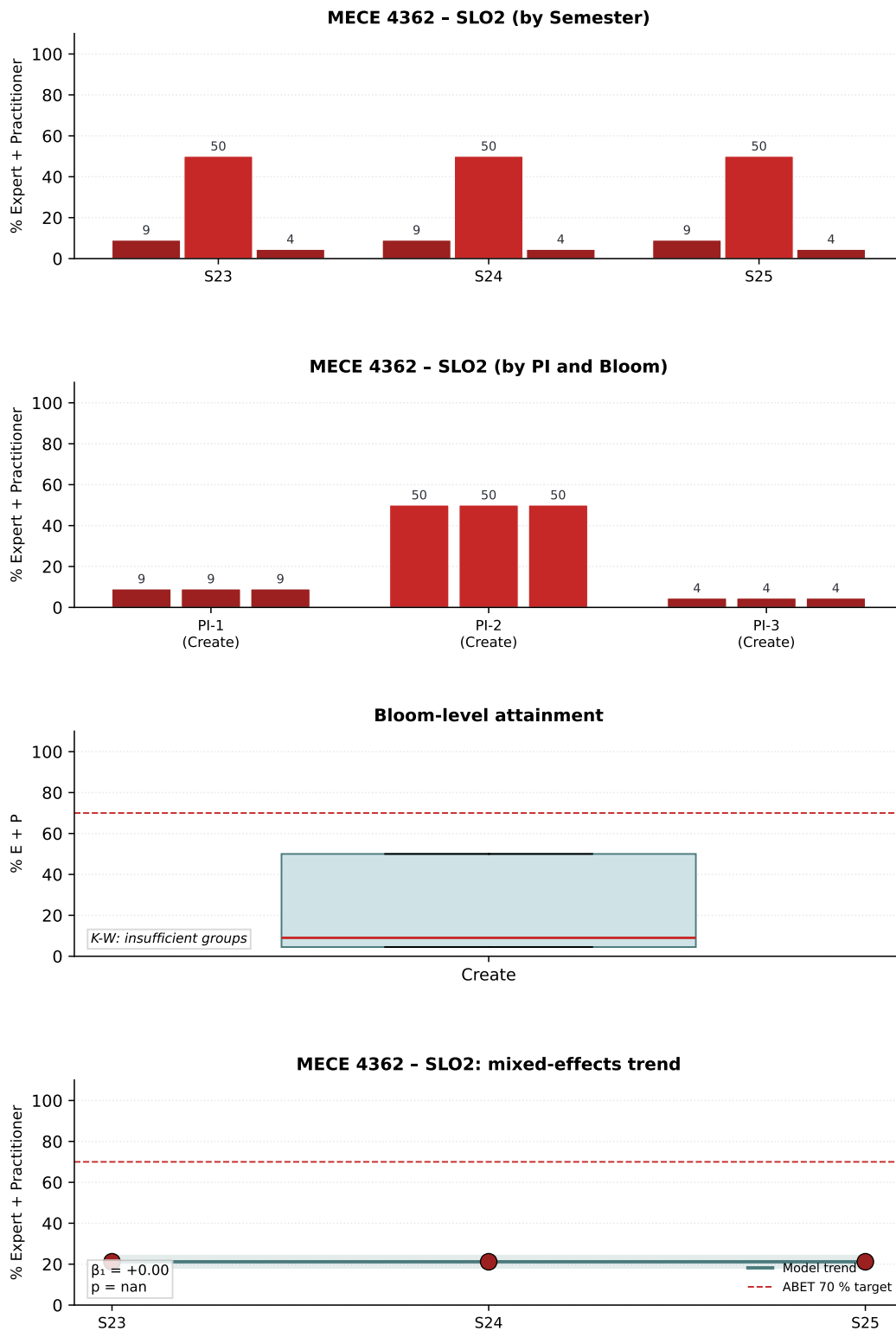


Figure 33: MECE 4362 (SD-II Capstone) SLO-2 portal analysis. Course-level average ~21% (PI-1=9%, PI-2=50%, PI-3=4.5%), stable across S 2023, S 2024, S 2025, and F 2025; genuinely below-threshold SD-II PIs reviewed under NX-01 (Section 4.B.3.8).

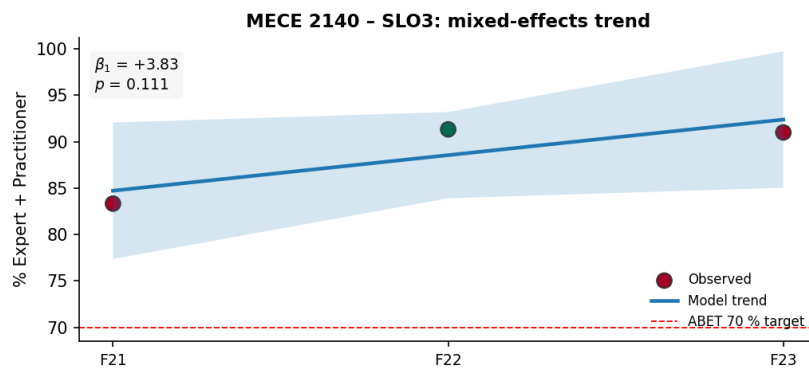
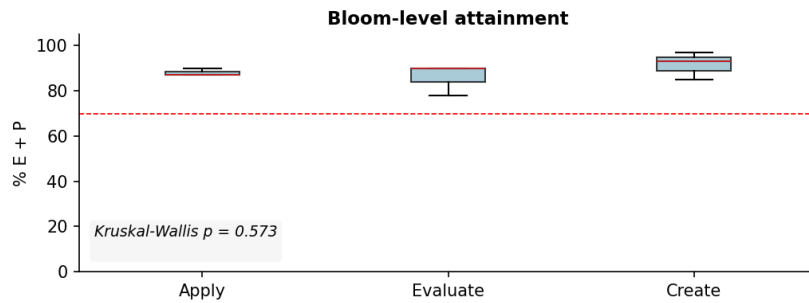
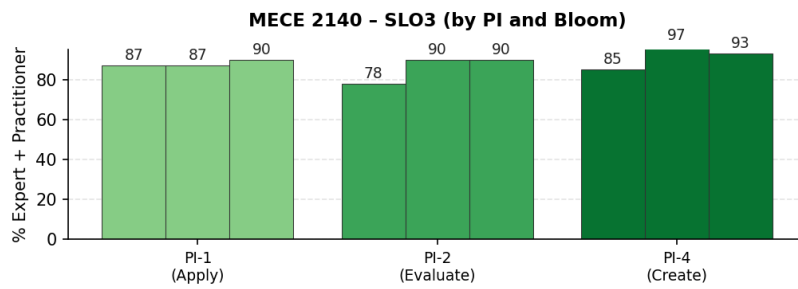
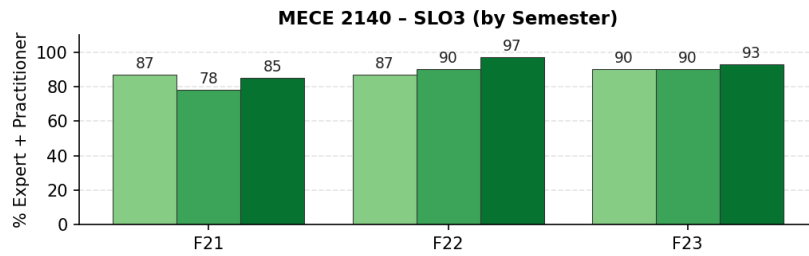


Figure 34: MECE 2140 SLO-3 portal analysis. Strong attainment sustained across F 2021–F 2023 (term averages 83%, 91%, 91% respectively) across PI-1 (graphics, Apply), PI-2 (presentation, Evaluate), and PI-4 (contextualization, Create).

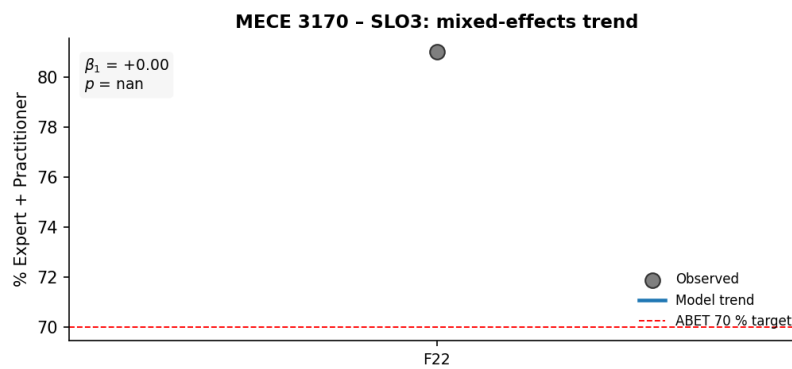
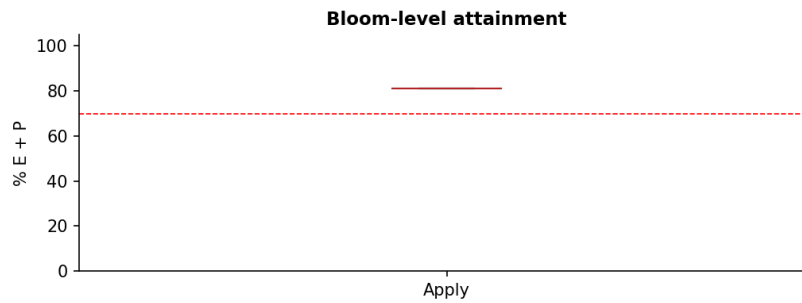
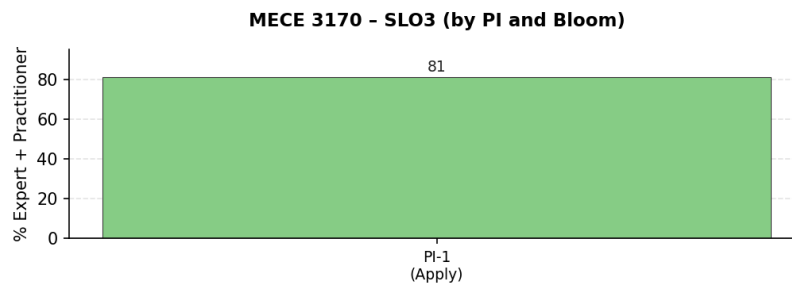
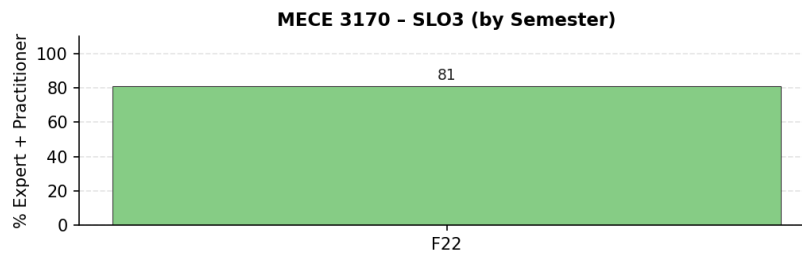


Figure 35: MECE 3170 (Thermal-Fluids Lab) SLO-3 portal analysis. Single Fall 2022 data point at 81% (PI-1 graphics, Apply level). Coverage expansion across additional terms is part of next-cycle item NX-05 (Section 4.B.5).

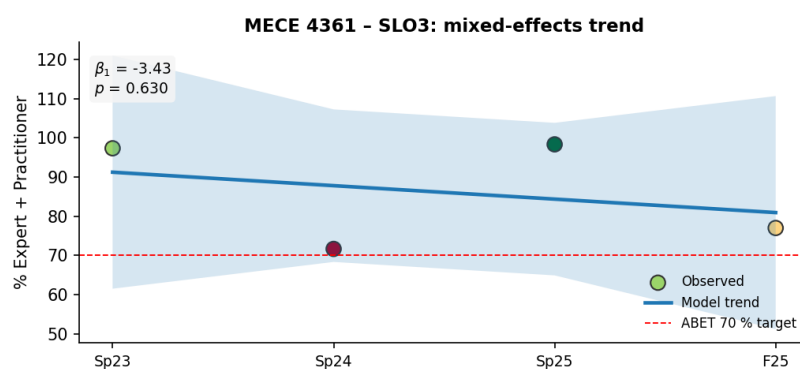
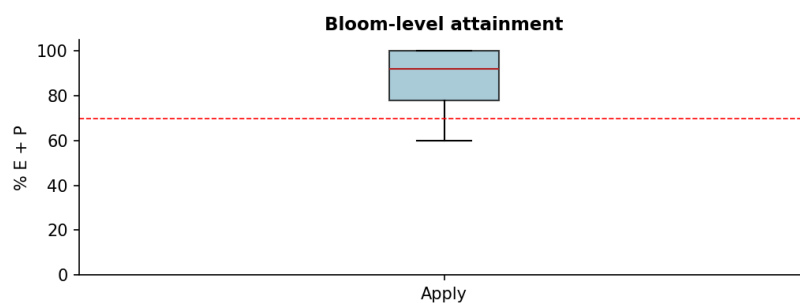
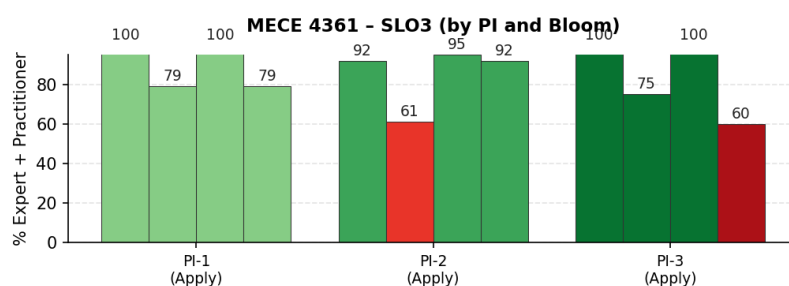
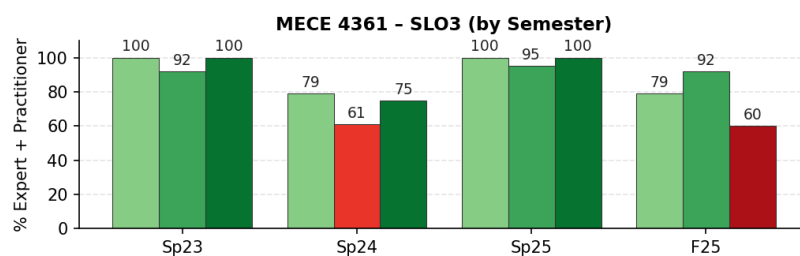


Figure 36: MECE 4361 (SD-I) SLO-3 portal analysis. Strong attainment in S 2023 (97%) and Sp 2025 (98%); Sp 2024 dip to 72% driven by PI-2 Presentation skills (61%); Fall 2025 first post-deployment term shows PI-2 recovery to 92% but PI-3 Technical Writing remaining at 60%.

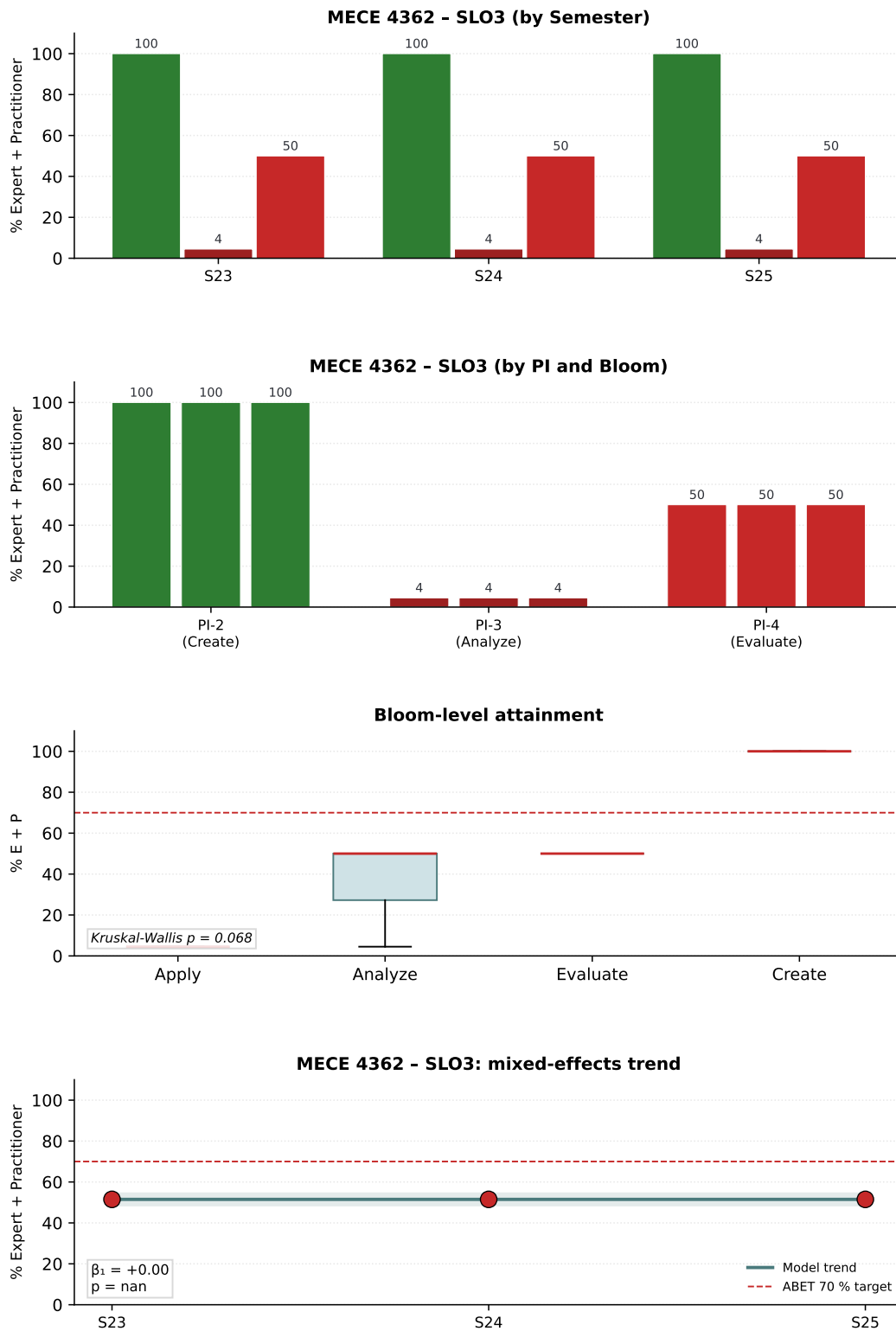


Figure 37: MECE 4362 (SD-II) SLO-3 portal analysis. Course-level average ~52% (PI-2=100%, PI-3=4.5%, PI-4=50%, stable across S 2023, S 2024, S 2025, F 2025)—genuinely low SD-II PIs reviewed under NX-01 (Section 4.B.3.8).

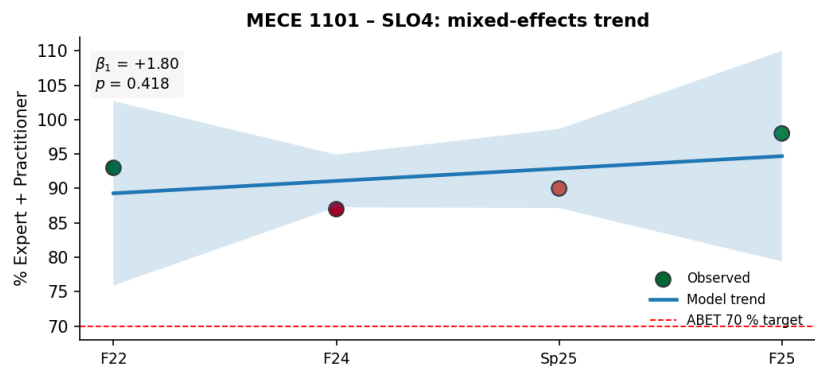
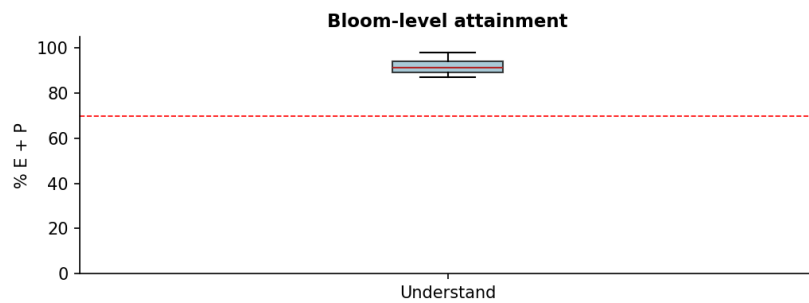
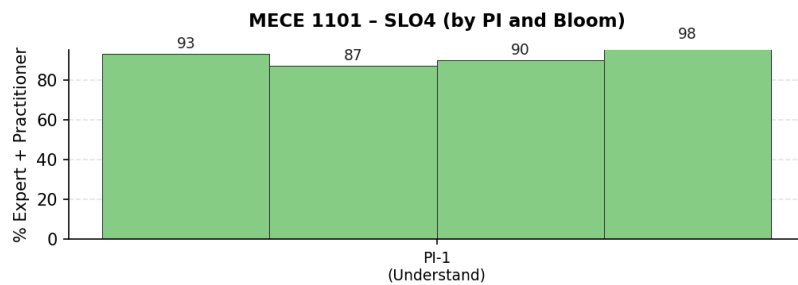
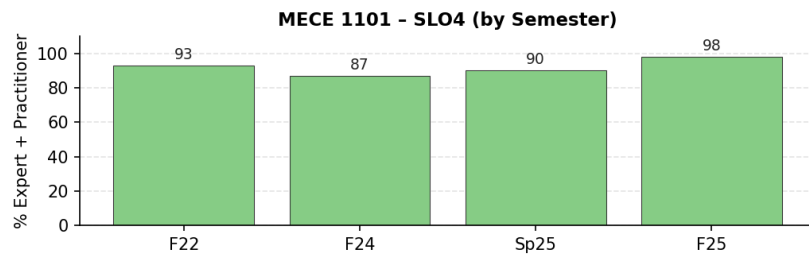


Figure 38: MECE 1101 (Intro to ME) SLO-4 portal analysis. Strong ethics-foundation attainment across four terms (F 2022 93%, F 2024 87%, Sp 2025 90%, F 2025 98%). PI-1 (Recognize ethical and professional responsibilities) assessed via two final-exam multiple-choice questions in F 2022/F 2024/Sp 2025; expanded to three questions in F 2025.

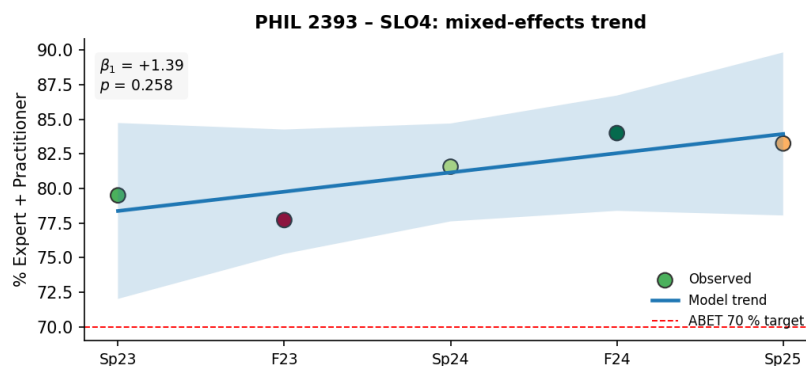
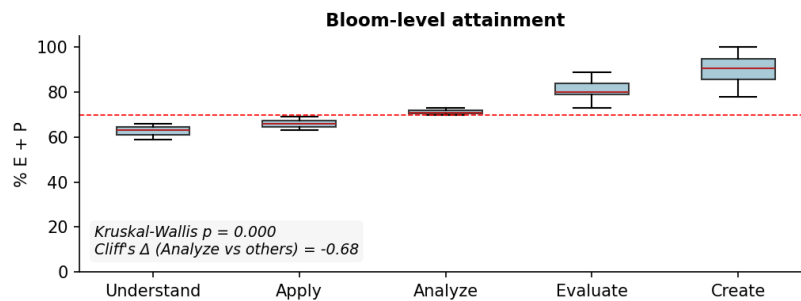
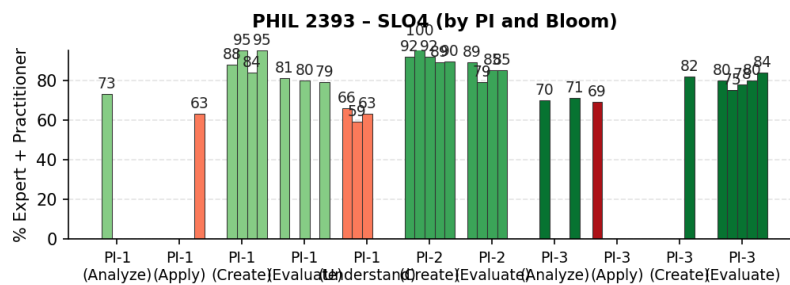
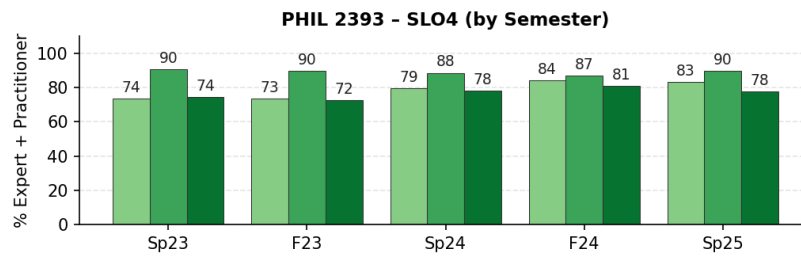


Figure 39: PHIL 2393 (labeled Philosophy in the portal; Ethics, Technology & Society in the catalog) SLO-4 portal analysis. Stable disaggregated BSME-student ethics-instrument evidence across F 2023/S 2024/F 2024/S 2025 (term averages 78–84%); cycle average ~81%. Instructor explanation fields are brief (“Excellent”, “Very good”, “Below 70%”); per-PI evidence covers PI-1, PI-2, and PI-3.

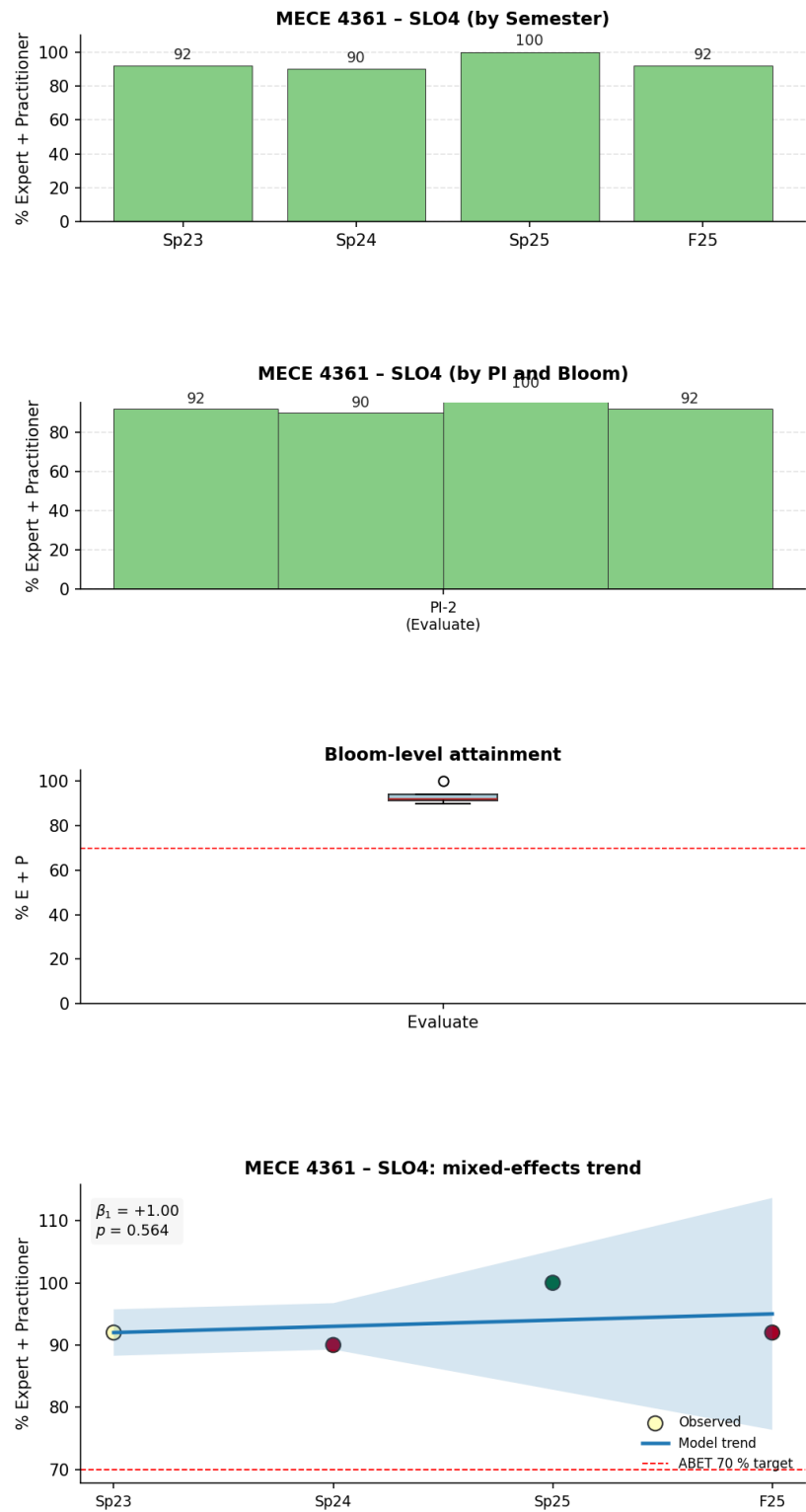


Figure 40: MECE 4361 (SD-I) SLO-4 portal analysis. Capstone ethics attainment (~95% averaged across S 2023/S 2025/F 2025; PI-2 Make informed ethical and professional judgments, assessed via Midterm Poster Q&A at Evaluate level).

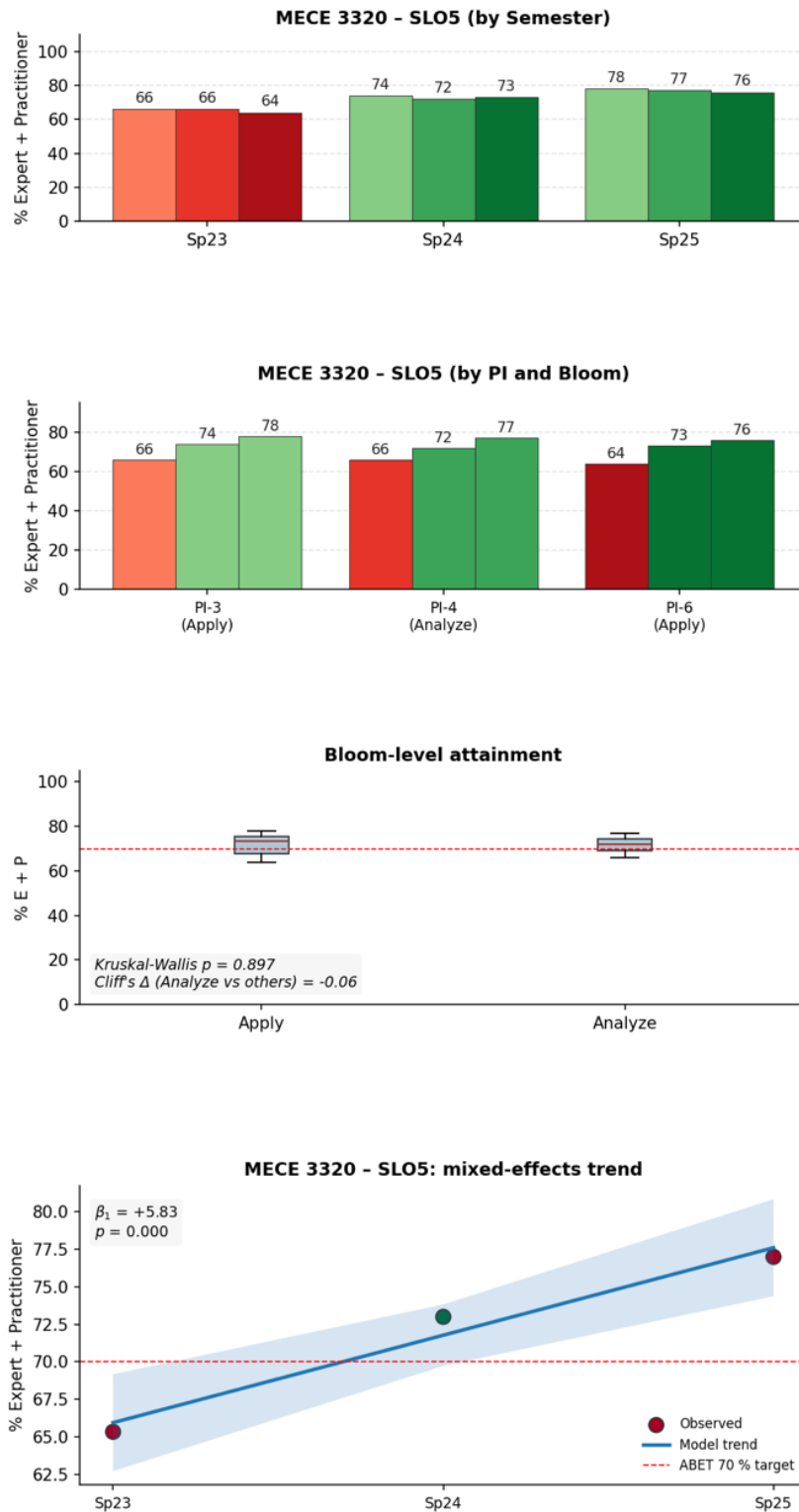


Figure 41: MECE 3320 SLO-5 portal analysis. Three-term rising trajectory (Sp 2023 ~65%, Sp 2024 ~73%, Sp 2025 ~77%) with all three assessed PIs (PI-3 Apply, PI-4 Analyze, PI-6 Apply) rising in parallel. Mixed-effects trend $\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$. The trajectory parallels the MECE 3320 SLO-6 closed loop (Section 4.B.3.6) and reflects the same progressive-artifact-complexity sequence.

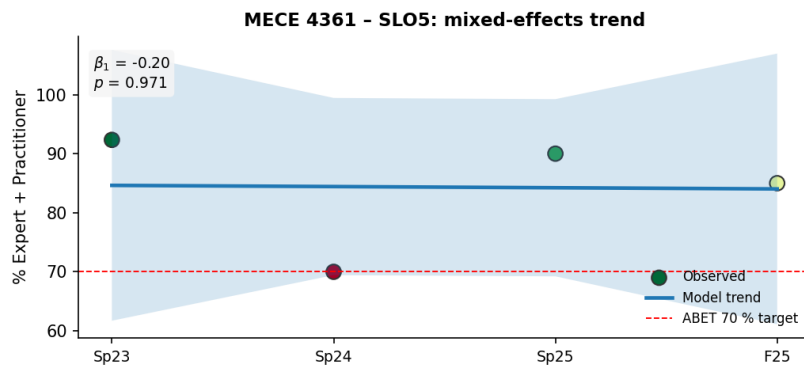
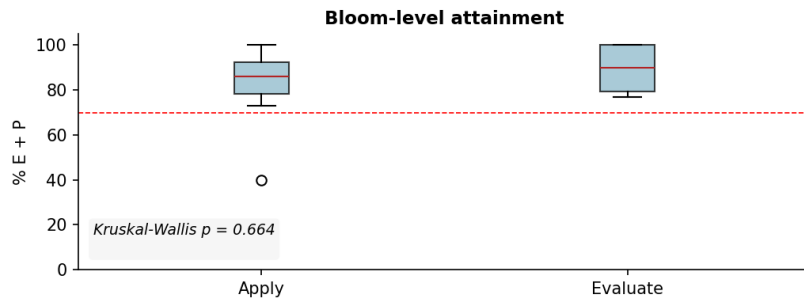
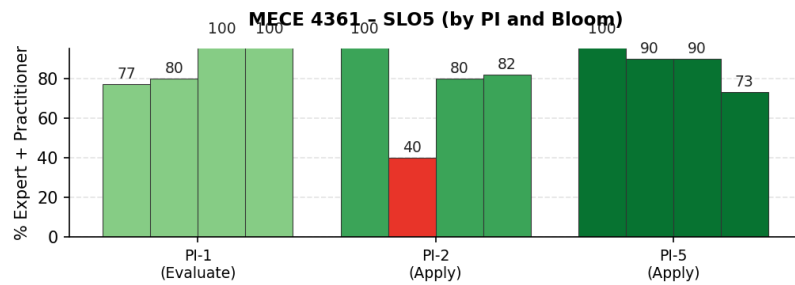
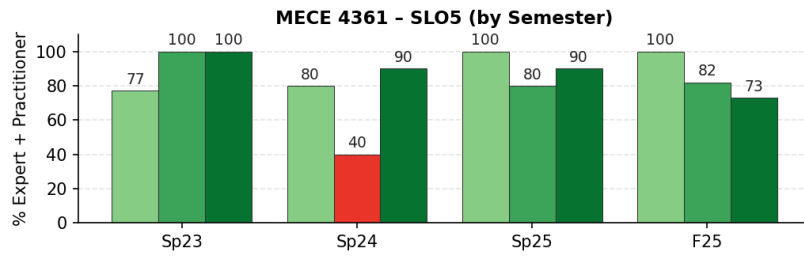


Figure 42: MECE 4361 (SD-I) SLO-5 portal analysis. Capstone teamwork cycle average ~84% across S 2023 (92%), S 2024 (70%), S 2025 (90%), F 2025 (85%). The Sp 2024 dip is driven by PI-2 (Plan tasks and meet deadlines, 40% in that single term); the F 2025 partial dip is driven by PI-5 (Communicates effectively in a team setting, 73%).

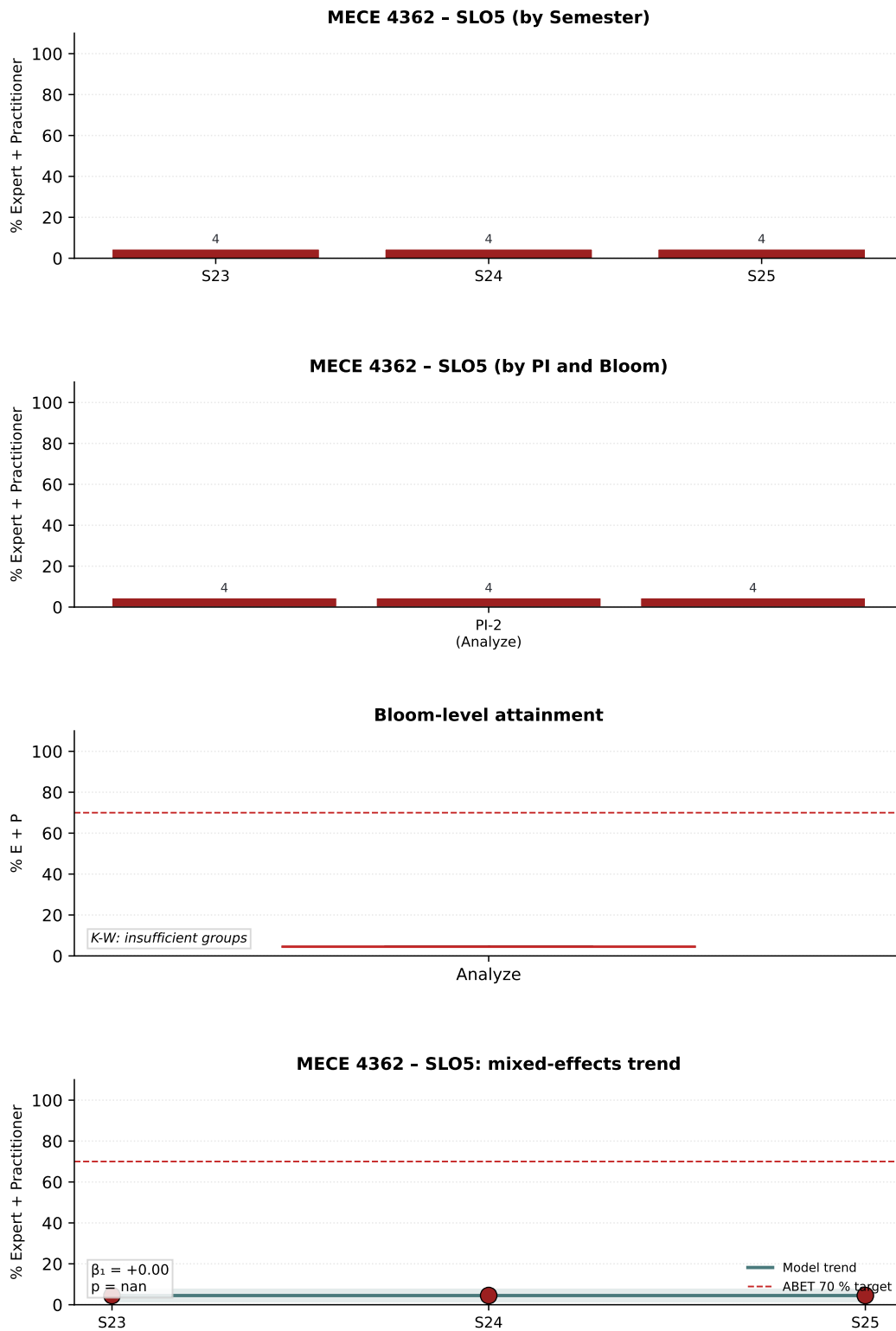


Figure 43: MECE 4362 (SD-II) SLO-5 portal analysis. Course-level PI-2 attainment at ~4.5%, stable across S 2023, S 2024, S 2025, and F 2025; genuinely below-threshold SD-II PI reviewed under NX-01 (Section 4.B.3.8).

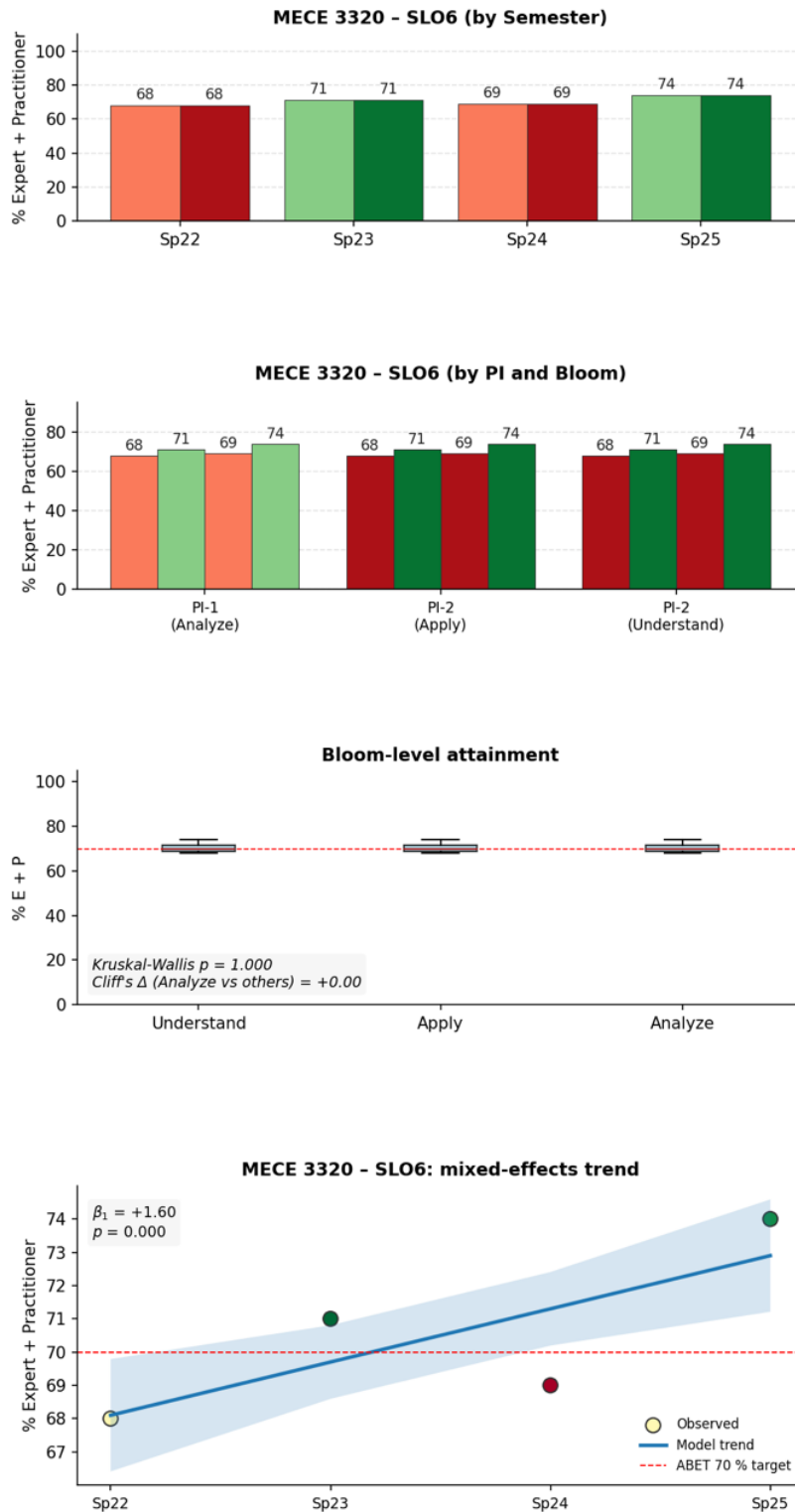


Figure 44: MECE 3320 (Measurements & Instrumentation) SLO-6 portal analysis—the course-level closed-loop case. Top panel: per-term EP% (68/71/69/74 across Sp 2022–Sp 2025). Second panel: by PI and Bloom level, all three PI/Bloom pairs (PI-1 Analyze, PI-2 Apply, PI-2 Understand) showing identical per-term values—uniform improvement across cognitive levels. Third panel: Bloom-level distribution essentially flat (Kruskal-Wallis $p = 1.000$). Bottom panel: mixed-effects trend $\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$.

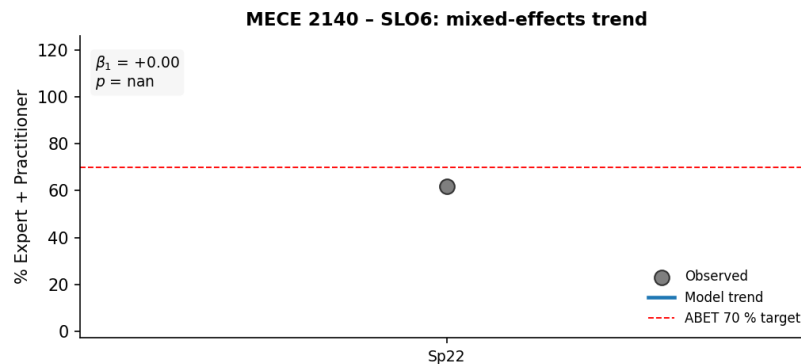
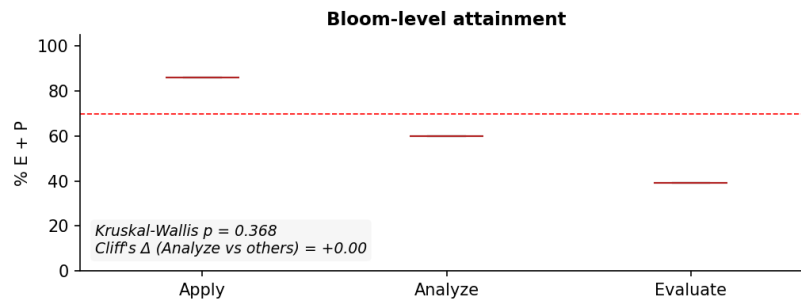
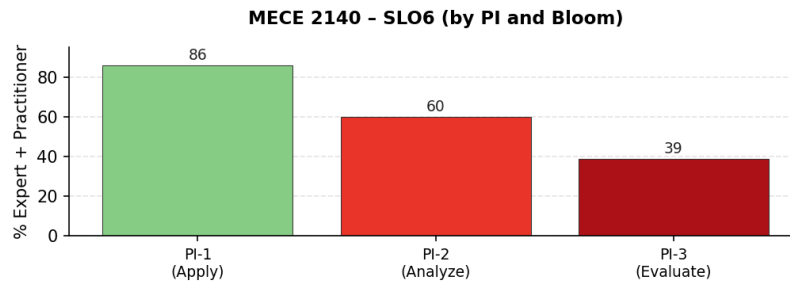
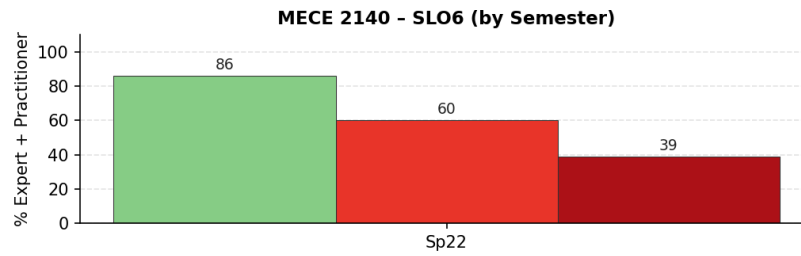


Figure 45: MECE 2140 (Engineering Materials Lab) SLO-6 portal analysis. Single Sp 2022 data point at 62% with cognitive-level gradient (Apply 86%, Analyze 60%, Evaluate 39%). The single observation makes course-level trend analysis impossible; the cognitive-level gradient suggests strength at Apply and weakness at Evaluate. Coverage gap noted in Section 4.B.3.8, closed by NX-03.

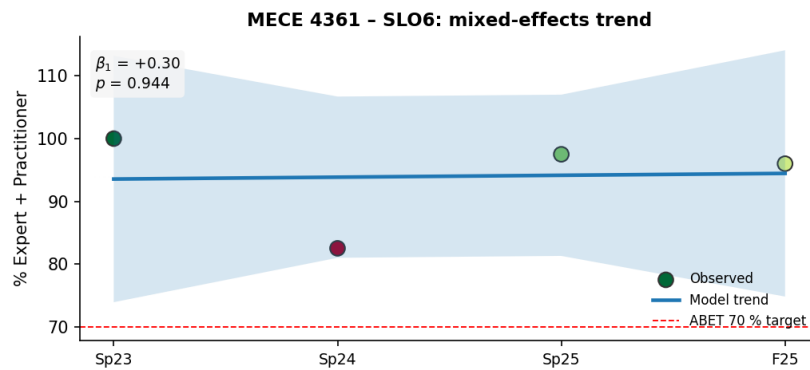
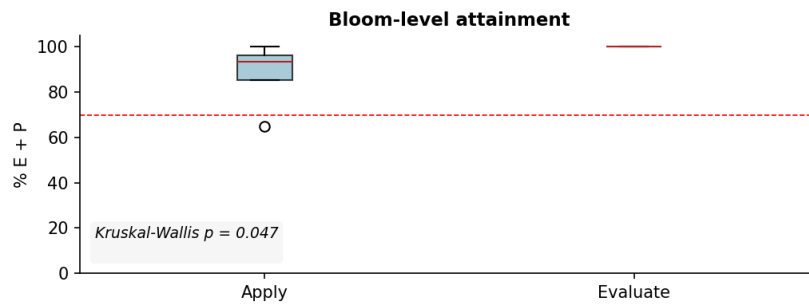
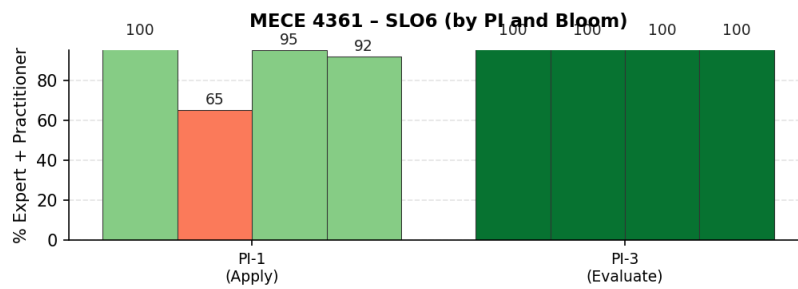
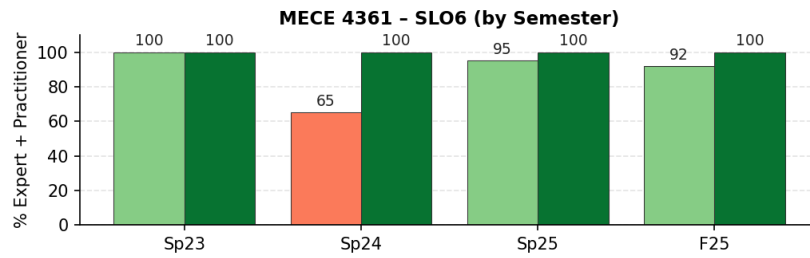


Figure 46: MECE 4361 (SD-I) SLO-6 portal analysis. Strong capstone experimentation attainment (~94% averaged across the four cycle terms).

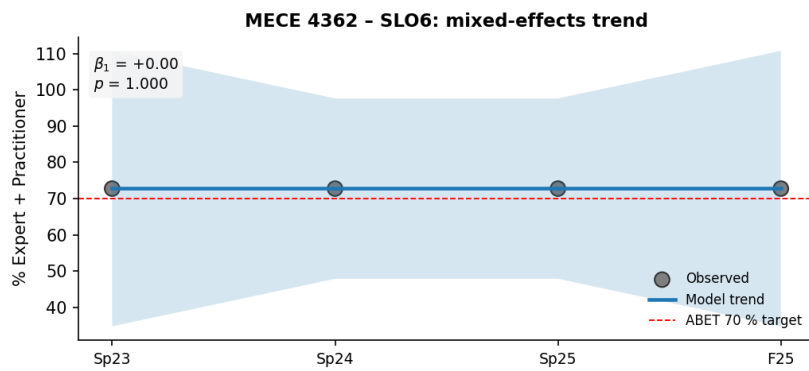
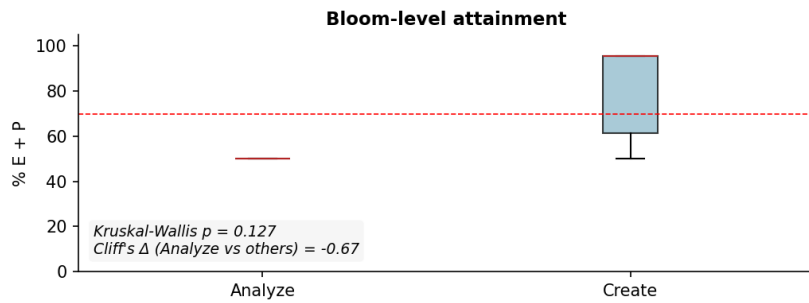
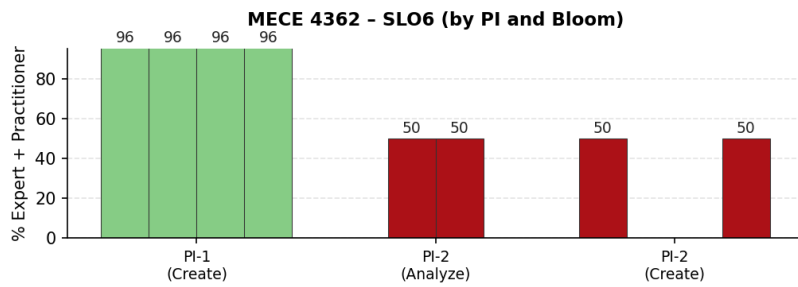
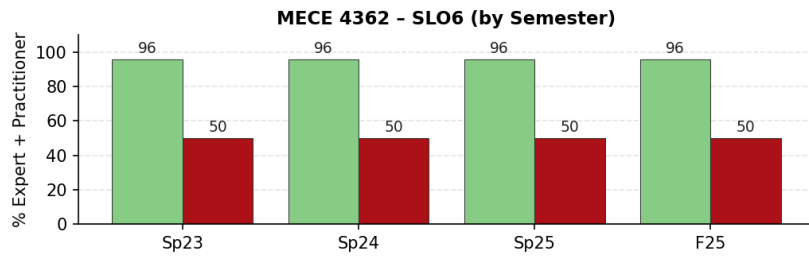


Figure 47: MECE 4362 (SD-II) SLO-6 portal analysis. Course-level ~73%, with PI-1 $\approx$ 95.5% and PI-2 $\approx$ 50%, stable across S 2023, S 2024, S 2025, and F 2025 (Section 4.B.3.8; SD-II reviewed under NX-01).

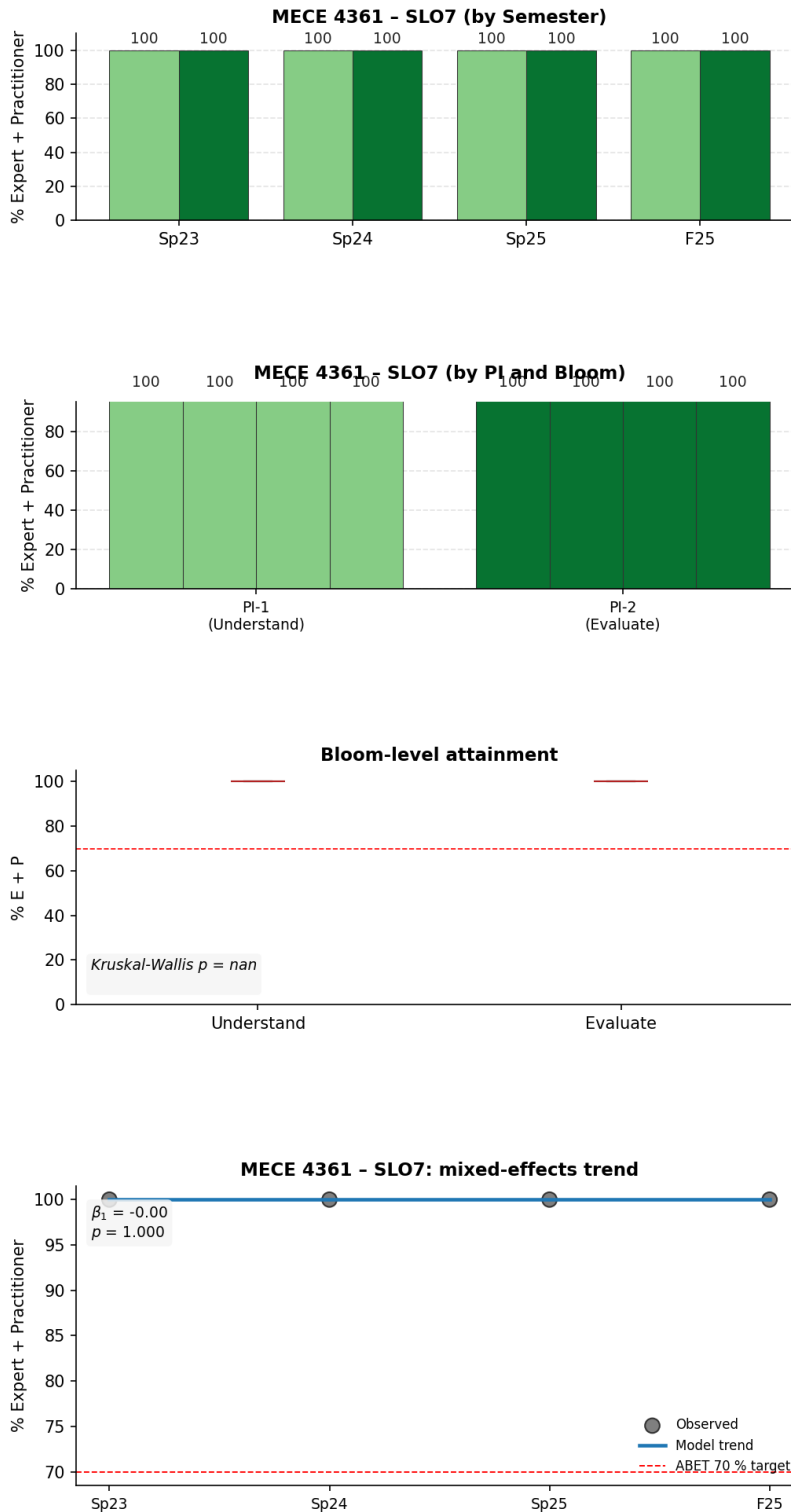


Figure 48: MECE 4361 (SD-I) SLO-7 portal analysis. All three PIs (PI-1 Recognize ongoing need to acquire new knowledge, *Understand* level via SD-1 Final Oral Reflection; PI-2 Choose appropriate learning strategies, *Evaluate* level via SD-1 Final Oral Future Work; PI-3 Apply new knowledge appropriately, *Apply* level via SD-1 design-deliverable integration) at 100% attainment across S 2023, S 2024, S 2025, and F 2025—strong above-threshold capstone evidence on all three Performance Indicators.

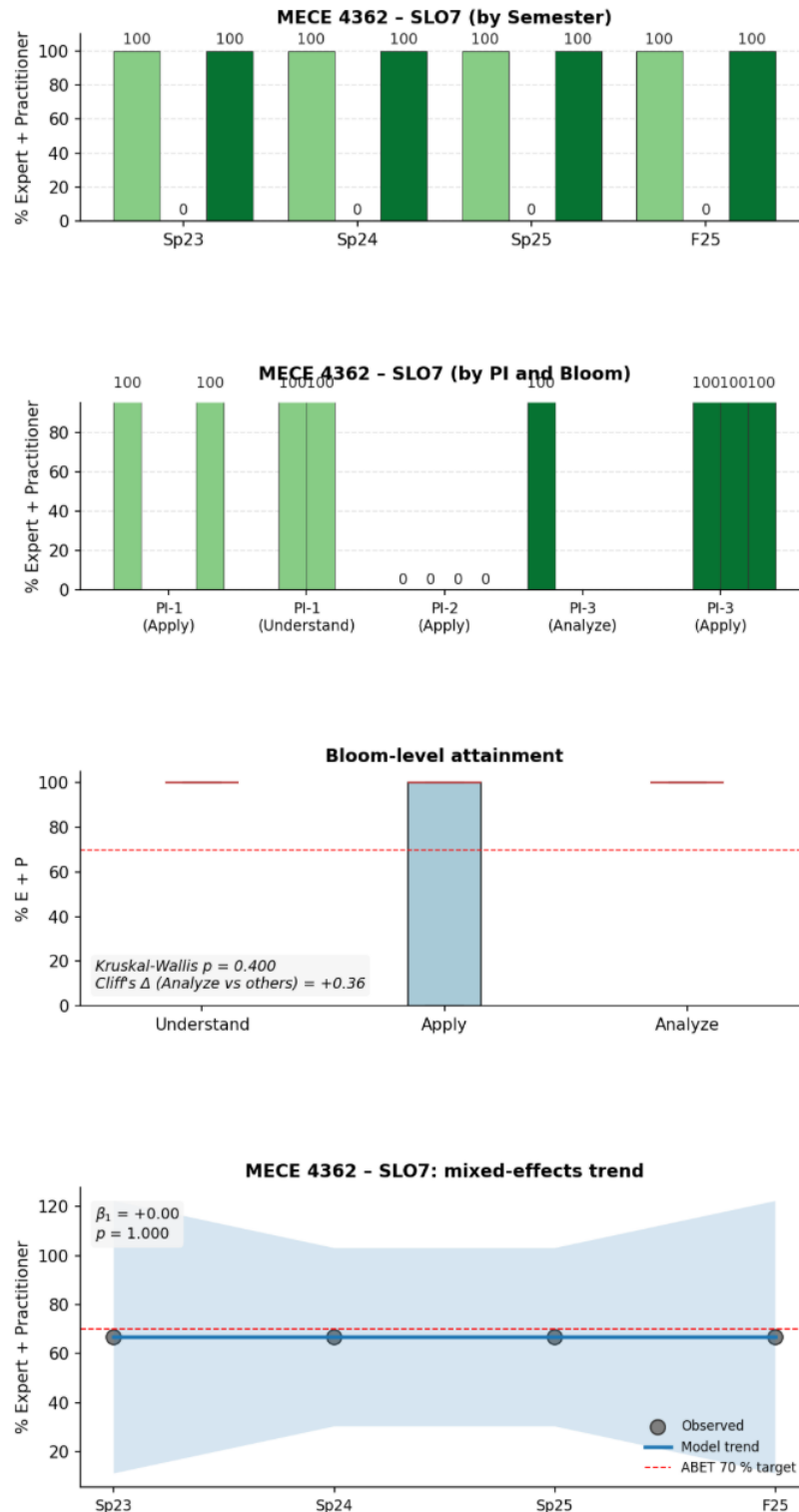


Figure 49: MECE 4362 (SD-II) SLO-7 portal analysis. PI-1 (Apply) = 100%, PI-1 (Understand) = 100%, PI-2 (Apply) = 0%, PI-3 (Analyze) = 100%, PI-3 (Apply) = 100% stable across S 2023, S 2024, S 2025, and F 2025—the PI-2 Apply=0% value is the most prominent below-threshold signal in the MECE 4362 SLO-7 dataset, reviewed at the course level under NX-01 (Section 4.B.3.8). Mixed-effects trend $\beta_1 = +0.00$, $p = 1.000$ (no term-to-term variance). Bloom-level Kruskal-Wallis $p = 0.400$, Cliff's $\Delta = +0.36$.

BSME Student Learning Outcomes — At-a-Glance Attainment Summary

Reconciled dataset (F~2020–Sp~2025); MECE 4362 F25 entries excluded per data-quality precaution (Sec. 4.B.3.8)

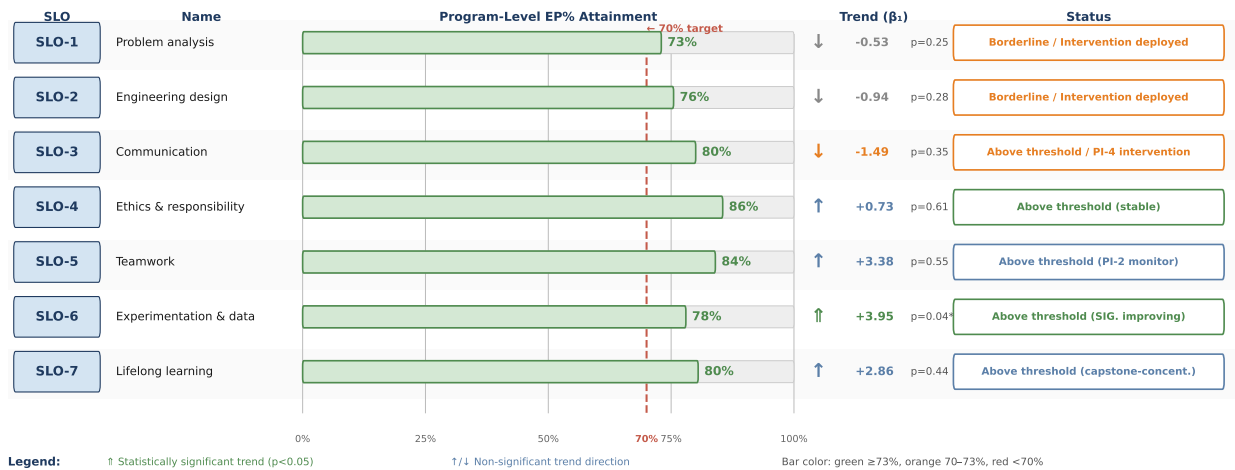


Figure 50: BSME Student Learning Outcomes at-a-glance attainment summary (F 2020–Sp 2025). Each SLO shows program-level EP% (bar with 70% threshold marked), multi-term trend direction β_1 and p -value (no SLO reaches statistical significance at $\alpha = 0.05$ at the program level; SLO-6 is closest at $p = 0.085$; MECE 3320 reaches significance at the course level on both SLO-6 and SLO-5), and current status badge. SLO-4 and SLO-6 are stably above threshold with no formal AFI; SLO-7 is above threshold on the program-level aggregate (80% per term) and on PI-1 and PI-3 (100%) but the program-level PI-2 average is 50% reflecting the MECE 4362 PI-2 Apply=0% value (reviewed under NX-01); SLO-5 is above threshold with mixed within-cycle behavior and joint PI-level monitoring under NX-04; SLO-1, SLO-2, and SLO-3 are at or above threshold with department-level Overlay AFIs deployed Fall 2025 in response to documented within-SLO signals.

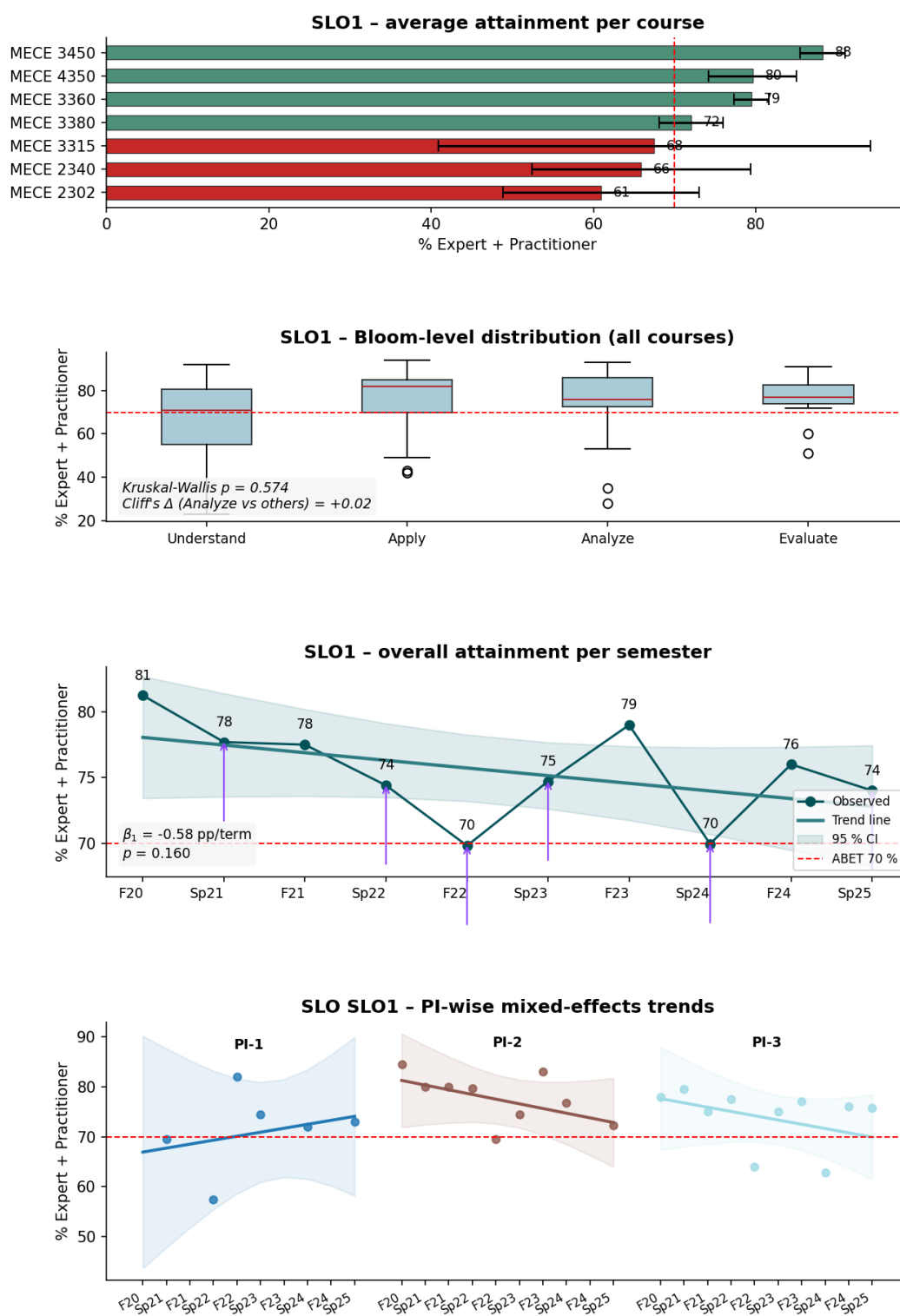


Figure 51: SLO-1 program-level portal summary. Panel A: course-level average EP% distribution; Panel B: Bloom-level distribution; Panel C: semester-by-semester program-level trend with OLS fit and 95% confidence band ($\beta_1 = -0.53$ pp/term, $p = 0.248$); Panel D: PI-wise mixed-effects trends.

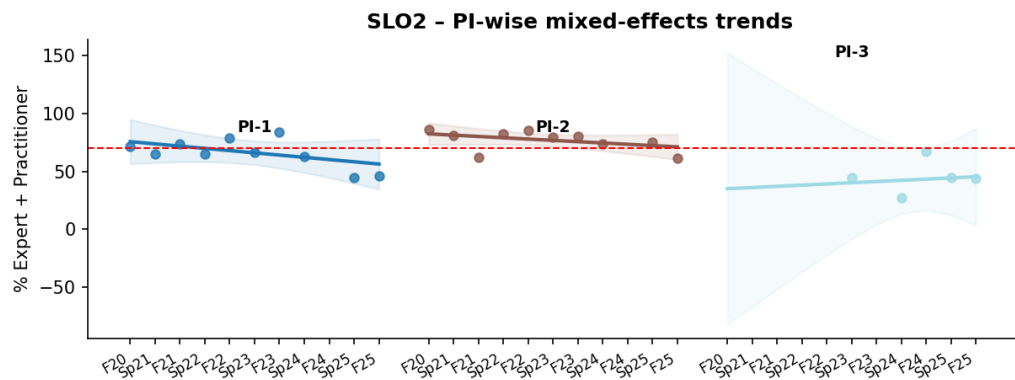
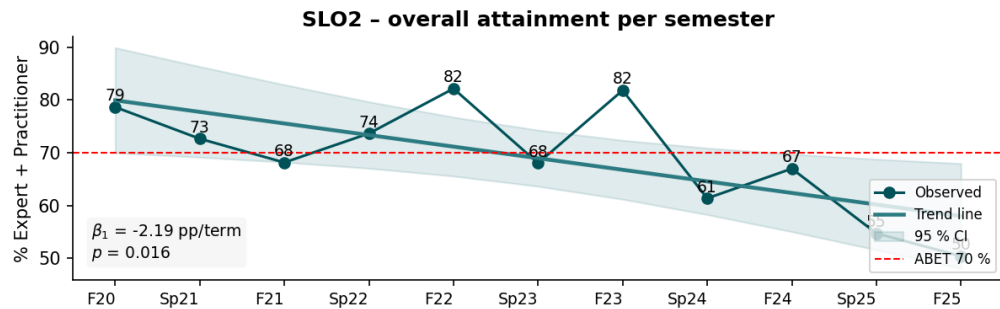
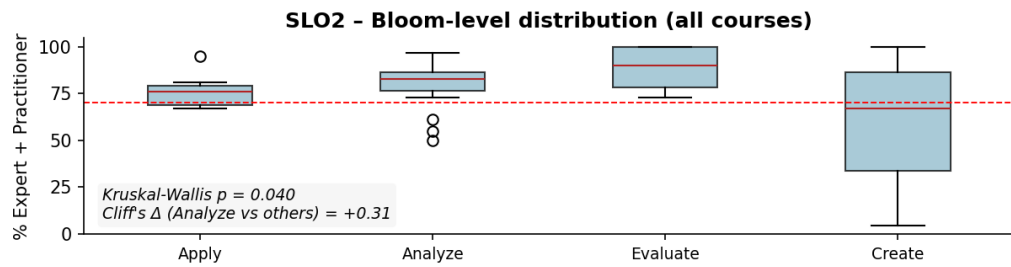
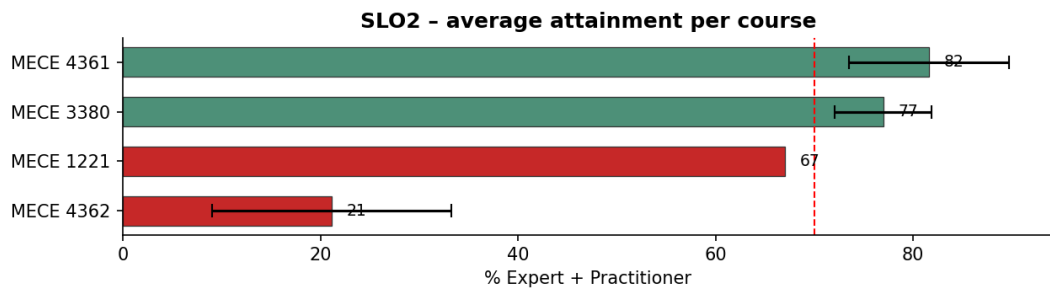


Figure 52: SLO-2 program-level portal summary. Panel A: course-level average EP% distribution; Panel B: Bloom-level distribution; Panel C: semester-by-semester program-level trend ($\hat{\beta}_1 = -0.94$ pp/term, $p = 0.276$); Panel D: PI-wise mixed-effects trends.

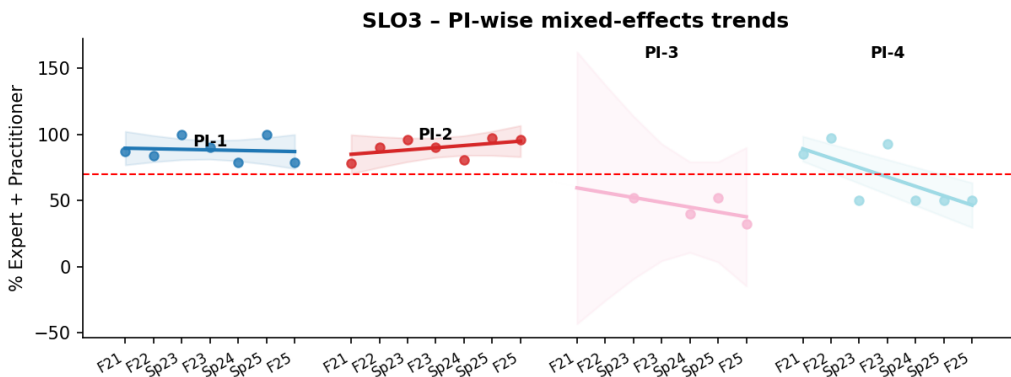
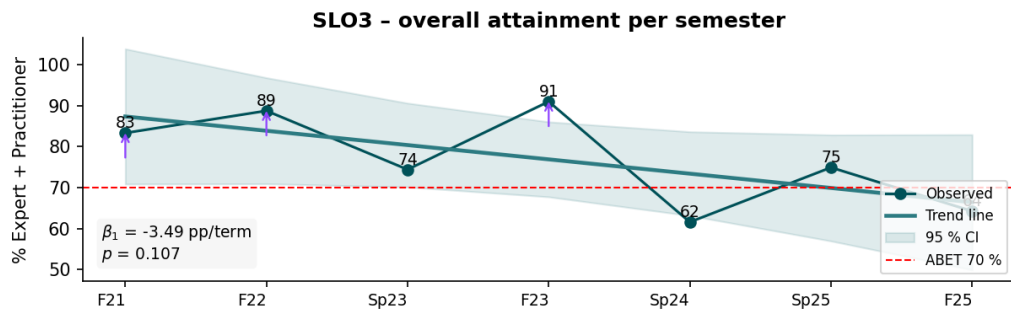
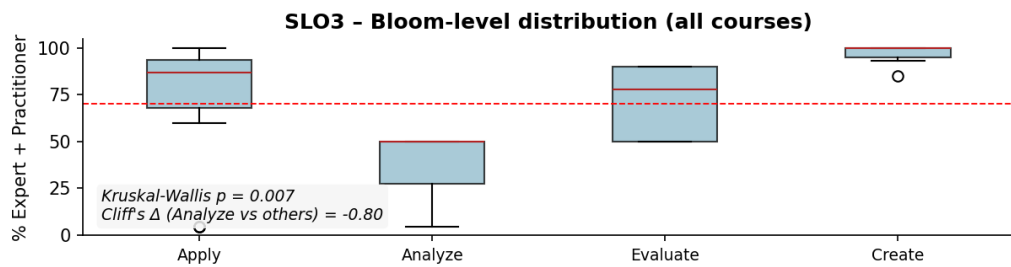
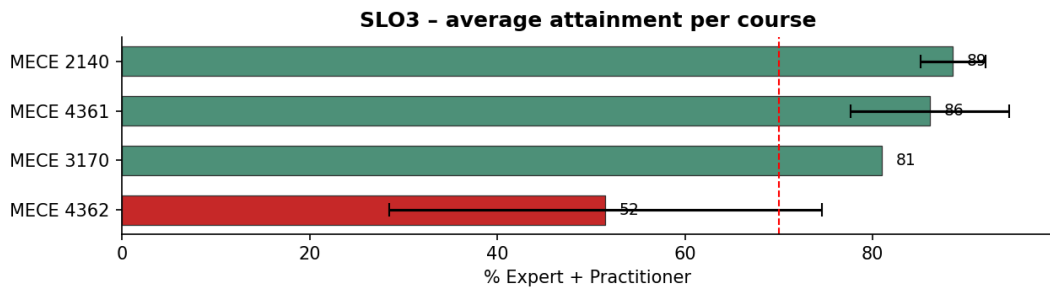


Figure 53: SLO-3 program-level portal summary (all assessing courses). Panel A: course-level average EP% distribution; Panel B: Bloom-level distribution; Panel C: semester-by-semester program-level trend ($\beta_1 = -1.49$ pp/term, $p = 0.355$); Panel D: PI-wise mixed-effects trends.

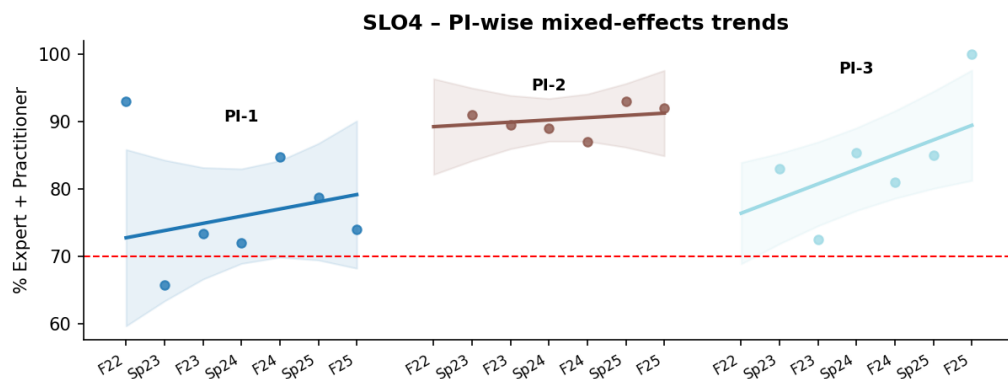
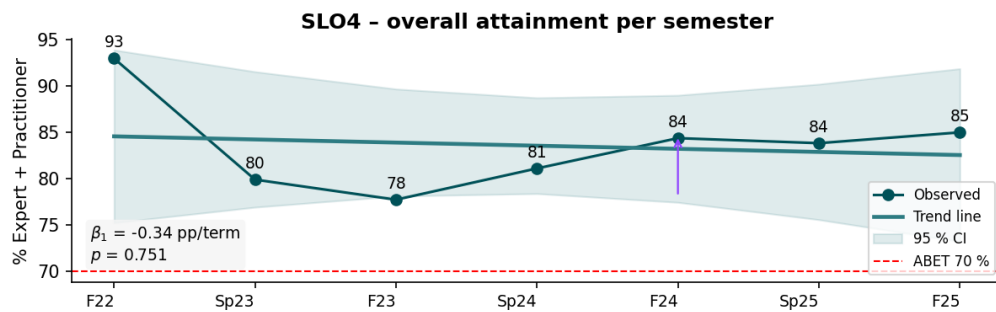
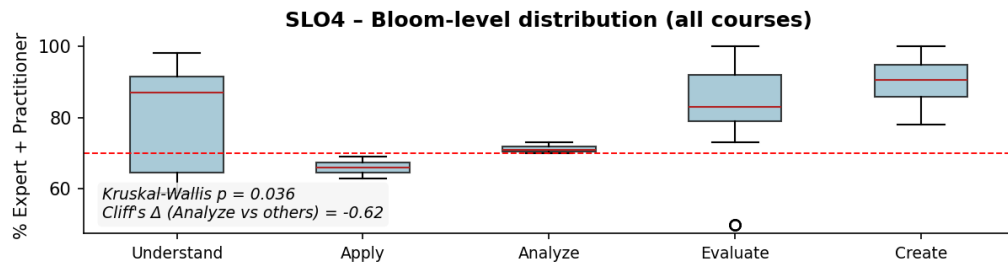
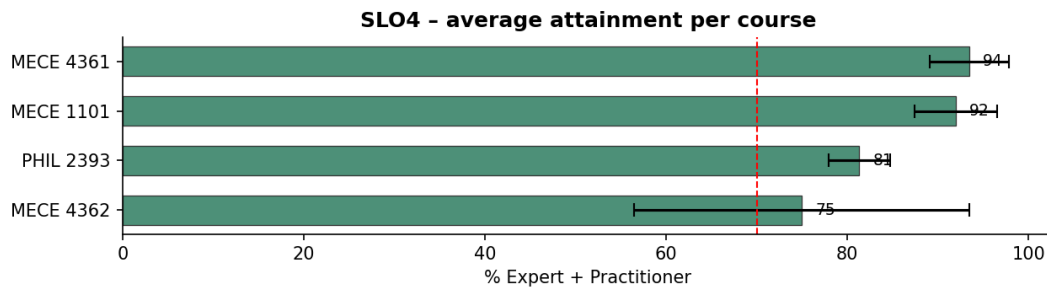


Figure 54: SLO-4 program-level portal summary (all assessing courses). Panel A: course-level average EP% distribution; Panel B: Bloom-level distribution; Panel C: semester-by-semester program-level trend ($\beta_1 = +0.73$ pp/term, $p = 0.609$); Panel D: PI-wise mixed-effects trends.

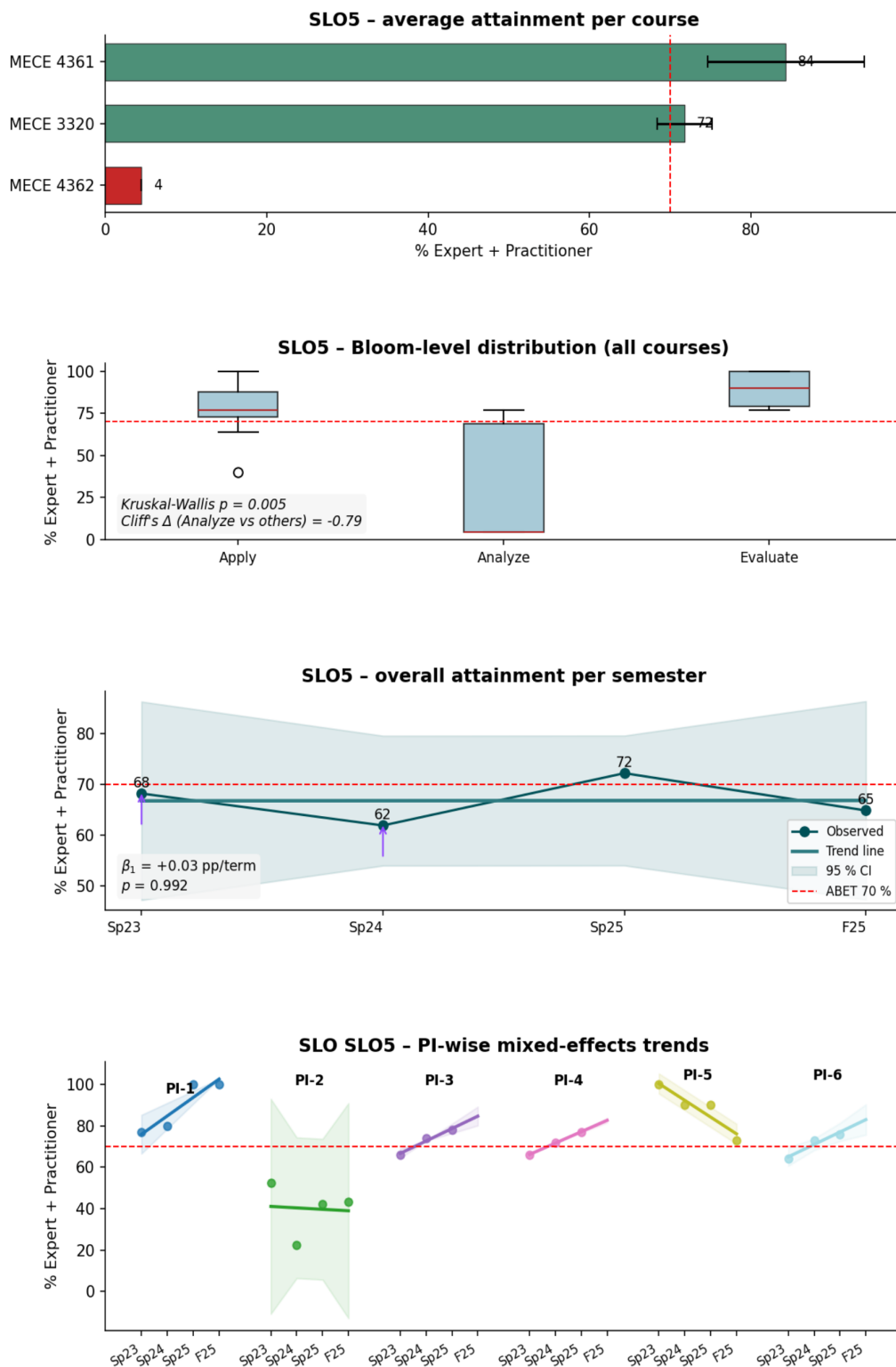


Figure 55: SLO-5 program-level portal summary (all assessing courses). Panel A: course-level average EP% (MECE 4361 84%, MECE 3320 72%, MECE 4362 4%, stable across terms); Panel B: Bloom-level distribution showing Apply and Evaluate above threshold but Analyze with substantial variability (Kruskal-Wallis $p = 0.005$); Panel C: program-level semester trend essentially flat ($\beta_1 = +0.03$ pp/term, $p = 0.992$); Panel D: PI-wise mixed-effects trends including the rising MECE 3320 PI-3/PI-4/PI-6 and the declining MECE 4361 PI-5.

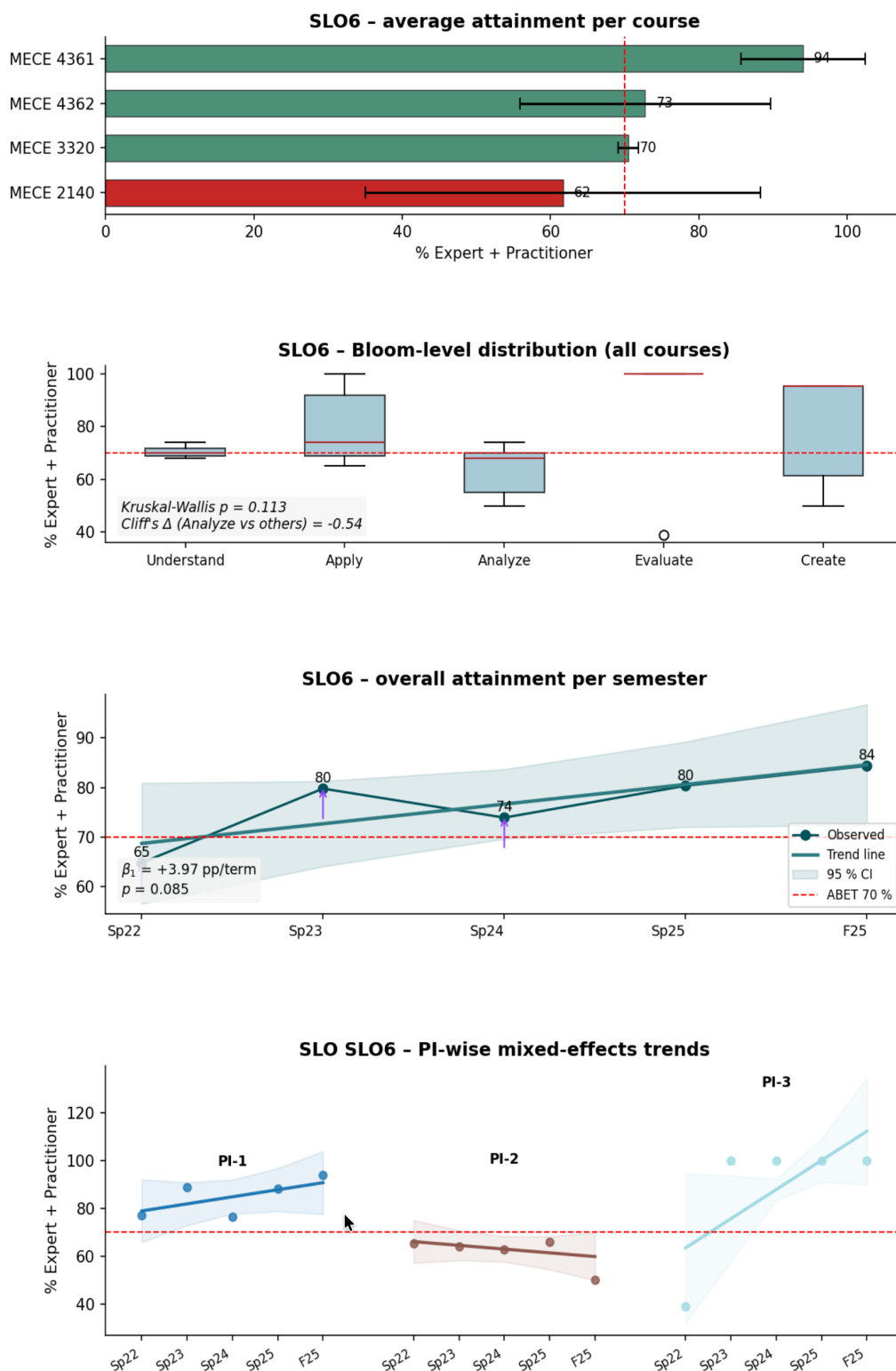


Figure 56: SLO-6 program-level portal summary (all four assessing courses). Panel A: course-level EP% (MECE 4361 94%, MECE 4362 73%, stable across terms, MECE 3320 70%, MECE 2140 62% below threshold on a single Sp 2022 observation); Panel B: Bloom-level distribution with Analyze below threshold (Kruskal-Wallis $p = 0.113$, Cliff's $\Delta = -0.54$); Panel C: semester-by-semester program-level trend with $\beta_1 = +3.97$ pp/term, $p = 0.085$ (positive but not statistically significant at $\alpha = 0.05$ given five program-level observations); Panel D: PI-wise mixed-effects trends.

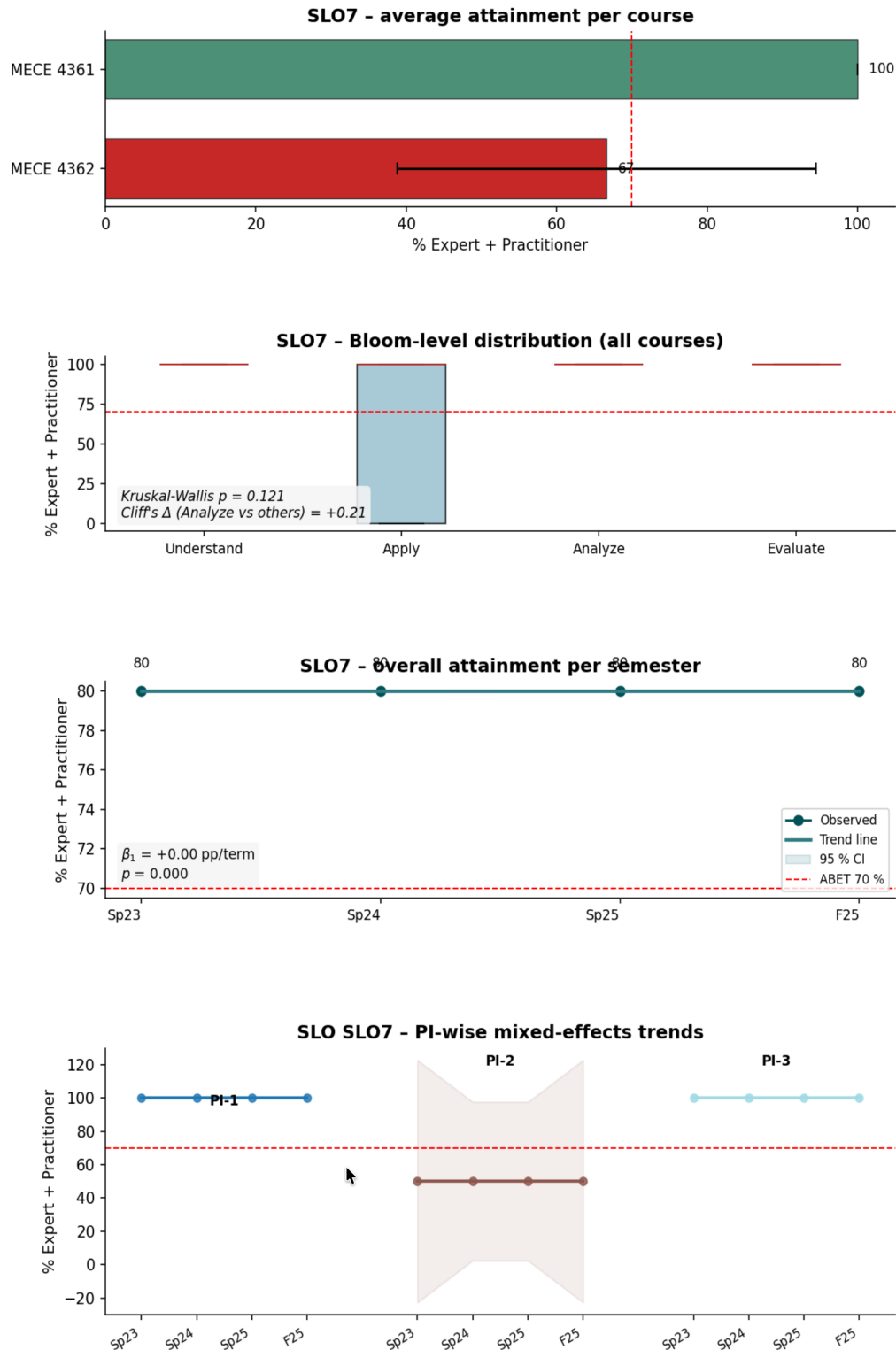


Figure 57: SLO-7 program-level portal summary (both assessing courses). Panel A: course-level cycle averages (MECE 4361 100%, MECE 4362 67%, stable across terms). Panel B: Bloom-level distribution showing PI-2 Apply variability (50–100%) driven by the MECE 4362 PI-2=0% value; Understand/Analyze/Evaluate at 100% single-value indicators (Kruskal-Wallis $p = 0.121$, Cliff's $\Delta = +0.21$). Panel C: program-level per-term trajectory perfectly flat at 80% ($\hat{\beta}_1 = +0.00$ pp/term, $p = 0.000$). Panel D: PI-wise mixed-effects fits showing PI-1 and PI-3 at 100% across all four terms but PI-2 at 50%—the 50% reflects the MECE 4362 PI-2 Apply=0% value, a genuine within-SLO signal reviewed under NX-01.

Appendix item D.14: Process, governance, and reference tables (relocated from the main response)

Table 32: Evidence Map for Criterion 4: repositories, evidence types, ownership, and retention.

Repository	Evidence Types (Examples)	Owner	Retention
ABET Portal (web-based)	Per-record E/P/A/N rubric data with instructor narrative; computed EP% attainment; SLO summary tables; trend figures (course-level and SLO-level analysis pages); Capstone Exit Survey responses and analytics; AFI/Intervention logs; administrative CSV/HTML exports	ABET/Assessment Coordinator; SLO Chairs	Full review window (2020–2026) plus archived prior-cycle summaries; historical data preserved across rubric updates
SLO Binders (7, physical)	SLO/PI rubrics (current + prior versions); artifact descriptions; representative graded exemplars with rubric evidence; semester results printouts; AFI documentation supporting each SLO	SLO Chairs / SLO Committees	Current cycle + prior-cycle exemplars as needed for longitudinal evidence; rubric version history retained
Course Binders (physical, key MECE courses)	ABET syllabi; instructional materials; assignments; quizzes/exams; projects; selected samples supporting assessed artifacts in key courses	Instructors of Record; Course Coordinators (as applicable)	Maintained for visit readiness each cycle; representative samples retained; full instructional records per departmental practice
Brightspace (course shells)	Course artifacts and rubric scoring used for assessment (embedded rubrics, graded assignments/exams/projects); supporting materials used to generate E/P/A/N distributions	Instructors of Record	Per standard instructional recordkeeping; program retains curated exemplars in binders and aggregated results in portal

Continued on next page

Repository	Evidence Types (Examples)	Owner	Retention
Department Records Archive	Faculty meeting minutes; retreat agendas/minutes; SWOT tables; policy approvals; curriculum proposals; governance records supporting evaluation and decisions	Department Chair; Administrative Staff	Permanent departmental records (minutes, policies, approvals retained long-term)
On-site Displays (visit)	Senior Design reports/presentations; prototypes; competition projects; research posters (evidence of culminating experiences)	Senior Design Faculty; Faculty Advisors	Current cycle artifacts; selected exemplars retained as needed for documentation

Table 33: Evolution of the BSME assessment system across the 2020–2026 review window. Items marked “pre-cycle” were already in place before 2020 and have been carried forward unchanged.

Period	Element	Description / Change	Status in current SSR
Pre-2020	Seven SLOs (1–7), one-to-one with ABET SOs	Adopted in the previous ABET cycle, replacing the legacy (a)–(k) framework	Unchanged; in continuous use across 2020–2026
Pre-2020	E/P/A/N four-level rubric	Adopted prior to 2020; common rubric scale across all PIs	Unchanged
Pre-2020	EP% $\geq$ 70% attainment threshold	Single program-wide threshold with 67–70% borderline band	Unchanged
Pre-2020	Capstone Exit Survey	Administered to graduating seniors as the program’s primary indirect measure	Instrument retained; administration migrated to portal Fall 2024
Pre-2020	SLO-committee governance	Seven SLO committees; faculty-retreat review; full-faculty AFI approval	Unchanged
Sep 2, 2022	EAC meeting	PEOs/SLOs reviewed and approved by the council; written EAC feedback received	Input to evaluation (Section 4.B.4.1)

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Period	Element	Description / Change	Status in current SSR
2022–2024 catalog	Math/analysis sequence revision	MECE 2350/3449 replaced by MECE 3440 and MATH 2415; CSCI 1380 added	Detail in Criterion 5; context for SLO-1 assessment
During cycle	PC → PI terminology realignment	Following BSME ABET Coordinator's attendance at the ABET Assessment Workshop (Baltimore, MD); adopts ABET-current "Performance Indicator" language; one-to-one mapping preserved	Implemented; rubric criterion text reviewed at PI level
During cycle	PI-level rubric criterion text review	Conducted alongside the PC→PI relabeling; cognitive level (Bloom's), artifact courses, and the four-level scale preserved	Implemented
2023–2025	PEO three-year review	Comprehensive PEO revision adopted in 2025 (Criterion 2); SLO wording reviewed in parallel and retained	Implemented (PEO detail in Criterion 2)
Dec 4, 2024	EAC meeting	Revised entrance requirement, new technical electives, and student-success initiatives reviewed	Input to evaluation
During cycle	Calculus I attempt-limit policy	Maximum attempts reduced from 3 to 2 as BSME entrance requirement	Implemented; reviewed with EAC December 2024
Fall 2024	ABET Assessment Portal — launch	Web-based, role-controlled, PI-grained data store with embedded analysis engines; replaces Excel/email workflow; Capstone Exit Survey module integrated	Live as the system of record; in continuous use through Spring 2026

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Period	Element	Description / Change	Status in current SSR
Fall 2024 onward	Embedded analysis (course-level and SLO-level)	On-demand mixed-effects/OLS trend modeling, course-wise attainment bars, Bloom-level distributions, PI-wise mixed-effects trends, automatic intervention-arrow annotation	Used to generate the figures in Section 4.A.6
Mar 6, 2026	EAC meeting	Continuation of periodic-review cycle	Scheduled

Table 34: Inputs to the BSME continuous improvement process. Direct measures generate PI-level attainment evidence; indirect measures, stakeholder voice, and institutional metrics provide triangulation and “other available information” per ABET Criterion 4.

Input	Category	Form / How Collected	Role in Improvement
PI-level EP% attainment (per course / term / PI)	Direct	ABET Portal: rubric E/P/A/N counts → EP% (Section 4.0.4.3)	Identifies threshold breaches and trend movement at the smallest measurable unit
Course-level attainment summaries	Direct	Portal /analyze_course pages (Section 4.0.4.4)	Surfaces course-level bottlenecks for targeted AFIs
SLO-level attainment summaries (4-panel figures)	Direct	Portal /analyze_slo pages (Section 4.0.4.5)	Provides the outcome-level view for SLO-committee analysis and full-faculty review
Multi-semester trend statistics ($\hat{\beta}_1$, 95% CI, p -value)	Direct	Portal embedded mixed-effects / OLS models	Distinguishes drift from variability; prioritizes outcomes/PIs with negative or stalled trends
PI-wise mixed-effects trends	Direct	Portal SLO analysis Panel D (Section 4.0.4.5)	Allows PI-targeted (not blanket SLO-level) action

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Input	Category	Form / How Collected	Role in Improvement
Instructor narrative observations	Direct (qualitative)	Portal required “Observations” field per record	Surfaces context, root-cause hypotheses, and proposed local actions
Bloom-level distribution and Kruskal–Wallis / Cliff’s Δ	Direct	Portal SLO analysis Panel B (Section 4.0.4.5)	Surfaces cognitive-level patterns (e.g., systematically weaker performance at higher Bloom levels)
Capstone Exit Survey (Likert items & open response)	Indirect	Portal Capstone Exit Survey module (Section 4.0.4.6); separate survey database	Triangulates direct measures with graduating-student perception of curriculum effectiveness, faculty guidance, and impact of recent interventions
EAC industry input	Stakeholder voice	Council meetings (e.g., Sept 2, 2022; Dec 4, 2024; Mar 6, 2026); written member feedback	Industry perspective on curriculum currency, graduate competencies, and emerging needs
Alumni and employer feedback (where collected)	Stakeholder voice	Department-level outreach; PEO/SLO review cycle inputs	Longitudinal validation that graduates are succeeding in practice (Criterion 2 links)
Academic success indicators	Institutional metrics	Departmental tracking (pass rates, retention, graduation rates, GPA trends, time to degree)	Flags upstream issues (gateway course bottlenecks; sequencing problems) that affect SLO attainment
Faculty SWOT analysis	Internal	Biennial faculty-retreat deliverable, every two years (Section 4.0.3.4)	Synthesizes priorities across all the above into a program-level priority list
Curriculum/policy change reviews	Internal	Faculty meetings; college/university governance traces (Criterion 5, Criterion 6.E.1)	Captures effects of policy changes (e.g., Calculus I attempt limit) on the upstream pipeline

Table 35: Mapping of the cycle-level AFI catalog to ABET-aligned action categories.

Category	AFI IDs	Summary of Actions
Course-level / Pedagogy	SLO-01, SLO-02, SLO-03	Three department-level overlay-rubric AFIs targeting Bloom-level weaknesses in SLO-1 problem analysis, SLO-2 concept development (capstone), and SLO-3 audience adaptation (mid-program communication).
Curriculum-level	CR-01, CR-02, CR-03	Math/analysis sequence revision; CSCI 1380 programming addition; three new technical electives (CFD, Advanced Alloys, Smart Materials).
Student-support / Policy	SS-01, SS-02	Calculus I attempt-limit policy (3→2); career-readiness content enhancement following EAC member feedback.
Program policy (carry-forward)	POL-01, POL-02	BSME GPA graduation requirement (≥ 2.5 in major; Fall 2015 catalog, in continuous enforcement); BSME Two-Peat Rule limiting repeat enrollment in gatekeeper courses (prior cycle, in continuous enforcement). These triangulate with pass-rate data (Section 4.B.4.5) and operationalize faculty oversight of student progression.
Facilities / Lab resources	—	No formal facilities-level AFI was required during the current cycle. Lab equipment is maintained through standard departmental and college budget processes; lab needs identified in instructor portal observations are addressed in normal operations.
Assessment-system	AS-01, AS-02, AS-03	PC→PI terminology realignment; ABET Portal deployment (Fall 2024); three-year PEO/SLO review cycle (2023–2025).

Table 36: EAC Table D-1, BSME row block (reproduced verbatim). FT = full-time; PT = part-time. Per Table D-1 Note 2, the AY 25–26 Bachelor’s count reflects Fall 2025 conferrals only. Per Note 3, 5th-year enrollment counts post-baccalaureate students.

Year	AY	Status	1st	2nd	3rd	4th	5th	Total UG	Total Grad	Assoc.	Bach.	Mast.	Doc.
Current	25–26	FT	274	182	212	261	7	936	25	0	50	11	0
Current	25–26	PT	84	56	60	127	3	330	26				
1	24–25	FT	277	154	180	277	8	896	44	0	123	28	0
1	24–25	PT	79	56	42	93	1	271	20				

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Year	AY	Status	1st	2nd	3rd	4th	5th	Total UG	Total Grad	Assoc.	Bach.	Mast.	Doc.
2	23–24	FT	142	144	146	257	5	694	32	0	153	16	0
2	23–24	PT	73	36	43	119	4	275	17				
3	22–23	FT	182	115	153	259	5	714	26	0	128	22	0
3	22–23	PT	50	33	43	104	1	231	26				
4	21–22	FT	139	117	162	268	6	692	35	0	115	30	0
4	21–22	PT	66	34	51	92	2	245	29				

Table 37: CECS engineering programs: Fall 2025 FT-UG enrollment, AY 21–22 baseline, cycle-over-cycle growth, and Fall 2025 share of CECS engineering FT-UG. BSME is the largest program and grew faster than the CECS engineering aggregate. Source: EAC Table D-1.

Program	F21 FT-UG	F25 FT-UG	Growth	F25 Share
Civil Engineering	483	597	+23.6%	25.1%
Computer Engineering	306	271	–11.4%	11.4%
Electrical Engineering	273	473	+73.3%	19.9%
Manufacturing Engineering	87	103	+18.4%	4.3%
Mechanical Engineering	692	936	+35.3%	39.3%
CECS engineering total	1,841	2,380	+29.3%	—

Table 38: BSME priority actions for the next assessment cycle (AY 2026–27 onward). Continuing actions (AS-04, AS-05, AS-06) carry forward AFIs SLO-01, SLO-02, and SLO-03 from the current-cycle catalog (Section 4.B.2.1). Each action has a documented owner; re-assessment timing is aligned with the faculty-retreat review cycle.

ID	Action	Target Start	Re-assessment	Owner	Source Evidence
AS-04	SLO-1 “Problem Analysis” Overlay re-assessment (Given–Find–Assumptions–Approach–Units framework, program-wide)	Sp 2026 deployed; first post-deployment data accruing	Sp 2027	SLO-1 Committee Chair; instructors of record	AFI SLO-01 deployed Sp 2026 (Spring-only trigger courses); first program-wide post-deployment cohort F 2026; closed-loop comparison Sp 2027 (Section 4.B.2.2.1)

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ID	Action	Target Start	Re-assessment	Owner	Source Evidence
AS-05	SLO-2 "Concept Development" Overlay re-assessment (MECE 4361 capstone)	F 2025 deployed; first post-deployment data in portal	Sp 2027	SLO-2 Committee Chair; MECE 4361 instructor	AFI SLO-02 in early re-assessment: F 2025 MECE 4361 shows all three PIs above threshold (PI-1 Create=83%, PI-2 Evaluate=73%, PI-3 Analyze=83%); PI-1 Create-level trajectory 65% Sp 2024 → 80% Sp 2025 → 83% F 2025 sustains above threshold at the Bloom level the Overlay targets (Section 4.B.2.2.2, 4.B.3.2)
AS-06	SLO-3 "Audience Adaptation" Overlay re-assessment (two-audience deliverables in lab and project courses)	F 2025 deployed; first post-deployment data in portal	Sp 2027	SLO-3 Committee Chair; instructors of record	AFI SLO-03 in early re-assessment: F 2025 shows MECE 4361 PI-2 Presentation recovery to 92%, PI-3 Technical Writing remaining at 60% (Section 4.B.2.2.3, 4.B.3.3)
NX-01	MECE 4362 (SD-II) course-level assessment review (examine SD-II artifacts and rubric application for the genuinely below-threshold PIs, most prominently SLO-7 PI-2; scope a course-level AFI where warranted)	F 2026	Sp 2027	SLO-2/3/5/7 Chairs; MECE 4362 instructor of record; ABET Coordinator	Sections 4.B.3.5, 4.B.3.7, 4.B.3.8
NX-01a	Portal reporting enhancements: cross-term comparison view, a dashboard tile summarizing course/SLO pairs flat or below threshold across multiple terms, and a source=backfill provenance flag for pre-Fall-2024 records; plus MECE 3380 narrative-field back-fill	F 2026	F 2026 (deployment); continuous thereafter	ABET Coordinator (portal maintenance)	Section 4.B.3.8

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ID	Action	Target Start	Re-assessment	Owner	Source Evidence
NX-02	MECE 1221 SLO-2 additional data collection: extend portal record beyond the single F 2024 data point (67%) by collecting two-to-three additional terms (Sp 2026, F 2026, Sp 2027) before deciding whether the F 2024 result reflects a persistent pattern or a single-term effect; assess against Concept Development Overlay diffusion to introductory-design courses	F 2026	Sp 2028	SLO-2 Committee Chair; MECE 1221 instructor of record	Section 4.B.3.2 (single F 2024 below-threshold data point at 67%)
NX-03	MECE 2140 SLO-6 coverage gap closure (collect additional MECE 2140 SLO-6 evidence in F 2026 and Sp 2027 to expand the single Sp 2022 data point; reassess whether course-level intervention is needed)	F 2026	F 2027	SLO-6 Committee Chair; MECE 2140 instructor of record	Section 4.B.3.6, 4.B.3.8 (single Sp 2022 data point at 62%)
NX-04	SLO-5 MECE 4361 PI-2 and PI-5 follow-up investigation (analyze Sp 2024 PI-2 dip to 40% and F 2025 PI-5 decline to 73% jointly; determine whether the signals share a common cohort/grading/rubric-application cause; review rubric calibration across instructors)	Sp 2027	Sp 2028	SLO-5 Committee Chair; MECE 4361 instructor of record	Section 4.B.3.5 (joint PI-level signals)
NX-05	Earlier-pipeline coverage expansion for SLO-7 (add evidence points in MECE 2140 and MECE 3170 to broaden the SLO-7 evidence base beyond capstone-only; SLO-5 coverage is no longer narrow given the MECE 3320 SLO-5 data discovered in this cycle's reconciliation)	Sp 2027	Sp 2028	SLO-7 Committee Chair; instructors of record in target courses	Section 4.B.3.7 (capstone-only concentration)

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ID	Action	Target Start	Re-assessment	Owner	Source Evidence
NX-06	Alumni survey instrument design and pilot deployment (develop a 1-, 3-, and 5-year post-graduation alumni survey instrument with PEO-aligned questions; pilot with the F 2025–Sp 2026 graduating cohort)	F 2026	Sp 2028	ABET Coordinator; Department Chair; EAC liaison	Section 4.B.4.2 (indirect-measure breadth gap)
NX-07	Employer satisfaction instrument design (develop a stand-alone employer satisfaction survey instrument distinct from EAC industry feedback; coordinate with EAC member companies for distribution)	Sp 2027	Sp 2028	ABET Coordinator; Department Chair; EAC liaison	Section 4.B.4.2 (indirect-measure breadth gap)
NX-08	SLO-4 current-issues-and-trends dimension monitoring (track the program-quality dimension flagged in the Capstone Exit Survey at ~86%, and identify PI-level signal sources within SLO-4 for next-cycle attention)	F 2026	Continuous through Sp 2029	SLO-4 Committee Chair; ABET Coordinator	Section 4.B.4.2 (Capstone Exit Survey program-quality dimension)

Table 39: Index of assessment instruments and supporting materials. “Portal” refers to the BSME ABET Assessment Portal (Section 4.0.4); “SLO Binder” refers to the physical outcome-level binders maintained by SLO chairs (Section 4.0.5); “Records Archive” refers to the department’s records archive maintained by the ABET Coordinator. All items are available for on-site review.

Category	Items	Location / Access Path
SLO Definitions and PI Maps	Current SLO statements (SLO-1 through SLO-7); current PI lists per SLO; prior-cycle PC→PI realignment mapping (AFI AS-01)	Section 4.A.1 of this chapter; Portal Admin view; archived in SLO Binders 1–7
Rubrics (current cycle)	Master E/P/A/N rubric (used across all SLOs); per-PI scoring criteria; rubric calibration notes	SLO Binders 1–7 (physical); Brightspace embedded rubrics (per-course); Portal upload module
Rubrics (prior cycle, for cross-cycle reference)	Pre-2020 PC-labeled rubrics; pre-PI relabeling documents; rubric-version change log (AFI AS-01 records)	SLO Binders 1–7 (archived section); Records Archive
Direct-Measure Records	Course–SLO–PI assessment entries (F 2020–Sp 2025); E/P/A/N distributions; EP% calculations; instructor narrative justifications; Bloom-level designations	Portal database (system of record); CSV export available via Admin module (Section 4.0.4.7); paper copies of per-course summaries in Course Binders

Category	Items	Location / Access Path
Direct-Measure Artifacts (scored student work)	Representative graded exemplars (Expert, Practitioner, Apprentice, Novice) for each SLO-assessing course; redacted to protect student identity	SLO Binders 1–7; Course Binders (per-course); selected samples available on-site display
Indirect-Measure Instruments	Capstone Exit Survey instrument (master form); per-term response data (F 2020–Sp 2025; $N = 504$); EAC subcommittee meeting agendas and notes	Portal Survey module (/survey/analyze); Records Archive; Section 4.B.4.2 reproduces key tables; Section 4.B.4.1 indexes EAC items
Statistical Reports	Per-SLO four-panel portal figures (used in Sections 4.A.6 and 4.B.3); per-course per-SLO portal panels; OLS trends ($\hat{\beta}_1$, p -value); mixed-effects course-level trends	Portal Analysis pages (/slo_analysis, /course_analysis); regenerable on demand from current portal records
Capstone Evaluation Materials	MECE 4361 (SD-I) and MECE 4362 (SD-II) ABET Stats spreadsheets; sponsor evaluation forms; mid-term poster grading rubrics; final-report grading rubrics; sample sponsor letters	Senior Design Coordinator's files; Spring 2025 SD-I and SD-II ABET Stats workbooks are part of the chapter's evidence base (referenced in Sections 4.B.3.2, 4.B.3.5, 4.B.3.7)
AFI Documentation	AFI Catalog (Section 4.B.2.1, thirteen actions); per-AFI Analysis–Intervention–Comparison (AIC) entries with owner, term, status; SLO Overlay rubrics deployed Fall 2025 (POL-01/POL-02 carry-forward policy text)	Portal Interventions log; Records Archive (faculty meeting/retreat minutes referenced); Section 4.B.2.2 reproduces the three SLO Overlay rubrics
Course Materials	ABET-format syllabi for all MECE courses; assignments, quizzes, projects supporting assessed artifacts; instructional materials for SLO-assessing courses	Course Binders (physical, key MECE courses); Brightspace course shells; Criterion 5 (curriculum) references
Governance Records	Faculty meeting agendas and minutes (cycle window); faculty retreat agendas and minutes; SLO Committee meeting notes; ABET Coordinator's working files	Records Archive (Department Chair's office); Section 4.C.2 below indexes specific items by date
Policy Documents (continuing enforcement)	BSME GPA Graduation Requirement (POL-01; F2015 catalog onward); BSME Two-Peat Rule (POL-02; prior-cycle adoption, continuous enforcement); Calculus I attempt-limit policy (AFI SS-01)	Department Catalog (UTRGV); Records Archive; Section 4.B.2.1 catalog rows reproduce policy text
Cross-Cycle Reference Data	Prior-cycle SSR Section 4.2 (August 2020) and supporting tables; prior-cycle pass-rate Table 4.1.36; prior-cycle GPA, retention, and graduation tables	Records Archive (digital and paper copies); Sections 4.B.4.5–4.B.4.9 reproduce key tables in range form for cross-cycle context

Table 40: Index of meeting minutes and action logs referenced in this chapter. Records are maintained in the department's Records Archive (Department Chair's office) and the ABET Coordinator's working files. Items are listed in approximate chronological order. PEVs may request the full minutes for any item during the visit.

Approximate Date	Meeting Type	Topic / Reference in This Chapter
AY 2020–21	Faculty meeting (approval)	PC→PI terminology realignment (AFI AS-01) following ABET Workshop attendance; rubric labels updated across SLO binders (Section 4.0.6.2)
AY 2021–22	Faculty retreat (review)	SLO Committee structure confirmed; outcome-level ownership and analysis cadence reviewed (Section 4.0.3)
AY 2021–22	Faculty meeting (approval)	Math sequence revision (AFI CR-01) proposed; MECE 2350 + MECE 3449 to be replaced by MECE 3440 + MATH 2415 in subsequent catalog (Section 4.B.2.1)
AY 2022–23	Faculty meeting (approval)	CSCI 1380 programming requirement added (AFI CR-02); math sequence revision (AFI CR-01) advances through university governance
Sept 2022	EAC meeting + subcommittee	Career-readiness feedback from EAC member (Lizcano, NASA Glenn Research Center, per Section 4.0.6.5); triggers AFI SS-02 (career-readiness content enhancement); see Section 4.B.4.1
AY 2022–23	Faculty meeting (approval)	Career-readiness content enhancement (AFI SS-02) integrated into co-curricular activities and senior design sequence
AY 2023–24	Faculty retreat (review)	Initial review of three-year PEO review process (AFI AS-03); SLO wording review begins
AY 2023–24	Faculty meeting (approval)	New technical electives proposed (AFI CR-03): MECE 4306 (CFD), MECE 4334 (Advanced Alloys), MECE 4335 (Smart Materials)
Spring 2024	ABET Portal development review	Portal architecture, data model, and analysis-engine design reviewed (Section 4.0.4); deployment scheduled for Fall 2024
Fall 2024	Faculty meeting (approval)	ABET Assessment Portal deployed (AFI AS-02); becomes system of record for direct-measure data (Section 4.0.4)
Dec 2024	EAC meeting + subcommittee	Discussion of Calculus I attempt-limit policy (AFI SS-01); new electives (CR-03) reviewed with industry input; see Section 4.B.4.1
AY 2024–25	Faculty meeting (approval)	Calculus I attempt-limit policy revised from 3 to 2 attempts (AFI SS-01); reviewed with EAC December 2024
Spring 2025	Faculty retreat (approval)	Three-year PEO/SLO review completed (AFI AS-03); PEO revisions adopted; SLO wording reviewed and retained
Aug/Sept 2025	Faculty meeting (approval)	Three SLO Overlay AFIs approved (SLO-01 Problem Analysis Overlay; SLO-02 Concept Development Overlay; SLO-03 Audience Adaptation Overlay); SLO-02 and SLO-03 deployed Fall 2025; SLO-01 deployed Spring 2026 (primary trigger courses Spring-only) (Section 4.B.2.2)
Dec 2025	Faculty meeting (review)	Status review of Fall 2025 SLO-02 and SLO-03 Overlay AFI deployment; first post-deployment portal data discussed; SLO-01 Spring 2026 deployment confirmed (Section 4.B.3)
May 2026	Faculty retreat (approval, scheduled)	Next-cycle AFI Catalog extensions (AS-04 through AS-06 and NX-01 through NX-08) scheduled for formal approval; see Section 4.B.5

Appendix item D.15: Process and timeline figures

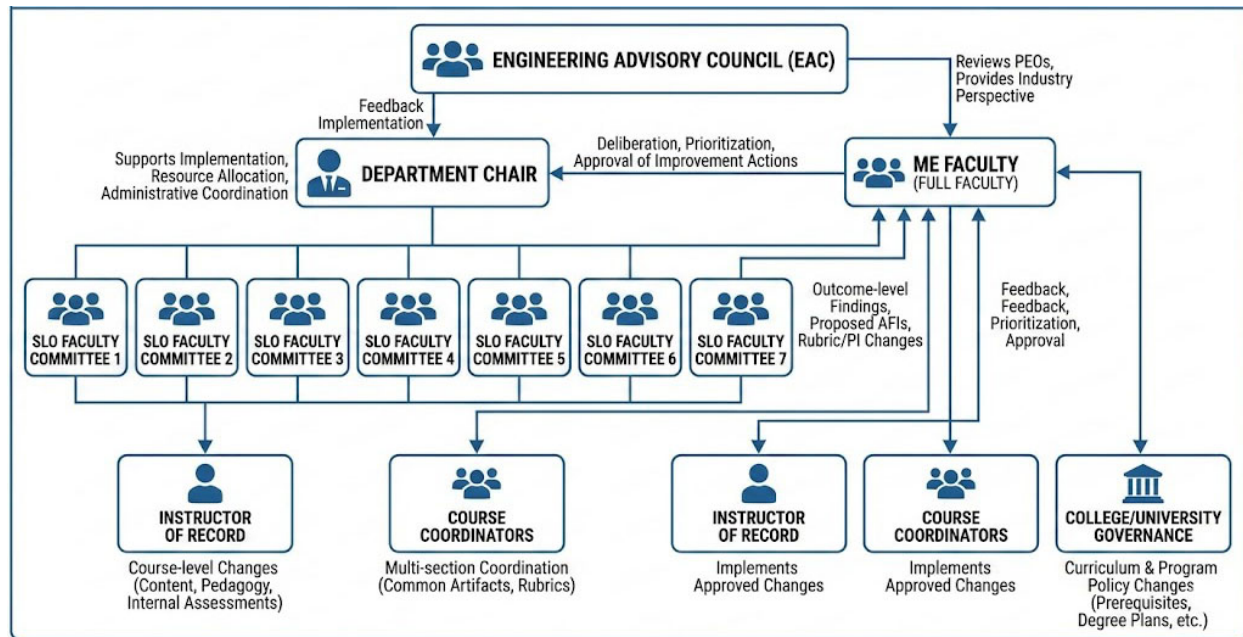


Figure 58: Governance structure and decision authority.

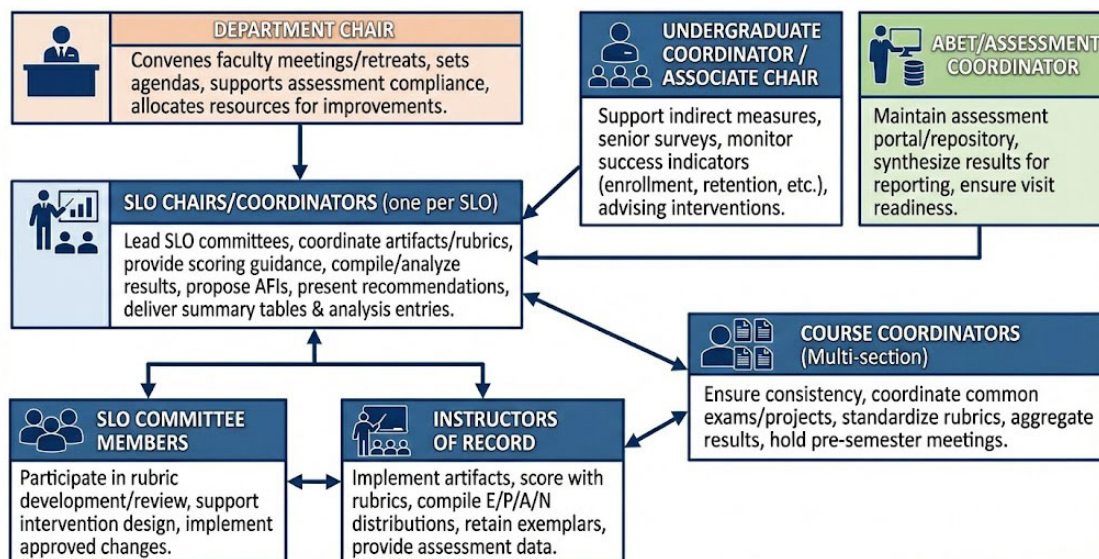


Figure 59: Roles and responsibilities in the continuous improvement system.

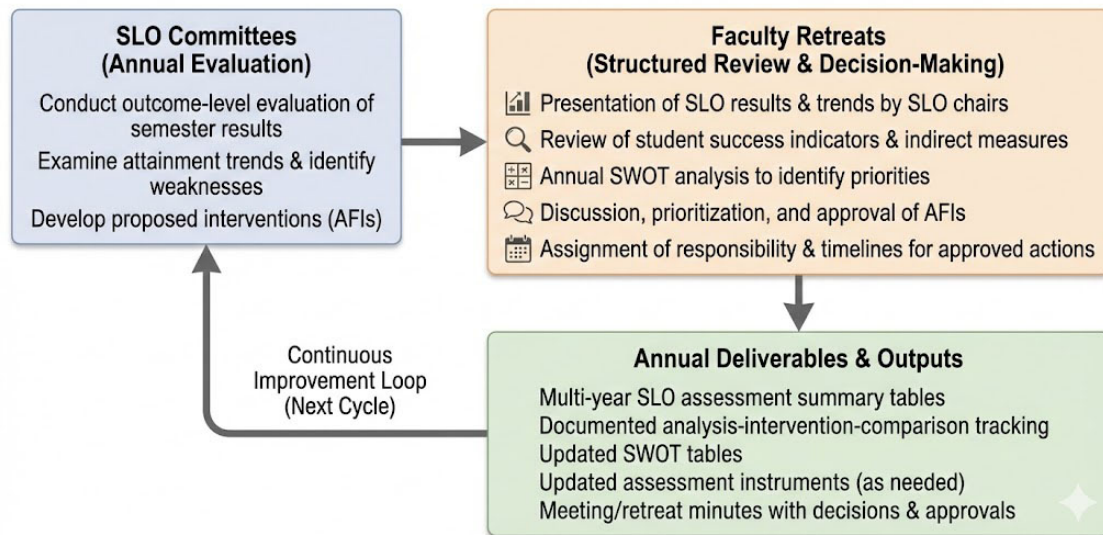


Figure 60: Annual cycle overview.

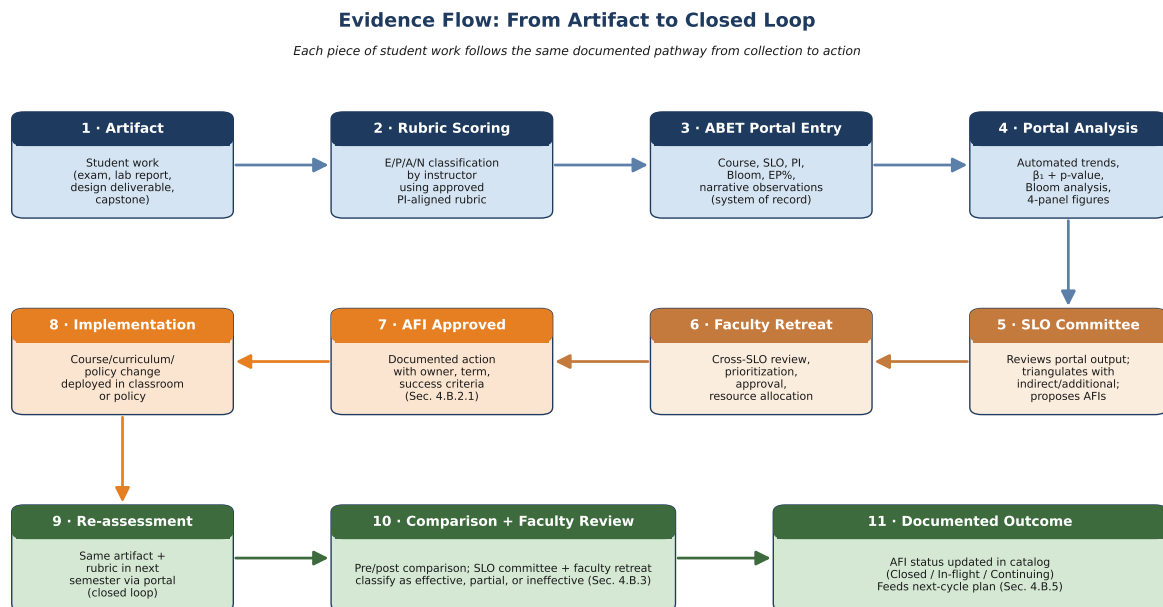


Figure: BSME evidence flow. Three layers (Collection, Evaluation, Re-assessment) form a single auditable pathway from individual artifacts to documented continuous-improvement actions.

Figure 61: BSME evidence flow from artifact to closed loop. Three layers (Collection in blue, Evaluation in orange, Re-assessment in green) form a single auditable pathway from individual student work to documented continuous-improvement actions. The portal serves as system of record for the Collection layer; the SLO committee and faculty retreat operate the Evaluation layer; the Re-assessment layer feeds back into the Collection layer for the next cycle.

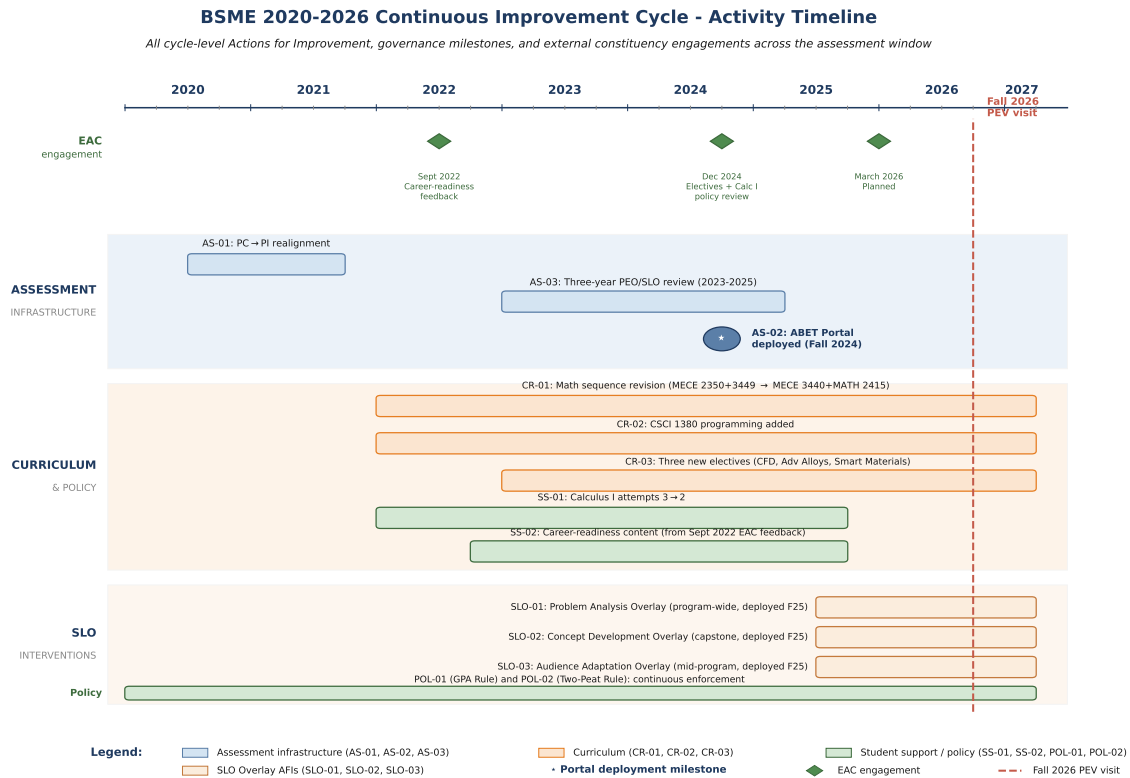


Figure 62: BSME 2020–2026 continuous improvement cycle activity timeline. Three swimlanes show: (i) Assessment Infrastructure including the PC→PI realignment (AS-01), the three-year PEO/SLO review (AS-03), and the Fall 2024 ABET Portal deployment milestone (AS-02, marked *); (ii) Curriculum and Policy including the math sequence revision (CR-01), CSCI 1380 addition (CR-02), three new electives (CR-03), Calculus I attempt-limit policy (SS-01), and career-readiness enhancement (SS-02); and (iii) SLO Interventions including the three Overlay AFIs (SLO-02/03 deployed Fall 2025; SLO-01 deployed Spring 2026 due to Spring-only trigger courses) and the continuous-enforcement policies (POL-01 GPA Rule, POL-02 Two-Peat Rule). Green diamonds mark documented EAC engagement events; the dashed red line marks the Fall 2026 PEV visit.

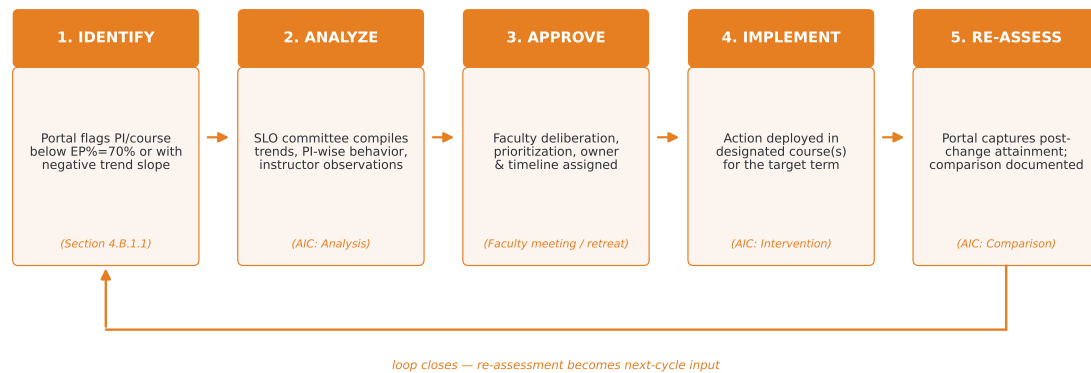


Figure 63: The BSME Analysis–Intervention–Comparison (AIC) workflow for Actions for Improvement. The portal supports steps 1, 2, and 5 directly; faculty governance owns steps 3 and 4.

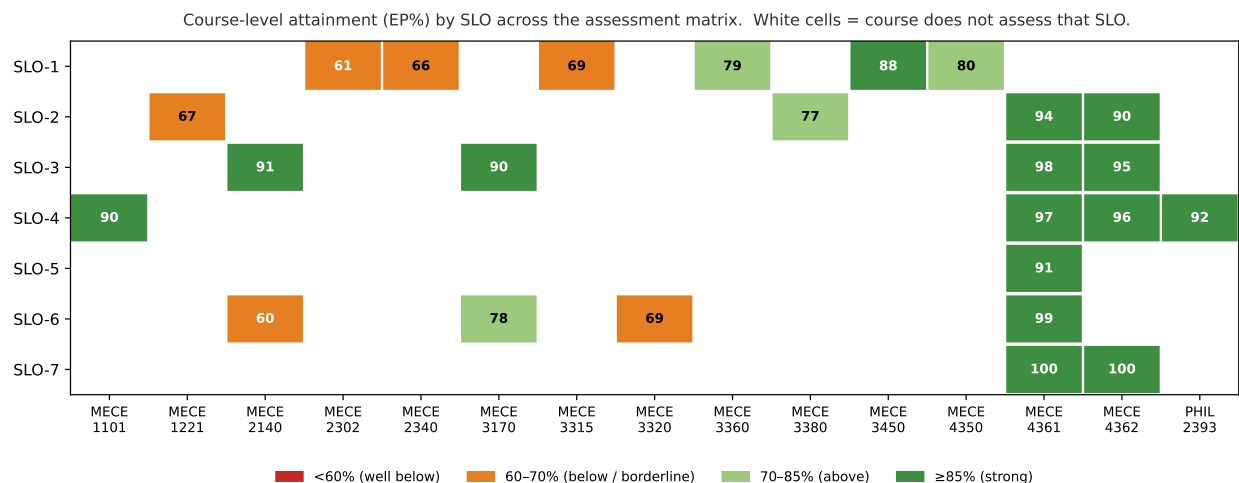


Figure 64: Course-level attainment (EP%) by SLO across the BSME assessment matrix. White cells indicate courses that do not formally assess that SLO. Below-threshold cells concentrate in SLO-1 lower/mid-level analytical courses (MECE 2302/2340/3315), the SLO-2 introductory design course (MECE 1221), and the SLO-6 measurements/lab courses (MECE 2140, MECE 3320). These bottlenecks motivated the three department-level interventions documented below.

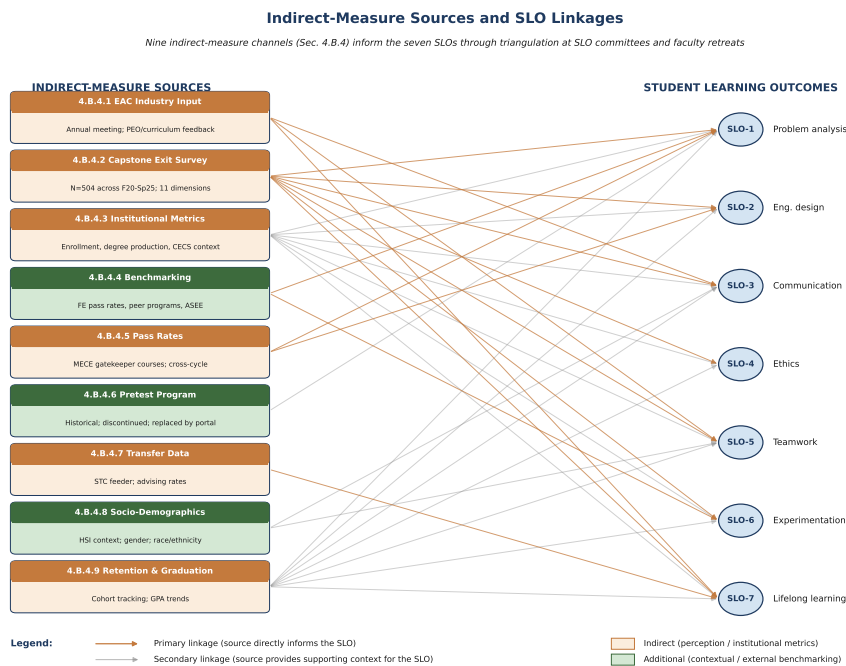


Figure: Indirect-measure source map. Capstone Exit Survey and Retention/Graduation streams inform all seven SLOs broadly; other channels have more targeted linkages.

Figure 65: Indirect-measure source map. The nine indirect-measure channels documented in Sections 4.B.4.1–4.B.4.9 are shown on the left, with primary linkages (solid orange arrows) and secondary linkages (lighter grey arrows) to the seven SLOs on the right. The Capstone Exit Survey and Retention/Graduation streams inform all seven SLOs broadly; other channels have more targeted linkages.

4.B.6 Cycle-Lessons Synthesis (2020–2026)

Section 4.B.6 synthesizes what the BSME program has learned across the 2020–2026 review cycle, drawn from the evidence presented in Sections 4.A.6, 4.B.3, and 4.B.4. Six cycle-level lessons emerge; each is tied to specific evidence in this chapter and to a documented next-cycle action where appropriate.

Lesson 1 – Infrastructure modernization pays off, but the controls layer must grow with the system. The cycle's most consequential assessment-system change was the Fall 2024 deployment of the ABET Assessment Portal (Section 4.0.6.3, AFI AS-02), which replaced spreadsheet-and-email workflows with a PI-grained, role-based, analysis-integrated system of record. The portal made the rest of this chapter possible: every program-level trend, per-course panel, and triangulation table in Section 4.A.6 and Section 4.B.3 is generated from portal records. The cycle also showed where the portal's launch configuration can be strengthened: the MECE 3380 narrative-completeness gap (placeholder explanation fields across 66 records; Appendix D.4) persisted because the portal's launch validation focused on quantitative structural integrity (every record has the right E/P/A/N fields) rather than on narrative-content completeness. The portal-reporting subtask (NX-01a) and a narrative back-fill address this explicitly. The general lesson is that the assessment infrastructure and the documentation and reporting layer on top of it must evolve together; the next cycle adds the reporting and back-fill work that the current cycle's experience identified as useful.

Lesson 2 – Proactive intervention before threshold breach is a viable operating mode. The three SLO Overlay AFIs (Problem Analysis, Concept Development, Audience Adaptation) were each approved while program-level attainment was at or above threshold; the faculty chose to act on documented within-SLO signals (MECE 2302's declining trajectory; the MECE 4361 SD-I Sp 2024 dip with partial recovery; the SLO-3 MECE 4361 Sp 2024 PI-2 and F 2025 PI-3 dips) rather than wait for program-level failure. The Overlays were sequenced to align with course offerings: SLO-2 and SLO-3 deployed Fall 2025 (their primary intervention sites, MECE 4361, offered in Fall and Spring); SLO-1 deployed Spring 2026 (its primary trigger courses, MECE 2302 and MECE 2340, offered Spring only). This operating mode is feasible because the portal surfaces PI-level signals at a level of detail that pre-portal workflows could not. The cycle-level evidence supporting it is the SLO-2 SD-I Sp 2025 informal recovery to 88% *before* the formal AFI was approved (Section 4.B.3.2), which corroborated that the underlying signal was addressable through normal instructor response and that the department-level AFI was reinforcing a real trajectory rather than imagining one. The first post-deployment term (F 2025) for SLO-2 provides confirming evidence: the MECE 4361 PI-1 Create-level trajectory 65% Sp 2024 → 80% Sp 2025 → 83% F 2025 sustains above threshold at the Bloom level the Overlay specifically targets. The lesson is that the $EP\% \geq 70\%$ threshold functions as a floor, not as the only trigger; the program's three-tier response framework (Section 4.B.2.3) operationalizes this distinction.

Lesson 3 – Diagnostic certainty, not signal severity, determines the right response. Three current-cycle signals show that the same signal magnitude can warrant three different responses depending

on what the data can tell the faculty: MECE 2302's persistent declining trend ($\hat{\beta}_1 = -11.03$ pp/term, $p = 0.025$) supported a formal AFI because the diagnosis was clear (lower-division analytical bottleneck) and the intervention was identifiable (Problem Analysis Overlay); MECE 3320's SLO-6 trajectory rose under sustained instructor-of-record action without any formal AFI ($\hat{\beta}_1 = +1.60$ pp/term, $p = 0.000$ at the course level) because the closed-loop case operated through normal system operation—and the same instructor's intervention drove a parallel statistically significant SLO-5 trajectory ($\hat{\beta}_1 = +5.83$ pp/term, $p = 0.000$); and the MECE 4361 SLO-5 PI-2 Sp 2024 dip to 40% and PI-5 F 2025 decline to 73% (with surrounding terms above threshold in both cases) were deferred to joint structured next-cycle investigation (NX-04) because the diagnosis admitted multiple competing interpretations (cohort effect, grading-calibration drift, or shared underlying cause). The lesson is that the program's response calibrates to what is known about the cause of a signal, not to the headline number; this is documented in the three-tier framework (Section 4.B.2.3) and is the substantive reason some signals produced AFIs while others did not.

Lesson 4 – Triangulation works; direct and indirect measures track the same underlying signals. The Capstone Exit Survey (Section 4.B.4.2, $N = 504$ across ten terms) provides graduating-senior perception across all seven SLOs and eleven program-quality dimensions. The cycle-level pattern is consistent with the direct-measure picture: all seven SLOs sustain agreement above 85% in every term with most terms above 95%; the Sp 2025 dip ($\sim 87\text{--}93\%$) coincides with the direct-measure dips that motivated the Fall 2025 SLO-2 and SLO-3 AFIs; the lowest-rated program-quality dimensions (written communication 88.6%, current issues and trends 86.0%) correspond to specific within-SLO concerns the direct measure already identified. The lesson is that the indirect measure does not contradict the direct measure on any SLO, and the two together provide stronger evidence than either alone. The next-cycle alumni and employer survey instruments (NX-06, NX-07) broaden the indirect-measure base for the next cycle.

Lesson 5 – Honest reporting is structurally easier when the data model retains everything. The genuinely below-threshold SD-II (MECE 4362) performance indicators and the MECE 3380 narrative-completeness gap are reported openly in Section 4.B.3.8. The reporting is structurally easier because the portal's data model retains every record at the PI grain: the program reports the SD-II below-threshold values as recorded rather than averaging them away, documents them term-by-term and SLO-by-SLO, and scopes next-cycle actions (NX-01, NX-01a) to review and act on the underlying results. The lesson is that the portal architecture (PI-grained data + narrative justification + auditable record retention) makes honest reporting compatible with continued program operation; the program does not need to mask weaker results to function and does not benefit from doing so.

Lesson 6 – The 2020–2026 cycle measurement framework held, deliberately. The SLO framework (seven outcomes aligned with ABET SO 1–7), the four-level E/P/A/N rubric, the $EP\% \geq 70\%$ threshold, the SLO-committee governance, and the Capstone Exit Survey instrument were all stable across the entire cycle (Section 4.0.6). The two substantive system-level changes (PC $\rightarrow$ PI terminology realignment, AFI AS-01; portal deployment, AFI AS-02) were deliberately non-disruptive to the underlying measurement framework. The result is that the multi-year attainment results in Section 4.A.6 and the per-SLO case studies in Section 4.B.3 can be interpreted as genuine longitudinal trends rather than artifacts of instrument change. The lesson is that comparability across an accreditation cycle is a deliberate engineering choice; the program made it and preserved it.

What the next cycle inherits. The eleven priority actions in Section 4.B.5 carry forward three things from the current cycle: (i) the closed-loop completion of the three SLO Overlay AFIs (AS-04, AS-05, AS-06) on the same direct-measure infrastructure that motivated them; (ii) the course-level-review and coverage actions (NX-01 through NX-05) that the current-cycle honest accounting flagged; and (iii) the indirect-measure breadth expansion (NX-06, NX-07) that the EAC and the Capstone Exit Survey both pointed toward. Each is scoped to existing departmental resources, sequenced against the May 2027 mid-cycle and May 2028 annual retreats, and grounded in specific current-cycle evidence cited in Section 4.B.5.2. The program enters the next cycle with the assessment-system improvements of the current cycle (portal, PI terminology, three-year PEO/SLO review process) in continuous operation and with a documented next-cycle work plan ready for formal approval at the May 2026 retreat.

Appendix item D.17: September 2024 Program SWOT Analysis Summary

This is the summary of the program-level SWOT analysis conducted at the September 2024 faculty retreat—the most recent SWOT in the program’s biennial (every-two-year) review cadence (Sections 4.0.3.4 and 4.B.1.2). It records the faculty’s strategic self-assessment across four dimensions and feeds the AFI prioritization process described in Section 4.B.1.2; the findings most relevant to student outcomes are linked to current-cycle actions there. No SWOT was conducted in September 2025; the next is scheduled for the September 2026 retreat.

Strengths. The program exhibits a robust institutional foundation characterized by a devoted, collaborative, and highly awarded faculty, whose youthful energy is complemented by an excellent support staff and effective, distributive leadership. This academic community maintains positive relationships with upper administration and local institutions like STC, which helps fuel impressive student population growth and a stellar reputation that drives high demand for its graduates. Cultivating a deeply caring environment, the program excels in student mentoring, hands-on learning via state-of-the-art instrumentation and strong lab rubrics, and a diverse curriculum that now includes specialized fields like CFD. Furthermore, the department stands out for its high productivity in technical and engineering education research, notably integrating a large number of undergraduate students into active research, design competitions, and academic co-authorship, while successfully providing pathways for Hispanic students into graduate programs and maintaining impactful K-16 outreach.

Weaknesses. To fully maximize its potential, the program is navigating several structural and operational considerations that stem from current resource allocations. Key areas for strategic investment include enhancing faculty compensation to align with market standards, increasing operational and teaching assistant (TA) funding, optimizing unrestricted resources and indirect cost recovery, and expanding support staff to manage growing administrative needs. The program is restricted by limited research and teaching space, outdated configured classrooms, aging computer labs and software, and limited maintenance funding for its equipment. On the student level, minimal enrollment controls have led to a high student-to-faculty ratio, placing an excessive service burden on faculty; this influx also presents academic challenges, including a significant population of students with low GPAs, poor incoming college preparedness, gaps in prerequisite knowledge, and issues surrounding student technical writing and academic integrity. Finally, despite its undergraduate growth, the graduate program remains hindered by the lack of course offerings in important specialized fields such as heat transfer.

Opportunities. The program stands at a pivotal juncture to capitalize on numerous strategic opportunities that align with institutional growth, particularly through cultivating collaborative external funding proposals across university colleges. A central focus is the elevation of academic standards by refining student recruitment, implementing coordinated exam schedules, and increasing student engagement, while simultaneously establishing robust mechanisms to help at-risk students succeed or smoothly transition into alternative pathways. Finally, by clarifying the department’s long-term vision, the program can substantially enhance its relationship with upper administration, increase active interactions with the Engineering Advisory Council (EAC) and capital campaigns, and deepen alumni engagement to foster a thriving, well-supported engineering ecosystem.

Threats. The program faces a complex array of interconnected risks, primarily driven by rapid

student growth that has led to oversized classes, laboratory security vulnerabilities, and a bottleneck in the math/science sequence. This rapid expansion, combined with a high level of faculty research activity, high service loads and uncompetitive faculty/staff salaries, severely threatens faculty retention and risks straining the department's historically collegial environment. On the student side, academic quality is challenged by rising issues with academic integrity, a perceived sense of entitlement, low engagement, and a tendency for students to deviate from prescribed advising tracks, all of which are compounded by, for example, inefficient student hiring processes. These operational strains are further amplified by insufficient staff support and by broader institutional challenges.